

Proposal for Agent/Banca Portal & Back-office Solution

SalesVerse 2.0, configured for IIFT as the IIFT Agent/Banca Portal & Back-office

Technical and commercial proposal in response to IIFT's Request for Proposal



PREPARED FOR

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SUBMITTED BY

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Commercial-in-Confidence. This proposal contains confidential commercial information of iorta TechNXT and is submitted solely for the evaluation of the above RFP by Insurans Islam Family Takaful Sendirian Berhad. It may not be disclosed to third parties without prior written consent.

Cover Letter

[Day] October 2026

Our ref: [Proposal reference no.]

Your ref: [IIFT RFP / tender reference no.]

The Management

Insurans Islam Family Takaful Sendirian Berhad

[Street address]

Bandar Seri Begawan [Postcode]

Brunei Darussalam

Dear Sir/Madam,

Re: Proposal for Agent/Banca Portal & Back-office Solution

iorta TechNXT is pleased to submit its proposal for the Agent/Banca Portal and Back-office Solution. We offer to configure, integrate, test, deploy and support the solution, with training, documentation, a six-month warranty and five years of maintenance and support.

The solution is SalesVerse 2.0, iorta TechNXT's distribution platform for Takaful and insurance, configured for IIFT as the IIFT Agent/Banca Portal & Back-office. A working version already runs the seven products in Appendix 3 of the RFP, the B\$150,000 referral rule, the seven-day grace-period agency block, maker-checker approvals, AML watch-list screening, e-signature, e-Policy and e-Receipt issuance and the end-of-day FIN interface file. The remaining work (integration with IIFT's systems, IIFT's rates and templates, testing, migration and training) is set out in this proposal at a fixed fee. We offer two commercial models and one add-on:

- **Option A, on-premise perpetual licence (recommended):** one-time B\$ 190,000 (licence B\$ 60,000; implementation services B\$ 130,000); annual maintenance B\$ 33,990 in Year 1, being 22% of licence and customisation, +5% a year.
- **Option B, subscription hosted and managed by iorta on cloud:** one-time B\$ 145,000 (implementation services B\$ 130,000; cloud set-up B\$ 15,000); subscription B\$ 3,400 a month in Year 1, +5% a year, minimum term 60 months; managed services B\$ 1,100 a month; cloud infrastructure recharged at cost.
- **Option C, source code handover:** the core platform source code after go-live and hypercare, with an 8-week knowledge transfer and transition plan, for B\$ 88,000.
- Fees are inclusive of withholding tax. Out-of-pocket expenses (estimated at B\$ 31,440) and cloud infrastructure are recharged at cost as disbursements, and third-party and infrastructure charges, listed in the Bill of Materials, are excluded.
- Go-live in **week 24**, then four weeks of hypercare.

The proposal contains the items requested in your invitation, in the order requested:

Item	Content	Where
(a)	Cover letter	This letter
(b)	Corporate profile and relevant experience	Enclosure B, attached separately
(c)	Business registration certificate	Enclosure C, attached separately
(d)	Detailed scope of work and methodology	Sections 5, 6 and 15
(e)	Detailed fee and pricing structure, stating WHT, OPE and third-party charges	Section 17 and the pricing workbook
(f)	Delivery and implementation timeline with key milestones	Section 16
(g)	Quotation validity period	Section 22: 90 days
(h)	Other supporting information	Sections 7, 14 and 23; technical documents
(i)	Terms and conditions	Section 24

We confirm that this proposal follows the RFP and would welcome the opportunity to demonstrate the working application. For clarification, please contact [Name of contact person], [Designation], at [e-mail address] or [telephone number].

Yours faithfully,

[Name of authorised signatory]

[Designation]

For and on behalf of **iorta TechNXT**

[Company stamp]

Document Control

Version history

Version	Date	Author	Description
0.9	[Date]	iorta TechNXT	Internal review draft
1.0	October 2026	iorta TechNXT	Submission version

Review and approval

Role	Name	Signature / date
Prepared by (Bid Manager)	[Name]	
Reviewed by (Solution Architect)	[Name]	
Approved by (Authorised Signatory)	[Name, designation]	

Distribution

Recipient	Organisation	Format
Tender / Evaluation Committee	Insurans Islam Family Takaful Sendirian Berhad	Signed PDF and editable DOCX; pricing workbook and Bill of Materials (XLSX)
Bid file	iorta TechNXT	Archive copy

Companion documents

- IIFT-Commercial-Pricing-iorta-TechNXT.xlsx: the pricing workbook (options summary, fees in IIFT's format, Options A, B and C, RFP section 10 breakdown, OPE, rate card, payment schedules and assumptions).
- IIFT-Bill-of-Materials-iorta-TechNXT.xlsx: the Bill of Materials (on-premise infrastructure, cloud services, software licences and third-party services, with indicative costs and who pays).
- Enclosure B: corporate profile and relevant experience (attached separately).
- Enclosure C: business registration certificate (attached separately).
- Technical document pack in docs/technical: solution architecture, data dictionary, code standards and quality report, security assessment report, production support handover.

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Response Checklist

The invitation asked for items (a) to (i). The checklist shows where each is answered and whether it is in this document or attached separately.

Item	Invitation requirement	Where addressed	Status
(a)	Cover letter	Cover Letter, after the cover page	Included
(b)	Corporate profile and relevant experience	Enclosure B (Section 3)	To be attached by iorta
(c)	Business registration certificate	Enclosure C (Section 4)	To be attached by iorta
(d)	Detailed scope of work and methodology	Sections 5 (scope), 6 (fitment), 15 (methodology)	Included
(e)	Detailed fees and pricing structure, stating WHT, OPE and third-party charges	Section 17; pricing workbook; Bill of Materials workbook; Annex C	Included
(f)	Delivery and implementation timeline with key milestones	Section 16	Included
(g)	Quotation validity period	Section 22: 90 days	Included
(h)	Other supporting information	Sections 7, 14 and 23; technical document pack	Included
(i)	Terms and conditions	Section 24	Included
Additional chapters			
	Personas, screens, journeys and navigation	Section 7 (journeys in 7.14, screen index in 7.15)	
	Functional rules and products	Sections 8 and 9	
	Architecture, security, deployment, infrastructure	Sections 10 to 13	
	Technical document pack	Section 14	
	Five-year maintenance and support, SLA	Section 18	
	Team, assumptions and risks	Sections 19 to 21	
	Compliance and fitment matrix (every RFP requirement ID)	Annex A	
	Bill of Materials summary, software components, network flows	Annex C	

Abbreviations

Takaful and technical terms are explained in the glossary in Annex D.

Abbreviation	Meaning
AD / LDAP	Active Directory / Lightweight Directory Access Protocol, used for single sign-on
AMBD	Autoriti Monetari Brunei Darussalam, the central bank and financial regulator of Brunei Darussalam
AMC	Annual maintenance charge
AML / KYC	Anti-Money Laundering / Know Your Customer screening and due diligence
API	Application Programming Interface
ASVS	OWASP Application Security Verification Standard
BRR	The issuance register / report referenced in RFP Appendix 3 against which daily issuance and FIN postings are tallied
BOM / SBOM	Bill of materials / software bill of materials (open-source components and licences)
BRS / FRS	Business Requirement Specification / Functional Requirement Specification
B\$ / BND	Brunei Dollar
CI/CD	Continuous Integration / Continuous Delivery
CR	Change Request
DR	Disaster Recovery
EOD	End-of-Day processing
FIN	IIFT financial / accounting system referenced in the RFP
FFR	Functional Flow and Requirements (RFP Appendix 3)
HA	High Availability
IC	Identity Card (Brunei national identity card)
IIFT	Insurans Islam Family Takaful Sendirian Berhad
IIGT	Insurans Islam General Takaful, the sister company that operates the IIGT Agent Portal
IITH	Insurans Islam TAIB Holding, the group holding company
KT	Knowledge transfer
L1 / L2 / L3	First-, second- and third-line support
MFA / OTP / TOTP	Multi-Factor Authentication / One-Time Password / Time-based One-Time Password
OPE	Out-of-Pocket Expenses (travel, accommodation, per diem, insurance, transfers, visas)
OWASP	Open Worldwide Application Security Project
PDPO	Personal Data Protection Order 2025 (Brunei Darussalam)
PITR	Point-in-Time Recovery
PMO	Project Management Office
QC	Quality Check
RACI	Responsible, Accountable, Consulted, Informed
RBAC	Role-Based Access Control
RPO / RTO	Recovery Point Objective (maximum data loss) / Recovery Time Objective (maximum downtime)
RTM	Requirements Traceability Matrix
SAD / TDD	Solution Architecture Document / Technical Design Document
SAST / SCA	Static Application Security Testing / Software Composition Analysis
SIT / UAT	System Integration Testing / User Acceptance Testing
SLA	Service Level Agreement
SSO	Single Sign-On
TCO	Total Cost of Ownership
VAPT	Vulnerability Assessment and Penetration Testing
WAF	Web Application Firewall
WHT	Withholding Tax

1. Executive Summary

IIFT will receive SalesVerse 2.0, configured as the IIFT Agent/Banca Portal & Back-office, live in 24 weeks at a fixed implementation price. Banca submissions, approvals, payments and reporting move from paper and e-mail to one controlled system. We recommend Option A, an on-premise perpetual licence in the IITH data centre, which keeps data in Brunei and has the lowest five-year cost.

Solution	SalesVerse 2.0, configured as the IIFT Agent/Banca Portal & Back-office: a portal for agents and bank officers and a back-office for IIFT and IITH staff.
Scope	All 29 in-scope components of RFP section 3.1, the seven Appendix 3 products, and integration with IIFT's core system, FIN, AML provider, Active Directory, SMS gateway and e-mail.
Timeline	Go-live in week 24. Hypercare in weeks 25 to 28. Warranty for six months from go-live.
Option A (recommended)	On-premise perpetual licence: B\$ 190,000 (USD 146,154) one-time; AMC B\$ 33,990 in Year 1 (22% of licence and customisation), +5% a year.
Option B	Subscription on iorta-managed cloud: B\$ 145,000 (USD 111,538) one-time (including cloud set-up B\$ 15,000); B\$ 3,400 a month in Year 1, +5% a year, minimum 60 months; managed services B\$ 1,100 a month; cloud recharged at cost.
Option C (add-on)	Core source code handover with 8 weeks of knowledge transfer after hypercare: B\$ 88,000.
Five-year cost	Option A B\$ 409,256 (USD 314,812); Option B B\$ 556,144 (USD 427,803), including estimated OPE and cloud charges.
Fee basis	Inclusive of WHT. OPE and cloud infrastructure recharged at cost as disbursements. Third-party and infrastructure charges excluded (Bill of Materials).
Licence and users	Unlimited named users for IIFT under both options; no per-user or per-policy fees.
Source code	IIFT-specific source code at go-live and every release under every option; core platform source through Option C or escrow.
Compliance	234 of 234 RFP requirement IDs addressed (Annex A); fitment in Section 6.

Table 1: What IIFT receives

1.1 Why iorta TechNXT

The platform already works. IIFT's seven products, the B\$150,000 referral, the seven-day agency block, maker-checker with amount thresholds, watch-list screening, e-signature, e-Policy and e-Receipt PDFs, the EOD report and FIN interface file all run today. Discovery workshops start from working screens, which is why requirements and design take five weeks.

The price follows IIFT's scale. The RFP describes 26 named users and about 600 policies a year. There is no per-user or per-policy fee, so more bank branches or agents do not change the price.

Controls sit in the data layer. The API enforces maker-checker, a database trigger makes the audit table append-only, and IC and passport numbers are encrypted field by field with AES-256-GCM. These controls cannot be bypassed from a screen.

IIFT keeps its options open. It can own a perpetual licence or subscribe, convert from subscription to licence, and take the platform source code with a structured handover once the solution is stable.

Delivery risk is shared. Implementation is a fixed price, 85% of the services fee is paid against signed deliverables, and yearly increases are capped at 5%.

1.2 About iorta TechNXT and SalesVerse 2.0

iorta TechNXT builds and supports software for insurance and Takaful distribution. SalesVerse 2.0 is its platform for agency and bancassurance portals, back-office administration and the integrations that connect

them to core, finance and compliance systems. The corporate profile and relevant experience are in Enclosure B.

2. Understanding of IIFT's Requirements

2.1 Background

IIFT sells Mortgage Takaful and Annual (Individual) Family Takaful plans through agents and bank partners. Banca work today relies on paper or e-mailed submissions and manual follow-up with the back-office. Status is hard to track and management has little real-time visibility.

The IIGT Agent Portal went live on 17 July 2026 for Motor Takaful. It covers agent onboarding, e-quotation, e-Cover Notes, AML screening, documents, e-signature, payment proof verification, audit and reporting. IIFT needs the same kind of capability for Family Takaful products and the Banca channel, plus a back-office in which IIFT staff review, approve and report on every Banca transaction.

2.2 Project objectives and our response

#	RFP objective	How the solution meets it
1	Digitalise Agent/Banca operations	Agent/Banca Portal for onboarding, quotation, issuance, servicing, payments, claims notification and issues.
2	Improve operational efficiency	Data captured once at source; contribution calculated by the system; e-Policy, e-Receipt and EOD files produced automatically.
3	Strengthen Agent/Banca management	Agency and bank hierarchy, main/sub-agent links, five agent statuses and document expiry tracking.
4	Strengthen governance and controls	Role-based access with data scoping, maker-checker with configurable levels and thresholds, append-only audit trail.
5	AML/KYC management	Watch-list screening with fuzzy name matching at onboarding and on change; Compliance review queue.
6	Improve data quality	Field validation, controlled master data, duplicate checks on IC and registration numbers, daily reconciliation.
7	Improve integration	Transactional outbox with retry and dead-letter; adapters for Core, FIN, AML, Active Directory, SMS and e-mail.
8	Improve reporting	Dashboards and standard reports with Excel, CSV and PDF export and scheduled distribution.
9	Improve user experience	Step-by-step quotation wizards, one design system themed to Insurans Islam TAIB colours, WCAG 2.1 AA target.
10	Strengthen security	Controls aligned with OWASP ASVS Level 2 and an independent VAPT before go-live.
11	Ensure scalability	Sized for 100 concurrent and 500 named users, more than ten times the projected policy volume.
12	Long-term sustainability	Five years of support with yearly increases capped at 5%, maintained documentation, IIFT-specific source code with every release and an option to take the platform source code.

2.3 Users and volumes

Appendix 1 lists 26 named users and Appendix 2 projects 628 policies a year within five years. These volumes are small for a relational database and two application servers, so we size for 100 concurrent and 500 named users and ten times the projected volume (NFR-02, NFR-03, NFR-06, NFR-07).

Company	Application	Expected users
Bank	Portal	12
IIFT	Back-office (Admin)	5
IIFT	Portal	5
Agent	Portal	4

Company	Application	Expected users
Total		26

Table 2: Expected users (RFP Appendix 1)

Line of business	Product	2025	2026 (to July)	Projected in 5 years
Mortgage Takaful	Financing Takaful Plan – Hire Purchaser	59	16	75
Mortgage Takaful	Financing Takaful Plan – Non-Participating	158	65	205
Mortgage Takaful	Property Financing Takaful Plan	88	60	115
Annual (Individual)	Personal Home Assistant Takaful Plan	102	41	132
Annual (Individual)	Professional Takaful Plan	40	19	52
Annual (Individual)	Khairat Takaful Plan	31	16	40
Annual (Individual)	Overseas Student Assist Takaful Plan	7	2	9
Total		485	219	628

Table 3: Policy volumes (RFP Appendix 2)

2.4 Business rules that shape the design

- Financing above **B\$150,000** is the high-risk limit. The agent or banker cannot complete it; the case is referred to IIFT Sales (FFR01, AP-60).
- A **seven-day payment grace period** applies to issued policies. If payment is not submitted in time, every agent in that agency is blocked from issuing until it is settled (AP-42).
- **Maker-checker** covers agent registration and profile changes, participant updates, endorsements, cancellations, payment verification and AML review (BO-10, BO-17).
- **End-of-day processing** produces the daily issuance report and FIN entries, and issuance is reconciled with BRR for every product (FFR01 to FFR05).
- Product eligibility differs by plan: Professional is limited to Occupational Class I, Khairat has no renewal, and Overseas Student Assist requires a Brunei citizen registered as a student and aged 65 or under.

3. Enclosure B – Corporate Profile and Relevant Experience

Attached separately. iorta TechNXT's latest corporate profile and relevant experience (invitation item b) are submitted as a separate document with this proposal.

Document	[File name of the corporate profile]
Contents	Company overview, leadership, delivery capability and relevant reference projects
Contact	[Name, designation, e-mail, telephone]

Table 4: Enclosure B

4. Enclosure C – Business Registration Certificate

Attached separately. A certified copy of iorta TechNXT's business registration certificate (invitation item c) is submitted with this proposal.

Document	[File name of the certificate]
Business registration no.	[Business registration no.]

Table 5: Enclosure C

5. Scope of Work

We will configure, integrate, test, deploy and support SalesVerse 2.0 as the IIFT Agent/Banca Portal & Back-office, with training, documentation, a six-month warranty and five years of maintenance. This chapter follows the RFP structure. For each item it states what we deliver, how, and the evidence IIFT uses to accept it. Fitment against the working platform is in Section 6; the per-ID matrix is in Annex A.

5.1 In-scope components (RFP 3.1)

#	Component	What we deliver	How	Acceptance evidence
1	Agent/Banca Portal	Web portal (/portal) for agents, main/sub-agents and bank officers	Configure the SalesVerse 2.0 portal for IIFT roles, products and branding	UAT sign-off of portal scenarios
2	Back-office (Admin)	Internal application (/backoffice) for IIFT and IITH staff	Configure back-office roles, queues and dashboards; publish only on the internal network	UAT sign-off; network test showing no internet exposure
3	Agent/Banca management	Registration, profile, status, agent code, hierarchy	Load agent master data; set numbering pattern and status rules	SIT cases; migrated agent reconciliation
4	Agency/Banca management	Agencies, banks, branches and their relationships	Load agency and bank structure; map hierarchy to data scopes	Hierarchy review signed by Banca/Sales
5	Agent/Banca onboarding	Online application, KYC data, documents, AML, approval	Configure document checklist and approval levels	Journey 1 executed in UAT
6	Participant management	Shared participant record, search, updates with approval, AML	Set search criteria, duplicate rules and update workflow	SIT and UAT cases for AP-11 to AP-16
7	Policy management	Seven products, quotation, drafts, issuance, statuses, servicing	Load IIFT rates, questionnaires and templates for all seven products	Rate verification sheet signed by IIFT actuarial; UAT
8	Billing management	Payment dashboard, single and bulk payment, verification, e-Receipt, 7-day block	Configure payment rules, thresholds and receipt numbering	Journey 4 in UAT; grace-period test evidence
9	Claims management	Claim notification with policy validation and documents	Configure claim types and document lists	UAT cases for AP-43
10	AML/KYC	Watch-list screening with fuzzy matching, Compliance review	Connect to IIFT's AML provider or lists; set match threshold	Screening test pack agreed with Compliance
11	Workflow management	Configurable workflows for all approval transactions	Load IIFT approval matrix as workflow definitions	Approval matrix sign-off; SIT cases
12	Maker-checker	Segregation of duties on every configured transaction	Assign maker and checker permissions per role	Negative tests: maker cannot approve own request
13	Document management	Encrypted document store; e-Policy, schedule and e-Receipt PDFs	Load IIFT templates and wording; deploy ClamAV in each environment	Template sign-off; upload security tests
14	Dashboard	Agent, bank, back-office and management dashboards	Agree KPI definitions and targets	Dashboard review in UAT
15	Reporting	Report catalogue with filters, schedules, Excel/CSV/PDF export	Configure the 12 existing reports; build the remaining catalogue items	Report sign-off against sample data
16	Issue management	Issue reporting, assignment, priorities, SLA timers	Set categories, assignment rules and SLA targets	Journey 7 in UAT
17	Notification	E-mail, SMS and in-portal notifications	Connect SMTP and SMS gateway; agree templates	Notification test log
18	Audit trail	Append-only audit with search and export	Set retention; optional SIEM export	Audit tamper test (UPDATE/DELETE rejected)
19	Administration	Users, roles, permissions, master data, parameters	Load IIFT roles and parameter values	Role matrix sign-off
20	Integration	Core, FIN, AML, AD, SMS, e-mail adapters with monitoring and reconciliation	Build adapters against agreed interface specifications	End-to-end interface tests with each system owner
21	Security	Controls in Section 11; MFA; independent VAPT	Apply IITH policy values; remediate VAPT findings	VAPT and re-test report (DEL-17)
22	Data migration	Agent, agency, bank, participant and reference data	Scripted extract, cleanse, load, reconcile; two rehearsals	Signed reconciliation report
23	Testing	Unit, API, SIT, regression, performance, security; UAT support	Automated suites in CI; SIT and UAT plans	SIT and UAT results (DEL-14, DEL-16)

#	Component	What we deliver	How	Acceptance evidence
24	Deployment	DEV, SIT, UAT, PROD and DR environments; CI/CD; cut-over	Container deployment on IIFT infrastructure (Option A) or IORTA-managed cloud (Option B)	Deployment plan and go-live checklist signed
25	Training	Agents, bank officers, back-office, administrators, IT	Train-the-trainer, hands-on sessions, manuals	Attendance records and feedback
26	Documentation	User, administrator, technical and operations manuals	Update the SalesVerse 2.0 document pack for IIFT	Document acceptance by IIFT
27	Warranty	Six months of defect correction from go-live	Fixes under the SLA at no charge	Monthly service report
28	Maintenance	Five years of support, patches and upgrades	Service desk, monthly patch cycle, releases	Monthly service report; quarterly review
29	Transition	Knowledge transfer, handover and exit support	Year 5 exit plan and shadow support	Signed transition acceptance

Table 6: In-scope components 1 to 29

5.2 Agent/Banca Portal functions (RFP 4.1)

RFP area and IDs	What we deliver	How	Acceptance evidence
Authentication (AP-01–04)	Secure login, password change and reset, session control, lockout	Server-side sessions; policy values set to IITH standards	Security test cases; VAPT
Agent/Banca profile (AP-05–06)	Profile view and change requests with approval	Profile change workflow with before and after values	UAT case; audit record of approved change
Agent/Banca management (AP-07–12)	Online registration, agent code, hierarchy, agency information, shared participants	Registration journey; data scoping by agency and bank	Journey 1 in UAT
Participant management (AP-13–16)	Individual and corporate registration, search, approved updates, AML screening	Duplicate checks on encrypted IC numbers via blind index	SIT cases; AML test pack
Quotation (AP-17–20)	Authorised product list, product wizards, automatic contribution, drafts	Seven configured products with IIFT rate tables	Rate verification sheet; UAT
Issuance and status (AP-21–25, AP-32–36)	Policy list, search, details, issuance, validation, routing, restrictions, notifications, audit	Server-side validation and workflow routing	Journeys 2 and 3 in UAT
Policy servicing (AP-26–31)	Renewal, endorsement, cancellation, documents, payment status, history	Servicing requests through maker-checker	Journey 5 in UAT
Billing (AP-37–42)	Payment dashboard, bulk payment per policy, proof upload, verification, e-Receipt, 7-day block	Billing module and nightly grace-period job	Journey 4; grace-period test evidence
Claims (AP-43)	Claim notification with policy validation and documents	Claims module	Journey 6 in UAT
Documents (AP-44–47)	Policy schedule, e-Policy and e-Receipt PDFs; download, e-mail, upload, status and expiry	IIFT templates; encrypted store; ClamAV scan	Template sign-off; upload tests
AML information (AP-48)	AML/KYC data captured at onboarding	Configurable KYC fields	Compliance review of form
Workflow (AP-49–51)	Save, submit, track; status list; rejection remarks and resubmission	My requests screen over the workflow engine	Journey 9 in UAT
Dashboard (AP-52–53)	Agent dashboard and pending actions	Role-based dashboard widgets	UAT review
Notifications (AP-54)	Alerts on status changes, approvals, rejections, actions	Notification templates per event	Notification test log
Issue management (AP-55–57)	Report issues with reference number, tracking and attachments	Issues module with SLA timers	Journey 7 in UAT
Reporting (AP-58)	Role-scoped reports for agency, agent and bank	Report catalogue with data scoping	Report sign-off
Security and approvals (AP-59–61)	Role-based access, authority limits, approval history	Permissions and data scopes; amount thresholds	Role matrix sign-off; negative tests
Digital signature (AP-62)	On-screen signature, single-use e-mail link, upload of signed documents	e-Signature module with evidence record	UAT case; evidence sample reviewed by IIFT Legal

5.3 Back-office functions (RFP 4.2)

RFP area and IDs	What we deliver	How	Acceptance evidence
Authentication and access (BO-01–02)	Internal login, MFA, Active Directory sign-on, role-based access	MFA and AD adapter built during implementation	Security tests; VAPT
User management (BO-03–04)	User administration, roles and permissions	Role builder; IIFT role matrix loaded	Role matrix sign-off
Agent management (BO-05–08, BO-10)	Agent records, search, status lifecycle, hierarchy, approval of registrations and changes	Agents module with maker-checker	SIT cases; Journey 1
Agency management (BO-09)	Agency and bank records with linked agents	Agencies and banks screens	Hierarchy review
Documents (BO-11–12)	Validate, approve, reject documents; completeness and expiry	Document checks queue	UAT case
AML/KYC (BO-13–15)	Screening, results with dates, Compliance review	AML cases and watch-list screens	AML test pack
Workflow (BO-16–19)	Configurable approvals, maker-checker, status tracking, rejection remarks	Workflow definitions for the IIFT matrix	Journey 9; negative tests
Dashboards (BO-20–21)	Management KPIs and pending actions	Back-office dashboard	UAT review
Reporting (BO-22–25)	Standard reports, filters, Excel/CSV/PDF export, schedules	12 existing reports plus the agreed catalogue	Report sign-off
Audit (BO-26–28)	Audit of activity and changes with before and after values; search	Append-only audit table protected by trigger	Tamper test; audit search demo
Issue management (BO-29–31)	Assignment, priorities, SLA monitoring	Issue rules and SLA targets	Journey 7
Administration (BO-32–33)	Master data and configurable rules, statuses, levels, parameters	Parameters and master data screens	Configuration workbook signed

5.4 Integration (RFP 4.3)

RFP area and IDs	What we deliver	How	Acceptance evidence
Core system (INT-01–03)	Agent, agency and bank master synchronisation; agent information exchange	Adapter over the outbox; API or file per IITH capability	End-to-end test with core owner
Financial system (INT-04–05)	EOD FIN interface file, receipts, commission and referral fee data	FIN file format agreed; SFTP or API delivery	Journey 8; FIN reconciliation sign-off
Identity (INT-06)	Active Directory single sign-on for back-office	LDAPS adapter	Login test with IITH accounts
Notifications (INT-07–08)	E-mail through IIFT SMTP; SMS through IIFT gateway	SMTP and SMS adapters	Delivery test log
APIs (INT-09–11)	Secure REST APIs, authentication, monitoring	OpenAPI 3; mTLS or OAuth2; integration monitor	API security tests; monitor demo
Data (INT-12–15)	Validation, error handling and retry, logging, reconciliation	Schema validation; outbox retry and dead-letter; daily reconciliation	Failure-injection tests; reconciliation report

5.5 Shared platform (RFP 4.4)

RFP area and IDs	What we deliver	How	Acceptance evidence
Security (COM-01, 02, 10)	Protection of sensitive data, access control, encryption	Controls in Section 11	VAPT; encryption check
Audit (COM-03)	Common audit service	Append-only audit	Tamper test
Workflow (COM-04)	Configurable workflows by type, role, level, rule	Workflow engine	Approval matrix tests
Notification (COM-05)	E-mail, SMS, portal notifications	Notification service and templates	Notification test log
Documents (COM-06)	Secure document repository	Encrypted storage; access through API only	Access tests
Search (COM-07)	Global search across records	Search with configurable criteria	UAT case
Reporting (COM-08)	Common reporting framework	Report definitions, exports, schedules	Report sign-off

RFP area and IDs	What we deliver	How	Acceptance evidence
Configuration (COM-09)	Parameters maintained without code changes	Parameters and master data screens	Configuration workbook signed

5.6 Non-functional requirements (RFP 5)

RFP area and IDs	What we deliver	How	Acceptance evidence
Performance (NFR-01–03)	Agreed response times for 100 concurrent users and projected volume	k6 load, stress and soak tests on production-like SIT	Performance test report (DEL-18)
Availability (NFR-04–05)	99.5% overall, 99.9% in business hours; no single point of failure	Two app nodes, database standby, load balancer pair	Monthly availability report
Scalability (NFR-06–07)	Growth in users and volume without redesign	Stateless API nodes; indexed schema	Capacity statement in SAD
Security (NFR-08–14)	Encryption, authentication, authorisation, audit, vulnerability management, VAPT, secure SDLC	Section 11	VAPT report; scan results
Reliability (NFR-15–16)	Controlled errors; transaction integrity	Generic messages with correlation id; ACID transactions	Negative tests
Backup and DR (NFR-17–19)	Backups, DR to RPO 15 minutes and RTO 4 hours, DR tests	WAL archiving; warm standby; annual DR test	Restore test and DR test reports
Maintainability (NFR-20–21)	Modular design; maintained documentation	Modular monolith; document pack updated per release	Document acceptance
Usability and accessibility (NFR-22–24)	Consistent UI, browser support, WCAG 2.1 AA target	One design system; browser test matrix; accessibility check	UAT feedback; accessibility check report
Compatibility (NFR-25)	Standard APIs	REST/JSON, OpenAPI 3	API specification
Monitoring and logging (NFR-26–28)	System monitoring, alerting, central logs	Prometheus, Grafana, central log store	Alert test during SIT
Data (NFR-29–30)	Retention and data quality	Retention rules; validation and constraints	Retention configuration signed

5.7 Deliverables (RFP 7)

Documents are delivered in Word or Excel and PDF. IIFT reviews within five business days and we issue the final version within three business days of comments.

ID	Deliverable	Content and approach	Due	Acceptance evidence
DEL-01	Project charter	Objectives, scope, governance, stakeholders, assumptions, milestones	Wk 2	Steering Committee approval
DEL-02	Project plan	Activities, dependencies, resources, timeline; updated fortnightly	Wk 2	IIFT PM approval
DEL-03	Business requirement specification	Agreed processes from fit-gap workshops on the working system	Wk 4	Business owner sign-off
DEL-04	Functional requirement specification	Rules, screens, reports and configuration per product	Wk 5	Business owner sign-off
DEL-05	Requirements traceability matrix	Every RFP ID traced to design, test and acceptance	Wk 6, kept current	Final RTM at closure
DEL-06	Solution architecture	Application, infrastructure, security, integration, deployment	Wk 6	Design Authority sign-off
DEL-07	Technical design	Components, database, APIs, interfaces, configuration	Wk 6	Design Authority sign-off
DEL-08	UI/UX design	Screen designs and journeys based on the working screens	Wk 5	Business owner sign-off
DEL-09	Interface specification	APIs, mappings, methods, security, error handling per interface	Wk 6	System owner sign-off
DEL-10	Configured solution	SalesVerse 2.0 configured and integrated for IIFT	Wk 16	Build-complete demo (M3)

ID	Deliverable	Content and approach	Due	Acceptance evidence
DEL-11	Source code	Source code of IIFT-specific components (configuration, adapters, reports, templates), build scripts, migrations; core platform via Option C or escrow	Wk 24 and each release	Repository handover checklist
DEL-12	Test strategy	Levels, roles, environments, entry and exit criteria	Wk 6	IIFT approval
DEL-13	SIT test cases	Scenarios traced to requirements	Wk 14	IITH IT review
DEL-14	SIT results	Execution evidence, defect log, closure	Wk 19	SIT exit sign-off (M4)
DEL-15	UAT test cases	Business scenarios per product and role	Wk 18	Business owner review
DEL-16	UAT results	Execution evidence and business sign-off	Wk 22	UAT certificate (M5)
DEL-17	Security assessment	Independent VAPT, remediation, re-test	Wk 19	No open Critical or High findings
DEL-18	Performance test report	Load, stress and soak results against targets	Wk 19	Targets met
DEL-19	Data migration plan	Extraction, transformation, validation, reconciliation	Wk 8	Data owner sign-off
DEL-20	Deployment plan	Sequence, roles, rollback and contingency	Wk 21	IITH IT approval
DEL-21	Go-live checklist	Technical, operational, security, business, support readiness	Wk 23	Go/no-go decision
DEL-22	User training	Agents, bank officers, back-office users	Wk 21–23	Attendance and feedback
DEL-23	Administrator training	Administrators and support staff	Wk 22–23	Attendance and hands-on check
DEL-24	User manual	Portal and back-office guides with screenshots	Wk 21	Document acceptance
DEL-25	Administrator manual	Configuration, user administration, operations support	Wk 21	Document acceptance
DEL-26	Technical manual	Architecture, database, API, integration, deployment	Wk 23	Document acceptance
DEL-27	Operations manual	Monitoring, backup, incidents, routine operations	Wk 23	Document acceptance
DEL-28	Production deployment	Deployment per approved plan	Wk 24	Go-live confirmation (M6)
DEL-29	Go-live support	Four weeks of hypercare	Wk 25–28	Hypercare exit criteria met
DEL-30	Project closure report	Status, open items, lessons, acceptance	Wk 28	Closure sign-off (M7)

Table 7: Deliverables DEL-01 to DEL-30

5.8 Maintenance and support (RFP 8)

ID	Requirement	What we deliver and how	Acceptance evidence
MNT-01	Helpdesk support	Service desk via issue module, e-mail and phone for five years	Ticket records
MNT-02	Incident management	Record, classify, investigate, resolve, close	Monthly incident statistics
MNT-03	Problem management	Root-cause analysis for P1, P2 and recurring incidents	RCA reports within 5 business days
MNT-04	Service requests	User support, configuration and operational help	Request log
MNT-05	Critical incident SLA	P1: 30-minute response, 4-hour restoration, 24x7	SLA report
MNT-06	High priority SLA	P2: 2-hour response, 1 business day	SLA report
MNT-07	Medium/Low/Cosmetic SLA	P3: 4 business hours / 3 business days; P4: 1 business day / next release	SLA report
MNT-08	Bug fixing	Defects fixed under warranty, then under maintenance	Release notes
MNT-09	Corrective maintenance	Changes needed to keep functions working	Release notes
MNT-10	Preventive maintenance	Monthly health check of logs, jobs, capacity, queries, certificates	Health check in monthly report
MNT-11	Security patches	Monthly patch cycle; emergency patches as needed	Patch log
MNT-12	Vulnerability remediation	Critical 7 days, High 14, Medium 30, Low 90	Vulnerability register
MNT-13	Certificate management	Inventory, expiry alerts, renewal support	Certificate register
MNT-14	OS/platform compatibility	Compatibility with supported OS, Node.js, PostgreSQL	Compatibility statement per release
MNT-15	Browser compatibility	Current and previous browser versions	Browser test record
MNT-16	Database maintenance	Health checks, vacuum and index tuning, upgrades	Health check results
MNT-17	Performance monitoring	Monthly review and tuning advice	Performance section of monthly report

ID	Requirement	What we deliver and how	Acceptance evidence
MNT-18	System monitoring	Availability and component monitoring with alerts	Availability figures
MNT-19	Backup verification	Quarterly restore test	Restore test record
MNT-20	Disaster recovery	DR planning and annual DR test	DR test report
MNT-21	Release management	Tested, approved releases in agreed windows	Release records and approvals
MNT-22	Version upgrade	Application and platform upgrades, including SalesVerse 2.0 updates	Upgrade release notes
MNT-23	Minor enhancement	60 hours a year included; more at rate card	Enhancement log
MNT-24	Change request	Impact assessment, quotation, approval, delivery	Signed CR forms
MNT-25	Monthly service report	Incidents, SLA, availability, changes, open issues	Report issued monthly
MNT-26	Quarterly review	Service performance and improvement plan	Minutes and action log
MNT-27	Documentation updates	Documents updated after material change	Document version history
MNT-28	Knowledge transfer	Sessions for IIFT/IITH support staff	Session records
MNT-29	Onsite support	Onsite for major incidents or upgrades (rate card plus OPE)	Visit reports
MNT-30	Transition support	Year 5 knowledge transfer, documents and technical help	Transition acceptance

Table 8: Maintenance requirements MNT-01 to MNT-30

6. Solution Fitment

Each functional and non-functional requirement is classified against SalesVerse 2.0 as it runs today. The four categories are:

Category	Meaning
Available in SalesVerse 2.0 today	Works in the current platform; IIFT can see it in the demonstration.
Configuration	Supported by the platform; needs IIFT values, rates, templates or rules.
Integration with IIFT systems	Adapter built on the platform's integration layer; completed once IIFT provides endpoints and credentials.
Delivered during implementation	Work or services performed during the project or, for MNT items, during maintenance.

RFP area	Available today	Configuration	Integration	During implementation	Total
Agent/Banca Portal (RFP 4.1)	49	12	1	0	62
Back-office (RFP 4.2)	27	6	0	0	33
Integration (RFP 4.3)	6	0	9	0	15
Shared platform (RFP 4.4)	8	2	0	0	10
Non-functional (RFP 5)	15	1	0	14	30
Deliverables (RFP 7)	0	0	0	30	30
Maintenance and support (RFP 8)	0	0	0	30	30
Total	105	21	10	74	210

Table 9: Fitment by RFP area (number of requirement IDs)

The 24 commercial requirements (RFP section 6) are contractual commitments rather than functions; they are answered in Section 17 and listed in Annex A without a fitment category.

6.1 What runs today and what is completed in the project

Capability	Available in SalesVerse 2.0 today	Completed during implementation
Products and rating	Seven Appendix 3 products (FTP-HP, FTP-NP, PFT, PHA, PRO, KHR, OSA) with configurable rating	IIFT actuarial rate tables replace the indicative rates; questionnaires and templates loaded
Underwriting referral	Financing above B\$150,000 referred to IIFT Sales	Further referral rules agreed in design
Payment control	Seven-day grace-period block on all agents of an agency	Bank-specific settlement rules, if any
Approvals	Maker-checker with configurable levels and amount thresholds, e.g. second approval from B\$300,000 sum covered	IIFT approval matrix and approver roles
AML/KYC	Watch-list screening with fuzzy name matching; Compliance review queue; a confirmed match rejects the subject's pending requests	Adapter to IIFT's AML provider or lists
e-Signature	On-screen signature and single-use e-mail link	Evidence format agreed with IIFT Legal
Documents	e-Policy and e-Receipt PDFs; encrypted document storage; ClamAV scan of uploads, mandatory in production	IIFT templates, wording and branding
End of day	EOD report, FIN interface file, receipt reconciliation; re-runs sent to FIN as numbered revisions	FIN file format sign-off; BRR reconciliation
Integration	Transactional outbox with retry and dead-letter	Adapters to IIFT core, FIN, AML provider, SMS gateway, SMTP and Active Directory
Reports	12 standard reports; Excel, CSV and PDF export; scheduling	Remaining reports from the agreed catalogue

Capability	Available in SalesVerse 2.0 today	Completed during implementation
Audit and data protection	Append-only audit enforced by a database trigger; AES-256-GCM encryption of IC and passport numbers	Retention and archival job; key rotation tooling; SIEM export
Access	Login, users, roles and permissions	IITH password and session values; MFA for back-office

Table 10: Capability status

6.2 Work that depends on IIFT

- Adapters to IIFT's core system, FIN, AML provider, SMS gateway, SMTP relay and Active Directory. Each needs IIFT's interface specification, test endpoint and credentials before it can be completed.
- IIFT's actuarial rate tables, questionnaires, product wording and document templates, replacing the indicative values in the working system.
- MFA for back-office users, production and DR infrastructure set-up, performance testing and the independent VAPT.
- Data migration, UAT, training and the IIFT-specific document set.

7. Personas, Screens, Journeys and Navigation

This chapter shows the IIFT Agent/Banca Portal & Back-office as each kind of user meets it: who they are, what their menu contains, which screens they work in and how the end-to-end journeys run across the portal, the back-office and the system's own steps (numbering, notifications, outbox messages, audit). Menus are taken from the role definitions shipped with the product; journeys are described as the services execute them today.

7.1 Who uses what

Eleven personas cover the RFP's users. Four work in the Agent/Banca Portal, seven in the Back-office. Each persona is a seeded role; IIFT can change the permissions of a role or add roles in the Back-office without a release.

Persona	Module	Seeded role	Data scope	Demo user
Main agent (agency principal)	Agent & Banca Portal	AGENCY_PRINCIPAL	Every record of the agency: own and sub-agents' quotations, policies, participants, payments, claims, requests and commission.	ag-000001
Sub-agent	Agent & Banca Portal	AGENT	Own records only (portal.agency.view_all not granted).	ag-000002
Bank officer (banca)	Agent & Banca Portal	BANCA_OFFICER	Own records only.	bk-000004
Bank supervisor (banca)	Agent & Banca Portal	BANCA_SUPERVISOR	Every record of the bank, all branches.	bk-000005
Operations officer (maker)	Back-office	OPERATIONS_OFFICER	All agencies and banks.	ops.maker
Operations supervisor (checker)	Back-office	OPERATIONS_SUPERVISOR	All agencies and banks.	ops.checker
Underwriter / quality check	Back-office	UNDERWRITER	All policies.	underwriter
Finance officer	Back-office	FINANCE_OFFICER	All payments and runs.	finance
Compliance officer	Back-office	COMPLIANCE_OFFICER	All screening cases and the audit trail.	compliance
Support officer	Back-office	SUPPORT_DESK	All issues; agents and agencies read-only.	support
System administrator	Back-office	SYSTEM_ADMINISTRATOR	Configuration and audit; no business records.	admin
Management (read-only)	Back-office	MANAGEMENT	KPIs, trends and reports; no approvals	manager
Internal auditor	Back-office	AUDITOR	Records and audit trail, read-only	–
Participant	Public e-signature page	no account	One quotation, through a single-use link	–

Table 11: Personas, roles and data scope

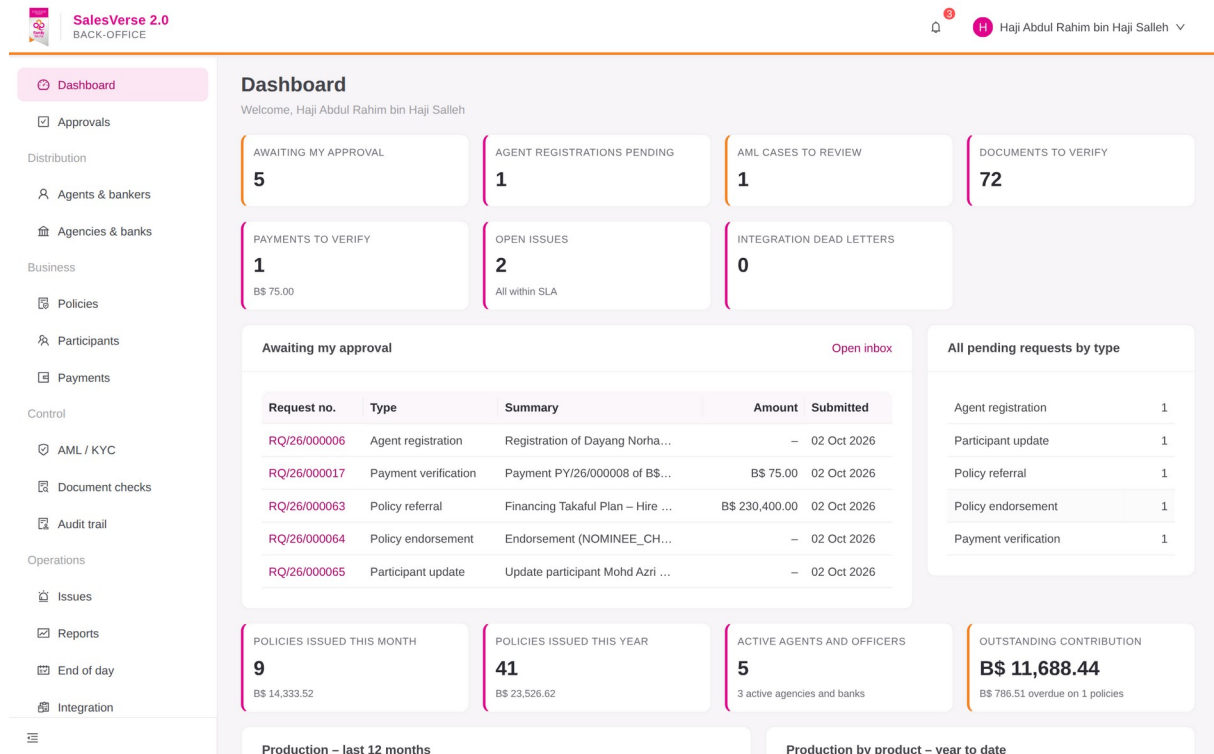
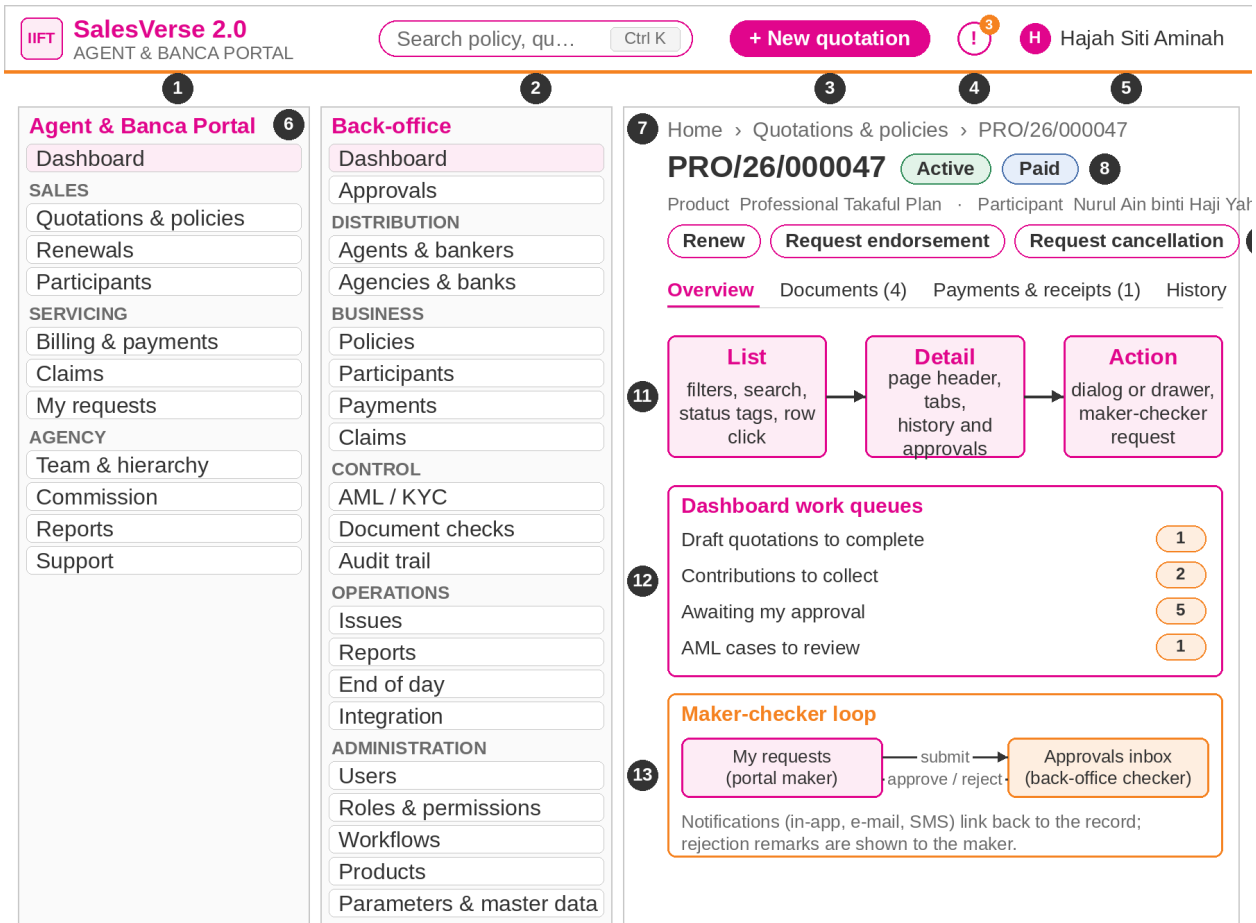


Figure 1: Management dashboard: Month and year-to-date issuance, outstanding contribution, production trend, split by product and channel.

7.2 Navigation model

Both modules share one shell: a header with the module name, global search, the module's primary action, notifications and the user menu; a side navigation of groups; and a content area that always runs list → detail → action. The figure names the parts; the table lists the conventions a user learns once and meets on every screen.



SalesVerse 2.0
AGENT & BANCA PORTAL

Search policy, qu... Ctrl K

+ New quotation

H Hajah Siti Aminah

Agent & Banca Portal

- Dashboard
- SALES
 - Quotations & policies
 - Renewals
 - Participants
- SERVICING
 - Billing & payments
 - Claims
 - My requests
- AGENCY
 - Team & hierarchy
 - Commission
 - Reports
 - Support

Back-office

- Dashboard
- Approvals
- DISTRIBUTION
 - Agents & bankers
 - Agencies & banks
- BUSINESS
 - Policies
 - Participants
 - Payments
 - Claims
- CONTROL
 - AML / KYC
 - Document checks
 - Audit trail
- OPERATIONS
 - Issues
 - Reports
 - End of day
 - Integration
- ADMINISTRATION
 - Users
 - Roles & permissions
 - Workflows
 - Products
 - Parameters & master data

Home > Quotations & policies > PRO/26/000047

PRO/26/000047 Active Paid

Product Professional Takaful Plan · Participant Nurul Ain binti Haji Ya

Renew Request endorsement Request cancellation

Overview Documents (4) Payments & receipts (1) History

List filters, search, status tags, row click

Detail page header, tabs, history and approvals

Action dialog or drawer, maker-checker request

Dashboard work queues

- Draft quotations to complete 1
- Contributions to collect 2
- Awaiting my approval 5
- AML cases to review 1

Maker-checker loop

My requests (portal maker) → submit → Approvals inbox (back-office checker)

approve / reject

Notifications (in-app, e-mail, SMS) link back to the record; rejection remarks are shown to the maker.

- Brand and module name; click returns to the dashboard
- Global search over policies, participants and agents (Ctrl+K)
- Primary action of the module (portal: New quotation)
- Notifications bell with unread count; opens the latest items
- User menu: profile, change password, sign out
- Side navigation: top entries, then groups; permitted items only
- Breadcrumb: Home > list > record
- Record reference as title, status tags beside it
- Page actions on the right; the primary action last
- Tabs for documents, payments, history, approvals, claims
- List → detail → action on every record type
- Dashboard queues link to the filtered list
- Requests and approvals are the same record in both modules

Figure 2: Navigation model: header, side navigation of both modules, record page and conventions

Convention	Behaviour
Ctrl+K	Focuses the global search in the header; results are grouped and a result opens the record.
Row click	Clicking a table row opens the record; links and buttons inside the row keep their own action.
Status tags	Statuses are colour tags everywhere (Draft, Pending approval, Active, Pending verification, Paid, Rejected, Cancelled); the same words are used in filters.
Breadcrumb	Home > list > record on every page; the record reference is the page title.
Page actions	Right-aligned; the primary action is last; dangerous actions (reject, cancel, discard) are red and confirmed.
Dialogs and drawers	A short change opens a dialog; a longer form (payment, role, product) opens a drawer from the right so the page stays visible.
Checklist	A draft application shows what remains before submission, with a button per item.
Maker-checker	A change that needs approval is submitted as a request; My requests (portal) and Approvals (back-office) show the same request from both sides; remarks are mandatory on rejection.
Work queues	Dashboards list queues with counts; each links to the filtered list.
Notifications	Bell with unread count; every notification links to the record; e-mail and SMS copies for the events configured.
Filters in the URL	Dashboard links carry the filter (for example status=DRAFT) so a list opens filtered.
Navigation state	Collapsed navigation and open groups are remembered per browser; on a phone the navigation is a drawer.

Convention	Behaviour
Session	Idle timeout and absolute lifetime from the Security parameters; signing in again shows why.

Table 12: Keyboard and interaction conventions

Screens shared by every persona:

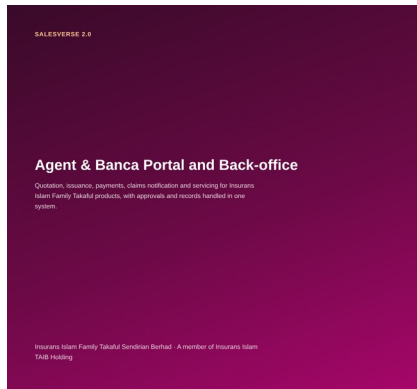


Figure 3: Sign in: Username and password; the account locks after the configured number of failed attempts.

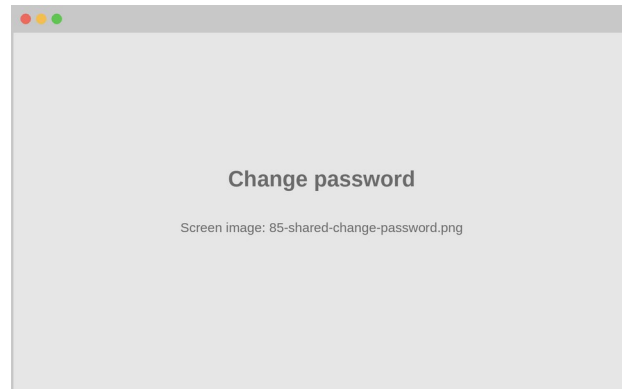


Figure 4: Change password: Password rules from the Security parameters; required at first sign-in and after a reset.

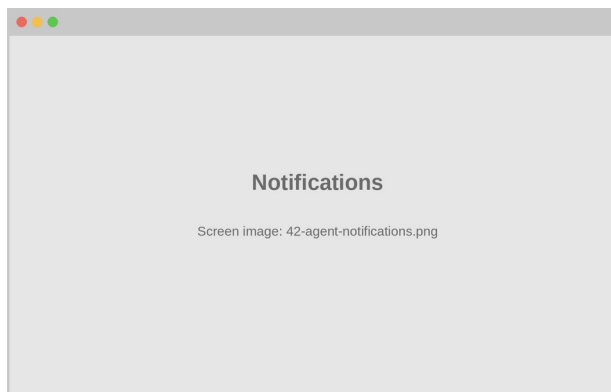


Figure 5: Notifications: Every in-app notification links to the record it is about; unread items are marked.

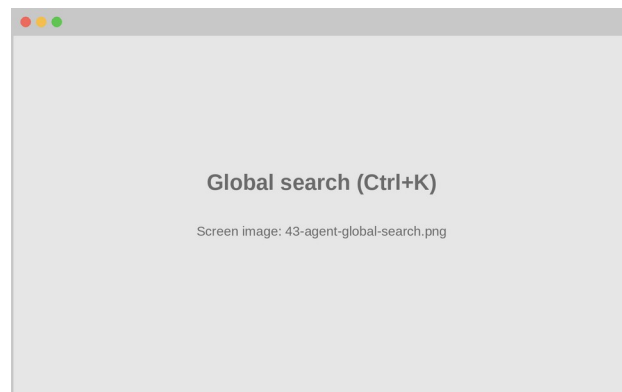


Figure 6: Global search (Ctrl+K): Results grouped by policies and quotations, participants and team; a result opens the record.

Insurans Islam Family Takaful Sendirian Berhad
A member of Insurans Islam TAB Holding

e-Policy Schedule
Policy No. PRO/26/000018 | Quotation No. QT/26/000021

Certificate	
Product	Professional Takaful Plan (PRO)
Line of business	Annual (Individual)
Plan	Plan C
Period of cover	02 Oct 2026 to 01 Oct 2027
Sum covered	RM 50,000.00
Date of issue	02 Oct 2026

Participant	
Name	Iskandar bin Haji Bakar
Identification	NRIC *****77
Participant no.	PT26/000021
Address	No. 24, Simpang 121, Kampong Rimba, BE3121, TUTONG
Contact	6738119131 / participant21@example

Contribution	
Item	Amount
Plan C - 1 year(s)	RM 210.00
Additional cover - Plan C	RM 140.00
of which Tabarru' (donation to the participants' risk fund)	RM 240.00
of which Wakalah fee (operator fee)	RM 195.00
Total contribution	RM 350.00

Nominees / Beneficiaries			
Name	Role	Relationship	Share
Hajah Mariam binti Haji Ahmad	NOMINEE	SPOUSE	60.00%
Muhammad Iqbal bin Hassan	NOMINEE	CHILD	40.00%

Particulars	
Additional Cover	Yes

Distribution	
Agent / Banker	Hajah Siti Aminah binti Haji Osman (AG-000001)
Agency / Bank	Seri Amanah Takaful Agency (AGY-SA)

This certificate is issued under the Takaful contract between the participant and Insurans Islam Family Takaful Sendirian Berhad, subject to the terms, conditions and exclusions of the product. Please read the Product Disclosure Sheet and keep this document safe.

Figure 7: e-Policy schedule (PDF): Generated on issuance and e-mailed to the participant.

7.3 Main agent (agency principal)

Profile

Who	Runs an IIFT agency; sells annual plans herself and is accountable for the production and conduct of her sub-agents.
Goals	Grow agency production; keep contributions collected inside the grace period so the agency is never blocked; bring new sub-agents on quickly.
How often	Daily; several times a day in campaign periods.
Device	Laptop in the office; phone or tablet when visiting participants (the portal reflows to a drawer menu).
Role and permissions	AGENCY_PRINCIPAL: All portal functions plus agency-wide visibility (portal.agency.view_all) and sub-agent registration (portal.agents.register).
Data scope	Every record of the agency: own and sub-agents' quotations, policies, participants, payments, claims, requests and commission.
Demo user	Hajah Siti Aminah binti Haji Osman, Seri Amanah Takaful Agency (ag-000001)
RFP requirements covered	AP-01–AP-10, AP-12–AP-47, AP-49–AP-58, AP-59, AP-62, INT-05

Table 13: Main agent (agency principal): profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.


SalesVerse 2.0
AGENT & BANCA PORTAL

Search policy, qu... Ctrl K

+ New quotation


 Hajah Siti Aminah

Dashboard

SALES

Quotations & policies

Renewals

Participants

SERVICING

Billing & payments

Claims

My requests

AGENCY

Team & hierarchy

Commission

Reports

Support

Data scope: Every record of the agency: own and sub-agents' quotations, policies, participants, payments, claims, requests and commission.

Where it leads

- Work queues, follow-ups, production chart, recent activity (whole agency or bank)
- List with filters → Policy detail: checklist, Declarations, Nominees, Signature, Send link, Submit; Renew, Request endorsement, Request cancellation, E-mail documents
- Policies ending inside the renewal window → Renew → new quotation
- List → Participant detail (policies, documents, update requests) · Register participant
- Outstanding → Submit payment (single or bulk, proof) · Payment history → Payment detail, e-Receipts
- List → Claim detail (documents, IIFT remarks) · Notify claim (validate policy first)
- Own requests and the team's → Request detail, remarks, Withdraw
- Agency or bank record, hierarchy, team members → Team member · Register agent / bank officer
- Commission and referral fee by period, whole agency
- Report catalogue → filters, preview, export (Excel, CSV, PDF)
- Own issues → Issue detail, conversation, attachments · Report an issue

The four portal roles share one menu. What differs is the data scope and two actions: the principal and the bank supervisor see the whole agency or bank and may register team members.

Figure 8: Main agent (agency principal): navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Dashboard	Tiles, work queues, production, recent activity	Starts the day from the queues; opens follow-ups	AP-37, AP-52, AP-53
New quotation (Product, Participant, Coverage, Review)	Four-step wizard with a live indicative contribution	Chooses the product, finds or registers the participant, sets the cover, saves the quotation	AP-17–AP-20, FFR01–05 step 1–3
Quotations & policies	List with search, status, payment status, product and date filters	Finds drafts, pending, rejected and active records	AP-21–AP-23
Policy detail	Checklist, overview, Documents, Payments & receipts, History, Approvals, Claims tabs	Completes declarations, nominees, documents and signatures; submits; downloads or e-mails the e-Policy and e-Receipt	AP-24, AP-25, AP-29–AP-36, AP-44–AP-47, AP-62
Renewals	Policies ending inside the renewal window	Creates the renewal quotation	AP-26
Participants and Register participant	List, detail with policies, documents and update requests	Registers or finds the participant; requests updates	AP-11, AP-13–AP-16, AP-48
Billing & payments and Payment detail	Outstanding, overdue, due soon, pending verification; payment history	Submits single or bulk payment with proof; downloads e-Receipts	AP-30, AP-37–AP-42
Claims and Notify claim	List, validation of the policy number, event details, documents	Notifies a claim and follows IIFT's remarks	AP-43

Screen	Purpose	What this persona does there	RFP IDs
My requests	Own (or team) maker-checker requests	Reads the decision and remarks; withdraws a pending request	AP-49–AP-51
Team & hierarchy and Register agent	Agency or bank record, hierarchy, members	Registers sub-agents or bank officers; follows their approval	AP-07–AP-10, AP-12, AP-46, AP-48
Commission	Accrued and paid commission by period	Checks the statement	INT-05
Reports	Role-scoped reports, preview, export	Runs production and payment reports	AP-58
Support and Report an issue	Own issues with SLA targets	Reports a problem with attachments; confirms or reopens the resolution	AP-55–AP-57
My profile	Profile, portal user name, documents, change requests	Requests contact changes (approved by IIFT)	AP-05, AP-06
Notifications	In-app notifications with links	Opens the record behind a notification	AP-34, AP-54

Table 14: Main agent (agency principal): screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

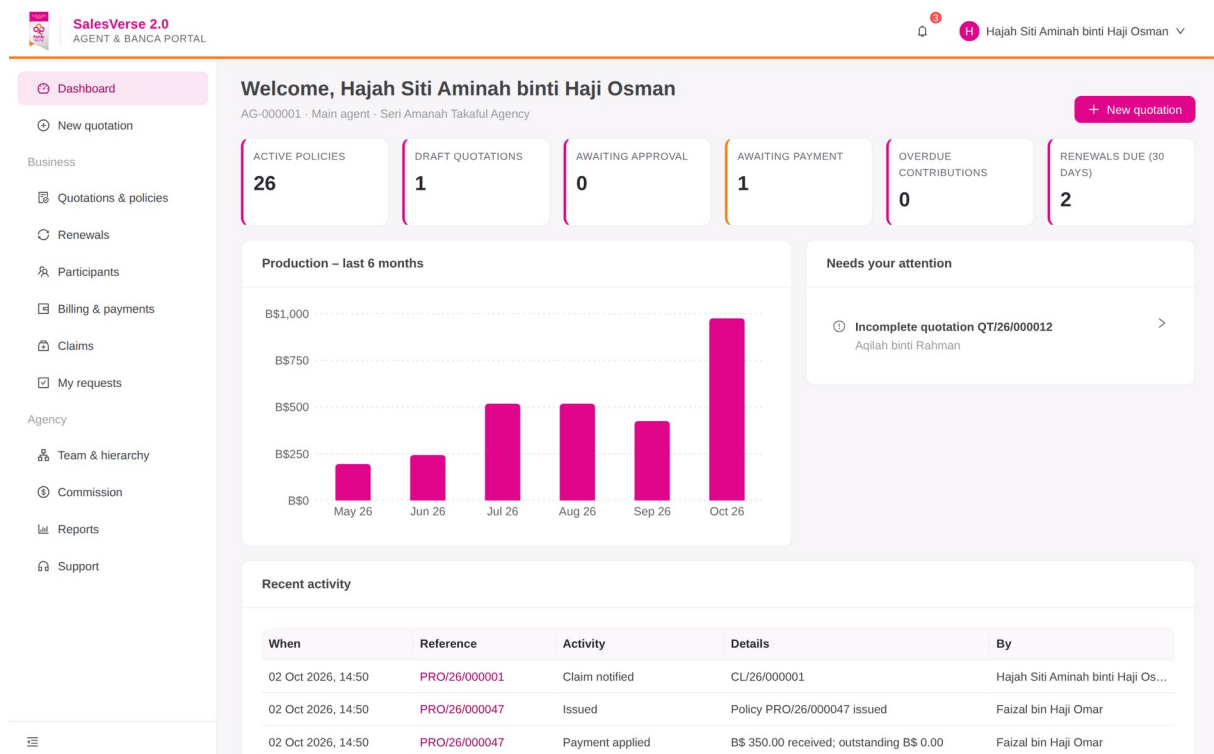


Figure 9: Dashboard: Agency-wide figures, work queues and follow-ups for the principal.


SalesVerse 2.0
AGENT & BANCA PORTAL



Hajah Siti Aminah binti Haji Osman

Dashboard

New quotation

Business

Quotations & policies

Renewals

Participants

Billing & payments

Claims

My requests

Agency

Team & hierarchy

Commission

Reports

Support

Home / Quotations & policies / PRO/26/000018

PRO/26/000018
Active
Paid

Professional Takaful Plan - Iskandar bin Haji Bakar - quotation QT/26/000021

Renew

Request endorsement

Request cancellation

E-mail documents

Overview

Documents (6)

Payments & receipts (1)

History

Approvals

Claims

Cover

Product: Professional Takaful Plan (PRO)

Line of business: Annual (Individual)

Plan: Plan C

Additional cover: Additional cover – Plan C

Term: 1 year

Period of cover: 02 Jul 2026 to 01 Jul 2027

Sum covered: **BS 50,000.00**

Contribution: **BS 350.00**

Outstanding: **BS 0.00**

Quotation no.: QT/26/000021

Policy no.: PRO/26/000018

Agent: Hajah Siti Aminah binti Haji Osman (AG-000001)

Agency / bank: Seri Amanah Takaful Agency

Quoted: 02 Jul 2026

Submitted: 02 Jul 2026

Issued: 02 Jul 2026

Contributions

Plan C – 1 year(s)

BS 210.00

Additional cover – Plan C

BS 140.00

Total contribution

BS 350.00

Tabarru' (participants' risk fund)

BS 245.00

Wakalah fee (operator)

BS 105.00

Participant

Name: Iskandar bin Haji Bakar

Participant no.: PT/26/000021

IC (NRIC): *****2877

Date of birth: 13 Apr 1991

Mobile: 6738119131

E-mail: participant21@mail.example

AML screening: Clear

Nominees

Name	ID	Relationship	Role	Share
Iskandar bin Haji Bakar				

Declarations

Declaration	Answer	Details
Are you currently suffering from any illness or medical condition, or receiving medical treatment?	No	–
In the past 5 years, have you been hospitalised or advised to undergo surgery?	No	–
Has any takaful or insurance application on your life been declined, postponed or accepted on special terms?	No	–

Figure 10: Policy detail (active): Overview, servicing actions, tabs for documents, payments and receipts, history and approvals.


SalesVerse 2.0
AGENT & BANCA PORTAL



Hajah Siti Aminah binti Haji Osman

Dashboard

New quotation

Business

Quotations & policies

Billing & payments

Renewals

Participants

Claims

My requests

Agency

Team & hierarchy

Commission

Reports

Support

Home / Billing & payments

Billing & payments

Contributions awaiting payment and payments submitted to IIFT

New business is blocked for your agency

1 policy(ies) with contribution not submitted within 7 days (BS786.51)

OUTSTANDING

BS 11,613.44

3 policies awaiting payment

OVERDUE

1

BS 786.51 past the due date

Outstanding (3)

Payment history

2 selected - BS 1,613.44

Policy / quotation no.	Product
☒ FTP-HP/26/000049	Financing Takaful Plan – Hire Purchase
☒ FTP-NP/26/000008	Financing Takaful Plan – Non-Participating
☐ PFT/26/000009	Property Financing Takaful Plan

Submit payment for 2 policies

Allocation

Policy / quotation no.	Participant	Outstanding	Amount paid
FTP-HP/26/000049	Dk Nur Syafiqah binti Pg Hashim	BS 786.51	BS 786.51
FTP-NP/26/000008	Ummi Kalsom binti Ahmad	BS 826.93	BS 826.93

Payment details

Method

Bank (optional)

Bank transfer

Bank reference

Payment date

TRX-260930-118

02 Oct 2026

Remarks (optional)

Proof of payment

Click or drag the file here

doc.pdf

Bank slip, transfer confirmation or cheque copy – PDF, PNG or JPEG

Total BS 1,613.44

Cancel

Submit for verification

Figure 11: Submit payment: Outstanding contributions; one payment allocated per policy with proof of payment.

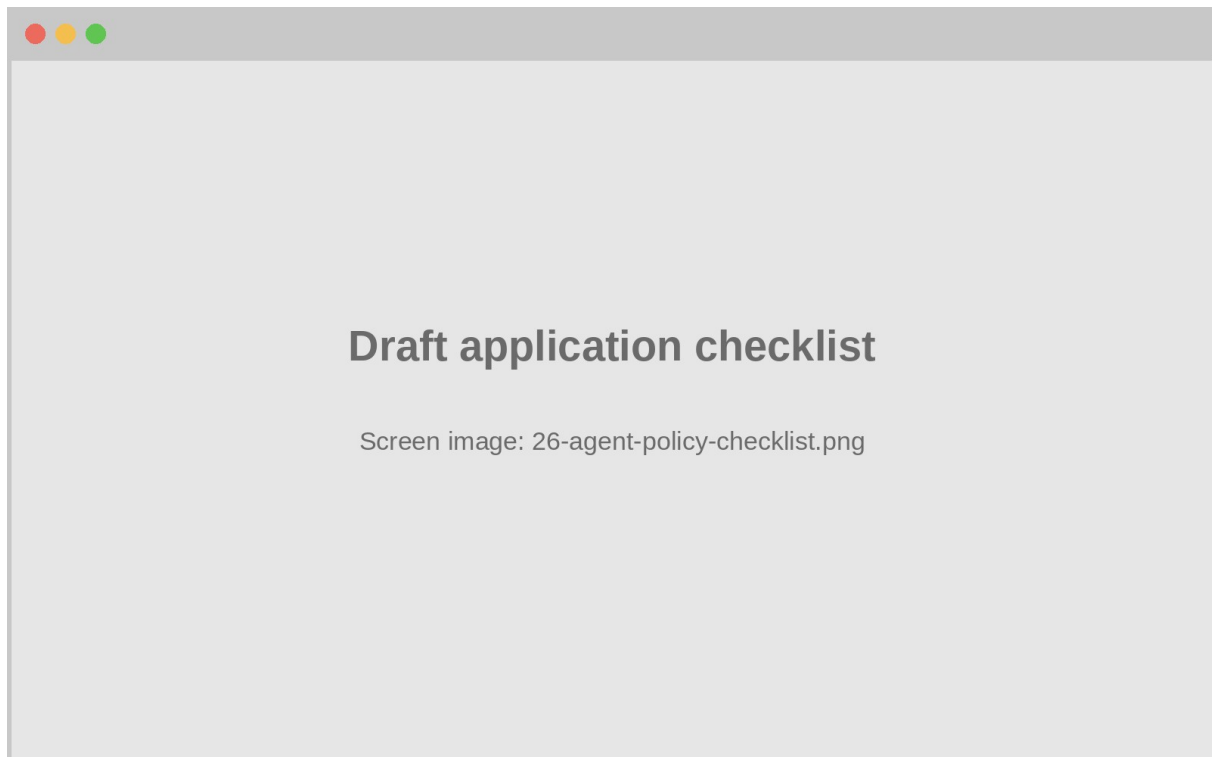


Figure 12: Draft application checklist: What remains before submission: declarations, nominees, documents and signatures.

SalesVerse 2.0
AGENT & BANCA PORTAL

Home / Quotations & policies

Quotations & policies
All business written by your agency

Policy / quotation no. or ... Status Payment status Product Created from ... Created to ...

Policy / quotation no.	Product	Participant	Agent	Status	Payment
Q726/000012	Overseas Student Assist Takaful Plan	Anglin bin Rahman	Hajah Sli Annisah binti Haj Osmen	Draft	Unpaid
Q726/000011	Professional Takaful Plan	Hji Rosnah bin Awg Tengah	Muhammad Firdaus bin Abdullah	Pending payment	Pending verification
PH426/000004	Personal Home Assistant Takaful Plan	Ok Nur Syafiqah binti Pg Nephin	Muhammad Firdaus bin Abdullah	Active	Paid
OS626/000003	Overseas Student Assist Takaful Plan	Hj Kamauddin bin Osman	Hajah Sli Annisah binti Haj Osmen	Active	Paid
KH426/000002	Khairol Takaful Plan	Siti Khadijah binti Saleh	Hajah Sli Annisah binti Haj Osmen	Active	Paid
PRD26/000001	Professional Takaful Plan	Muhammad bin Sulaiman	Hajah Sli Annisah binti Haj Osmen	Active	Paid
PRD26/000012	Professional Takaful Plan	Syazwan binti Kassim	Hajah Sli Annisah binti Haj Osmen	Active	Paid
KH426/000010	Khairol Takaful Plan	Mohd Rizal bin Hassan	Muhammad Firdaus bin Abdullah	Active	Paid
KH426/000015	Khairol Takaful Plan	Zuraidah binti Jusuf	Muhammad Firdaus bin Abdullah	Active	Paid
PRD26/000014	Professional Takaful Plan	Amrul bin Yusof	Hajah Sli Annisah binti Haj Osmen	Active	Paid
KH426/000018	Khairol Takaful Plan	Suhaila bin Hamzah	Muhammad Firdaus bin Abdullah	Active	Paid
PRD26/000016	Professional Takaful Plan	Iskandar bin Haj Bakar	Hajah Sli Annisah binti Haj Osmen	Active	Paid

Figure 13: Quotations & policies: Agency-wide list with the Agent column, status and payment filters.

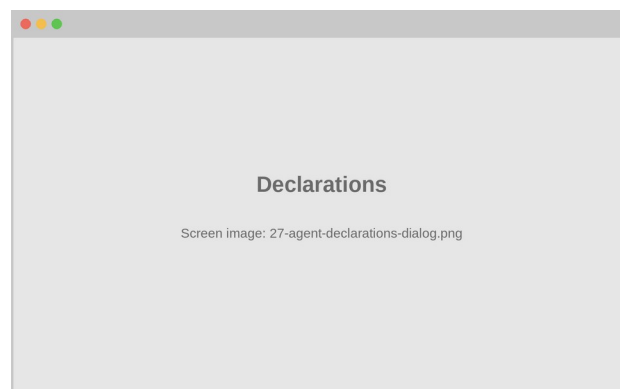


Figure 14: Declarations: Health and risk questions; a Yes answer needs details and refers the case to underwriting.

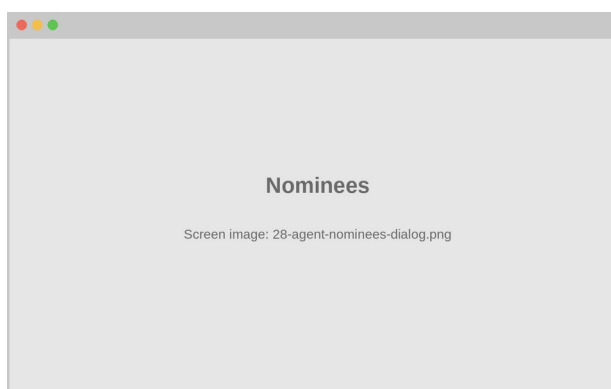


Figure 15: Nominees: Nominee, beneficiary or executor with relationship and share; shares must total 100%.

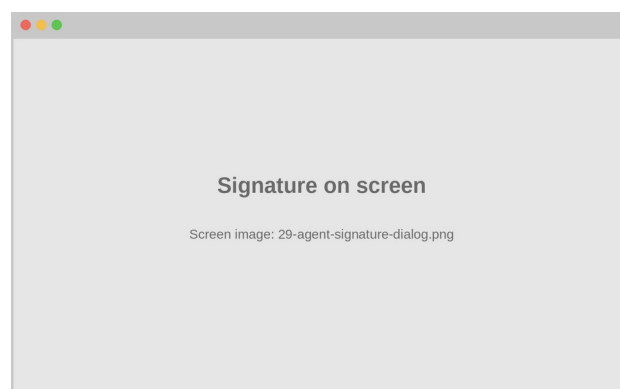


Figure 16: Signature on screen: Participant or agent signs on the agent's device; the image is stored against the application.

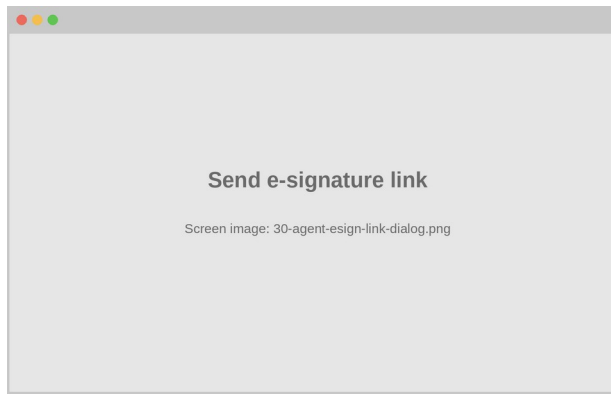


Figure 17: Send e-signature link: One-time link e-mailed to the participant; validity set by the e-signature parameter.

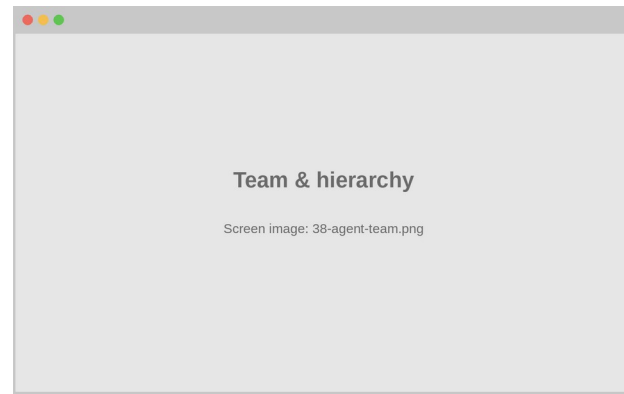


Figure 18: Team & hierarchy: Agency record, issuance block status, reporting hierarchy and team members.

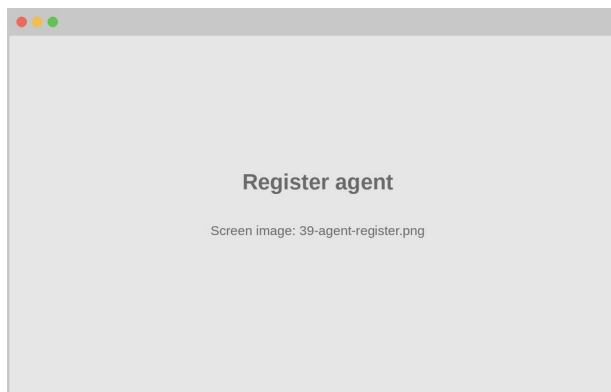


Figure 19: Register agent: Applicant details, documents and IIFT approval shown as three steps.

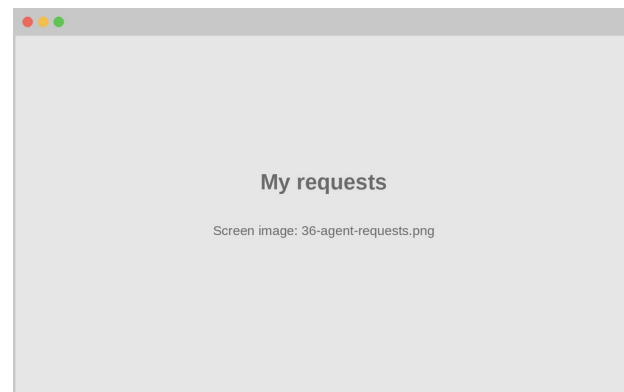


Figure 20: My requests: Every maker-checker request with its status, decision and remarks.

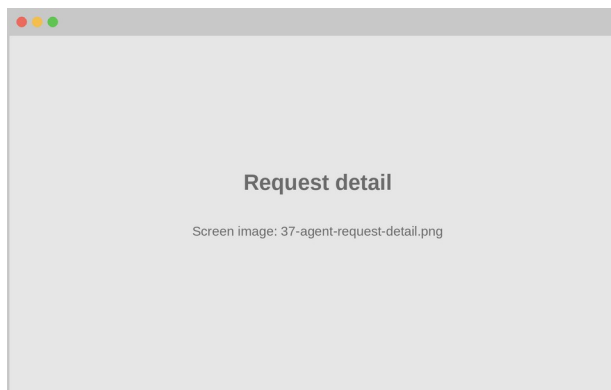


Figure 21: Request detail: Submitted details, approval history and a Withdraw action while pending.

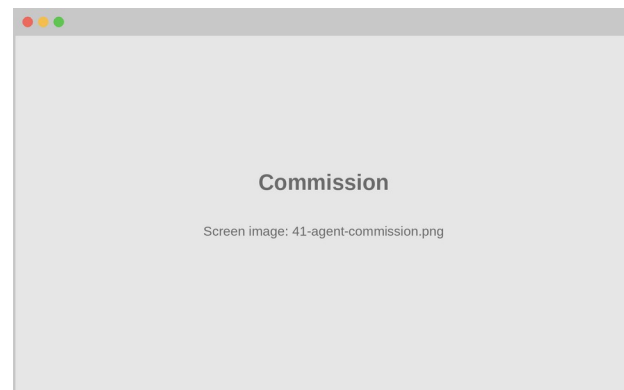


Figure 22: Commission: Accrued and paid commission by period, agency-wide for the principal.

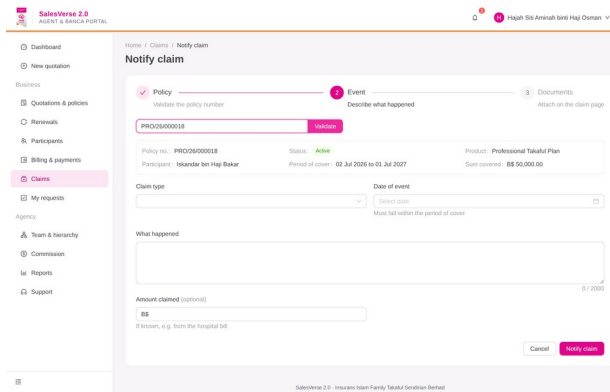


Figure 23: Claims: Claim notifications of the agency with status.

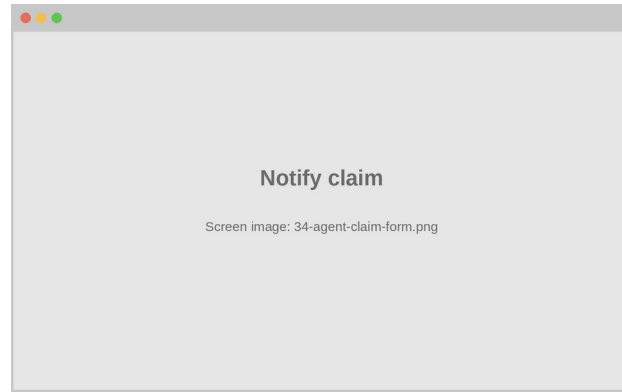


Figure 24: Notify claim: Policy validated first; then claim type, date of event within cover, description, amount.

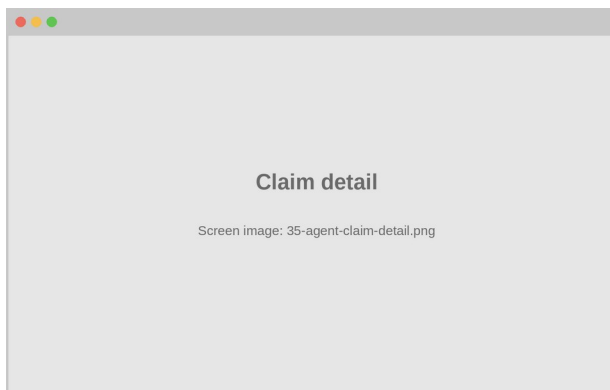


Figure 25: Claim detail: Claim, policy, IIFT remarks and supporting documents uploaded after notification.

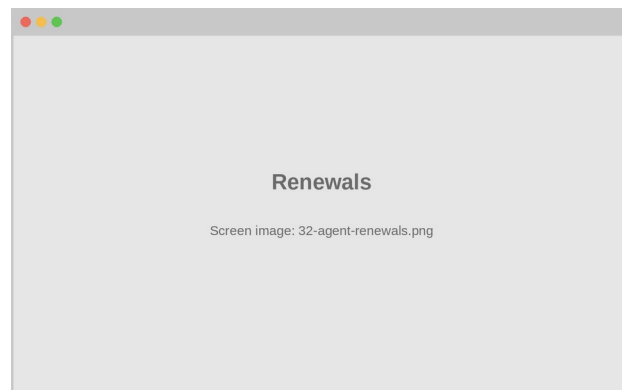


Figure 26: Renewals: Policies whose cover ends inside the renewal window, with days left and a Renew action.

7.4 Sub-agent

Profile

Who	Sells Professional, Khairat, Personal Home Assistant and Overseas Student plans under a main agent.
Goals	Quote and issue in one sitting with the participant; collect the contribution; earn commission; know at once when IIFT needs something.
How often	Daily during sales activity; weekly for renewals and payments.
Device	Phone or tablet in the field; laptop for document uploads.
Role and permissions	AGENT: Portal base permissions: quote, submit, service, bill, claim, report, support.
Data scope	Own records only (portal.agency.view_all not granted).
Demo user	Muhammad Firdaus bin Abdullah, reports to AG-000001 (ag-000002)
RFP requirements covered	AP-01–AP-06, AP-13–AP-47, AP-49–AP-58, AP-62

Table 15: Sub-agent: profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.


SalesVerse 2.0
AGENT & BANCA PORTAL

Search polic... Ctrl K

+ New quotation

 3

 Muhammad Firdaus bin

Dashboard

SALES

Quotations & policies

Renewals

Participants

SERVICING

Billing & payments

Claims

My requests

AGENCY

Team & hierarchy

Commission

Reports

Support

Data scope: Own records only
(portal.agency.view_all not granted).

Where it leads

- Work queues, follow-ups, production chart, recent activity (own records)
- List with filters → Policy detail: checklist, Declarations, Nominees, Signature, Send link, Submit; Renew, Request endorsement, Request cancellation, E-mail documents
- Policies ending inside the renewal window → Renew → new quotation
- List → Participant detail (policies, documents, update requests) · Register participant
- Outstanding → Submit payment (single or bulk, proof) · Payment history → Payment detail, e-Receipts
- List → Claim detail (documents, IIFT remarks) · Notify claim (validate policy first)
- Own requests → Request detail, remarks, Withdraw
- Agency or bank record, hierarchy, team members → Team member
- Commission and referral fee by period
- Report catalogue → filters, preview, export (Excel, CSV, PDF)
- Own issues → Issue detail, conversation, attachments · Report an issue

Figure 27: Sub-agent: navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Dashboard	Tiles, work queues, production, recent activity	Starts the day from the queues; opens follow-ups	AP-37, AP-52, AP-53
New quotation (Product, Participant, Coverage, Review)	Four-step wizard with a live indicative contribution	Chooses the product, finds or registers the participant, sets the cover, saves the quotation	AP-17–AP-20, FFR01–05 step 1–3
Quotations & policies	List with search, status, payment status, product and date filters	Finds drafts, pending, rejected and active records	AP-21–AP-23
Policy detail	Checklist, overview, Documents, Payments & receipts, History, Approvals, Claims tabs	Completes declarations, nominees, documents and signatures; submits; downloads or e-mails the e-Policy and e-Receipt	AP-24, AP-25, AP-29–AP-36, AP-44–AP-47, AP-62
Renewals	Policies ending inside the renewal window	Creates the renewal quotation	AP-26
Participants and Register participant	List, detail with policies, documents and update requests	Registers or finds the participant; requests updates	AP-11, AP-13–AP-16, AP-48
Billing & payments and Payment detail	Outstanding, overdue, due soon, pending verification; payment history	Submits single or bulk payment with proof; downloads e-Receipts	AP-30, AP-37–AP-42
Claims and Notify claim	List, validation of the policy number, event details, documents	Notifies a claim and follows IIFT's remarks	AP-43
My requests	Own (or team) maker-checker requests	Reads the decision and remarks; withdraws a pending request	AP-49–AP-51

Screen	Purpose	What this persona does there	RFP IDs
Team & hierarchy	Agency or bank record and reporting line	Sees own position and reporting line	AP-09, AP-10
Commission	Accrued and paid commission by period	Checks the statement	INT-05
Reports	Role-scoped reports, preview, export	Runs production and payment reports	AP-58
Support and Report an issue	Own issues with SLA targets	Reports a problem with attachments; confirms or reopens the resolution	AP-55–AP-57
My profile	Profile, portal user name, documents, change requests	Requests contact changes (approved by IIFT)	AP-05, AP-06
Notifications	In-app notifications with links	Opens the record behind a notification	AP-34, AP-54

Table 16: Sub-agent: screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

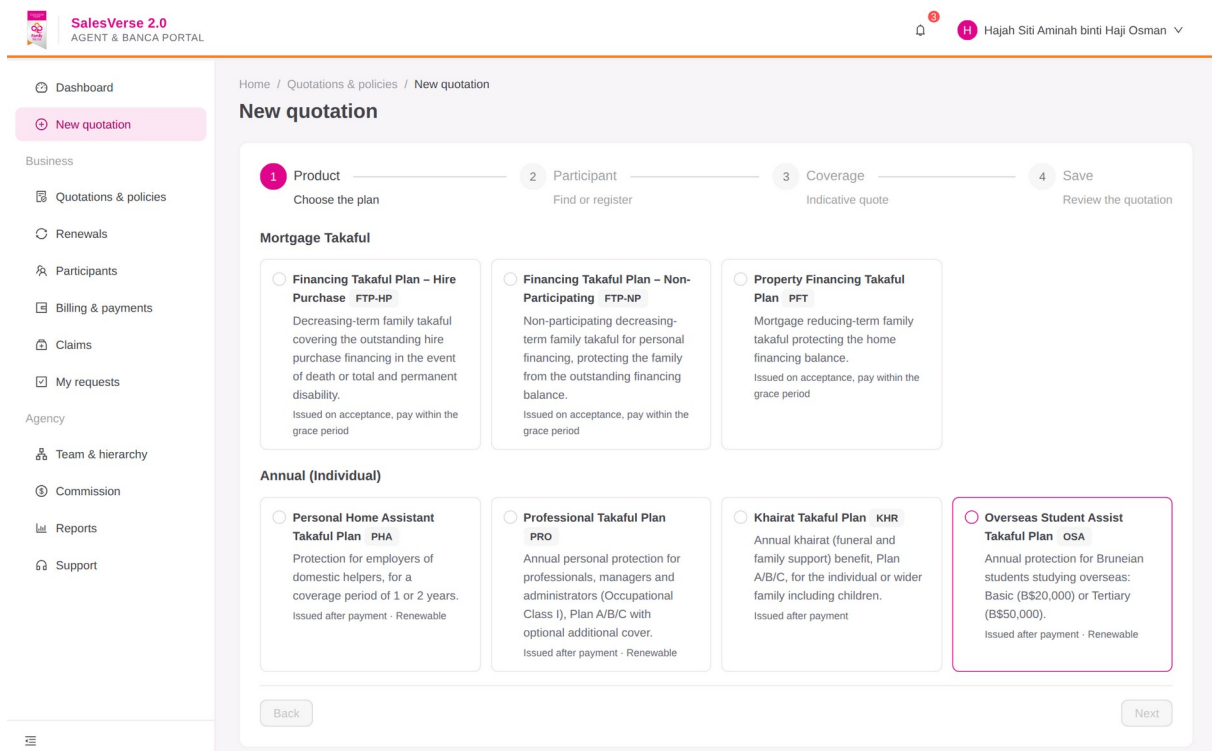


Figure 28: New quotation: product: Products grouped by line of business with pay-before or pay-after-issue tags.

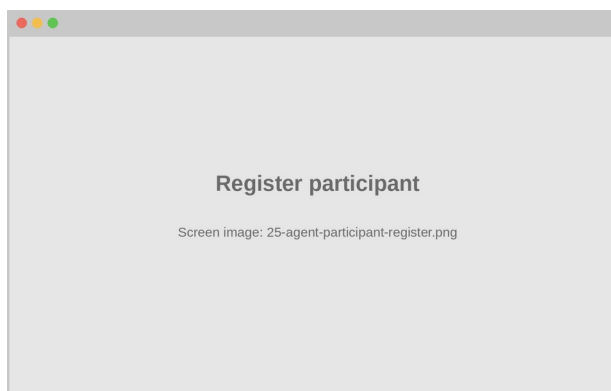


Figure 29: Register participant: Individual or corporate; ID, personal and contact details; AML screening runs on save.

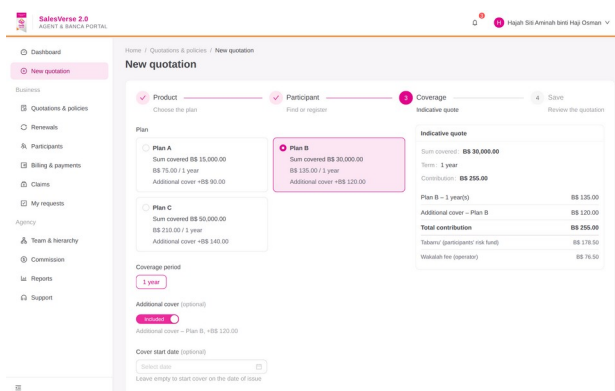


Figure 30: New quotation: coverage: Plan and options for the chosen product; the indicative contribution updates as fields change.

The screenshot shows a 'New quotation' review screen. It has a sidebar with navigation links like Dashboard, New quotation, Business, and Agency. The main content area is divided into three sections: Participant, Coverage, and Indicative quote. The Participant section shows details for 'Abdul Malik bin Ali'. The Coverage section shows 'Professional Takaful Plan' with a 1-year term. The Indicative quote section shows a total contribution of \$250.00, broken down into Plan B (\$125.00) and Additional cover (\$125.00). A pink banner at the bottom indicates the next step: 'Next: complete the application'.

Figure 31: New quotation: review: Participant, coverage and contribution breakdown before the quotation is saved.

The screenshot shows a 'Participants' list table. The table has columns for Participant no., Name, Type, ID, Date of birth, Mobile, AML, and Registered. The data includes several participants with their respective details and AML status (e.g., 'Clear', 'Flagged').

Participant no.	Name	Type	ID	Date of birth	Mobile	AML	Registered
PT026000034	Abdul Malik bin Ali	Individual	****4658	17 Aug 1979	6738130974	Clear	02 Oct 2025
PT026000054	Ahmad Zulkifli bin Harid	Individual	****7389	10 Jan 1974	6738149194	Flagged	02 Oct 2025
PT026000017	Amrul bin Yusof	Individual	****2329	18 Sep 1987	6738115487	Clear	02 Oct 2025
PT026000012	Aqilah binti Rahman	Individual	****1444	13 Apr 1982	6738110852	Clear	02 Oct 2025
PT026000036	Aqilah binti Rahman	Individual	****4352	10 Jan 1981	6738132796	Clear	02 Oct 2025
PT026000029	Ang Shahrul bin Matrusin	Individual	****9373	12 Mar 1974	6738126439	Clear	02 Oct 2025
PT026000023	Daniar bin Zainal	Individual	****3151	15 Jun 1993	6738120953	Clear	02 Oct 2025
PT026000004	Di Nur Syafiqah binti Pg Hashim	Individual	****0548	14 May 1974	6738103644	Clear	02 Oct 2025
PT026000028	Di Nur Syafiqah binti Pg Hashim	Individual	****8826	11 Feb 1973	6738125508	Clear	02 Oct 2025
PT026000043	Hafidz bin Latif	Individual	****5881	17 Aug 1988	6738139173	Clear	02 Oct 2025
PT026000003	Hj Kamarulhuda bin Osman	Individual	****0411	11 Feb 2004	6738102739	Clear	02 Oct 2025
PT026000001	Hjh Rosnah bin Ang Tengah	Individual	****1507	12 Mar 1981	6738133823	Clear	02 Oct 2025

Figure 32: Participants: Own participants with participant number, masked ID and AML status.

The screenshot shows a 'Payment detail' screen. It displays a large heading 'Payment detail' and a subtitle 'Screen image: 33-agent-payment-detail.png'.

Figure 33: Payment detail: Allocation, proof of payment, verification status and the e-Receipts once verified.

The screenshot shows a 'My profile' screen. It displays a large heading 'My profile' and a subtitle 'Screen image: 40-agent-profile.png'.

Figure 34: My profile: Registered profile, portal user name, documents and change requests.

The screenshot shows a 'Support' screen with a table of issues. The table has columns for Issue no., Title, Category, Priority, Status, Assigned to, Response due, and Resolution due. The data includes one issue: 'Unable to upload IC copy from mobile phone'.

Issue no.	Title	Category	Priority	Status	Assigned to	Response due	Resolution due
0206000001	Unable to upload IC copy from mobile phone	Documents	Medium	Assigned	Kenneth Lim	02 Oct 2026, 22:50 - incl	09 Oct 2026, 14:50 - 02

Figure 35: Support: Issues reported by the user with priority, status and SLA targets.

The screenshot shows a 'Report an issue' screen. It displays a large heading 'Report an issue' and a subtitle 'Screen image: 44-agent-report-issue-dialog.png'.

Figure 36: Report an issue: Title, category, priority with definitions and description; attachments after submission.

7.5 Bank officer (banca)

Profile

Who	Branch officer who attaches financing takaful (hire purchase, personal and property financing) to the bank's financing approvals.
Goals	Issue the e-Policy as soon as the financing is approved, so the drawdown is not held up; keep the branch inside the seven-day payment rule.
How often	Daily, in bursts around month-end drawdowns.
Device	Bank workstation on the branch network.

Role and permissions	BANCA_OFFICER: Portal base permissions; an authority limit on the agent record routes larger cases for approval.
Data scope	Own records only.
Demo user	Nurul Huda binti Hassan, Mutiara Islamic Bank, Gadong branch, authority limit B\$250,000 (bk-000004)
RFP requirements covered	AP-01–AP-06, AP-13–AP-47, AP-49–AP-58, AP-60, AP-62, FFR01 (FTPHP01–08), AP-42

Table 17: Bank officer (banca): profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.


SalesVerse 2.0
AGENT & BANCA PORTAL

Search policy, quot... Ctrl K

+ New quotation


 Nurul Huda binti

Dashboard

SALES

Quotations & policies

Renewals

Participants

SERVICING

Billing & payments

Claims

My requests

AGENCY

Team & hierarchy

Commission

Reports

Support

Data scope: Own records only.

Where it leads

→ Work queues, follow-ups, production chart, recent activity (own records)

→ List with filters → Policy detail: checklist, Declarations, Nominees, Signature, Send link, Submit; Renew, Request endorsement, Request cancellation, E-mail documents

→ Policies ending inside the renewal window → Renew → new quotation

→ List → Participant detail (policies, documents, update requests) · Register participant

→ Outstanding → Submit payment (single or bulk, proof) · Payment history → Payment detail, e-Receipts

→ List → Claim detail (documents, IIFT remarks) · Notify claim (validate policy first)

→ Own requests → Request detail, remarks, Withdraw

→ Agency or bank record, hierarchy, team members → Team member

→ Commission and referral fee by period

→ Report catalogue → filters, preview, export (Excel, CSV, PDF)

→ Own issues → Issue detail, conversation, attachments · Report an issue

Figure 37: Bank officer (banca): navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Dashboard	Tiles, work queues, production, recent activity	Starts the day from the queues; opens follow-ups	AP-37, AP-52, AP-53
New quotation (Product, Participant, Coverage, Review)	Four-step wizard with a live indicative contribution	Chooses the product, finds or registers the participant, sets the cover, saves the quotation	AP-17–AP-20, FFR01–05 step 1–3
Quotations & policies	List with search, status, payment status, product and date filters	Finds drafts, pending, rejected and active records	AP-21–AP-23

Screen	Purpose	What this persona does there	RFP IDs
Policy detail	Checklist, overview, Documents, Payments & receipts, History, Approvals, Claims tabs	Completes declarations, nominees, documents and signatures; submits; downloads or e-mails the e-Policy and e-Receipt	AP-24, AP-25, AP-29–AP-36, AP-44–AP-47, AP-62
Participants and Register participant	List, detail with policies, documents and update requests	Registers or finds the participant; requests updates	AP-11, AP-13–AP-16, AP-48
Billing & payments and Payment detail	Outstanding, overdue, due soon, pending verification; payment history	Submits single or bulk payment with proof; downloads e-Receipts	AP-30, AP-37–AP-42
Claims and Notify claim	List, validation of the policy number, event details, documents	Notifies a claim and follows IIFT's remarks	AP-43
My requests	Own (or team) maker-checker requests	Reads the decision and remarks; withdraws a pending request	AP-49–AP-51
Team & hierarchy	Agency or bank record and reporting line	Sees own position and reporting line	AP-09, AP-10
Commission	Accrued and paid commission by period	Checks the statement	INT-05
Reports	Role-scoped reports, preview, export	Runs production and payment reports	AP-58
Support and Report an issue	Own issues with SLA targets	Reports a problem with attachments; confirms or reopens the resolution	AP-55–AP-57
My profile	Profile, portal user name, documents, change requests	Requests contact changes (approved by IIFT)	AP-05, AP-06
Notifications	In-app notifications with links	Opens the record behind a notification	AP-34, AP-54

Table 18: Bank officer (banca): screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

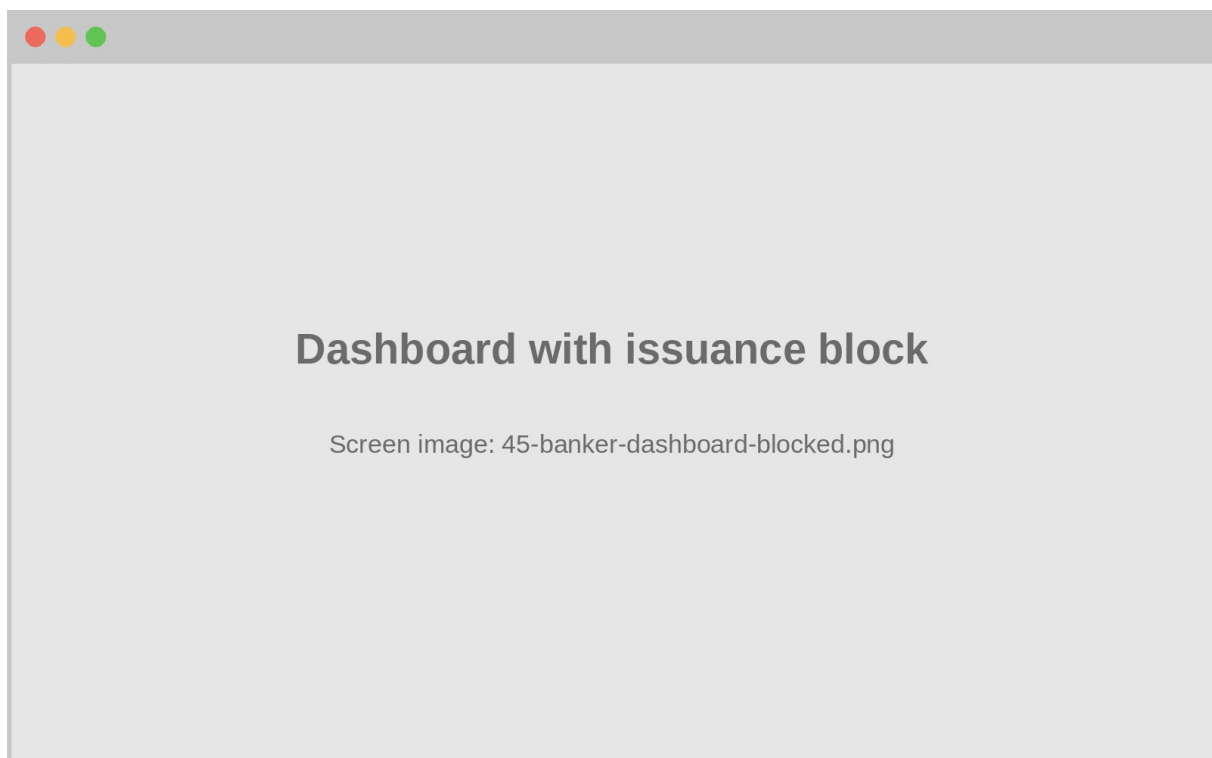


Figure 38: Dashboard with issuance block: The seven-day grace period has lapsed for the bank: new business is blocked until payment is submitted.

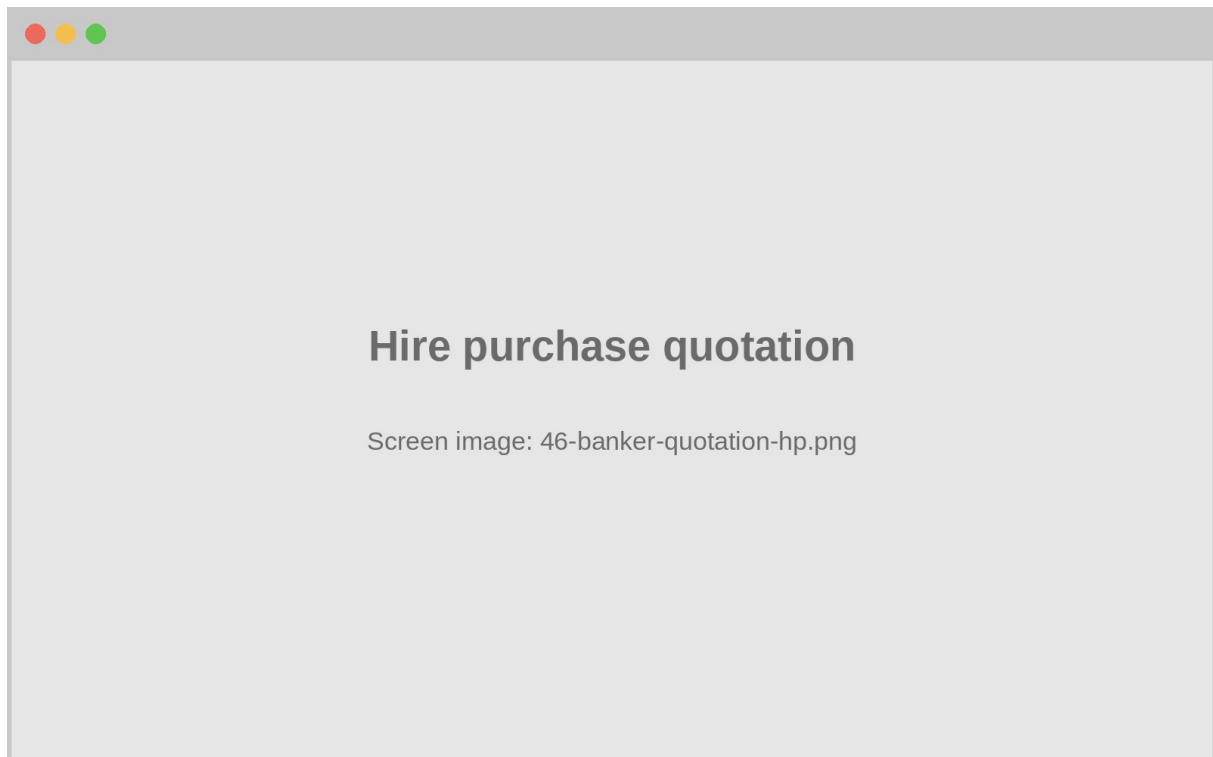


Figure 39: Hire purchase quotation: Financing amount, period, profit rate, financier and reference; above B\$150,000 the case is referred.

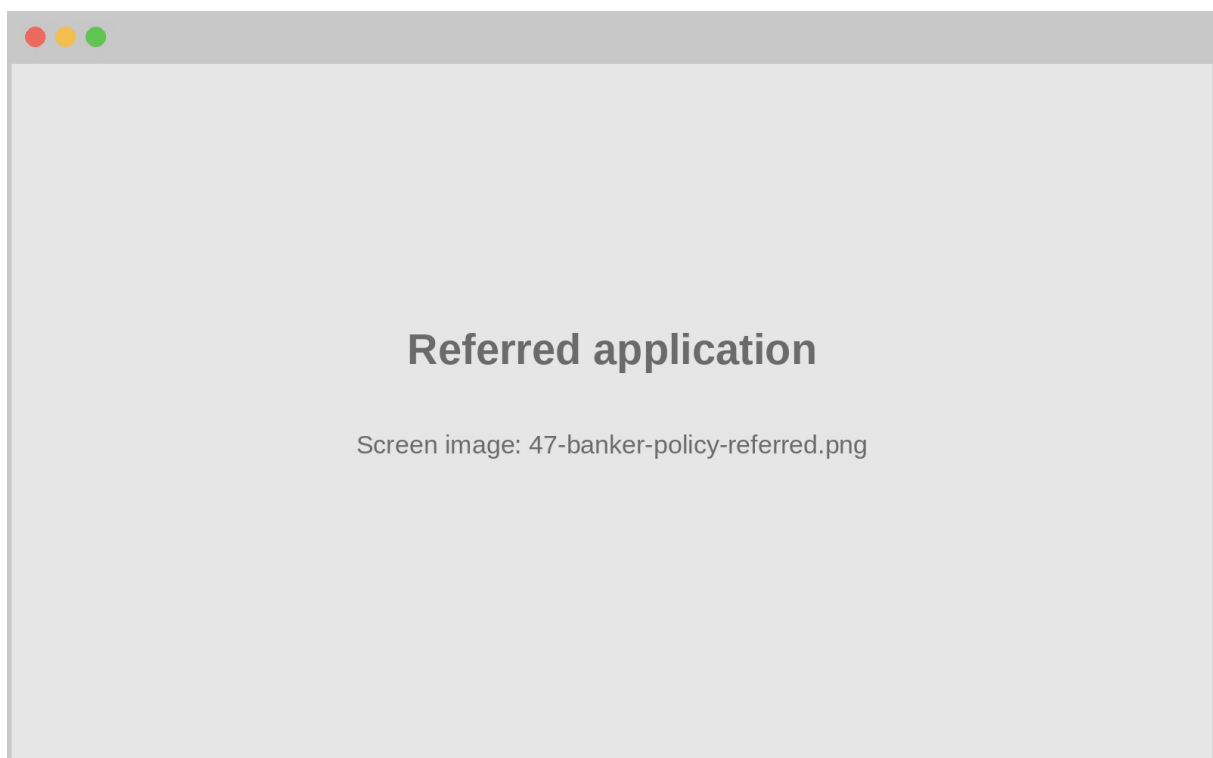


Figure 40: Referred application: Awaiting the IIFT underwriting decision; the referral reasons are shown on the overview.

7.6 Bank supervisor (banca)

Profile

Who	Head of bancassurance at the partner bank; registers officers, settles the bank's contributions in bulk and watches production per officer.
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Goals	One bank transfer for many policies; no issuance block; new officers active within a day.
How often	Daily review; weekly bulk payment.
Device	Bank workstation.
Role and permissions	BANCA_SUPERVISOR: Portal base permissions plus bank-wide visibility and officer registration.
Data scope	Every record of the bank, all branches.
Demo user	Pengiran Khairul Anwar bin Pengiran Ismail, Mutiara Islamic Bank, main branch (bk-000005)
RFP requirements covered	AP-07–AP-10, AP-12, AP-37–AP-42, AP-58, AP-59, plus the bank officer's set

Table 19: Bank supervisor (banca): profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.


SalesVerse 2.0
AGENT & BANCA PORTAL

Search polic... Ctrl K

+ New quotation

 3

 Pengiran Khairul Anwar

Dashboard

SALES

Quotations & policies

Renewals

Participants

SERVICING

Billing & payments

Claims

My requests

AGENCY

Team & hierarchy

Commission

Reports

Support

Data scope: Every record of the bank, all branches.

Where it leads

→ Work queues, follow-ups, production chart, recent activity (whole agency or bank)

→ List with filters → Policy detail: checklist, Declarations, Nominees, Signature, Send link, Submit; Renew, Request endorsement, Request cancellation, E-mail documents

→ Policies ending inside the renewal window → Renew → new quotation

→ List → Participant detail (policies, documents, update requests) · Register participant

→ Outstanding → Submit payment (single or bulk, proof) · Payment history → Payment detail, e-Receipts

→ List → Claim detail (documents, IIFT remarks) · Notify claim (validate policy first)

→ Own requests and the team's → Request detail, remarks, Withdraw

→ Agency or bank record, hierarchy, team members → Team member · Register agent / bank officer

→ Commission and referral fee by period, whole agency

→ Report catalogue → filters, preview, export (Excel, CSV, PDF)

→ Own issues → Issue detail, conversation, attachments · Report an issue

Figure 41: Bank supervisor (banca): navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Dashboard	Tiles, work queues, production, recent activity	Starts the day from the queues; opens follow-ups	AP-37, AP-52, AP-53

Screen	Purpose	What this persona does there	RFP IDs
New quotation (Product, Participant, Coverage, Review)	Four-step wizard with a live indicative contribution	Chooses the product, finds or registers the participant, sets the cover, saves the quotation	AP-17–AP-20, FFR01–05 step 1–3
Quotations & policies	List with search, status, payment status, product and date filters	Finds drafts, pending, rejected and active records	AP-21–AP-23
Policy detail	Checklist, overview, Documents, Payments & receipts, History, Approvals, Claims tabs	Completes declarations, nominees, documents and signatures; submits; downloads or e-mails the e-Policy and e-Receipt	AP-24, AP-25, AP-29–AP-36, AP-44–AP-47, AP-62
Participants and Register participant	List, detail with policies, documents and update requests	Registers or finds the participant; requests updates	AP-11, AP-13–AP-16, AP-48
Billing & payments and Payment detail	Outstanding, overdue, due soon, pending verification; payment history	Submits single or bulk payment with proof; downloads e-Receipts	AP-30, AP-37–AP-42
Claims and Notify claim	List, validation of the policy number, event details, documents	Notifies a claim and follows IIFT's remarks	AP-43
My requests	Own (or team) maker-checker requests	Reads the decision and remarks; withdraws a pending request	AP-49–AP-51
Team & hierarchy and Register agent	Agency or bank record, hierarchy, members	Registers sub-agents or bank officers; follows their approval	AP-07–AP-10, AP-12, AP-46, AP-48
Commission	Accrued and paid commission by period	Checks the statement	INT-05
Reports	Role-scoped reports, preview, export	Runs production and payment reports	AP-58
Support and Report an issue	Own issues with SLA targets	Reports a problem with attachments; confirms or reopens the resolution	AP-55–AP-57
My profile	Profile, portal user name, documents, change requests	Requests contact changes (approved by IIFT)	AP-05, AP-06
Notifications	In-app notifications with links	Opens the record behind a notification	AP-34, AP-54

Table 20: Bank supervisor (banca): screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

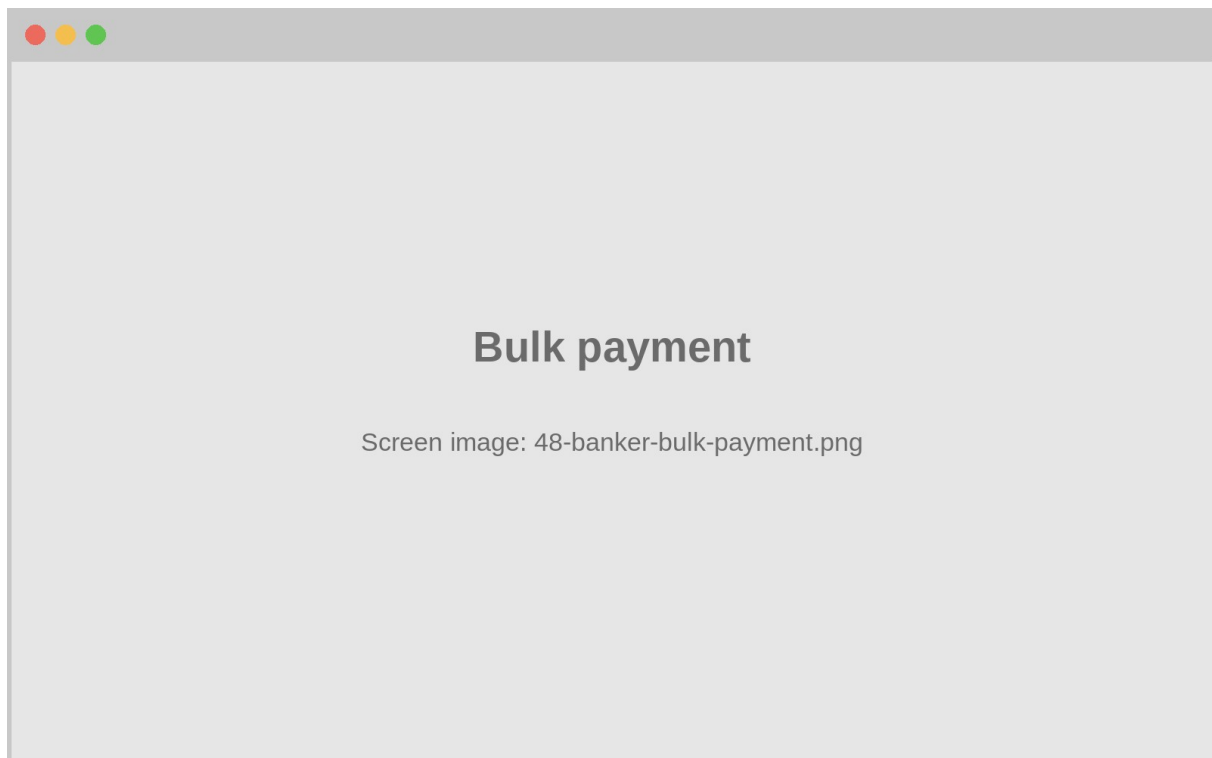


Figure 42: Bulk payment: Several policies settled with one bank transfer; amounts allocated per policy.

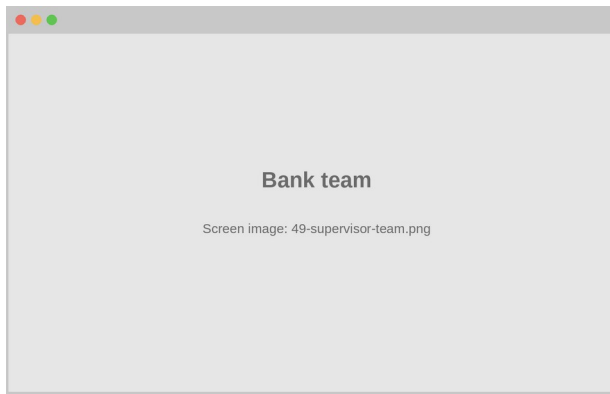


Figure 43: Bank team: Bank-wide hierarchy and officers, with the Register bank officer action.

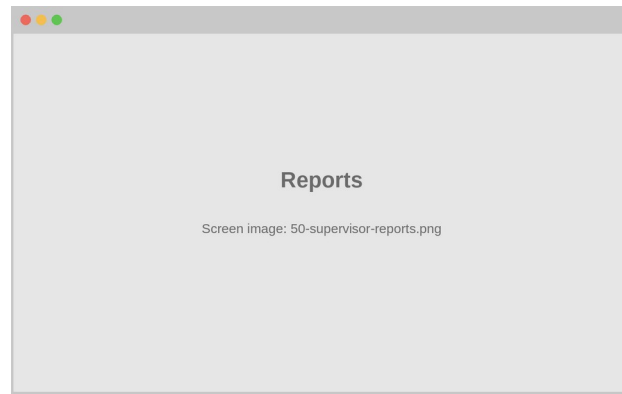


Figure 44: Reports: Role-scoped reports with filters, preview and export.

7.7 Operations officer (maker)

Profile

Who	IIFT Banca and agency administration: registers and maintains agents, agencies and banks, checks documents, handles claim notifications and portal issues.
Goals	Clean, complete records before a checker sees them; nothing waiting in the document queue overnight.
How often	All day.
Device	IIFT workstation on the internal network.
Role and permissions	OPERATIONS_OFFICER: Back-office read set plus agencies.manage, agents.manage, documents.verify, claims.manage, issues.manage. No approval permission: a maker never decides a request.
Data scope	All agencies and banks.
Demo user	Rosmawati binti Abdul Karim (ops.maker) (ops.maker)
RFP requirements covered	BO-05–BO-09, BO-11, BO-12, BO-21, BO-22–BO-24, BO-29, BO-30, AP-43

Table 21: Operations officer (maker): profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.

IIFT

SalesVerse 2.0

BACK-OFFICE

Ctrl K

3

R

Rosmawati binti Abdul

Dashboard

DISTRIBUTION

Agents & bankers

Agencies & banks

BUSINESS

Policies

Participants

Payments

Claims

CONTROL

Document checks

OPERATIONS

Issues

Reports

Data scope: All agencies and banks.

Where it leads

→ Work queues for this role; management tiles for Management

→ Search → Agent detail (documents, hierarchy, AML, approvals) · Register · Request profile update · Change status

→ List → Agency or bank detail, agents, issuance block · Add, Edit, Lift issuance block

→ Search across agencies → Policy record (read-only view of the portal page)

→ Search → Participant record, policies, documents, AML screening history

→ Search → Payment detail, proof, e-Receipts, verification request

→ List → Claim detail → Update status with remarks

→ Uploaded documents awaiting verification → Verify or Reject with reason

→ All issues → Issue detail: assign, priority, status, conversation

→ Run a report with filters and export

Figure 45: Operations officer (maker): navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Dashboard	Queues: agent registrations, documents to verify, payments, open claims, open issues	Picks the next piece of work	BO-21
Agents & bankers, Agent detail, Register agent or bank officer	Search, profile, documents, hierarchy, AML, approvals	Registers agents (maker), requests profile updates and status changes, uploads documents	BO-05–BO-08, BO-10, BO-11
Agencies & banks, Agency detail	Agency and bank records, agents, issuance block	Adds and edits agencies and banks; lifts a block with a reason	BO-09, AP-42
Document checks	Uploaded documents awaiting verification	Verifies or rejects with a reason; sees expiry	BO-11, BO-12
Policies, Participants, Payments	Read-only search and records	Looks up a record while handling a query	BO-06, AP-14, AP-23
Claims, Claim detail	Claim notifications and their status	Moves the claim to Under review, Acknowledged, Rejected or Closed with remarks	AP-43
Issues, Issue detail	Portal and back-office issues	Assigns, prioritises and answers issues	BO-29, BO-30
Reports	Operational reports	Agent register, document expiry	BO-22–BO-24

Table 22: Operations officer (maker): screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

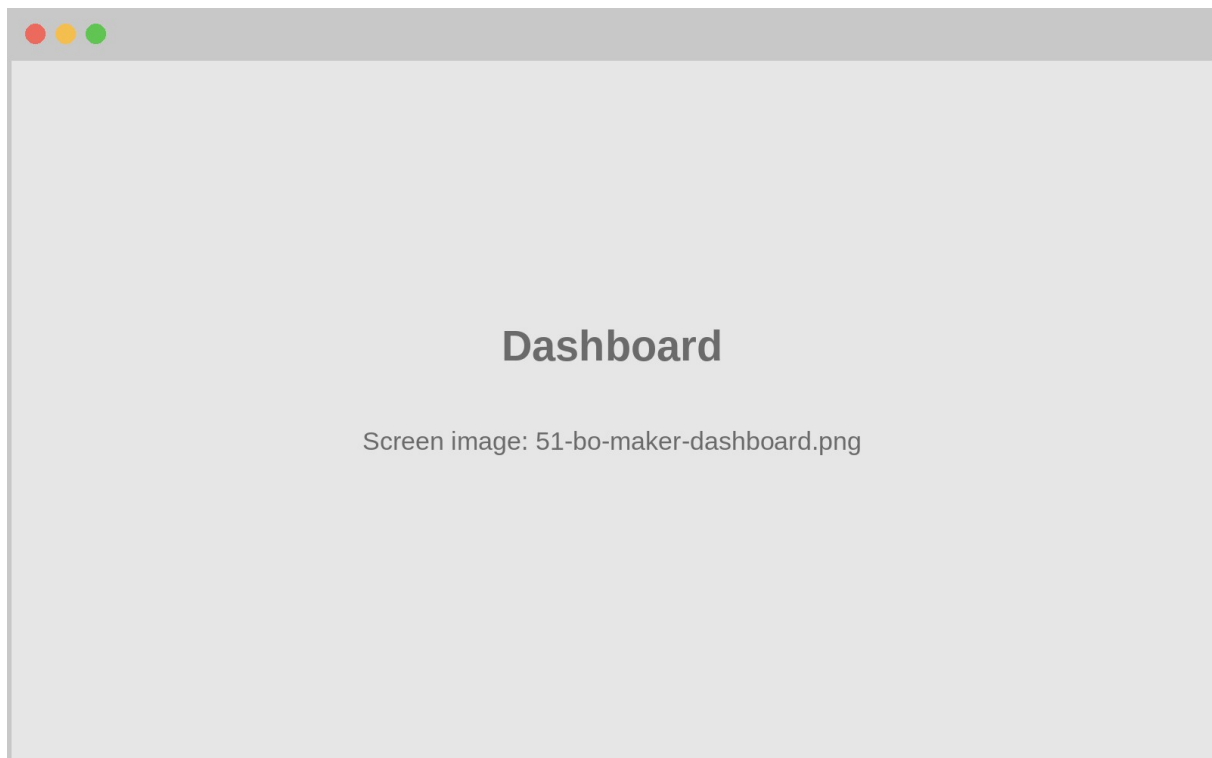


Figure 46: Dashboard: Work queues for registrations, documents, payments, claims and issues.

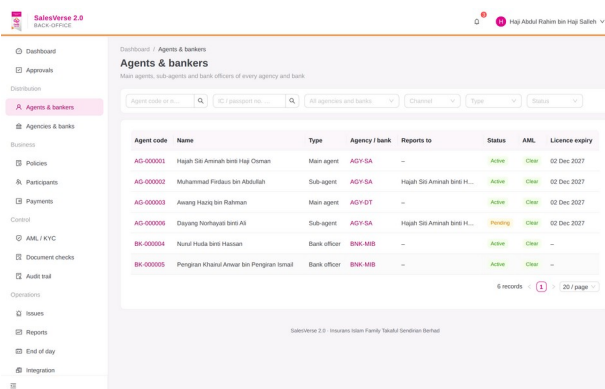


Figure 47: Agents & bankers: Search by code, name, ID number, agency or bank, channel, type and status.

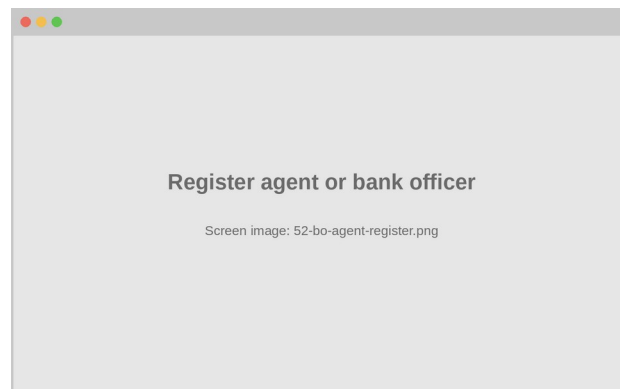


Figure 48: Register agent or bank officer: Registration under any agency or bank, with authority limit and reporting line; goes to approval.

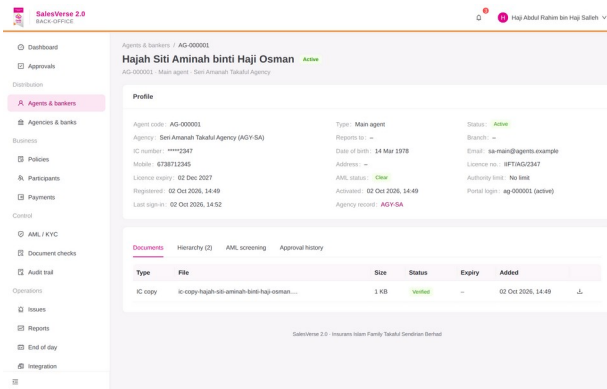


Figure 49: Agent detail (pending): Profile, documents, hierarchy, AML screening and approval history; maker actions in the header.

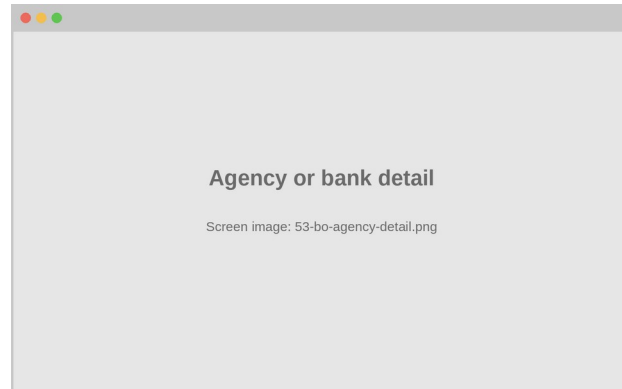


Figure 50: Agency or bank detail: Details, active agents and the issuance block with its reason.

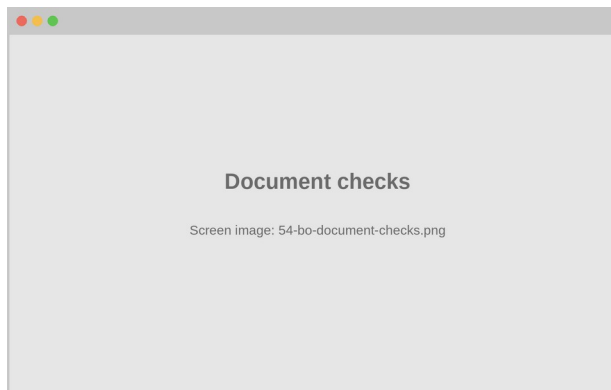


Figure 51: Document checks: Uploaded documents awaiting verification, oldest first, with expiry and reject reason.

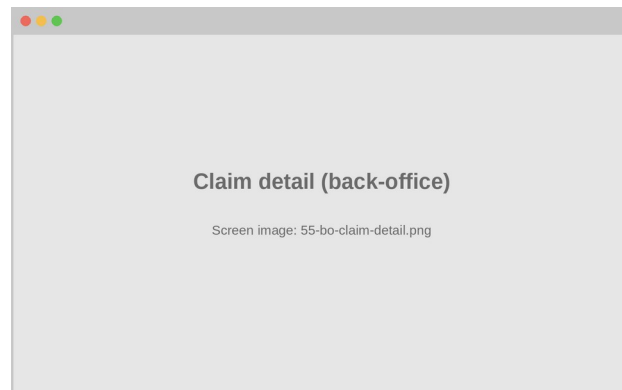


Figure 52: Claim detail (back-office): Status moves Submitted, Under review, Acknowledged or Rejected, Closed; remarks go to the agent.

7.8 Operations supervisor (checker)

Profile

Who	Approves what the maker and the portal submit: registrations, profile and status changes, participant updates, endorsements and cancellations.
Goals	Decide every request the same day, with the evidence on one page; reject with a reason the submitter can act on.
How often	Several times a day; the dashboard shows what is waiting.
Device	IIFT workstation.
Role and permissions	OPERATIONS_SUPERVISOR: Read set plus approve.agents, approve.participants, approve.servicing, documents.verify.
Data scope	All agencies and banks.
Demo user	Mohammad Hafiz bin Yusof (ops.checker) (ops.checker)
RFP requirements covered	BO-10, BO-16–BO-19, BO-21, BO-11, AP-15, AP-27, AP-28, AP-61

Table 23: Operations supervisor (checker): profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.

IIFT
SalesVerse 2.0
BACK-OFFICE

Search policy, participant or a...
Ctrl K

3
M
Mohammad Hafiz bin

Dashboard

Approvals

DISTRIBUTION

Agents & bankers

Agencies & banks

BUSINESS

Policies

Participants

Payments

CONTROL

Document checks

OPERATIONS

Reports

Data scope: All agencies and banks.

Where it leads

→ Work queues for this role; management tiles for Management

→ Awaiting my decision / All requests → Approval detail → Approve or Reject

→ Search → Agent detail (documents, hierarchy, AML, approvals)

→ List → Agency or bank detail, agents, issuance block

→ Search across agencies → Policy record (read-only view of the portal page)

→ Search → Participant record, policies, documents, AML screening history

→ Search → Payment detail, proof, e-Receipts, verification request

→ Uploaded documents awaiting verification → Verify or Reject with reason

→ Run a report with filters and export

Figure 53: Operations supervisor (checker): navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Dashboard	Awaiting my approval, pending requests by type	Opens the oldest request	BO-21
Approvals (Awaiting my decision, All requests)	Maker-checker inbox and history	Works the inbox; searches past requests by number, type, status	BO-16, BO-18
Approval detail	Request, submitted details, supporting evidence (AML, documents), approval history	Approves with optional remarks or rejects with a mandatory reason	BO-10, BO-17, BO-19, AP-61
Agents & bankers, Agencies & banks	Records behind a request	Cross-checks the registration	BO-05–BO-09
Document checks	Documents awaiting verification	Verifies the IC copy before approving a registration	BO-11, BO-12
Policies, Participants, Payments	Read-only records	Checks the policy before deciding an endorsement or cancellation	AP-27, AP-28
Reports	Approval turnaround	Reviews ageing of requests	BO-22

Table 24: Operations supervisor (checker): screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

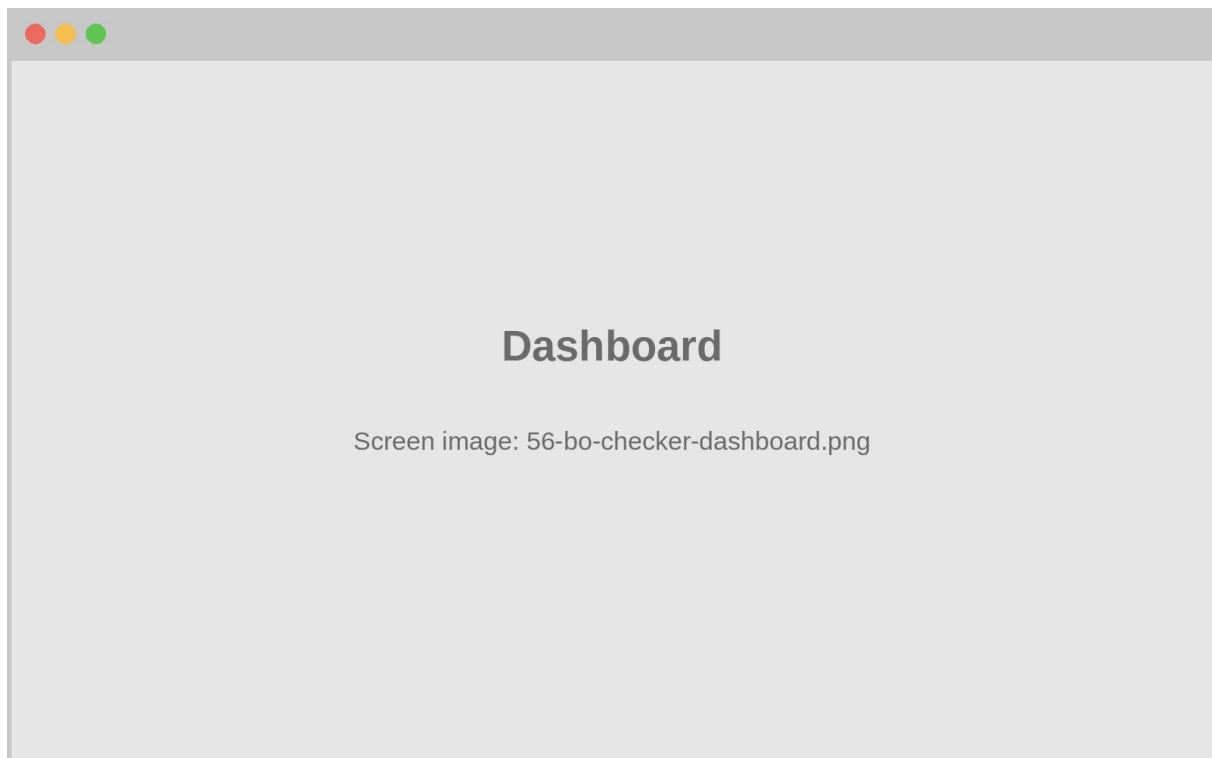


Figure 54: Dashboard: Requests awaiting the checker's decision and pending requests by type.

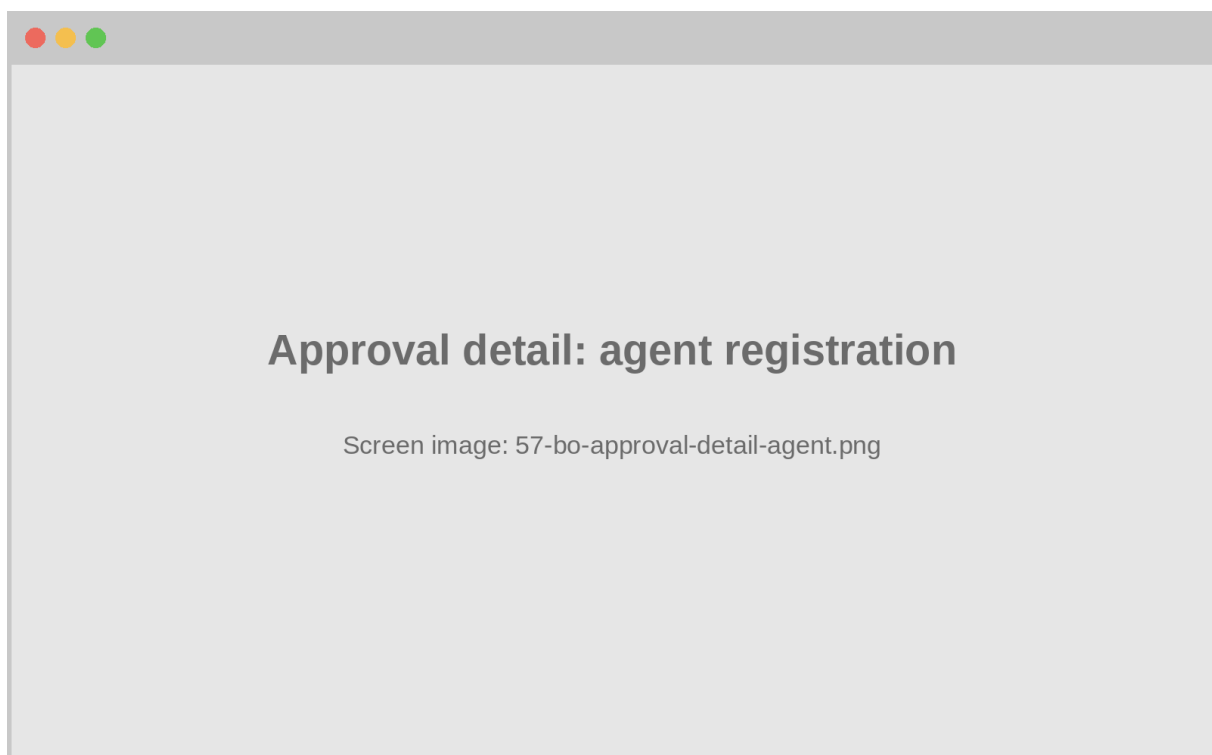


Figure 55: Approval detail: agent registration: Submitted details, AML outcome and documents as evidence, approval history on the right.

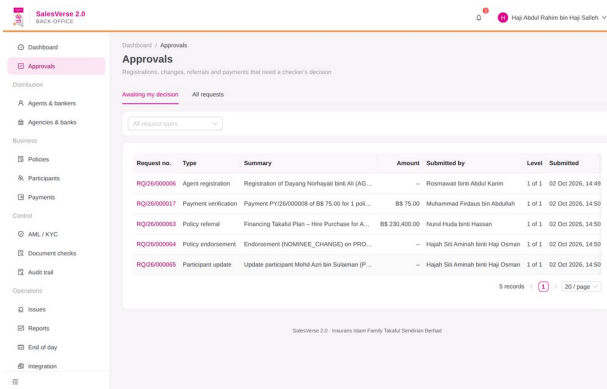


Figure 56: Approvals inbox: Awaiting my decision and All requests; never shows the checker's own requests.

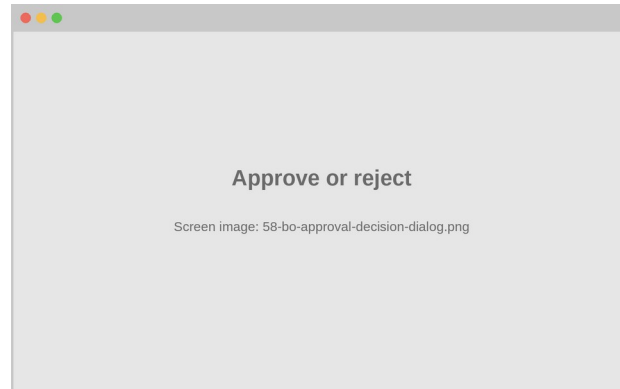


Figure 57: Approve or reject: Remarks optional on approval, mandatory on rejection; the reason is shown to the maker.

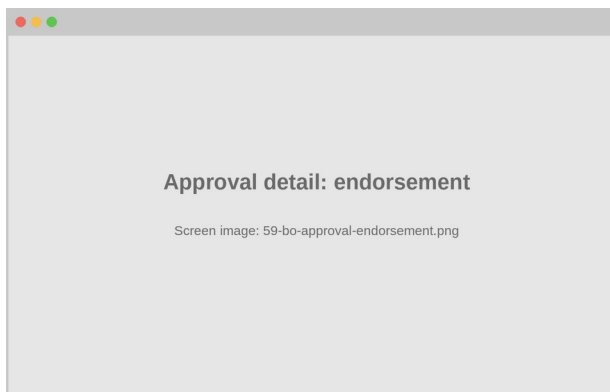


Figure 58: Approval detail: endorsement: Nominee change with the new nominees shown before it is applied to the policy.

7.9 Underwriter / quality check

Profile

Who	Decides referred quotations: health declarations answered Yes, financing above B\$150,000, cases above an officer's authority limit, and the quality check before a financing contract is issued.
Goals	Accept good risk quickly; see the declarations and documents without leaving the request.
How often	Daily.
Device	IIFT workstation.
Role and permissions	UNDERWRITER: Read set plus approve.policies and documents.verify.
Data scope	All policies.
Demo user	Dr. Liyana binti Ahmad (underwriter) (underwriter)
RFP requirements covered	AP-32, AP-33, AP-60, AP-61, BO-16–BO-19, FFR01 (FTPHP06)

Table 25: Underwriter / quality check: profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.


SalesVerse 2.0
BACK-OFFICE

Search policy, participant or a... Ctrl K



Dr. Liyana binti

Dashboard

Approvals

DISTRIBUTION

Agents & bankers

Agents & banks

BUSINESS

Policies

Participants

Payments

CONTROL

Document checks

OPERATIONS

Reports

Data scope: All policies.

Where it leads

→ Work queues for this role; management tiles for Management

→ Awaiting my decision / All requests → Approval detail → Approve or Reject

→ Search → Agent detail (documents, hierarchy, AML, approvals)

→ List → Agency or bank detail, agents, issuance block

→ Search across agencies → Policy record (read-only view of the portal page)

→ Search → Participant record, policies, documents, AML screening history

→ Search → Payment detail, proof, e-Receipts, verification request

→ Uploaded documents awaiting verification → Verify or Reject with reason

→ Run a report with filters and export

Figure 59: Underwriter / quality check: navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Dashboard	Referred quotations awaiting decision	Opens the queue	BO-21
Approvals, Approval detail	Referral reasons, sum covered, contribution, policy documents	Approves (level 1, and level 2 from B\$300,000 sum covered by a second underwriter) or rejects with remarks that the agent sees	AP-33, AP-60, AP-61, BO-16–BO-19
Policies, Policy record	Cover, declarations, risk details, nominees, documents, history	Reads the full application from the request	AP-24, AP-31
Document checks	Documents on policies	Verifies the proposal form or medical evidence	BO-11
Participants	Participant record and AML screening history	Checks the participant	AP-14
Reports	Referral and issuance reports	Reviews referrals by product and reason	BO-22

Table 26: Underwriter / quality check: screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

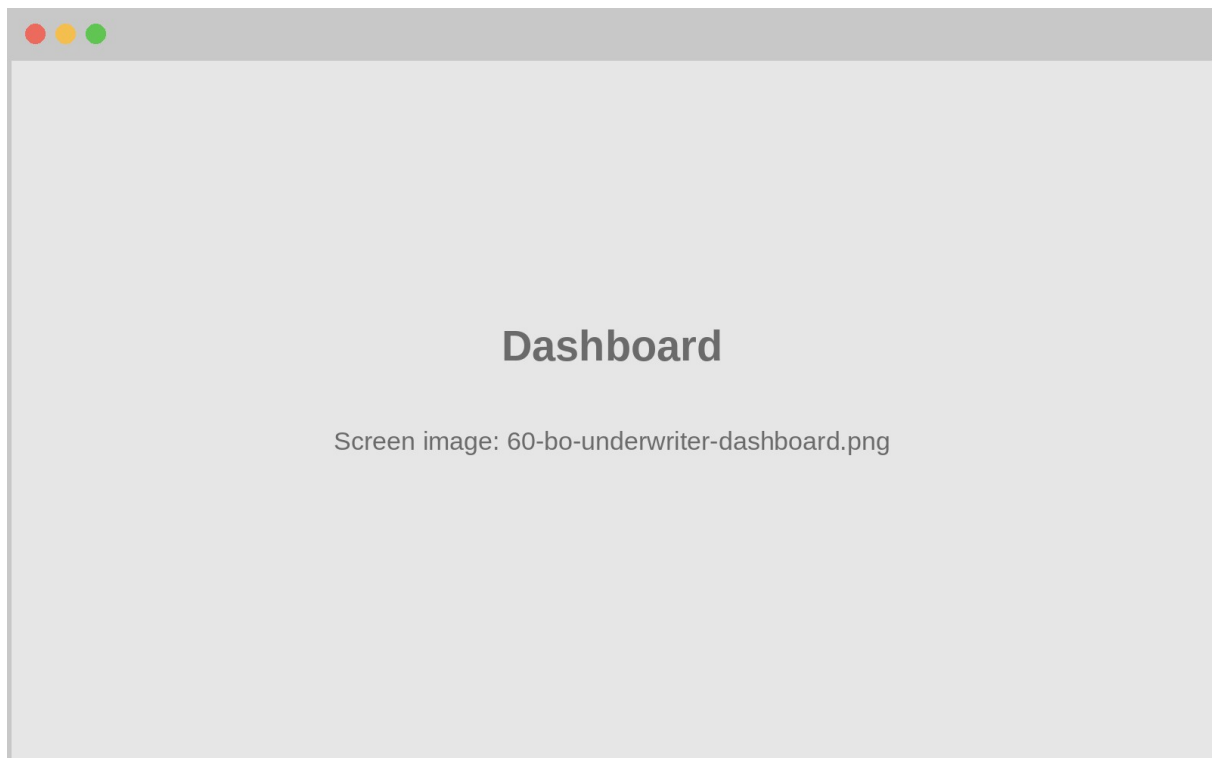


Figure 60: Dashboard: Referred quotations awaiting underwriting or quality check.

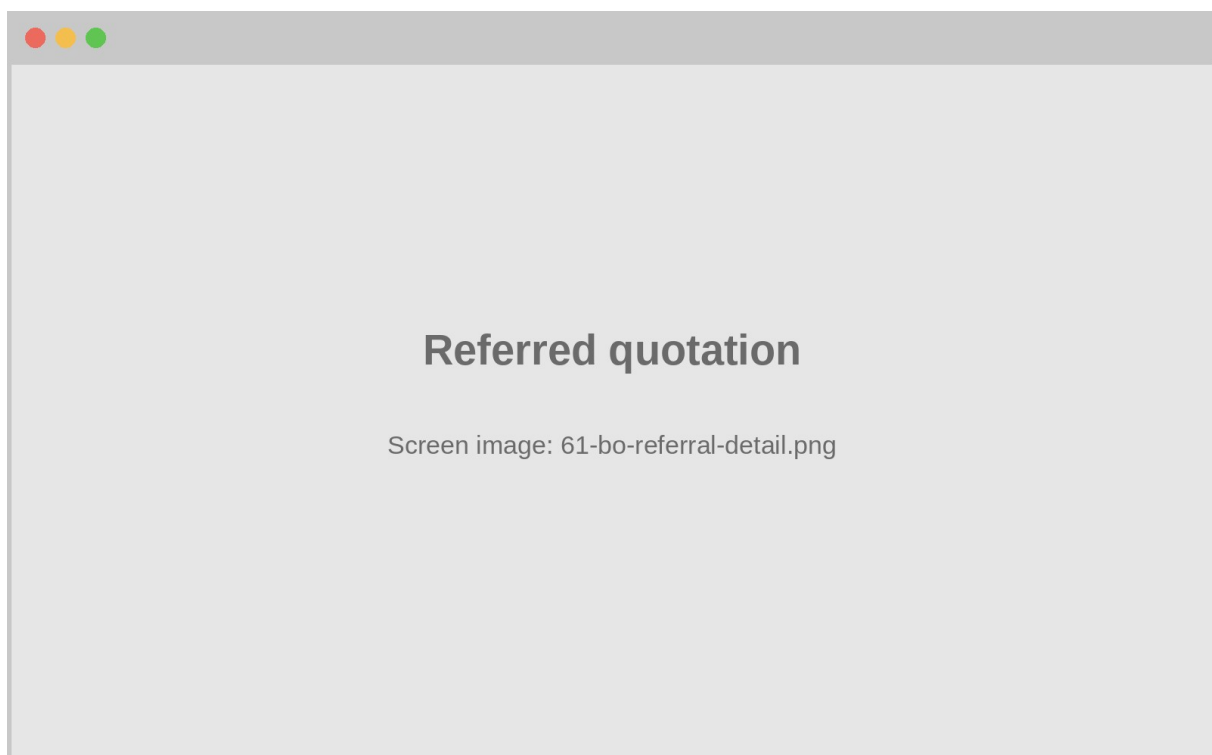


Figure 61: Referred quotation: Referral reasons, sum covered and contribution, policy documents and the second-level rule.

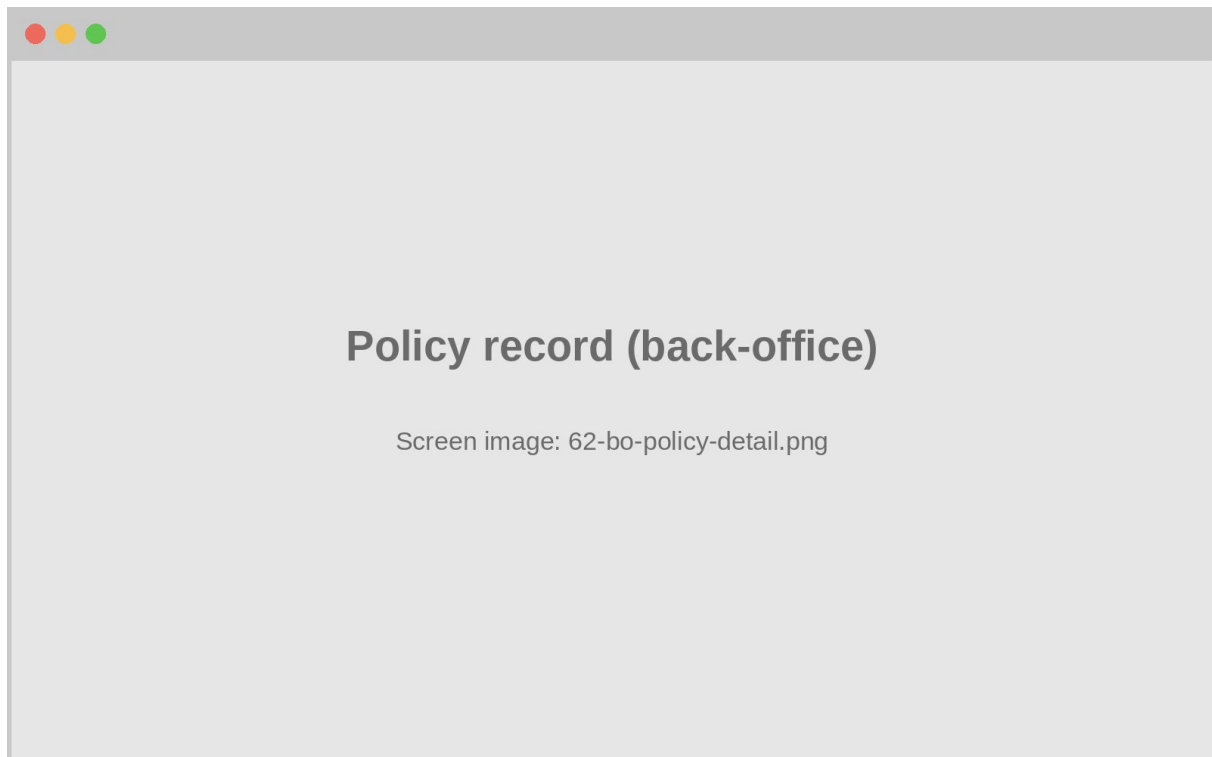


Figure 62: Policy record (back-office): Full record opened from the request: cover, participant, declarations, documents, history.

7.10 Finance officer

Profile

Who	Verifies payment proofs against the bank statement, issues e-Receipts, runs end of day and checks the FIN postings.
Goals	Every receipt matched; the day closed with a clean reconciliation; no agency blocked for a payment that has in fact arrived.
How often	Daily, with a fixed end-of-day routine.
Device	IIFT workstation.
Role and permissions	FINANCE_OFFICER: Read set plus approve.payments, eod.run, integration.manage, reports.schedule.
Data scope	All payments and runs.
Demo user	Faizal bin Haji Omar (finance) (finance)
RFP requirements covered	AP-40, AP-41, AP-42, AP-44, INT-04, INT-11, INT-13–INT-15, BO-25, FFR02 (APHA07), FFR03 (APPT08)

Table 27: Finance officer: profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.


SalesVerse 2.0
BACK-OFFICE

Search policy, participant or a... Ctrl K



Faizal bin Haji

Dashboard

Approvals

DISTRIBUTION

Agents & bankers

Agencies & banks

BUSINESS

Policies

Participants

Payments

OPERATIONS

Reports

End of day

Integration

Data scope: All payments and runs.

Where it leads

→ Work queues for this role; management tiles for Management

→ Awaiting my decision / All requests → Approval detail → Approve or Reject

→ Search → Agent detail (documents, hierarchy, AML, approvals)

→ List → Agency or bank detail, agents, issuance block

→ Search across agencies → Policy record (read-only view of the portal page)

→ Search → Participant record, policies, documents, AML screening history

→ Search → Payment detail, proof, e-Receipts, verification request

→ Run a report with filters and export · Scheduled reports

→ Runs by business date → Run end of day · EOD report · FIN file

→ Health per system · Message queue (re-submit) · Integration log · Reconciliation

Figure 63: Finance officer: navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Dashboard	Payments to verify with their total; integration dead letters	Starts with the oldest payment	BO-21
Approvals, Approval detail	Payment verification requests with proof of payment	Verifies (approves) or rejects; approval issues e-Receipts and posts to FIN	AP-40, AP-41, BO-16–BO-19
Payments, Payment detail	Payments by agency or bank, status, e-Receipts	Reviews a payment and its receipts; jumps to the request	AP-41, AP-44
End of day	Runs per business date, EOD report, FIN file	Runs or re-runs a date; downloads the files	FFR01 FTPHP08, FFR02 APHA07, FFR03 APPT08, FFR04 APKT07, FFR05 APAOS08, INT-04
Integration (Message queue, Integration log, Reconciliation)	Interface health, retries, dead letters, receipt reconciliation	Re-submits a message; runs reconciliation; investigates mismatches	INT-11, INT-13–INT-15
Reports, Scheduled reports	Finance reports, schedules with recipients	Schedules the receipts report to Finance	BO-22–BO-25
Policies, Participants, Agencies & banks	Read-only records	Checks a blocked agency or an outstanding policy	AP-30, AP-42

Table 28: Finance officer: screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

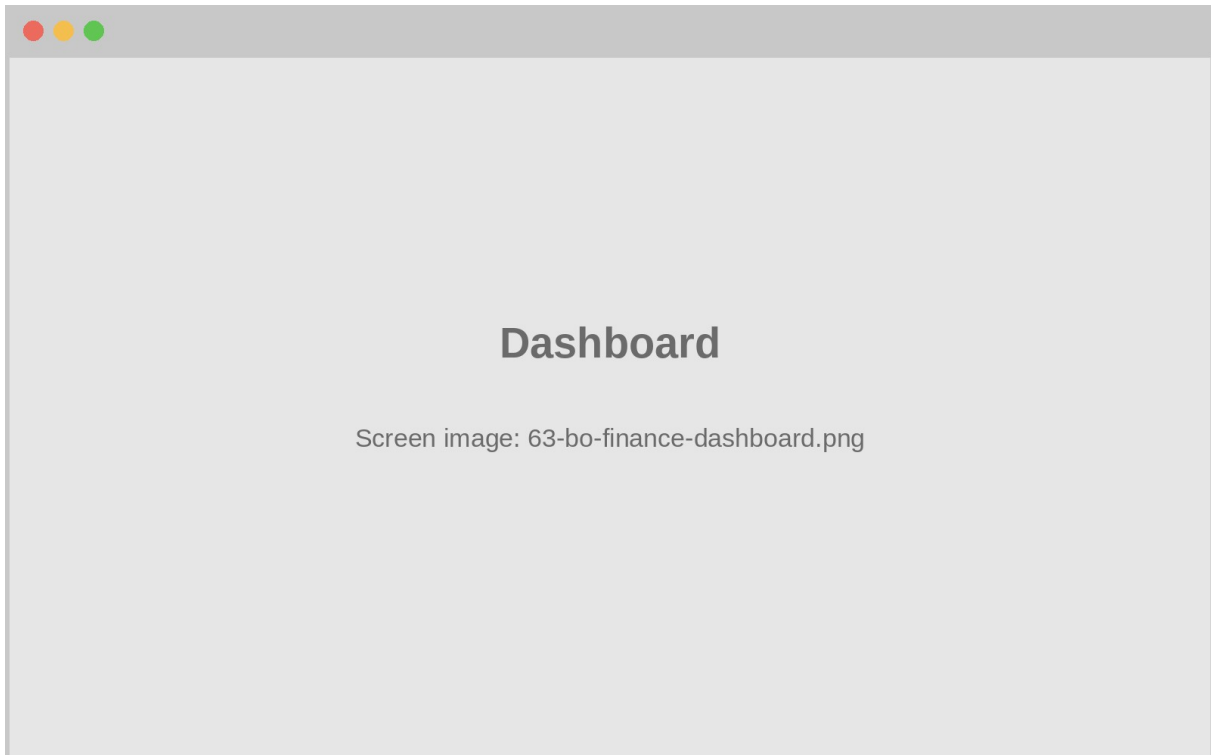


Figure 64: Dashboard: Payments to verify with their total, plus integration dead letters.

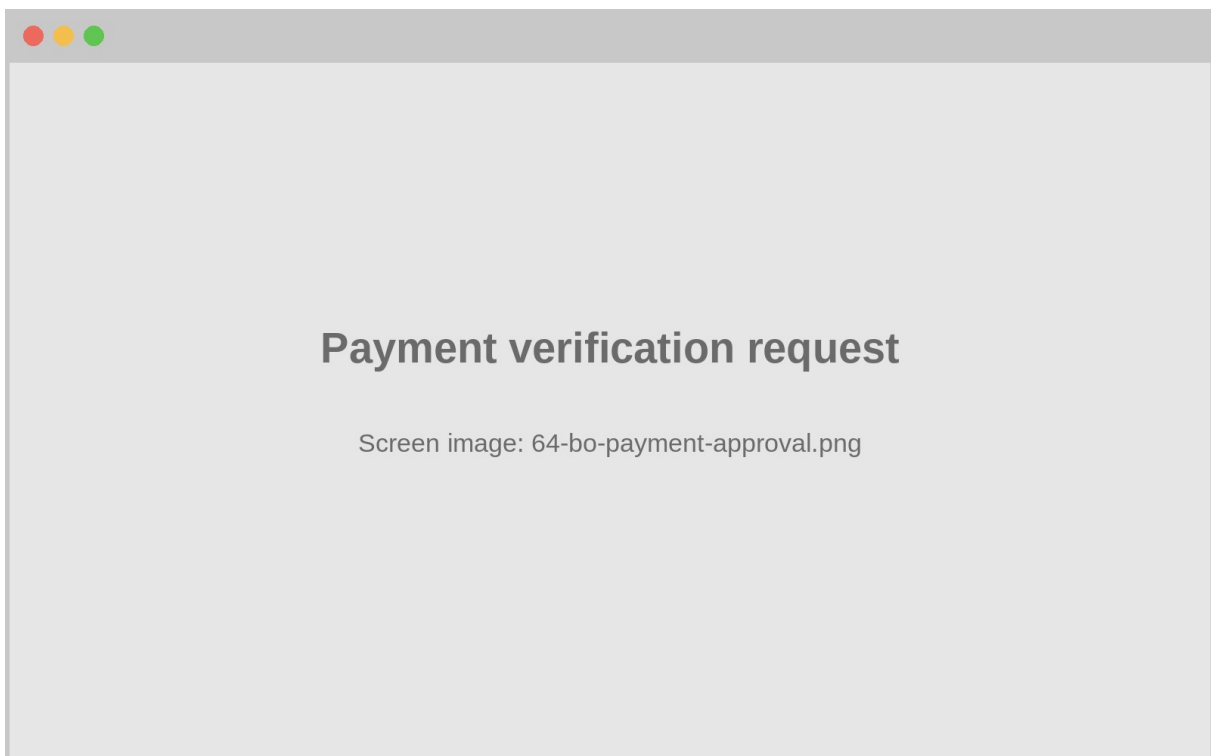


Figure 65: Payment verification request: Allocation, bank reference and the proof of payment under Supporting evidence.

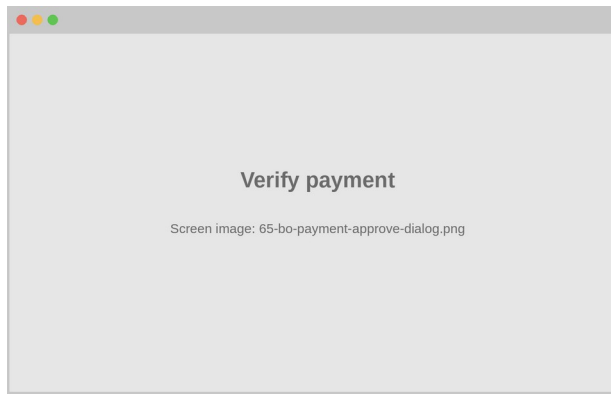


Figure 66: Verify payment: Approval issues one e-Receipt per policy and posts them to FIN.

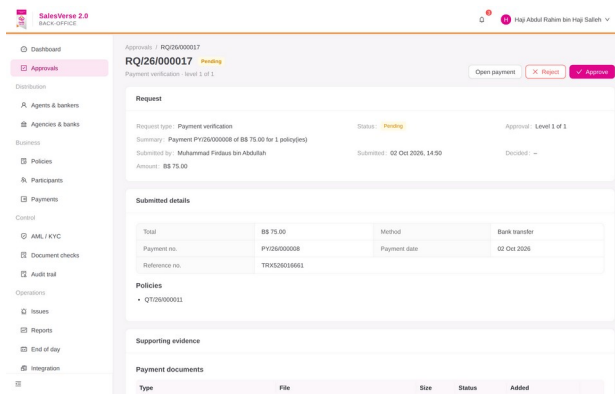


Figure 67: Payment detail (back-office): Payment, allocation, e-Receipts and a shortcut to the verification request.

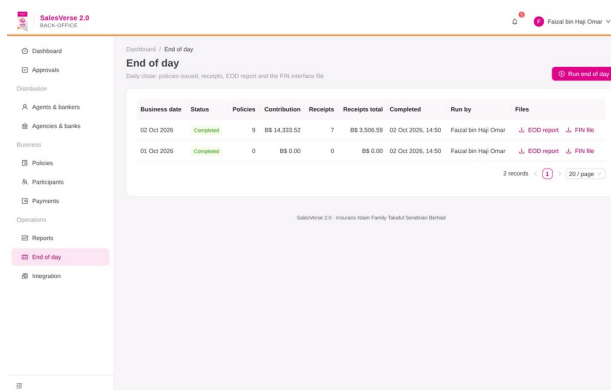


Figure 68: End of day: Runs per business date with policies, contribution, receipts and the EOD report and FIN file.

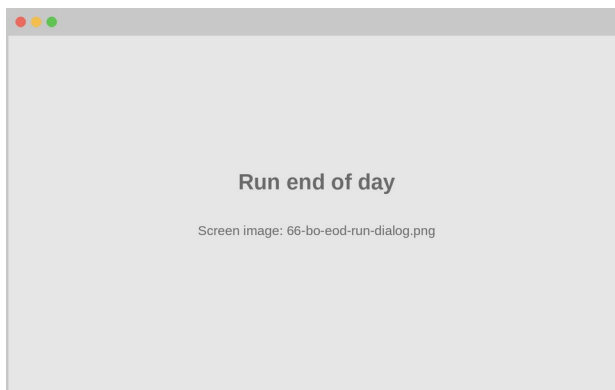


Figure 69: Run end of day: Business date chosen by Finance; a re-run replaces the day's report and FIN posting.

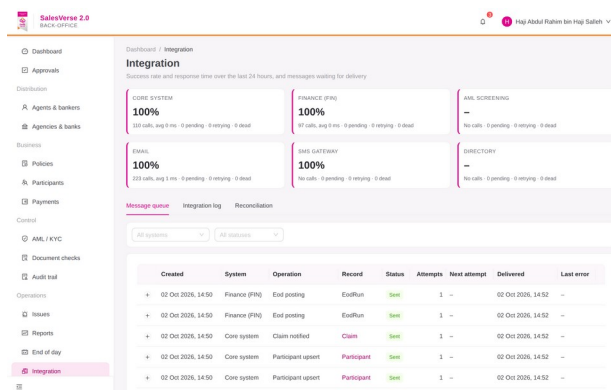


Figure 70: Integration: Health per system, message queue with re-submit, call log and reconciliation.

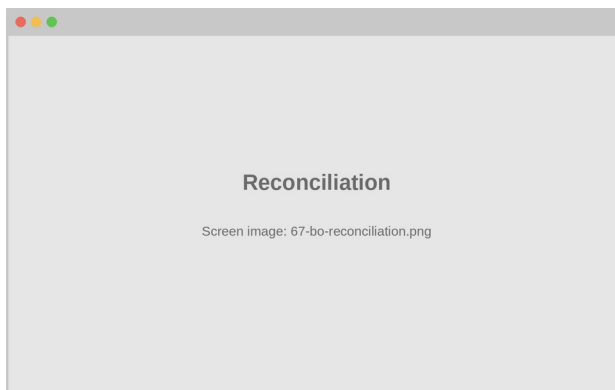


Figure 71: Reconciliation: Receipts issued matched against FIN postings; mismatches list the unmatched receipt numbers.

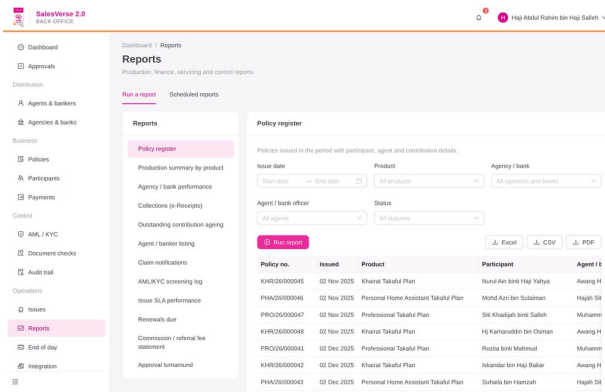


Figure 72: Reports: Report catalogue, filters, preview and export to Excel, CSV or PDF.

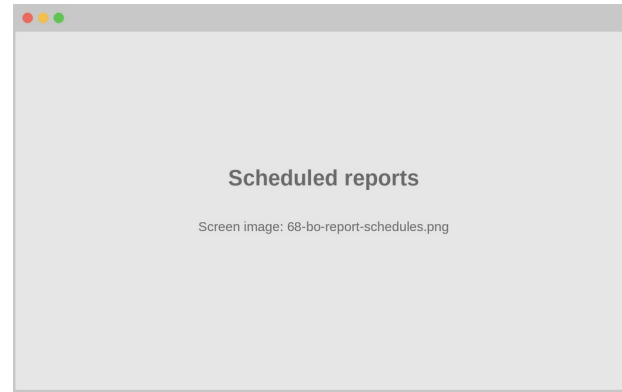


Figure 73: Scheduled reports: Frequency, data period, format and recipients; pause and resume.

7.11 Compliance officer

Profile

Who	Reviews possible watch-list matches on participants and agents and maintains the watch-lists.
Goals	Clear false positives the same day so sales are not held; confirm true matches so the record is closed everywhere at once.
How often	Daily, usually a few cases.
Device	IIFT workstation.
Role and permissions	COMPLIANCE_OFFICER: Read set plus aml.review and audit.view.
Data scope	All screening cases and the audit trail.
Demo user	Nur Azizah binti Mahmud (compliance) (compliance)
RFP requirements covered	AP-16, AP-48, BO-13–BO-15, BO-26–BO-28

Table 29: Compliance officer: profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.

IIFT
SalesVerse 2.0
BACK-OFFICE

Search policy, participant or a...
Ctrl K

!
N
Nur Azizah binti

Dashboard

DISTRIBUTION

Agents & bankers

Agencies & banks

BUSINESS

Policies

Participants

Payments

CONTROL

AML / KYC

Audit trail

OPERATIONS

Reports

Data scope: All screening cases and the audit trail.

Where it leads

→ Work queues for this role; management tiles for Management

→ Search → Agent detail (documents, hierarchy, AML, approvals)

→ List → Agency or bank detail, agents, issuance block

→ Search across agencies → Policy record (read-only view of the portal page)

→ Search → Participant record, policies, documents, AML screening history

→ Search → Payment detail, proof, e-Receipts, verification request

→ Screening cases → Review (Cleared / Confirmed match) · Watch-lists: add, import CSV

→ Search by period, user, action, record → before and after values

→ Run a report with filters and export

Figure 74: Compliance officer: navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Dashboard	AML cases to review	Opens pending cases	BO-21
AML / KYC: Screening cases	Pending, cleared, confirmed and auto-cleared cases with matches	Reviews a case: Cleared or Confirmed match with remarks	AP-16, BO-13–BO-15
AML / KYC: Watch-lists	Entries by list, add, import CSV, deactivate	Maintains sanctions, PEP and internal lists	BO-13
Participants, Agents & bankers	Records with AML screening history	Reads the subject's history before deciding	BO-14
Audit trail	Append-only record of actions with before and after values	Evidences who screened, reviewed and approved	BO-26–BO-28
Reports	AML screening and review report	Monthly compliance reporting	BO-22

Table 30: Compliance officer: screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

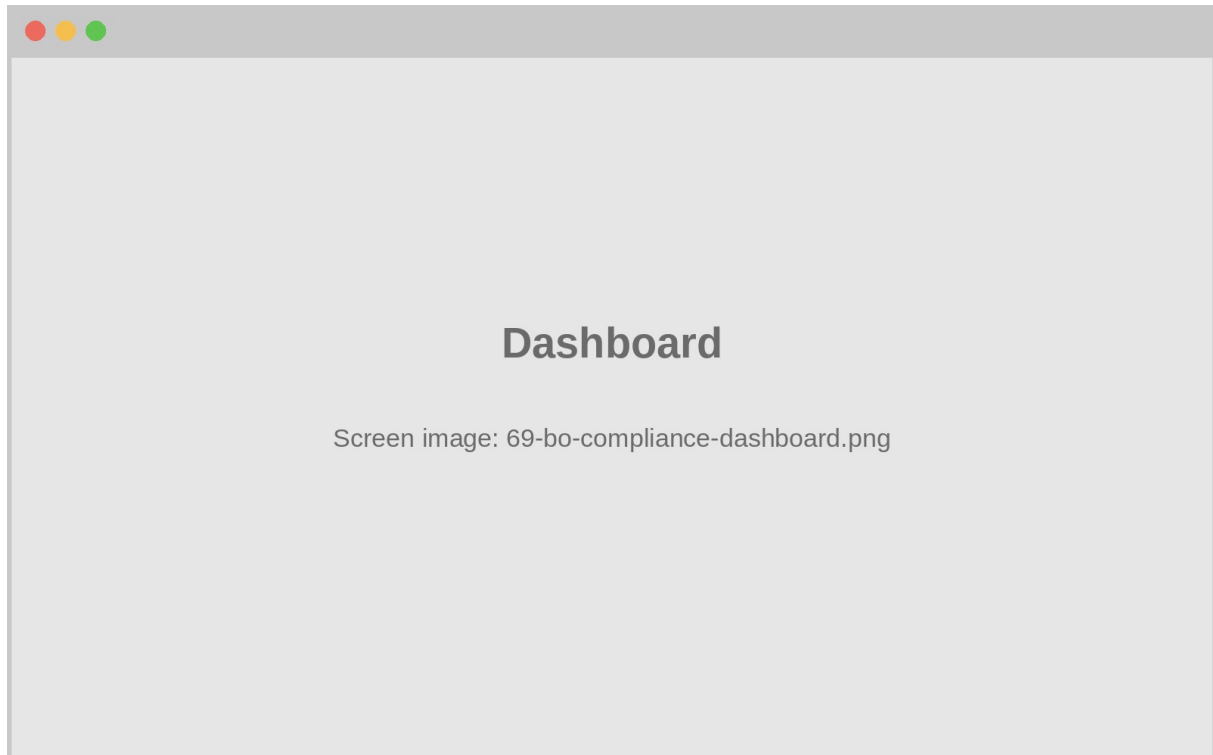


Figure 75: Dashboard: AML cases to review as the first queue.

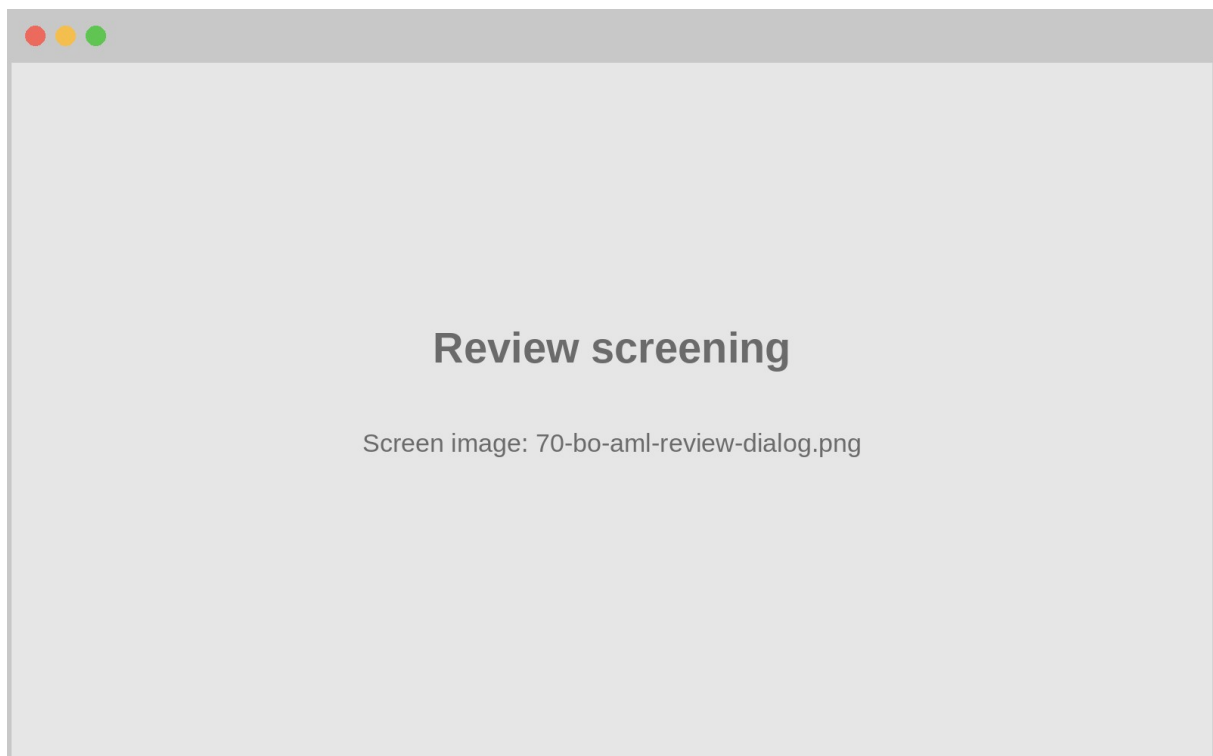


Figure 76: Review screening: Matches with list, score and reason; decision Cleared or Confirmed match with mandatory remarks.

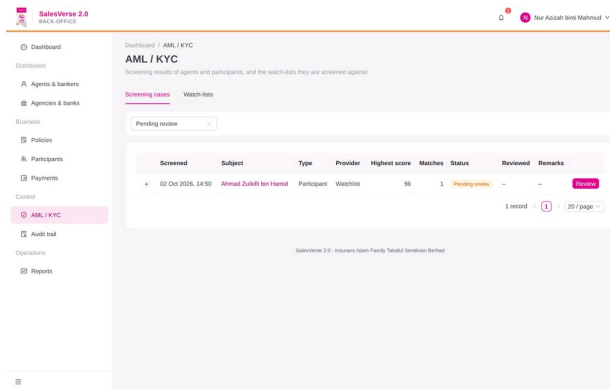


Figure 77: AML / KYC screening cases: Pending cases with highest score and match count; rows expand to the matches.

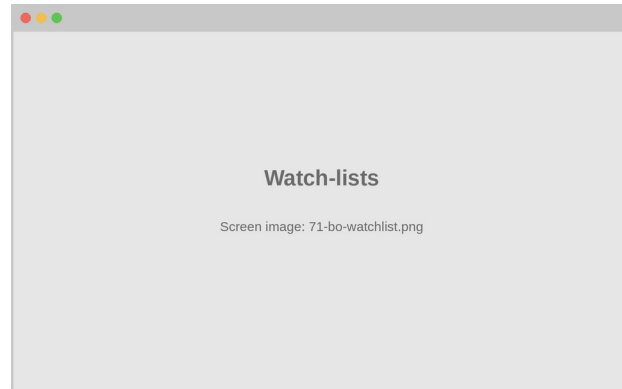


Figure 78: Watch-lists: Entries by list with add, CSV import and deactivation.

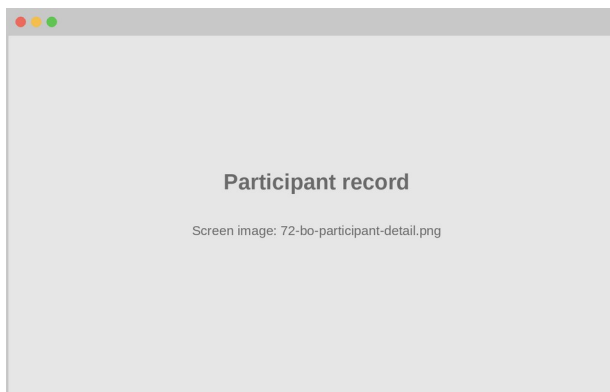


Figure 79: Participant record: Profile, policies, documents, update requests and the AML screening history.

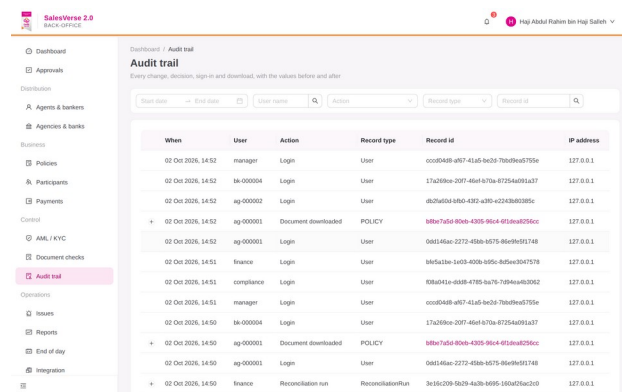


Figure 80: Audit trail: Filters by period, user, action, record type and id; rows expand to before and after values.

7.12 Support officer

Profile

Who	First-line support for portal and back-office users; owns issues until they are closed.
Goals	Respond inside the SLA for the priority; keep the reporter informed; no breached issue unassigned.
How often	All day.
Device	IIFT workstation.
Role and permissions	SUPPORT_DESK: dashboard, issues.manage, agents.view, agencies.view (to look up who is reporting).
Data scope	All issues; agents and agencies read-only.
Demo user	Kenneth Lim (support) (support)
RFP requirements covered	AP-55–AP-57, BO-29–BO-31

Table 31: Support officer: profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.


SalesVerse 2.0
BACK-OFFICE

Search policy, participant or a... Ctrl K


 Kenneth Lim

Dashboard

DISTRIBUTION

Agents & bankers

Agencies & banks

OPERATIONS

Issues

Data scope: All issues; agents and agencies read-only.

Where it leads

→ Work queues for this role; management tiles for Management

→ Search → Agent detail (documents, hierarchy, AML, approvals)

→ List → Agency or bank detail, agents, issuance block

→ All issues → Issue detail: assign, priority, status, conversation

Figure 81: Support officer: navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Dashboard	Open issues and how many are past SLA	Starts with breached issues	BO-21, BO-31
Issues	All issues with filters Assigned to me and SLA breached	Triages new issues	BO-29–BO-31
Issue detail	Details, SLA, attachments, conversation, assignment and priority	Assigns, changes priority (recalculates SLA), replies or adds internal notes, resolves with a resolution text	BO-29–BO-31, AP-56
Agents & bankers, Agencies & banks	Read-only lookups	Identifies the reporter's agency or bank	BO-06

Table 32: Support officer: screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

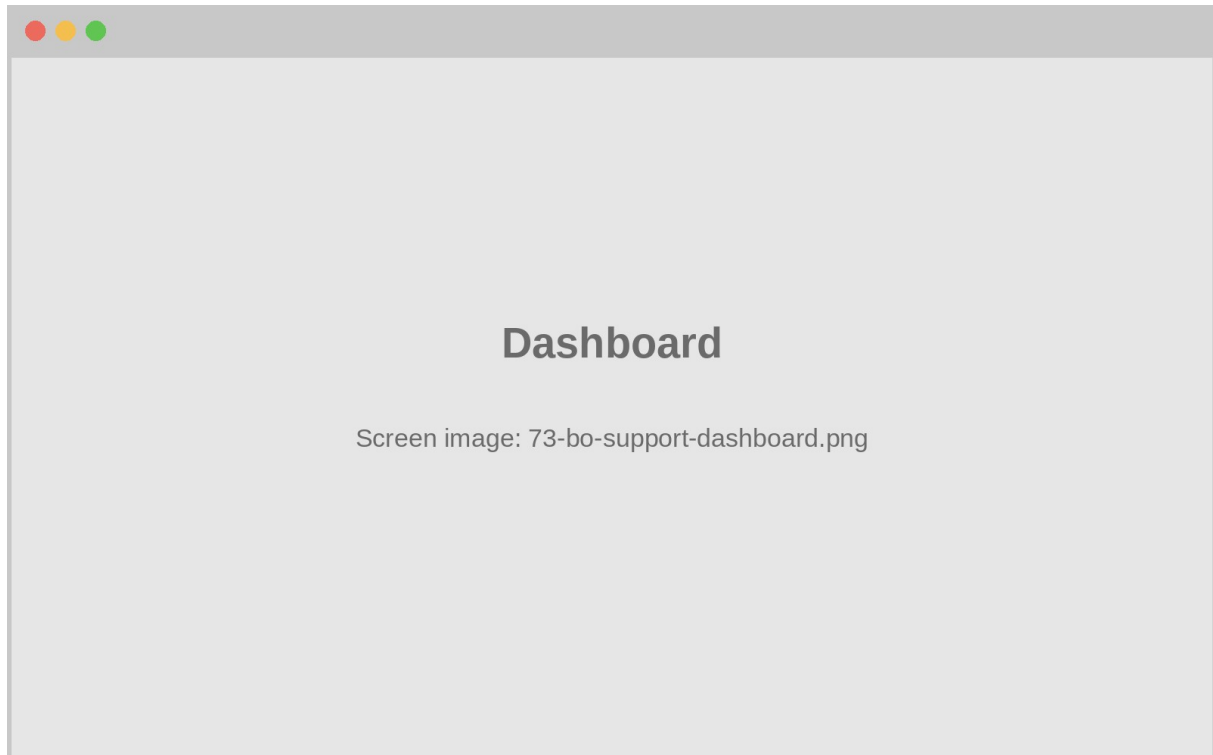


Figure 82: Dashboard: Open issues and how many are past their SLA.

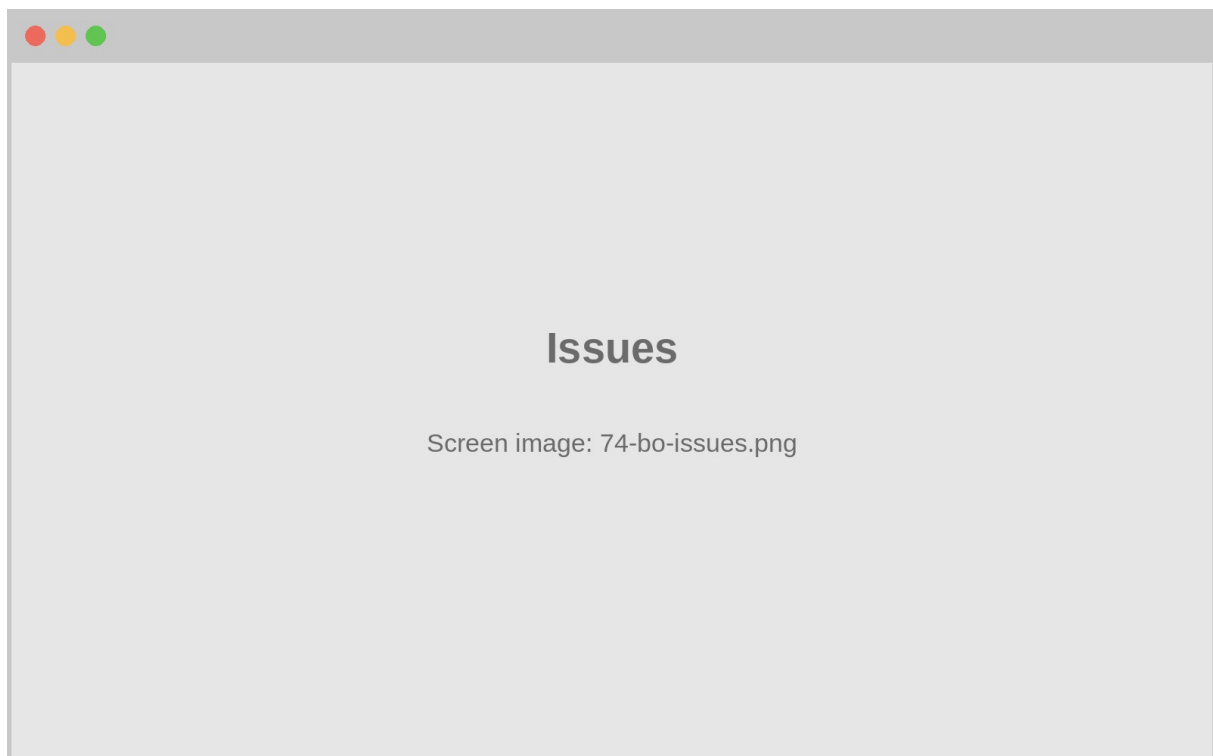


Figure 83: Issues: All issues with assignment, response and resolution targets; filters for mine and breached.

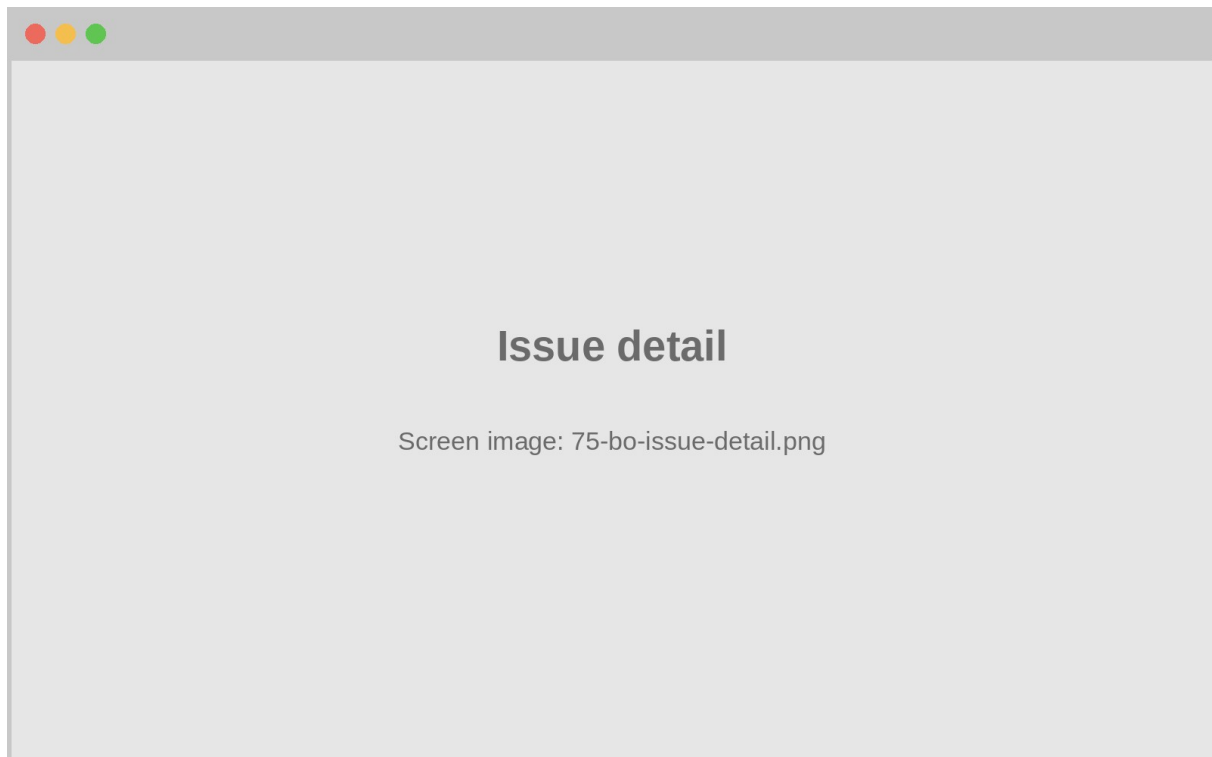


Figure 84: Issue detail: Details and SLA, attachments, conversation with internal notes, assignment and priority.

7.13 System administrator

Profile

Who	IIFT or IITH IT: users, roles, workflows, products, parameters, master data, integration health.
Goals	Change a rule, a threshold or a rate table without a release; prove who changed what.
How often	Weekly, and during product or rule changes.
Device	IIFT workstation.
Role and permissions	SYSTEM_ADMINISTRATOR: users.manage, roles.manage, config.manage, workflow.configure, products.manage, integration.manage, audit.view, agencies.view/manage, reports.view/schedule. No approval permission and no access to policies, participants or payments.
Data scope	Configuration and audit; no business records.
Demo user	admin (first administrator created by the seed) (admin)
RFP requirements covered	BO-02–BO-04, BO-09, BO-16, BO-18, BO-26–BO-28, BO-32, BO-33, COM-04, COM-09, INT-11, INT-13

Table 33: System administrator: profile

Navigation

The menu below is what this role sees after sign-in; items the role has no permission for are not rendered. The right-hand column lists the pages, dialogs and actions behind each item.


SalesVerse 2.0
BACK-OFFICE

No searchable records for this role



admin

Dashboard

DISTRIBUTION

Agencies & banks

CONTROL

Audit trail

OPERATIONS

Reports

Integration

ADMINISTRATION

Users

Roles & permissions

Workflows

Products

Parameters & master data

Data scope: Configuration and audit; no business records.

Where it leads

→ Work queues for this role; management tiles for Management

→ List → Agency or bank detail, agents, issuance block · Add, Edit, Lift issuance block

→ Search by period, user, action, record → before and after values

→ Run a report with filters and export · Scheduled reports

→ Health per system · Message queue (re-submit) · Integration log · Reconciliation

→ Users → New staff user, Edit, Disable, Unlock, Reset password

→ Roles → New role / Edit role (permission picker)

→ One card per transaction type: steps, approver, amount threshold

→ Products → Edit: flags, rating configuration, documents, questionnaire

→ Parameters by category · Master data code tables

Figure 85: System administrator: navigation map

Screens

Screen	Purpose	What this persona does there	RFP IDs
Users	Staff, agent and bank officer accounts	Creates staff users (local or directory), edits roles, disables, unlocks, resets passwords	BO-03, AP-04, INT-06
Roles & permissions	Roles per module with their permissions	Creates or edits roles from the permission catalogue	BO-02, BO-04, AP-59
Workflows	Approval definitions per transaction type	Switches approval on or off; sets levels, approvers and amount thresholds	BO-16, BO-18, COM-04
Products	Products, rating configuration, documents, questionnaire	Updates rates, plans, required documents and declarations	BO-32, BO-33
Parameters & master data	Parameters by category; code tables	Changes grace days, validity, SLA hours, thresholds; maintains banks, districts, reasons	BO-32, BO-33, COM-09
Agencies & banks	Agency and bank records	Creates a new bank partner	BO-09
Integration	Health, message queue, log, reconciliation	Watches interface health; re-submits	INT-11, INT-13, INT-14
Audit trail	All changes with before and after values	Evidences configuration changes	BO-26–BO-28
Reports, Scheduled reports	User access review report and schedules	Schedules the access review	BO-25

Table 34: System administrator: screens

Screens in sequence

The images follow the order in which this persona meets the screens. All data is demonstration data.

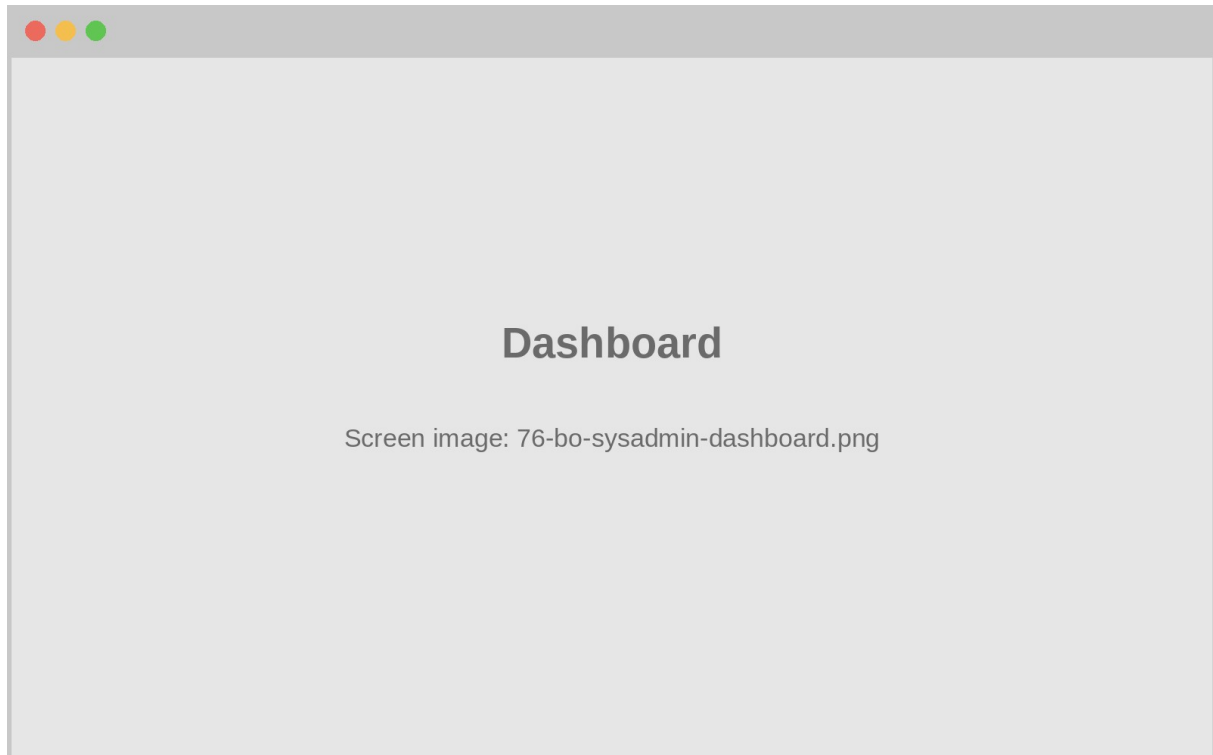


Figure 86: Dashboard: Integration queue and audit are the administrator's queues; no business records.

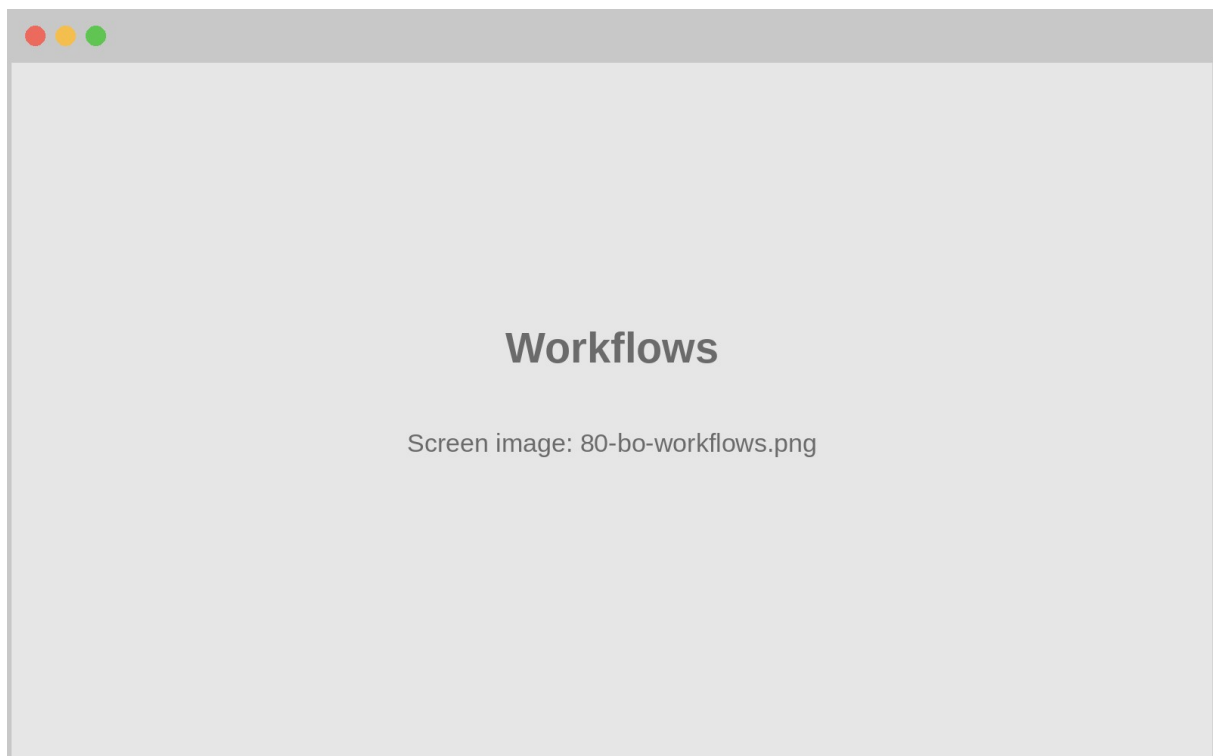


Figure 87: Workflows: Per transaction type: approval switch, steps with approver permission and amount threshold.

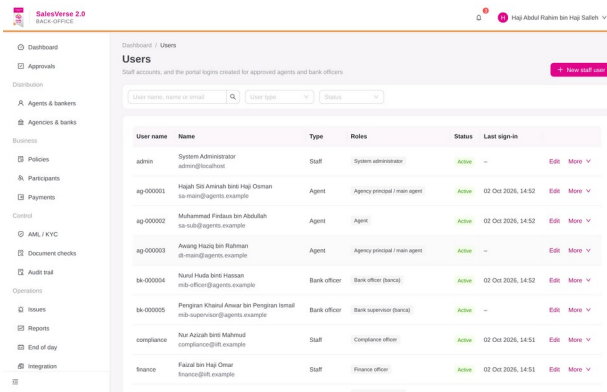


Figure 88: Users: Staff, agents and bank officers with roles, status and last sign-in; unlock and reset.

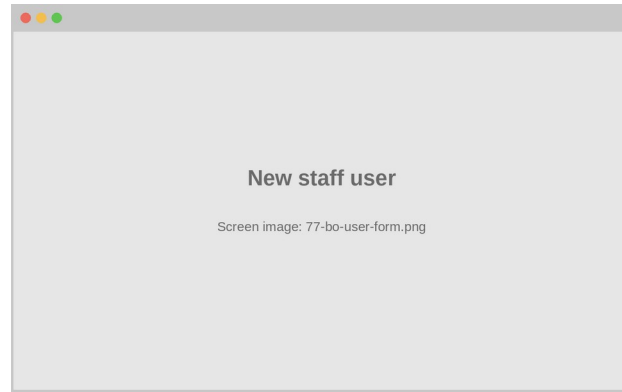


Figure 89: New staff user: Account, sign-in method (local or directory), details and roles.

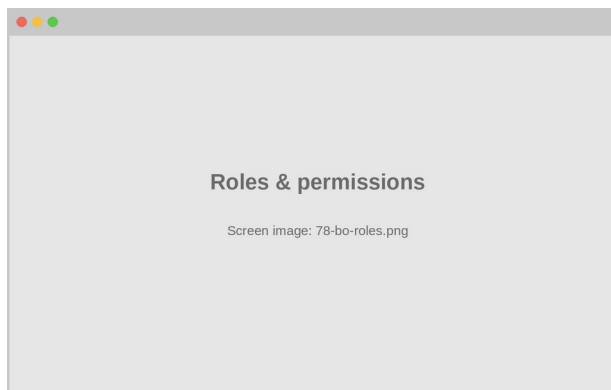


Figure 90: Roles & permissions: Roles per module with permission count and users; system roles cannot be deleted.

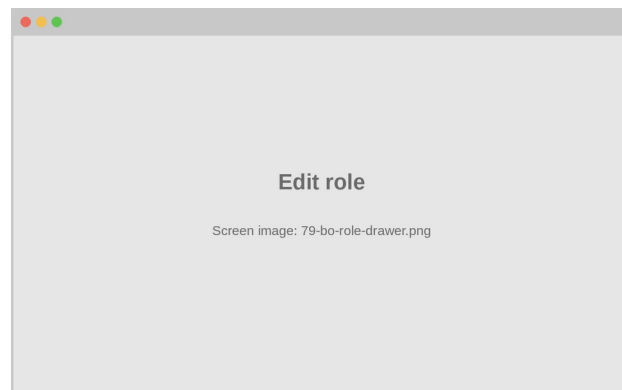


Figure 91: Edit role: Permission picker limited to the role's module.

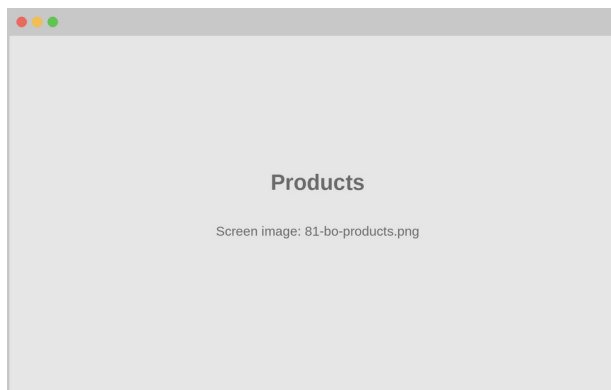


Figure 92: Products: Seven products with rating engine, payment rule, renewal flag and document count.

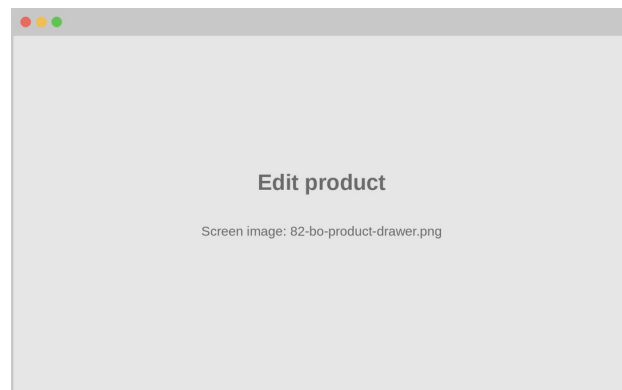


Figure 93: Edit product: Flags, rating configuration, required documents and questionnaire without code changes.

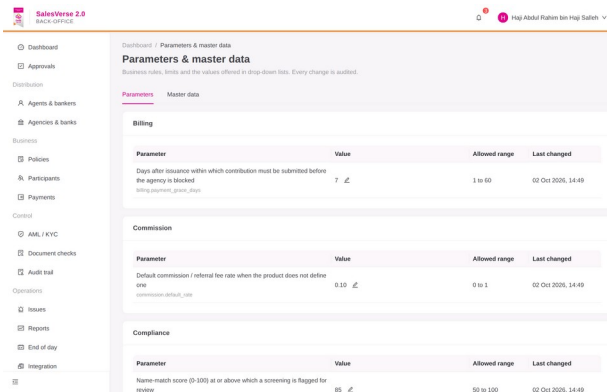


Figure 94: Parameters: Parameters by category with allowed range and last change.

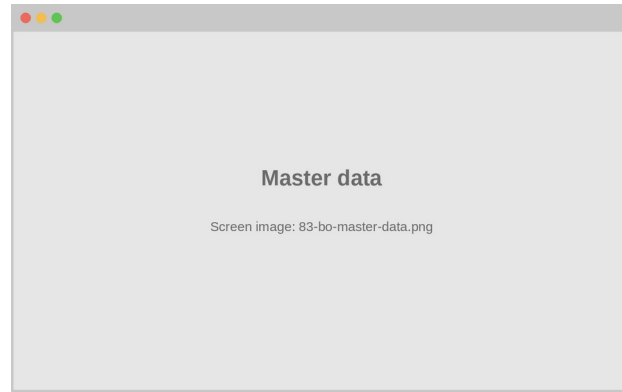


Figure 95: Master data: Code tables (banks, districts, claim types, reasons) with add, edit and deactivate.

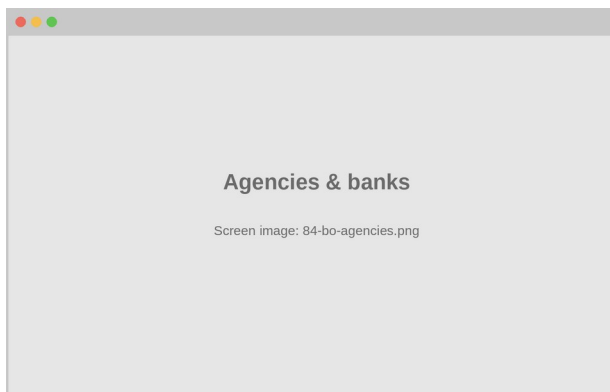


Figure 96: Agencies & banks: Agency and bank records with channel, active agents and new-business status.

7.14 End-to-end journeys

Ten journeys cover new business for both product families, onboarding, compliance, servicing, renewal, claims, support, end of day and administration. In each figure the columns are the people and the system, the boxes are numbered in order, orange boxes are automatic system steps, diamonds are the points where the system branches and orange arrows are hand-offs from one person to another. The table under each figure gives the detail of every step, including the numbers assigned, the messages sent and the audit records written. The times are what the steps take in a demonstration with the seeded data.

Journey 1: New business for an annual plan (Professional Takaful Plan)

Starts with	Sub-agent
Ends with	Policy ACTIVE and PAID; e-Policy and e-Receipt issued and e-mailed
Time in the demonstration	about 12 minutes including the Finance step
RFP requirements	AP-13–AP-20, AP-25, AP-32, AP-33, AP-36, AP-39–AP-41, AP-44, AP-45, AP-62, FFR03 (APPT01–07)

The sub-agent completes the application with the participant in one sitting. Payment is required before issuance for the annual plans, so the e-Policy follows Finance's verification.

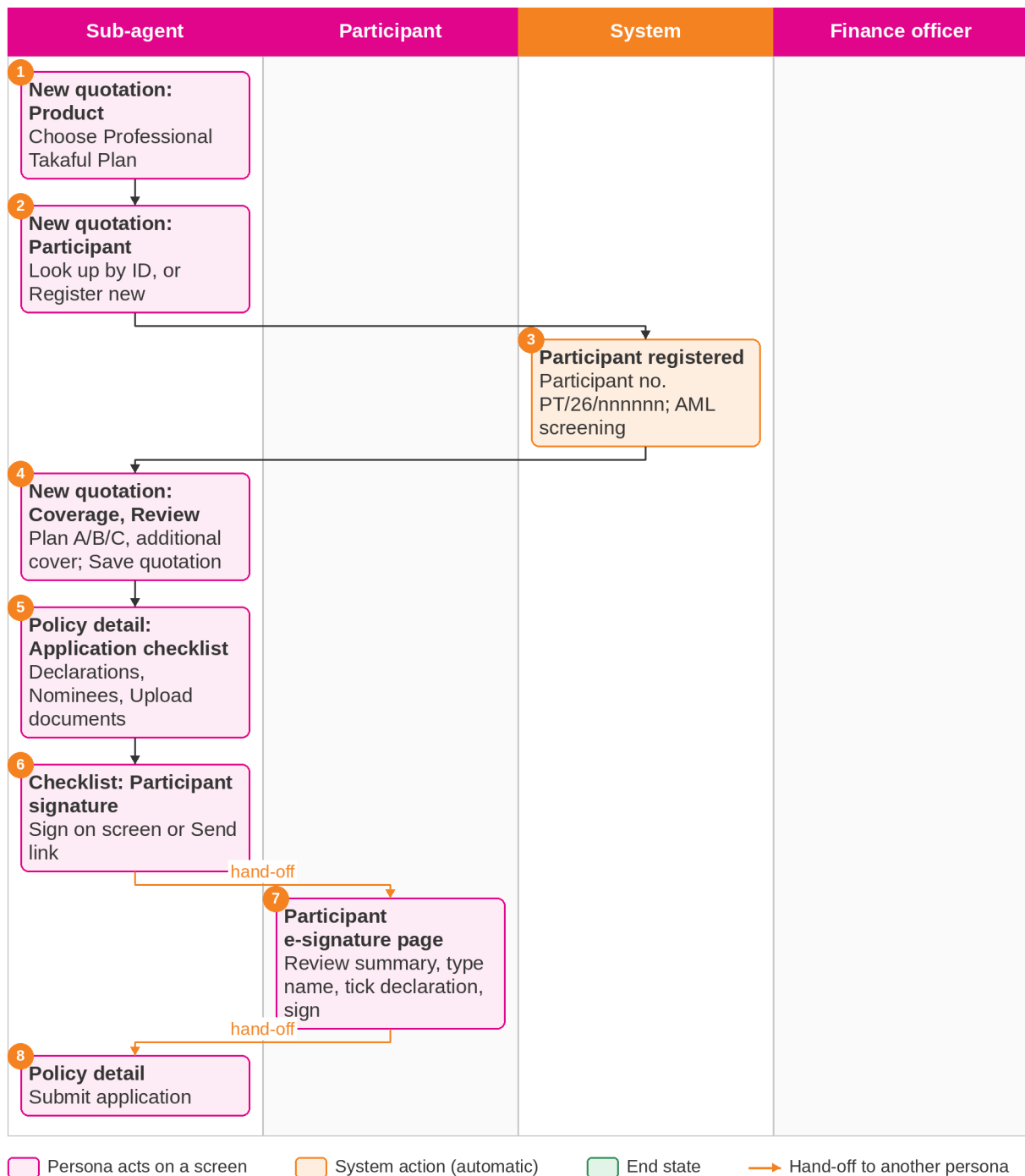


Figure 97: Journey 1: New business for an annual plan (Professional Takaful Plan) (part 1 of 2)

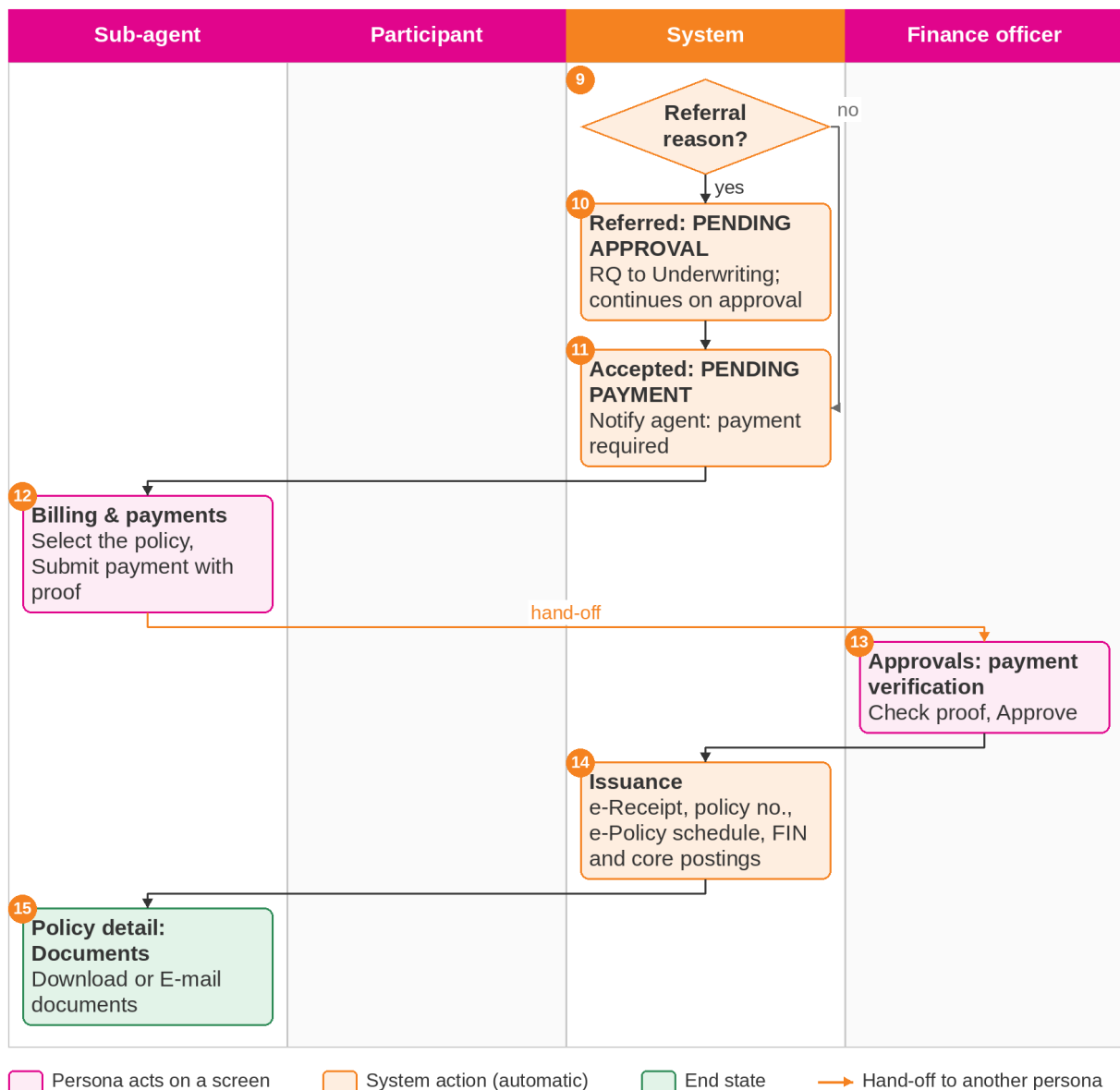


Figure 98: Journey 1: New business for an annual plan (Professional Takaful Plan) (part 2 of 2)

Step	Who	Screen or system step	What happens
1	Sub-agent	New quotation: Product	Header button New quotation. Products are grouped by line of business; the card shows Pay before issue and Renewable.
2	Sub-agent	New quotation: Participant	Find existing (name, number, mobile), Look up by ID (exact IC number across agencies, AP-11) or Register new: ID type and number, name, date of birth, nationality, occupation and class, mobile, e-mail, address.
3	System	Participant registered	The participant number is assigned and the name and IC number are screened against the watch-lists. A score at or above the threshold (85) flags the participant and e-mails Compliance; the agent sees the outcome on the Participant step.
4	Sub-agent	New quotation: Coverage, Review	The indicative contribution updates as the plan changes. Review shows participant and cover; Save quotation creates QT/26/nnnnnn in status DRAFT and opens the policy page.
5	Sub-agent	Policy detail: Application checklist	Declarations: three health questions (a Yes needs details and refers the case). Nominees: name, relationship, role, share (total 100%). Documents tab: questionnaire, IC copy, nominee IC.
6	Sub-agent	Checklist: Participant signature	On screen: the participant signs on the agent's device. Send link: a one-time link is e-mailed to the participant, valid for 72 hours (parameter).
7	Participant	Participant e-signature page	Public page without login. The typed name must match the participant; the signature image is stored, the policy history records the source address and the agent is notified.

Step	Who	Screen or system step	What happens
8	Sub-agent	Policy detail	The server checks quotation validity (30 days), the agency block, nominees, mandatory documents, participant signature and the AML status, then re-rates the quotation.
9	System	Decision: Referral reason?	Referral reasons: a Yes declaration, sum covered above the agent's authority limit, or a product quality check.
10	System	Referred: PENDING APPROVAL	Request RQ/26/nnnnnn goes to Underwriting with the reasons; the agent sees Awaiting IIFT underwriting decision. Journey 2 shows the approval; a rejection returns the quotation with the remarks and it can be reopened and resubmitted.
11	System	Accepted: PENDING PAYMENT	Without a referral the quotation is accepted and waits for the contribution. The agent receives an in-app notification with the amount.
12	Sub-agent	Billing & payments	Allocation per policy, method, bank, bank reference, payment date, proof file. The payment PY/26/nnnnnn is marked Pending verification and a Payment verification request is raised.
13	Finance officer	Approvals: payment verification	Finance opens the request from the inbox or dashboard, sees the allocation and the proof under Supporting evidence, and approves; a rejection needs a reason, which the agent sees.
14	System	Issuance	One e-Receipt RC/26/nnnnnn per policy; the policy is PAID and issued as PRO/26/nnnnnn with the e-Policy schedule PDF; commission accrues; outbox messages to the core system (policy issued) and FIN (receipt, commission); audit records; agent notified by e-mail with receipts; participant e-mailed the e-Policy.
15	Sub-agent	Policy detail: Documents	The agent downloads or re-sends the e-Policy and e-Receipt. The day's issuance is picked up by end of day (journey 9).

Journey 2: Banca financing policy above B\$150,000 with bulk payment by the bank

Starts with	Bank officer
Ends with	Policy ACTIVE; contribution settled by one bank transfer; receipts posted to FIN
Time in the demonstration	about 15 minutes across three users
RFP requirements	FFR01 (FTPHP01–07), AP-32, AP-33, AP-38–AP-42, AP-60, AP-61, BO-16–BO-19

Financing takaful is issued first and paid within the grace period. Every hire purchase case passes a quality check, and financing above B\$150,000 is a referral; both are decided on the same request. The supervisor settles several policies with one transfer.

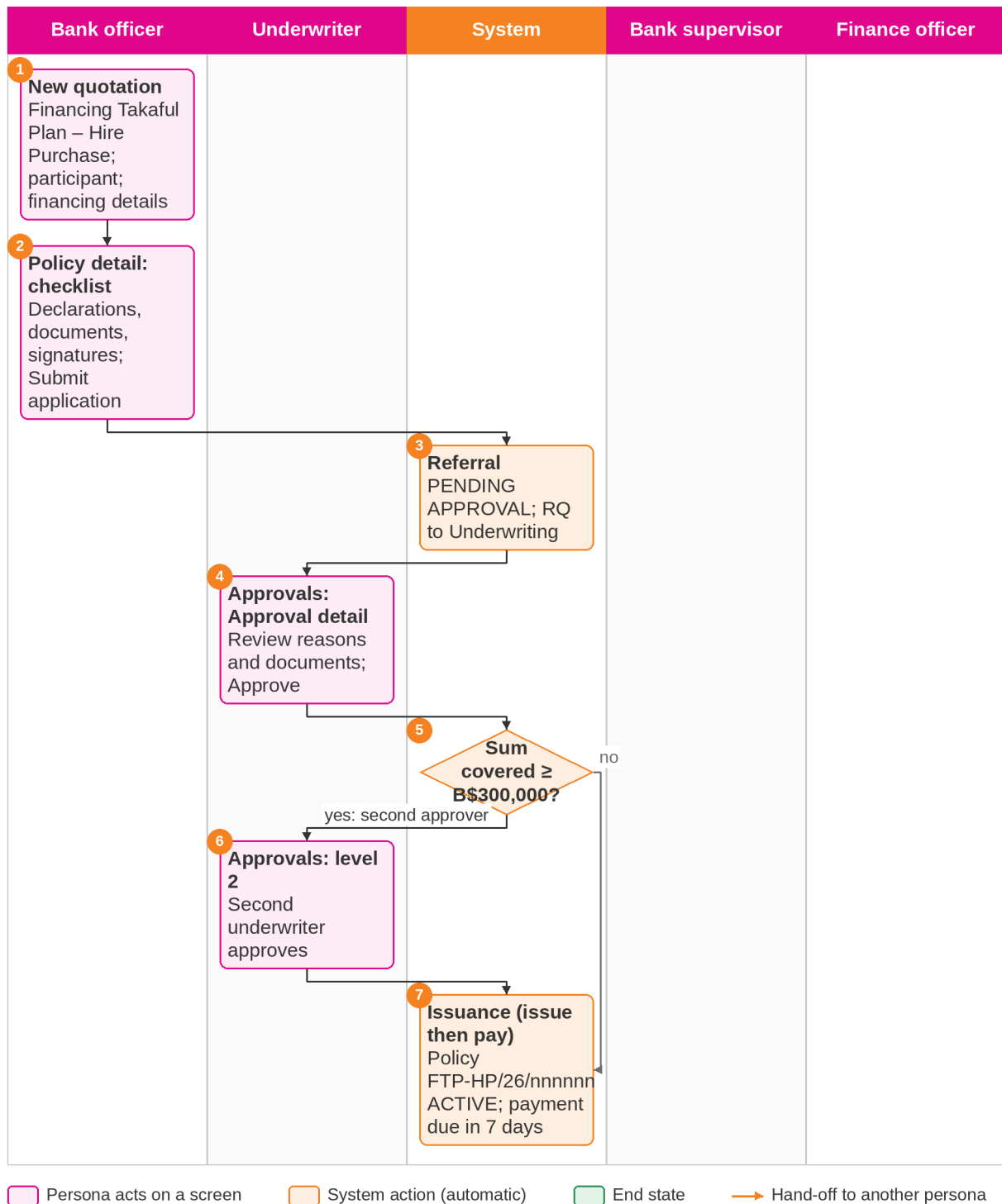


Figure 99: Journey 2: Banca financing policy above B\$150,000 with bulk payment by the bank (part 1 of 2)

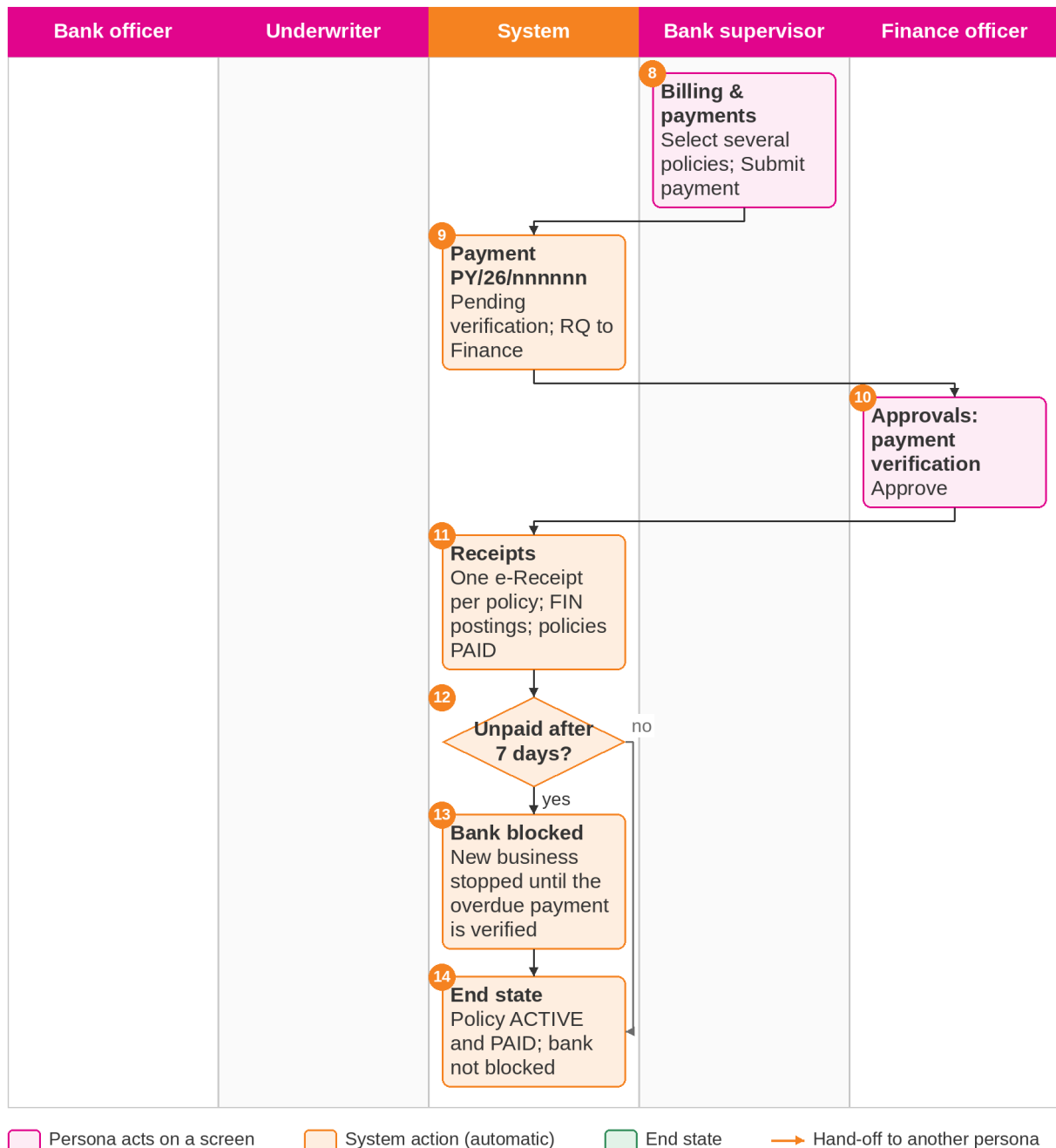


Figure 100: Journey 2: Banca financing policy above B\$150,000 with bulk payment by the bank (part 2 of 2)

Step	Who	Screen or system step	What happens
1	Bank officer	New quotation	Product tagged Pay after issue. Participant found by IC lookup or registered (AML screening). Coverage: financing amount B\$180,000, period 96 months, profit rate, financier, approval reference, vehicle registration. The indicative contribution shows the referral reason.
2	Bank officer	Policy detail: checklist	Mandatory documents for FTP-HP: proposal form, product disclosure sheet, IC copy, hire purchase approval letter, drawdown letter; the HP statement is optional. Participant signature on screen or by link.
3	System	Referral	Two referral reasons are recorded: financing above the B\$150,000 high-risk limit and the quality check before contract issuance. Request RQ/26/nnnnnn is created with the sum covered as its amount; approvers are notified; the officer sees Awaiting IIFT underwriting decision.
4	Underwriter	Approvals: Approval detail	The request shows the referral reasons, sum covered, contribution and the policy documents; Open policy shows the full record. Approve with remarks.
5	System	Decision: Sum covered ≥ B\$300,000?	The Referred quotation workflow has a second step from B\$300,000 sum covered. The same person cannot approve both levels.

Step	Who	Screen or system step	What happens
6	Underwriter	Approvals: level 2	A second underwriter decides; the request header shows Level 2 of 2.
7	System	Issuance (issue then pay)	e-Policy schedule generated and e-mailed; core system notified; commission accrued; the officer is notified with the due date; the policy appears in Billing & payments as outstanding.
8	Bank supervisor	Billing & payments	The supervisor sees the bank's outstanding list (overdue, due within 3 days). One drawer allocates the transfer across the selected policies, with the bank reference and proof.
9	System	Payment PY/26/nnnnnn	Each policy's payment status becomes Pending verification; the grace-period check runs for the bank.
10	Finance officer	Approvals: payment verification	Finance matches the reference and amount to the bank statement.
11	System	Receipts	Receipts RC/26/nnnnnn per policy, posted to FIN through the outbox; the supervisor receives the receipts by e-mail; any issuance block on the bank is lifted.
12	System	Decision: Unpaid after 7 days?	Nightly at 01:00 the grace-period job checks every agency and bank for policies whose payment due date has passed.
13	System	Bank blocked	Dashboard alert for every user of the bank, e-mail and SMS, audit record AGENCY_BLOCKED. The block lifts automatically once Finance verifies the payment, with a notification.
14	System	End state	The day's policy and receipts are in the EOD report and FIN file.

Journey 3: Sub-agent onboarding

Starts with	Main agent (or the Operations officer from the back-office)
Ends with	Agent ACTIVE with a portal account; first sign-in completed
Time in the demonstration	about 8 minutes across three users
RFP requirements	AP-07, AP-08, AP-46, AP-48, BO-05, BO-10, BO-11, BO-13, BO-17, BO-19, AP-01, AP-02

The principal registers the applicant in the portal; IIFT verifies the IC copy and approves. The portal account is created on approval, never before.

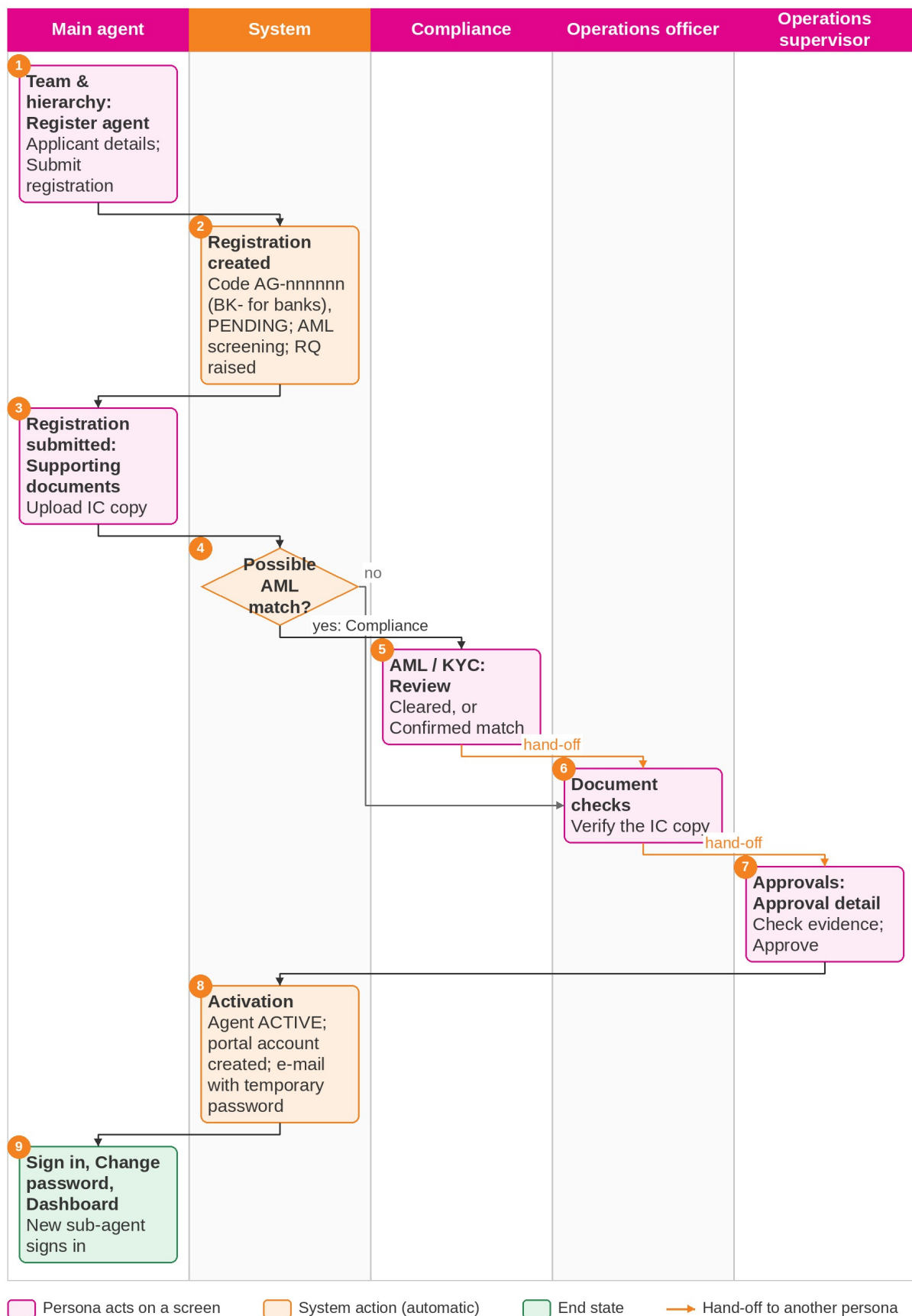


Figure 101: Journey 3: Sub-agent onboarding

Step	Who	Screen or system step	What happens
1	Main agent	Team & hierarchy: Register agent	Reports to (main agent), full name, ID type and number, date of birth, e-mail, mobile, address, licence number and expiry. Banks register bank officers on the same page.
2	System	Registration created	Duplicate ID numbers are refused. The agent is screened against the watch-lists; the Agent / banker registration workflow raises RQ/26/nnnnnn for an Operations supervisor; approvers are notified. The page shows the code, the AML outcome and the next steps.
3	Main agent	Registration submitted: Supporting documents	The IC copy is required before IIFT can approve; a validity date can be recorded.
4	System	Decision: Possible AML match?	A flagged applicant blocks approval (AML_NOT_CLEARED) until Compliance clears the case.
5	Compliance	AML / KYC: Review	Journey 4. A confirmed match rejects the registration automatically with the Compliance remarks.
6	Operations officer	Document checks	Documents awaiting verification, oldest first; Verify or Reject with a reason. A rejected document must be replaced by the principal.
7	Operations supervisor	Approvals: Approval detail	Supporting evidence shows the agent, status, AML status and documents. Missing documents or an uncleared screening stop the approval with guidance. The maker can never approve its own request.
8	System	Activation	User name is the agent code (lower case); the first sign-in forces a password change. The core system receives the agent record; the principal is notified.
9	Main agent	Sign in, Change password, Dashboard	Password rules come from the Security parameters. The sub-agent appears in the hierarchy and may quote at once.

Journey 4: AML review by Compliance, including a confirmed match

Starts with	System (screening) and Compliance officer
Ends with	Case CLEARED, or CONFIRMED MATCH with the subject rejected and its pending requests closed
Time in the demonstration	about 4 minutes
RFP requirements	AP-16, AP-48, BO-13–BO-15, BO-19, BO-26

Screening is automatic on every participant and agent registration and again at submission if a participant was never screened. Compliance decides flagged cases in one dialog.

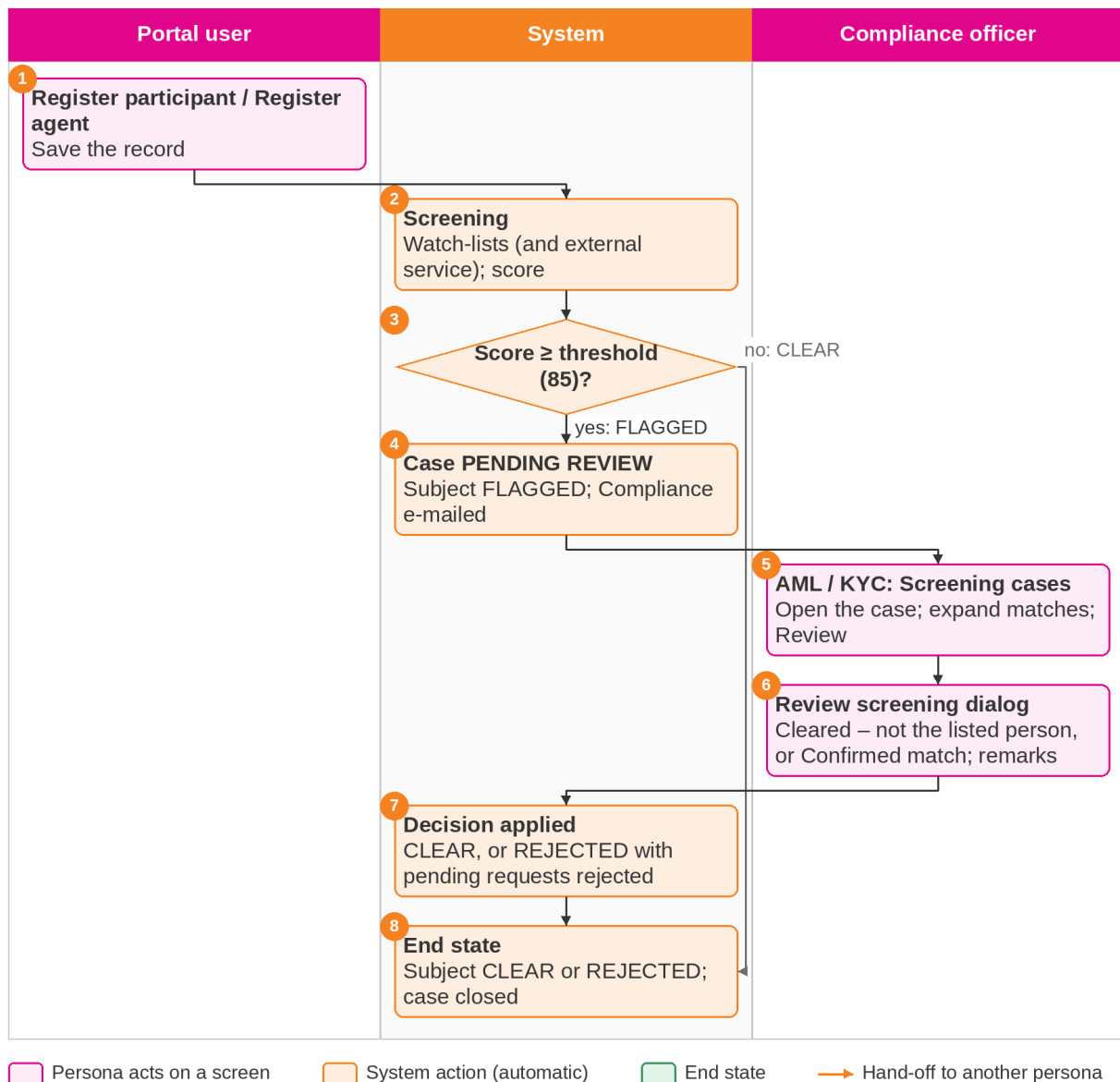


Figure 102: Journey 4: AML review by Compliance, including a confirmed match

Step	Who	Screen or system step	What happens
1	Portal user	Register participant / Register agent	The portal page reports the outcome: registered, or registered and referred to Compliance.
2	System	Screening	Exact ID number match scores 100; names are compared with fuzzy matching. An unavailable external service forces a manual review rather than silently passing.
3	System	Decision: Score $\geq$ threshold (85)?	Threshold is the Compliance parameter <code>aml.match_threshold</code> .
4	System	Case PENDING REVIEW	A flagged participant cannot be submitted; a flagged agent cannot be approved.
5	Compliance officer	AML / KYC: Screening cases	Matches show list, name, score and reason (name or ID number).
6	Compliance officer	Review screening dialog	Remarks are mandatory and record the evidence checked.
7	System	Decision applied	Cleared: the subject is CLEAR and the sale or registration continues. Confirmed match: the subject is REJECTED, every pending request on it (for example the agent registration) is rejected with the remarks, the maker is e-mailed, and the audit trail records <code>AML_REVIEWED</code> .
8	System	End state	The screening history stays on the agent and participant records.

Journey 5: Endorsement (nominee change) and cancellation through maker-checker

Starts with	Main agent
Ends with	Endorsement applied with a new e-Policy schedule; or policy CANCELLED with a pro-rata refund request
Time in the demonstration	about 6 minutes for both requests
RFP requirements	AP-27, AP-28, AP-31, AP-33, AP-49–AP-51, AP-61, BO-16–BO-19

Both changes are requests against an active policy. Nothing changes on the policy until the Operations supervisor approves; the agent follows the request under My requests.

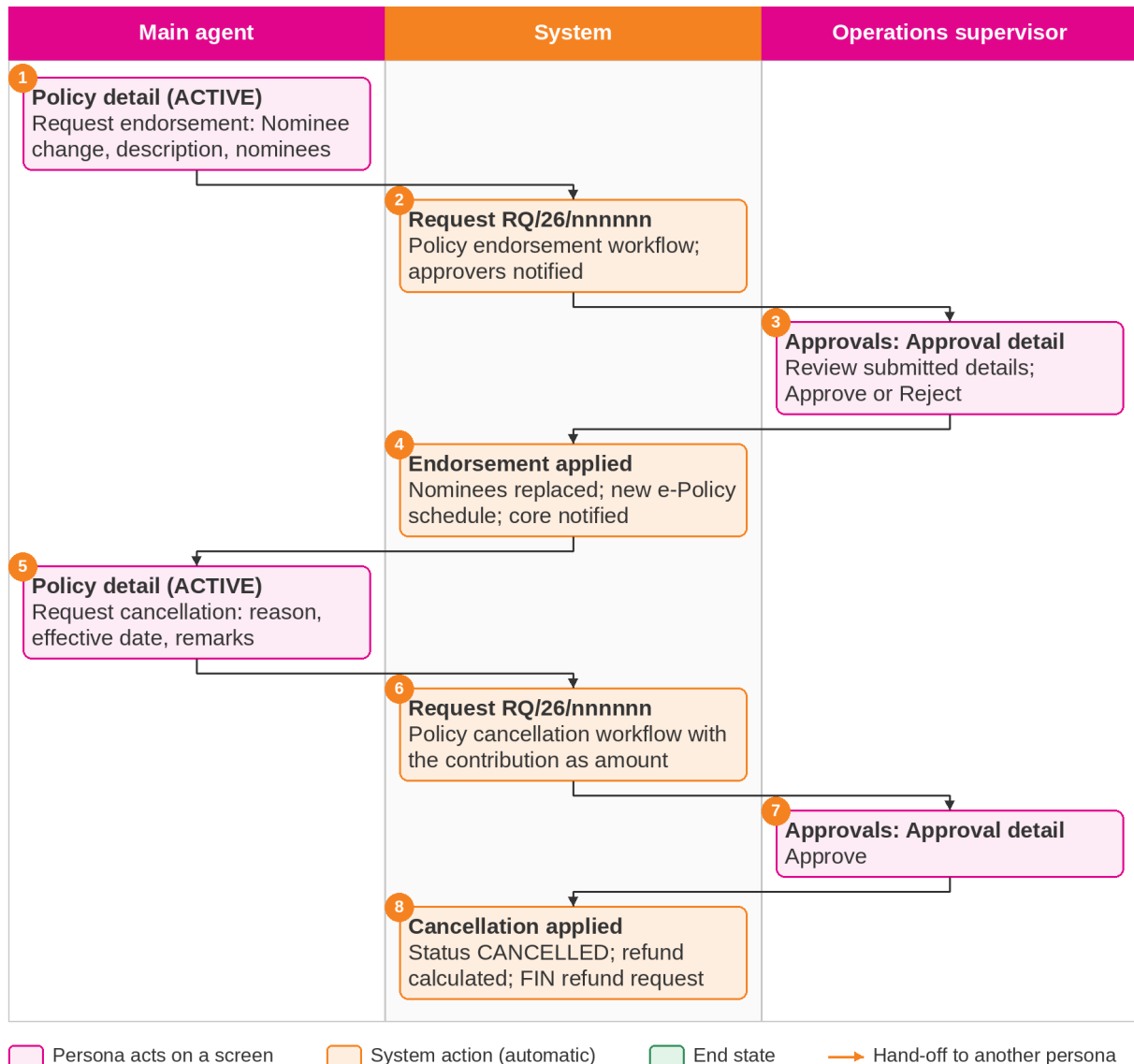


Figure 103: Journey 5: Endorsement (nominee change) and cancellation through maker-checker

Step	Who	Screen or system step	What happens
1	Main agent	Policy detail (ACTIVE)	The nominee editor pre-fills the current nominees; shares must total 100%. Submit request.
2	System	Request RQ/26/nnnnnn	The payload stores the new nominees with ID numbers encrypted; the portal shows the request as Pending under My requests.
3	Operations supervisor	Approvals: Approval detail	Rejection needs a reason, shown to the agent with a hint to correct and resubmit.

Step	Who	Screen or system step	What happens
4	System	Endorsement applied	Policy history records ENDORSED; an endorsed e-Policy schedule PDF is stored; audit POLICY_ENDORSED; the agent is notified.
5	Main agent	Policy detail (ACTIVE)	Reason codes come from master data; the effective date cannot precede the cover start. The policy stays in force until approval.
6	System	Request RQ/26/nnnnnn	An amount threshold can add a second level in Workflows.
7	Operations supervisor	Approvals: Approval detail	Same inbox; the maker-checker rule applies.
8	System	Cancellation applied	Refund is pro-rated on the unexpired days of what was actually paid; the core system is told; the agent is notified with the refund amount.

Journey 6: Renewal inside the window

Starts with	Sub-agent
Ends with	Renewal policy ACTIVE from the day after the old cover ends
Time in the demonstration	about 6 minutes including payment
RFP requirements	AP-26, AP-25, AP-34, AP-39–AP-41, FFR02 (APHA02), FFR03 (APPT02)

Renewable products (PHA, PRO, OSA) are listed from 45 days before the end of cover (parameter) until 30 days after it. KHR is configured as not renewable.

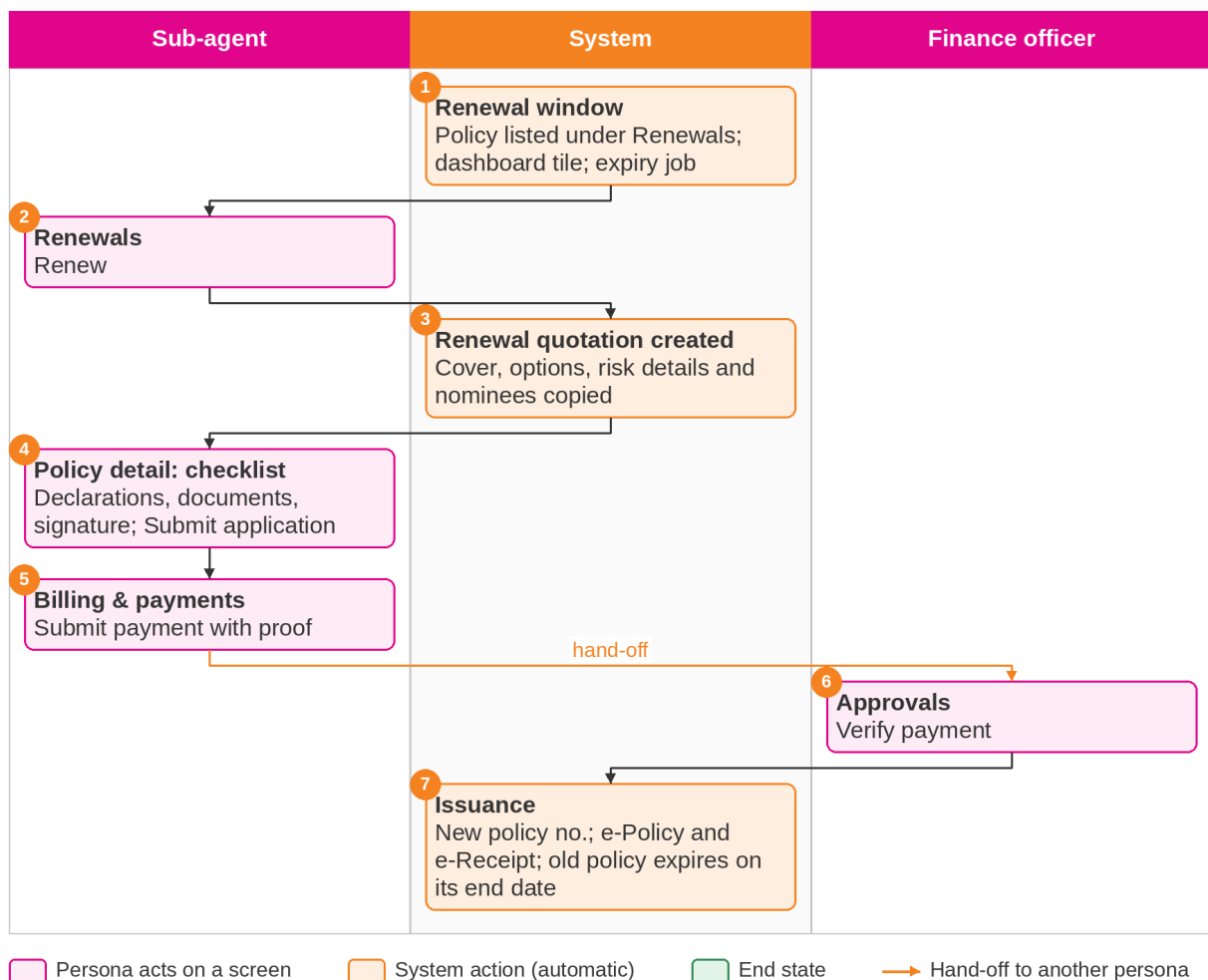


Figure 104: Journey 6: Renewal inside the window

Step	Who	Screen or system step	What happens
1	System	Renewal window	The nightly expiry job moves policies past their end date to EXPIRED and notifies the agent; an expired policy can still be renewed for 30 days.
2	Sub-agent	Renewals	Also available as Renew on the policy page. Days left are shown per policy.
3	System	Renewal quotation created	Start date is the day after the old end date; the quotation is linked as Renewal of the old policy; only one open renewal per policy.
4	Sub-agent	Policy detail: checklist	Declarations and the participant signature are collected again; documents can be re-uploaded.
5	Sub-agent	Billing & payments	Pay-before-issue product: the renewal waits for the contribution.
6	Finance officer	Approvals	As journey 1.
7	System	Issuance	Both policies show the link to each other in Policy record.

Journey 7: Claim notification and back-office review

Starts with	Main agent
Ends with	Claim CLOSED after acknowledgement or rejection; agent informed at each step
Time in the demonstration	about 4 minutes
RFP requirements	AP-43, AP-34, AP-54, INT-01

The portal records the notification and forwards it to IIFT; the claims team tracks the status here and the agent sees every remark.

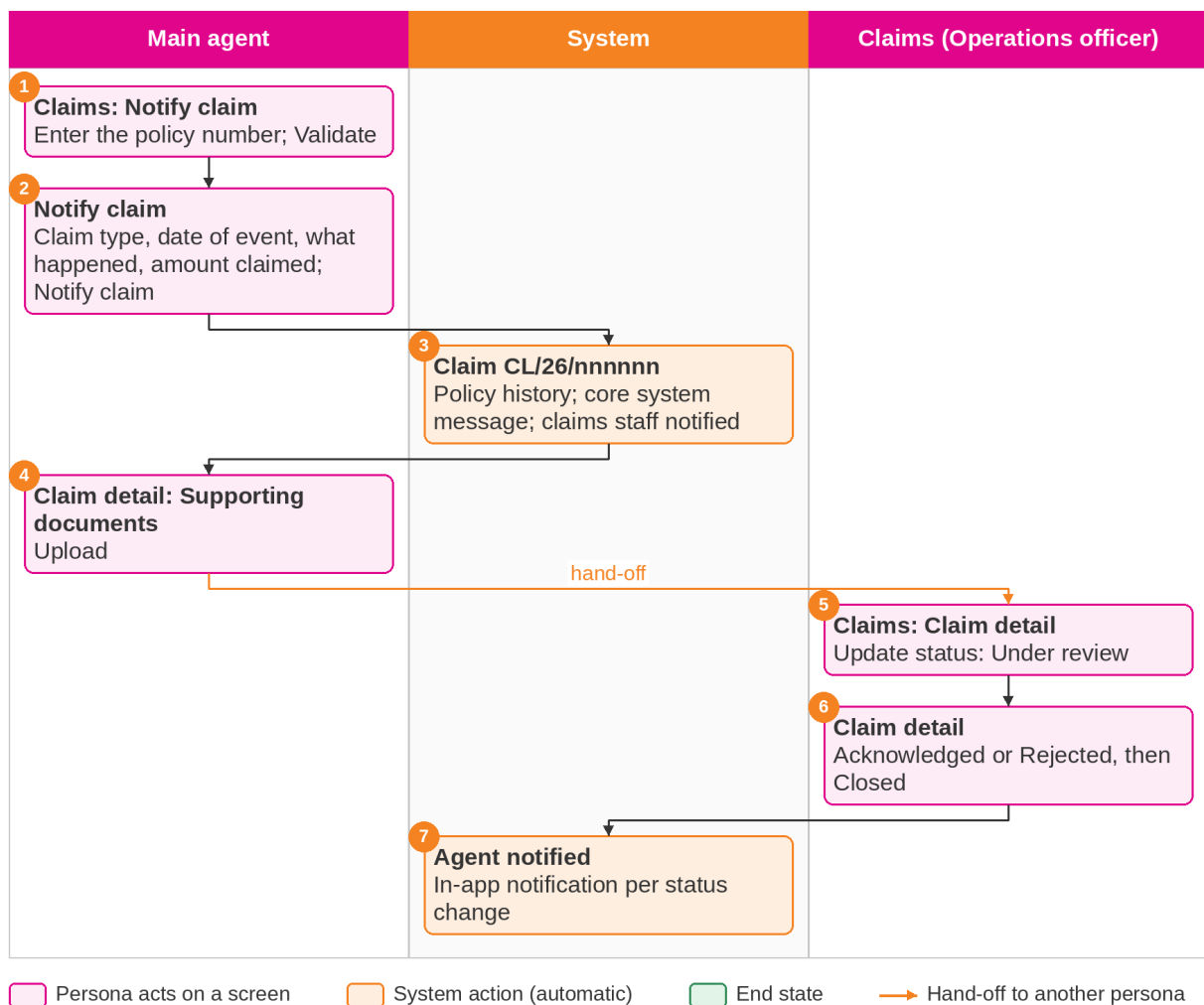


Figure 105: Journey 7: Claim notification and back-office review

Step	Who	Screen or system step	What happens
1	Main agent	Claims: Notify claim	Also from the policy page's Claims tab. The policy must be ACTIVE or EXPIRED and in the agent's scope; the card shows product, participant, period of cover and sum covered.
2	Main agent	Notify claim	The event date must fall within the period of cover and not in the future.
3	System	Claim CL/26/nnnnnn	Outbox message Claim notified to the core system; in-app notification to everyone with the claims permission; audit CLAIM_NOTIFIED.
4	Main agent	Claim detail: Supporting documents	Discharge summary, invoices, reports.
5	Claims (Operations officer)	Claims: Claim detail	Remarks are mandatory and shared with the agent.
6	Claims (Operations officer)	Claim detail	Allowed moves: Submitted → Under review or Rejected; Under review → Acknowledged or Rejected; Acknowledged or Rejected → Closed.
7	System	Agent notified	The agent opens the claim from the notification and reads IIFT's remarks.

Journey 8: Support issue with SLA

Starts with	Sub-agent (any user)
Ends with	Issue CLOSED by the reporter, or reopened
Time in the demonstration	about 5 minutes

RFP requirements	AP-55–AP-57, BO-29–BO-31, COM-05
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Response and resolution targets per priority are parameters (Critical 1 h / 4 h, High 2 h / 24 h, Medium 8 h / 72 h, Low 24 h / 168 h). A job checks them every five minutes.

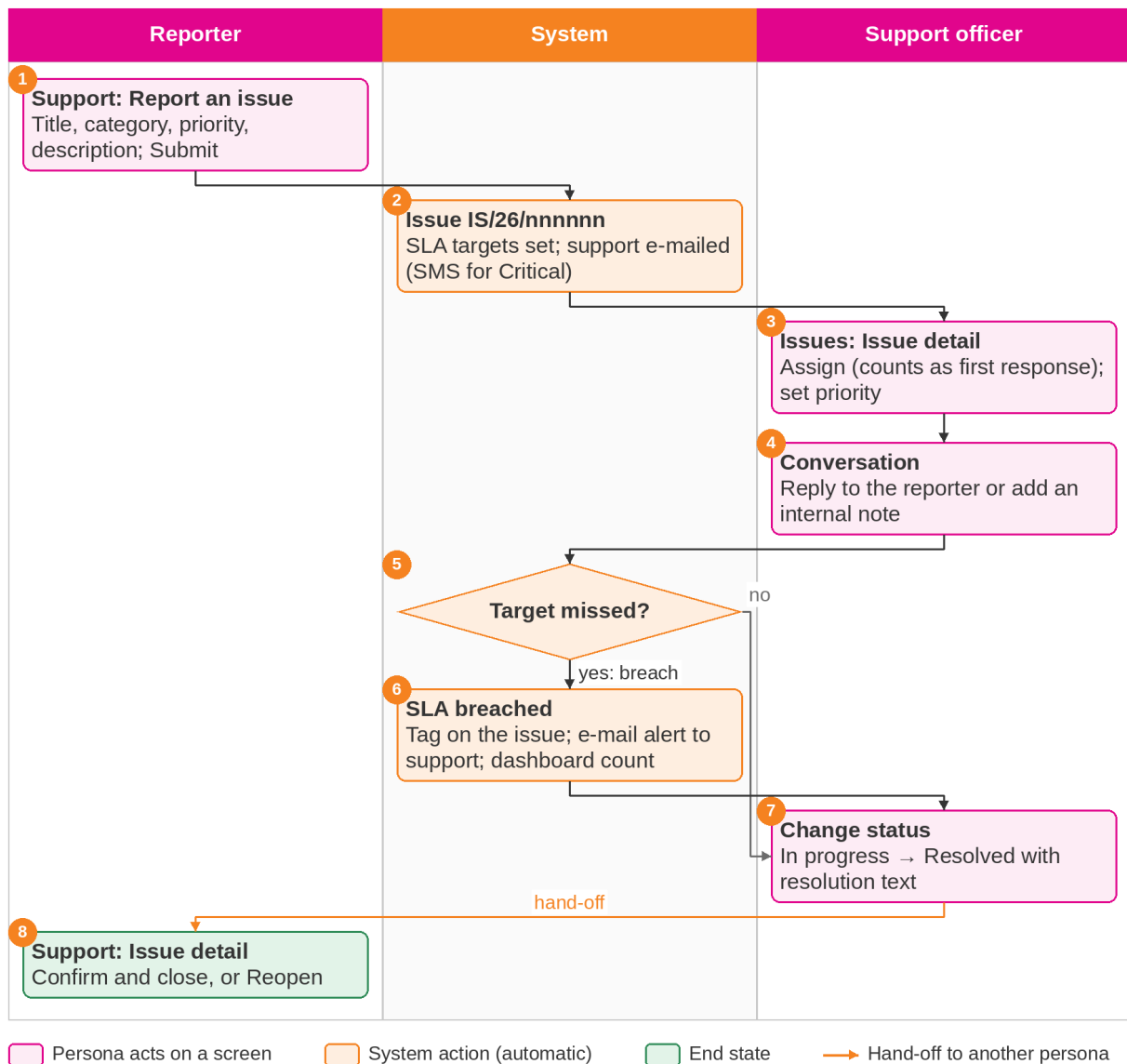


Figure 106: Journey 8: Support issue with SLA

Step	Who	Screen or system step	What happens
1	Reporter	Support: Report an issue	Priority definitions are shown on the form: Critical means business stopped.
2	System	Issue IS/26/nnnnnn	The confirmation shows the reference number and the response target; attachments can be added.
3	Support officer	Issues: Issue detail	Assignment to a support user and team; changing priority recalculates both targets from the reporting time.
4	Support officer	Conversation	Replies notify the reporter; internal notes stay in the back-office.
5	System	Decision: Target missed?	Every five minutes the monitor flags issues past their response or resolution target.
6	System	SLA breached	The dashboard queue links to the breached filter.
7	Support officer	Change status	The resolution is e-mailed to the reporter.
8	Reporter	Support: Issue detail	Reopening returns the issue to In progress.

Journey 9: End of day, FIN posting and reconciliation

Starts with	Scheduler (or Finance officer on demand)
Ends with	Run COMPLETED; FIN posting queued; reconciliation MATCHED, or MISMATCH with the unmatched receipts listed
Time in the demonstration	about 3 minutes
RFP requirements	FFR01 FTPHP08, FFR02 APHA07, FFR03 APPT08, FFR04 APKT07, FFR05 APAOS08, INT-04, INT-13–INT-15

End of day runs at the configured time and can be run again for a date; a re-run is a new revision that replaces the earlier FIN posting.

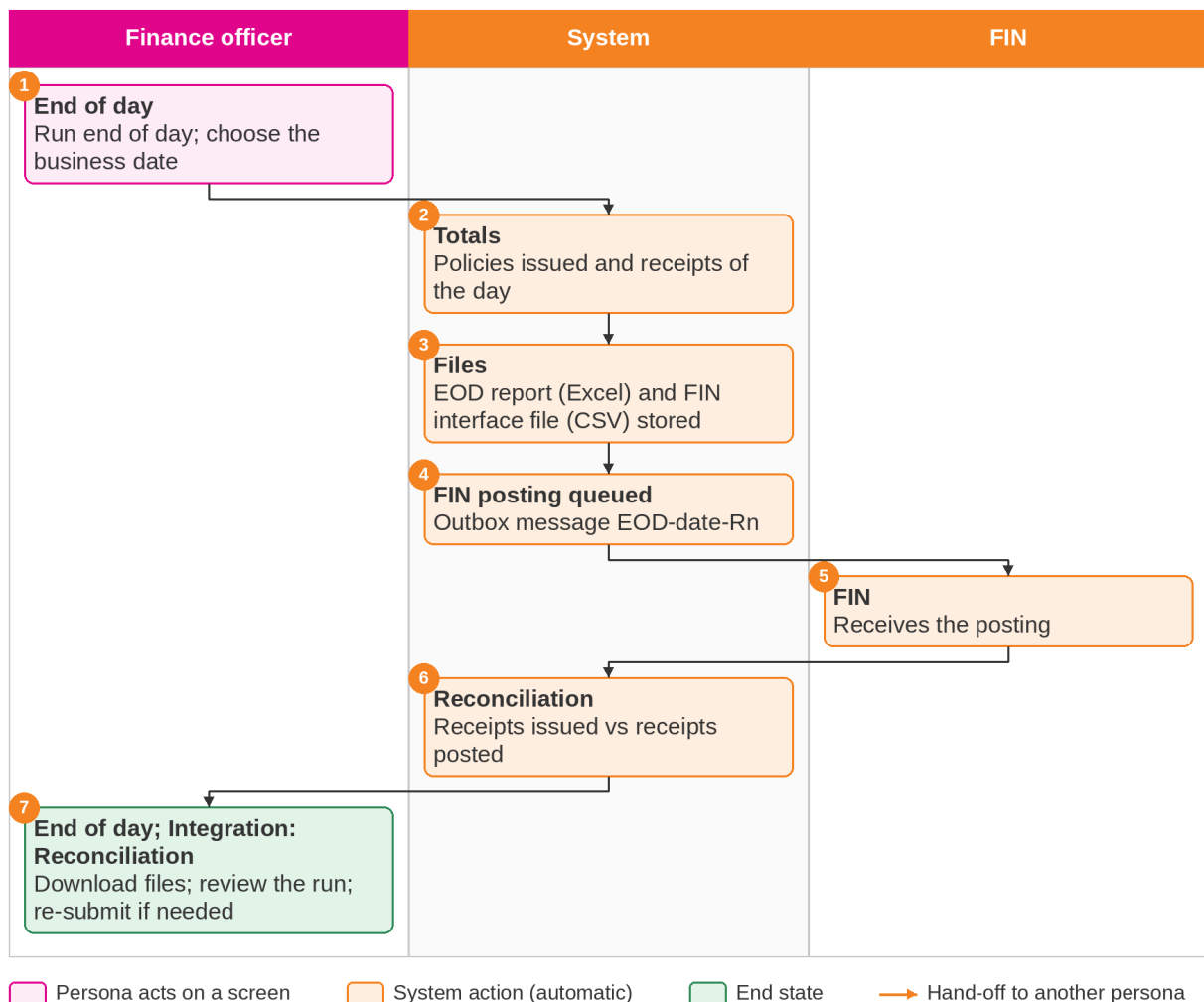


Figure 107: Journey 9: End of day, FIN posting and reconciliation

Step	Who	Screen or system step	What happens
1	Finance officer	End of day	The scheduler does the same at the EOD time. A run already in progress for the date is refused.
2	System	Totals	Counts and amounts by product.
3	System	Files	Report sheets: Summary, Policies issued, Receipts. FIN file: one PI line per policy, one RC line per receipt, agreed with Finance in the interface specification.
4	System	FIN posting queued	Idempotency key per revision so FIN ignores a duplicate; retries with back-off; dead letter after six attempts, visible on the Integration page.
5	FIN	FIN	In simulated mode the delivered outbox messages stand in for FIN.

Step	Who	Screen or system step	What happens
6	System	Reconciliation	Expected and matched counts and amounts; status MATCHED or MISMATCH with the unmatched receipt numbers; audit RECONCILIATION_RUN.
7	Finance officer	End of day; Integration: Reconciliation	A mismatch is investigated in the message queue and log, then Run reconciliation again for the date.

Journey 10: Administration: users, roles, workflows, products and parameters with audit

Starts with	System administrator
Ends with	Configuration changed with before and after values in the audit trail
Time in the demonstration	about 8 minutes
RFP requirements	BO-02–BO-04, BO-16, BO-18, BO-26–BO-28, BO-32, BO-33, COM-04, COM-09

Every maintenance action below writes an audit record with the previous and new values, the user, time, address and correlation id; the table is append-only.

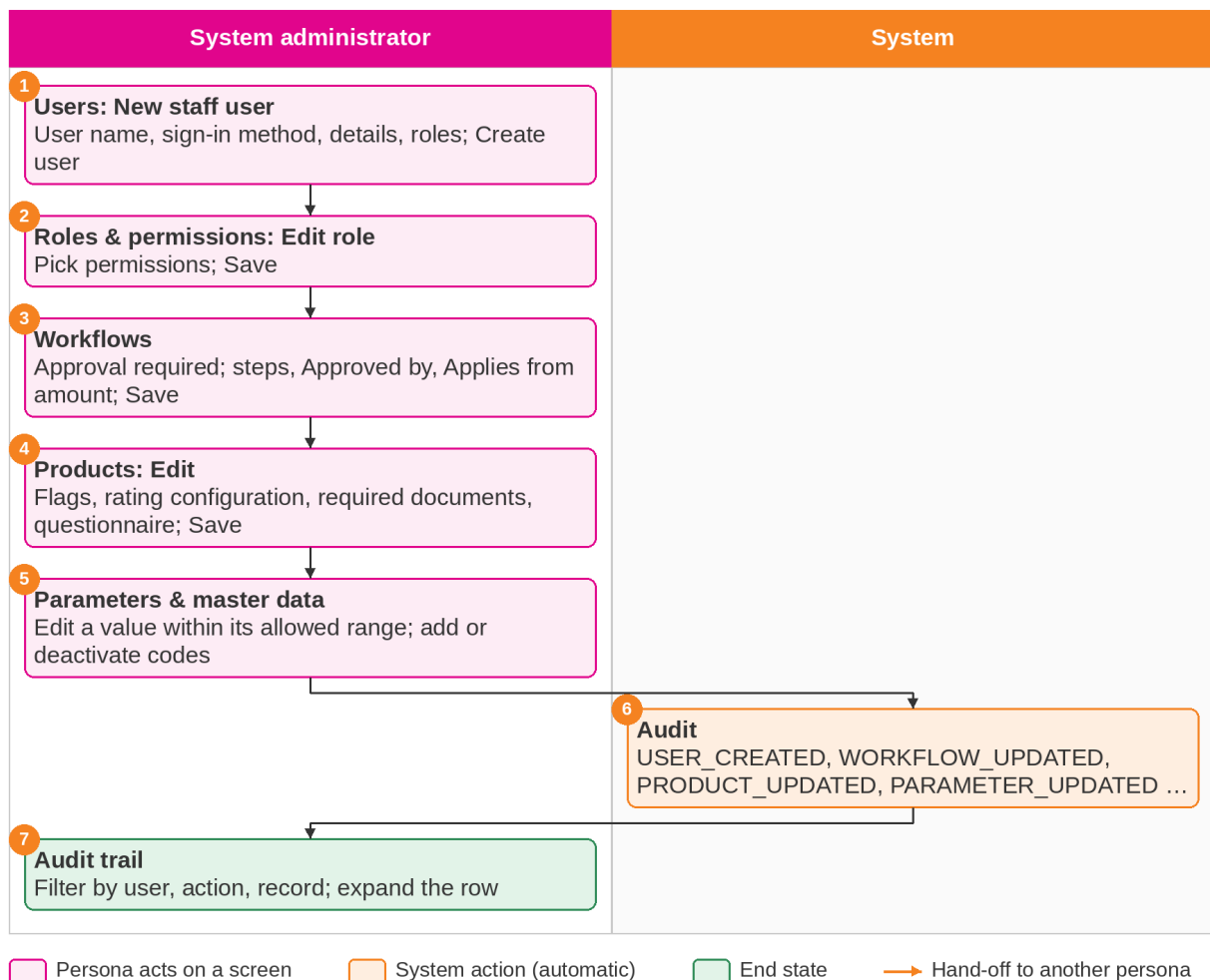


Figure 108: Journey 10: Administration: users, roles, workflows, products and parameters with audit

Step	Who	Screen or system step	What happens
1	System administrator	Users: New staff user	Local accounts get a one-time temporary password shown once; directory accounts sign in through AD. Edit, Disable, Unlock and Reset password are confirmed before they run.
2	System administrator	Roles & permissions: Edit role	Permissions are limited to the role's module; system roles cannot be deleted; a role in use cannot be deleted.

Step	Who	Screen or system step	What happens
3	System administrator	Workflows	Up to five steps per transaction type; only approval permissions are accepted; a step with an amount applies from that amount.
4	System administrator	Products: Edit	Payment before issuance, renewal allowed and offered for sale; JSON validated on save; the change applies to new quotations.
5	System administrator	Parameters & master data	Grace days, quotation validity, renewal notice, e-signature link hours, AML threshold, SLA hours, integration attempts; code tables for banks, districts, claim types, reasons.
6	System	Audit	Each with before and after values.
7	System administrator	Audit trail	Evidence for internal audit and the regulator.

The participant's only screen, reached from the e-mailed link in journeys 1 and 6:

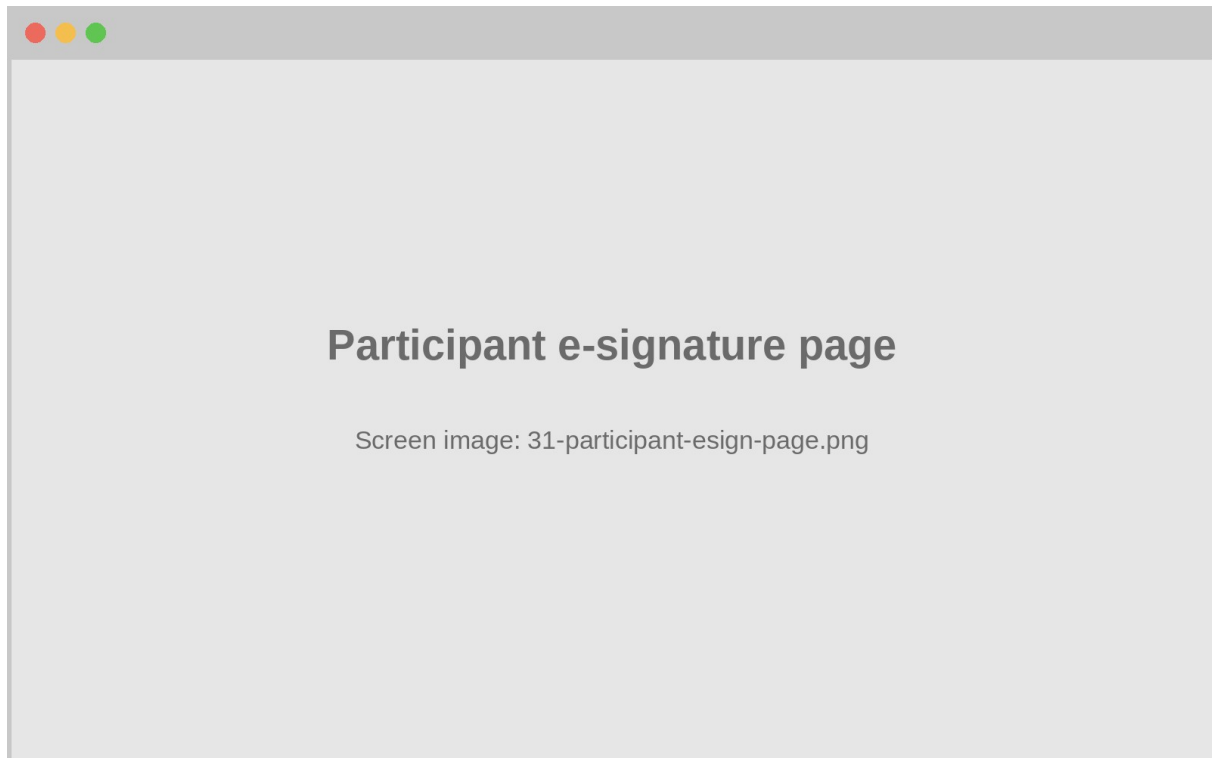


Figure 109: Participant e-signature page: Quotation summary, typed name, declaration and signature; the link works once.

7.15 Screen index

Every screen of the product with its path and the personas that use it. Paths are the browser routes of the single-page application; record pages take the record's identifier.

Screen	Path	Personas
Shared		
Sign in	/login	Everyone
Change password	/change-password	Everyone
Notifications	/portal/notifications, /backoffice/notifications	Everyone
Participant e-signature page	/esign/{token}	Participant (no account)
Agent & Banca Portal		
Dashboard	/portal	All portal personas
New quotation (Product, Participant, Coverage, Review)	/portal/quotations/new	All portal personas
Quotations & policies	/portal/policies	All portal personas
Policy detail (checklist, tabs, dialogs: Declarations, Nominees, Signature, Send e-signature link, Endorsement, Cancellation, E-mail documents)	/portal/policies/{id}	All portal personas
Renewals	/portal/renewals	All portal personas

Screen	Path	Personas
Participants, Register participant, Participant detail, Request update	/portal/participants, /participants/new, /participants/{id}	All portal personas
Billing & payments, Submit payment drawer, Payment detail	/portal/billing, /billing/payments/{id}	All portal personas; bulk payment typically the bank supervisor
Claims, Notify claim, Claim detail	/portal/claims, /claims/new, /claims/{id}	All portal personas
My requests, Request detail	/portal/requests, /requests/{id}	All portal personas
Team & hierarchy, Team member, Register agent / bank officer	/portal/team, /team/{id}, /team/register	All; registration by main agent and bank supervisor
Commission	/portal/commission	All portal personas
Reports	/portal/reports	All portal personas
Support, Report an issue, Issue detail	/portal/issues, /issues/{id}	All portal personas
My profile	/portal/profile	All portal personas
Back-office		
Dashboard	/backoffice	All back-office personas
Approvals, Approval detail	/backoffice/approvals, /approvals/{id}	Operations supervisor, Underwriter, Finance officer
Agents & bankers, Register agent or bank officer, Agent detail	/backoffice/agents, /agents/new, /agents/{id}	Operations officer (maker), checkers, Compliance, Support (read)
Agencies & banks, Agency detail	/backoffice/agencies, /agencies/{id}	Operations officer, System administrator, others read
Policies, Policy record	/backoffice/policies, /policies/{id}	Operations, Underwriter, Finance, Compliance
Participants, Participant record	/backoffice/participants, /participants/{id}	Operations, Underwriter, Finance, Compliance
Payments, Payment detail	/backoffice/payments, /payments/{id}	Finance officer; Operations read
Claims, Claim detail	/backoffice/claims, /claims/{id}	Operations officer
AML / KYC (Screening cases, Watch-lists, Review dialog)	/backoffice/aml	Compliance officer
Document checks	/backoffice/documents	Operations officer, Operations supervisor, Underwriter
Audit trail	/backoffice/audit	Compliance officer, System administrator, Internal auditor
Issues, Issue detail	/backoffice/issues, /issues/{id}	Support officer, Operations officer
Reports, Scheduled reports	/backoffice/reports	All back-office personas; schedules for Finance, Management, System administrator
End of day	/backoffice/eod	Finance officer
Integration (Message queue, Integration log, Reconciliation)	/backoffice/integration	Finance officer, System administrator
Users, New staff user	/backoffice/users	System administrator
Roles & permissions, Edit role	/backoffice/roles	System administrator
Workflows	/backoffice/workflows	System administrator
Products, Edit product	/backoffice/products	System administrator
Parameters & master data	/backoffice/settings	System administrator

Table 35: Screen index

8. Functional Solution

SalesVerse 2.0 is iorta TechNXT's distribution platform for Takaful and insurance. For IIFT it is configured as the IIFT Agent/Banca Portal & Back-office: one portal for agents and bank officers and one back-office for IIFT and IITH staff, sharing a single database, workflow engine, document store and audit trail. A transaction submitted in the portal appears in the back-office at once, and every action on it is recorded.

Functions are listed by RFP area in Section 5, users in Section 7, journeys in Section 7.14 and screens in Section 7.15. This chapter covers the rules that tie them together.

8.1 Policy lifecycle

Only the server changes a policy's status, after validation and any required approval. Validation covers product eligibility, mandatory documents, the AML result, the agency payment block and the authority limit. Each change is written to the policy history and the audit trail.

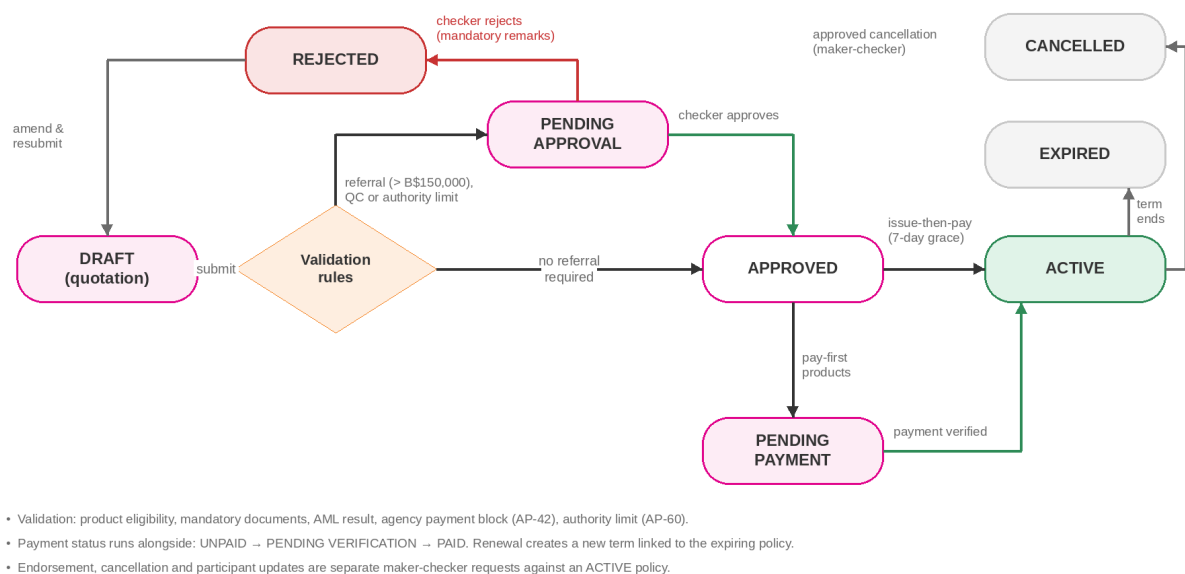


Figure 110: Policy lifecycle and status transitions

Status	Meaning	Next actions
DRAFT	Quotation saved; editable by the maker.	Submit
PENDING APPROVAL	Referral above the authority limit, quality check or other approval.	Approve or reject
PENDING PAYMENT	Approved; waiting for verified payment (pay-first products).	Payment verified
ACTIVE	In force; e-Policy and e-Receipt issued.	Endorse, cancel, renew, expire
REJECTED	Rejected with mandatory remarks.	Amend and resubmit
EXPIRED	Cover period ended.	Renew where enabled
CANCELLED	Cancellation approved.	None

8.2 Workflow engine and maker-checker

Approvals are data, not code. A workflow definition sets the transaction type, the number of levels, the permission needed at each level and amount thresholds that add a level, for example a second approval from B\$300,000 sum covered. On final approval a handler applies the change in the same database transaction as the audit record and the notification event. The API rejects any attempt by the maker to approve their own request. When Compliance confirms an AML match, the pending requests for that agent

or participant are rejected automatically with the Compliance remarks and the maker is notified. Journey 4 in Section 7.14 shows the flow.

Transaction	Level 1 checker	Level 2 checker	Rule
Agent / Banca registration	Banca/Sales officer	Banca/Sales manager	Always two levels
Agent profile update	Banca/Sales officer	None	None
Participant update	Underwriting officer	None	None
Quotation above B\$150,000 financing	IIFT Sales	Underwriting manager	Referral rule
New policy, sum covered from B\$300,000	Underwriting officer	Underwriting manager	Amount threshold
Quality check (Mortgage Takaful)	Underwriting / QC	None	Product rule
Endorsement	Underwriting officer	Underwriting manager	Contribution change threshold
Cancellation	Underwriting officer	Underwriting manager	Always two levels
Payment verification	Finance officer	Finance manager	Amount threshold
AML possible match	Compliance officer	Compliance manager	On escalation

Table 36: Default workflow definitions, confirmed in design

8.3 Seven-day grace period

A nightly job finds issued policies unpaid for more than seven days and blocks issuance for every agent of that agency. Agents see the reason on their dashboard. The block lifts automatically once Finance verifies the payment.

8.4 Notifications

Event	Recipients	Channels
Request submitted, approved or rejected (with remarks)	Maker; checker queue	Portal, e-mail
Policy issued with e-Policy and e-Receipt	Agent; participant	E-mail with PDF, portal
Payment overdue: reminder on day 5, block on day 8	Agent; agency principal; Finance	Portal, e-mail, SMS
Agency block lifted	All agents of the agency	Portal, e-mail
Renewal due (60, 30 and 7 days) and expiry	Agent	Portal, e-mail
Agent document expiring	Agent; Banca/Sales officer	Portal, e-mail
AML possible match	Compliance	Back-office, e-mail
Issue assigned, updated or SLA breached	Assignee; reporter; supervisor	Portal, e-mail
Dead-letter or EOD failure	Support team; IIFT IT	E-mail, monitoring alert

Table 37: Notification events (templates and channels configurable)

8.5 Reports

The platform has 12 standard reports with filters, Excel, CSV and PDF export and scheduled e-mail delivery. The go-live catalogue below is confirmed in design; reports not yet built are delivered during implementation within the fixed price.

Report	Primary users	Content
Daily issuance (EOD) report	Finance, Banca	Issued policies by product, bank and agent
FIN interface report	Finance	Entries sent to FIN with control totals
Receipt and BRR reconciliation	Finance	Matched and unmatched receipts and issuance
Policy register by status	Banca, Underwriting	Policies with status, dates and contribution
Production report	Management, Banca	Production by agent, agency, bank, product, period

Report	Primary users	Content
Outstanding payments and ageing	Finance, Banca	Unpaid policies by ageing bucket
Grace-period breaches and agency blocks	Finance, Banca	Policies unpaid over 7 days; blocked agencies
Renewals due and expiries	Banca, agents	Policies due for renewal or expired
Claims notification register	Claims, Banca	Notifications and their status
AML screening and review	Compliance	Screenings, possible matches, outcomes
Agent register and document expiry	Banca/Sales	Agents, hierarchy, status, licence and document validity
Approval turnaround	Management	Requests by type, ageing and turnaround
Issue log and SLA	IT, Management	Issues by priority, status and SLA result
User access review	IT security, Audit	Users, roles, last login, dormant accounts
Commission / referral fee statement	Agents, Finance	Statement retrieved from FIN (INT-05)

Table 38: Report catalogue for go-live

9. Products, Rating and Process Flows

All seven products in RFP section 1.1 and Appendix 3 are configured in the working application. Each product definition holds plans, sums covered, eligibility, required documents, questionnaires, renewal availability, payment mode (pay first, or issue then pay within the grace period) and its rating engine.

9.1 Product configuration

Code	Product	Line	Rating inputs	Key rules
FTP-HP	Financing Takaful Plan – Hire Purchaser	Mortgage	Financing amount, tenure, profit rate, age next birthday	Above B\$150,000 referred to IIFT Sales; QC before contract; issue-then-pay with 7-day grace
FTP-NP	Financing Takaful Plan – Non-Participating	Mortgage	As FTP-HP	Same engine and referral rule
PFT	Property Financing Takaful Plan	Mortgage	As FTP-HP (decreasing term)	Same engine; property financing documents
PHA	Personal Home Assistant Takaful Plan	Annual	Cover period 1 or 2 years	Employer IC, employee IC and details, passport, labour licence
PRO	Professional Takaful Plan	Annual	Plan A/B/C B\$15k/30k/50k; additional cover B\$90/120/140	Occupational Class I only; nominee, beneficiary or executor IC
KHR	Khairat Takaful Plan	Annual	Plan A/B/C B\$5k/10k/15k; Individual or Wider (Child)	Renewal switched off
OSA	Overseas Student Assist Takaful Plan	Annual	Basic B\$20k or Tertiary B\$50k	Brunei citizen, registered student, age 65 or under; student ID

Table 39: Product configuration

9.2 Contribution calculation

The Mortgage Takaful engine (FTP-HP, FTP-NP, PFT) calculates contribution from financing amount, financing period, profit rate and age next birthday using decreasing-term rate tables. Annual products use plan-based tables with riders and cover periods.

Rate tables, loadings, minimum contributions, rounding and authority limits are master data, changed in the back-office with every change audited. Maker-checker approval and effective-dated versions of rate tables, with the version recorded on each quotation, are added during implementation; until then a second person checks each rate change against IIFT's signed rate sheet. The working application holds indicative rates; IIFT's actuarial rates and wakalah fee and tabarru' split are loaded and verified during design and SIT.

9.3 Process flows (Appendix 3)

FFR01: Financing Takaful Plan, Hire Purchaser (FTPHP)

Step	RFP step	Portal / participant	Back-office / system
FTPHP01	Capture participant details	Full name, IC number, address, phone; duplicate check and AML screening.	Possible matches go to Compliance.
FTPHP02	Attach documents	Proposal form, product disclosure sheet, IC copy, HP agreement or financier approval; HP statement for existing HP.	Checklist enforces mandatory documents per case.
FTPHP03	Contribution	Calculated from financing amount, period, profit rate and age next birthday.	Above B\$150,000: referred to IIFT Sales.
FTPHP04	Drawdown letter	Agent uploads the drawdown letter and remaining documents and enters drawdown details.	Back-office validates documents.
FTPHP05	Issue receipt	e-Receipt issued for the contribution collected.	Receipt number assigned; FIN entry queued.
FTPHP06	Quality check	Status shows Pending Approval (QC).	Underwriting approves or rejects with remarks.

Step	RFP step	Portal / participant	Back-office / system
FTPHP07	e-Policy and e-Receipt	PDFs available to download and e-mailed to the participant.	Stored in the encrypted document store.
FTPHP08	EOD and final issuance	None	EOD report, master report, FIN file, reconciliation.

FFR02: Personal Home Assistant Takaful Plan (APHA)

Step	RFP step	Portal / participant	Back-office / system
APHA01	Capture participant details	Full name, IC number, address, phone.	Duplicate check, AML screening.
APHA02	Select cover	New or renewal; 1 or 2 years.	Plan rules from configuration.
APHA03	Quotation and questionnaire	Summary shown; Appendix II questionnaire completed.	Questionnaire version stored.
APHA04	Requirements	Employer IC, employee IC, employee details, passport, labour licence.	Checklist and expiry dates.
APHA05	Payment	Single or bulk payment with proof.	Finance verifies.
APHA06	e-Policy and e-Receipt	Issued on payment confirmation.	Policy becomes ACTIVE.
APHA07	FIN reporting	None	Daily report, FIN file, BRR reconciliation.

FFR03: Professional Takaful Plan (APPT)

Step	RFP step	Portal / participant	Back-office / system
APPT01	Capture participant details	Full name, IC number, address, phone.	Duplicate check, AML screening.
APPT02	Quotation, new or renewal	Plan A B\$15,000, Plan B B\$30,000, Plan C B\$50,000.	Occupational Class I only.
APPT03	Additional cover	Optional: Plan A B\$90, Plan B B\$120, Plan C B\$140.	Configured as rider.
APPT04	Summary and questionnaire	Summary shown; questionnaires completed.	Answers can trigger referral rules.
APPT05	Documents	Questionnaire, IC copy, IC of nominee, beneficiary or executor.	Checklist validation.
APPT06	Payment	Payment with proof.	Finance verifies.
APPT07	Policy and receipt	e-Policy and e-Receipt on confirmation.	Policy becomes ACTIVE.
APPT08	EOD reporting	None	EOD report, FIN entries, BRR reconciliation.

FFR04: Khairat Takaful Plan (APKT)

Step	RFP step	Portal / participant	Back-office / system
APKT01	Capture participant details	Full name, IC number, address, phone.	Duplicate check, AML screening.
APKT02	Quotation	Plan A B\$5,000, B B\$10,000, C B\$15,000; Individual or Wider (Child).	Renewal switched off.
APKT03	Summary and questionnaire	Summary shown; questionnaires completed.	Stored with quotation.
APKT04	Documents	Questionnaires, IC copy, nominee's IC copy.	Checklist validation.
APKT05	Payment	Payment with proof.	Finance verifies.
APKT06	Policy issuance	e-Policy and e-Receipt on confirmation.	Policy becomes ACTIVE.
APKT07	EOD reporting	None	EOD report to FIN; BRR reconciliation.

FFR05: Overseas Student Assist Takaful Plan (APAOS)

Step	RFP step	Portal / participant	Back-office / system
APAOS01	Capture participant details	Full name, IC number, address, phone.	Duplicate check, AML screening.
APAOS02	Select cover	Basic B\$20,000 or Tertiary B\$50,000.	Plan rules from configuration.
APAOS03	Eligibility	Brunei citizen, registered student, age 65 or under.	Ineligible cases cannot proceed.
APAOS04	Summary and questionnaire	Summary shown; Appendix IV questionnaire.	Stored with quotation.
APAOS05	Documents	Questionnaires, IC copy, student ID, IC of nominee, beneficiary or executor.	Checklist validation.
APAOS06	Payment	Payment with proof.	Finance verifies.
APAOS07	e-Policy and e-Receipt	Issued on confirmation.	Policy becomes ACTIVE.
APAOS08	EOD reporting	None	EOD report, FIN file, BRR reconciliation.

9.4 End-of-day processing

EOD runs on a schedule and can be re-run from the back-office. One locked job runs across application nodes. Each posting to FIN carries a revision number and its own idempotency key, and a re-run is marked as replacing the previous revision, so FIN can reject duplicates and keep only the latest figures for the day.

#	Step	Description
1	Cut-off	Configurable daily cut-off; later transactions roll to the next business day.
2	Issuance report	Policies issued, endorsed and cancelled and receipts approved, by product, bank and agent.
3	Final issuance	Master report for Mortgage Takaful final issuance (FFR01 step 8).
4	FIN interface file	Contribution and receipt entries, with wakalah fee and tabarru' split where applicable.
5	Reconciliation	Receipts reconciled today; BRR reconciliation added once IIFT confirms the BRR format.
6	Grace-period check	Policies unpaid for more than seven days found; agency blocks applied or lifted.
7	Sign-off	Finance reviews exceptions and signs off the run; run history kept.

10. Solution Architecture

10.1 SalesVerse 2.0 logical architecture

SalesVerse 2.0 is a modular monolith: one deployable application split into modules that follow the RFP functional areas. For 26 named users and about 600 policies a year this is simpler to run than microservices, and a module can be split out later without a rewrite (NFR-20).

Users reach two React applications through a reverse proxy and web application firewall. The back-office path is published only on the internal network. Both applications call one versioned REST API. The integration layer isolates IITH systems through a transactional outbox, and PostgreSQL is the system of record.

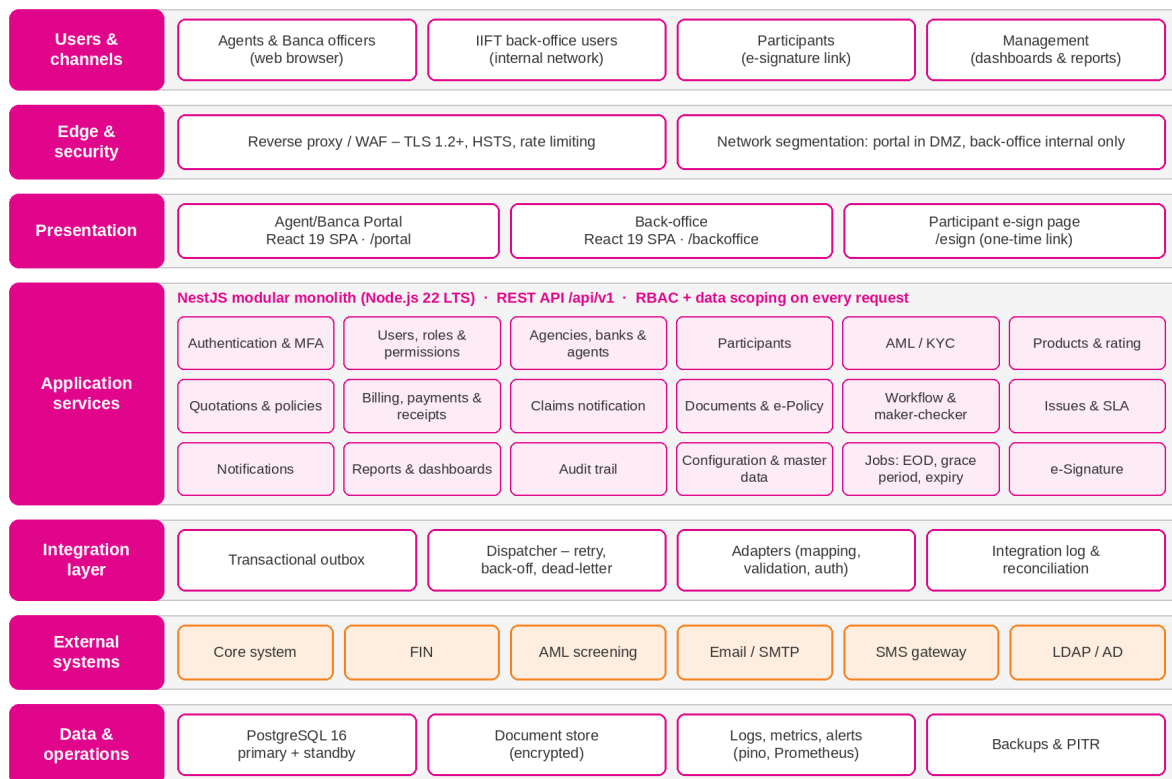


Figure 111: SalesVerse 2.0 logical architecture

10.2 SalesVerse 2.0 components

Module	Responsibilities	RFP requirements
auth, users-roles	Login, sessions, password policy, roles, permissions, data scopes	AP-01–04, AP-59, BO-01–04
agencies, agents	Agencies, banks, branches, agents, hierarchy, status, onboarding	AP-05–12, BO-05–12
participants, aml	Participant record, duplicate checks, watch-list screening, Compliance review	AP-11–16, AP-48, BO-13–15
products	Catalogue, plans, rate tables, rating engines, questionnaires	AP-17–19, BO-32
policies	Quotations, issuance, status, renewal, endorsement, cancellation, history	AP-20–36, AP-60
billing	Payments, allocation, proofs, verification, e-Receipts, grace-period rule	AP-37–42
claims, issues	Claim notifications; issues with priority and SLA timers	AP-43, AP-55–57, BO-29–31
documents, esign	Encrypted store, upload checks, PDF output, e-mail, on-screen and e-mail-link signatures	AP-44–47, AP-62, COM-06

Module	Responsibilities	RFP requirements
workflow	Workflow definitions, approval requests and actions, handlers	AP-33, AP-49–51, BO-16–19, COM-04
notifications	Templates; e-mail, SMS and in-portal delivery; delivery log	AP-34, AP-54, COM-05
reports, dashboard	Report definitions, exports, schedules, KPIs	AP-52–53, AP-58, BO-20–25, COM-08
audit, config	Append-only audit; parameters, master data, numbering	BO-26–28, BO-32–33, COM-03, COM-09
integration, jobs	Outbox, dispatcher, adapters, integration log; EOD, grace period, expiry, reminders	INT-01–15, AP-42

Table 40: Application modules

10.3 Data architecture

Entity group	Purpose
Agency, Bank, Branch, Agent	Distribution structure, agent code, type, status, parent agent, licence
User, Role, Permission, Session	Access control; sessions held on the server
Participant, Nominee	Shared participant record; IC and passport numbers encrypted
Product, Plan, RateTable, Questionnaire	Versioned product configuration
Quotation, Policy, PolicyTransaction	Policy lifecycle and history
Payment, PaymentAllocation, Receipt	Billing with exact NUMERIC amounts
Claim, Issue, Document	Claim notifications, issues with SLA timestamps, document metadata and checksums
WorkflowDefinition, ApprovalRequest, ApprovalAction	Maker-checker
AmlScreening, AuditEvent	Screening results; append-only audit with before and after values
OutboxEvent, IntegrationLog, EodRun, AgencyBlock	Integration delivery, EOD runs and grace-period blocks

Table 41: Key data entities

We chose PostgreSQL over MongoDB because IIFT's data is relational and financial. The comparison below summarises the reasons.

Criterion	PostgreSQL 16 (selected)	MongoDB
Data shape	Relational: agency, main agent, sub-agent; policy, participant, nominees, payments, receipts, claims	Suits self-contained documents with few relationships
Transaction integrity (NFR-16)	Issuance, receipt, audit and outbox commit in one ACID transaction; row locks for numbering	Multi-document transactions exist but fit this model less naturally
Financial accuracy	Exact NUMERIC plus foreign keys, CHECK and UNIQUE constraints	Decimal128; integrity rules mostly in application code
Reporting (BO-22–25)	Joins and aggregates written directly in SQL	Aggregation pipelines; joins via \$lookup
Flexible data	JSONB for questionnaires, product settings and audit values	Native
Operations	Streaming replication and point-in-time recovery; managed on every cloud and on-premise	Replica sets; a second skill set for IITH
Licence	PostgreSQL licence, no fee	SSPL; enterprise features are paid
Fit to volume	Under 5 GB of structured data in five years	Strength is very large schemaless data, not needed here

Table 42: Database selection

10.4 Integration architecture

Each outbound message is written to an outbox table in the same transaction as the business change. A policy therefore cannot be issued without its FIN entry being queued, and a failed call loses nothing. A

dispatcher delivers messages through adapters, retries with back-off, logs every attempt and moves exhausted messages to a dead-letter queue shown in the integration monitor. The outbox, retry and dead-letter handling run in the working application; the IIFT adapters are built during implementation against the interface specification (DEL-09).

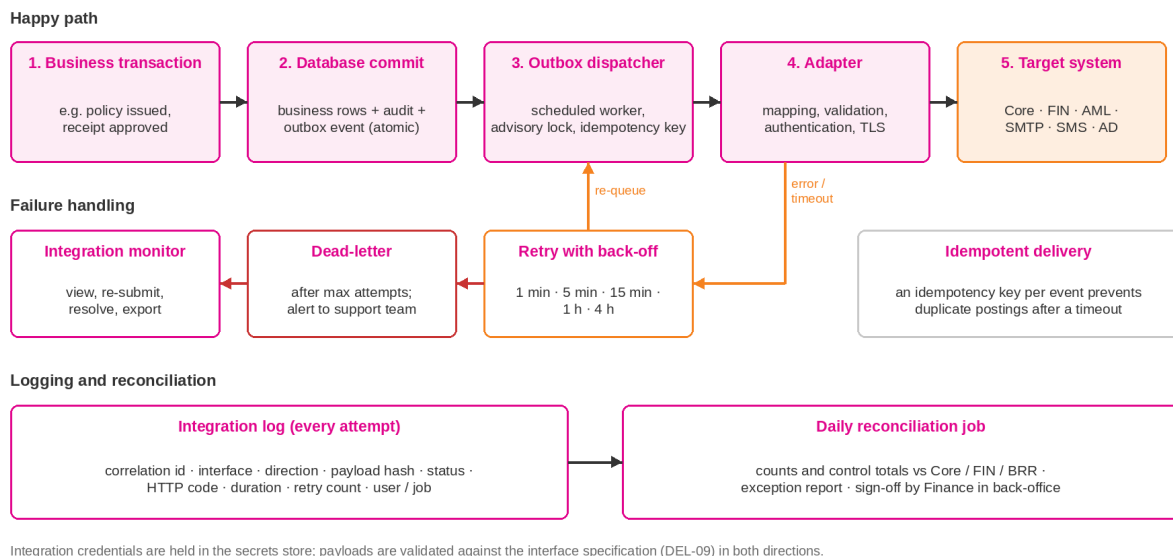


Figure 112: Integration flow with outbox, retry and reconciliation

Interface	Direction	Method	Frequency	Data	Security
Core system	Both ways	REST API or scheduled file	Event-driven plus daily sync	Agent, agency, bank master; status; policy summary	mTLS or API key; IP allow-list
FIN	Out (commission in)	Interface file over SFTP, or API	EOD; statements on demand	Contribution, receipts, allocations; commission and referral fees	SFTP keys or mTLS
AML screening	Request / response	REST API	On onboarding and change; scheduled re-screen	Name, IC, date of birth, nationality; match result	API key or OAuth2; TLS
Active Directory	Inbound	LDAPS	At login	Back-office authentication and groups	LDAPS service account
SMS gateway	Out	HTTPS API	Real time	OTP and alerts	API key; TLS
E-mail	Out	SMTP with STARTTLS	Real time	Notifications, documents, scheduled reports	Authenticated relay

Table 43: Interface catalogue

10.5 Technology stack

Layer	Technology and version	Reason
Front end	React 19, TypeScript, Vite, Ant Design 6, TanStack Query, React Router	Widely used; accessible components themed to Insurans Islam TAIB colours
API	Node.js 22 LTS, NestJS 12 (TypeScript), modular monolith	Module boundaries follow the RFP functional areas
Data access	Prisma 7 with the PostgreSQL driver adapter; versioned SQL migrations	Typed, parameterised queries only
Database	PostgreSQL 16	See data architecture
Documents and reports	pdfkit for PDFs, exceljs for Excel; encrypted local volume, MinIO or S3	No native dependencies
Jobs	NestJS scheduler with PostgreSQL advisory locks	One runner across replicas for EOD, grace period and outbox

Layer	Technology and version	Reason
Observability	pino JSON logs with correlation id; health endpoints; Prometheus metrics; Grafana	Central monitoring and alerting
Testing	Vitest (unit and API tests against PostgreSQL; web component tests with Testing Library), Playwright, k6	Run in the CI pipeline
Delivery	Non-root Docker images; docker compose on-premise; Kubernetes or ECS optional; GitHub Actions or GitLab CI	Same images in every environment

Table 44: SalesVerse 2.0 technology stack

All components are open source under permissive licences. The CI pipeline produces a CycloneDX software bill of materials of the production dependencies on every build; it is delivered with the technical manual (COM-17).

10.6 Non-functional requirements

Security requirements (NFR-08 to NFR-14) are covered in Section 11. The remaining targets are:

Area	Target and approach
Performance (NFR-01)	Page load under 3 s; standard API calls p95 under 1 s; reports on 12 months of data under 10 s. Proven by k6 tests (DEL-18).
Capacity (NFR-02, 03, 06, 07)	100 concurrent and 500 named users; ten times Appendix 2 volume. Stateless API nodes scale out.
Availability (NFR-04, 05)	99.5% overall and 99.9% in IIFT business hours; two application nodes and a PostgreSQL standby.
Backup and DR (NFR-17 to 19)	Daily full backup and continuous WAL archiving; 35 days online, monthly copies for 12 months; RPO 15 minutes, RTO 4 hours; annual DR test.
Integrity and errors (NFR-15, 16)	ACID transactions and idempotency keys; users see a generic message with a correlation id.
Monitoring and logs (NFR-26 to 28)	Health, metrics, job and integration monitors; alerts on downtime, error rate, latency, failed jobs and dead-letters; logs to a central store.
Retention and quality (NFR-29, 30)	Policy and financial records kept 7 years after expiry by default; validation on every input plus database constraints.
Usability (NFR-22 to 24)	One design system; WCAG 2.1 AA target; current and previous versions of Chrome, Edge, Firefox and Safari.
API standards (NFR-25)	REST and JSON over HTTPS, OpenAPI 3, ISO 8601 dates, versioned URLs (/api/v1).

11. Security Architecture & Controls

This chapter is the single description of security controls; other chapters refer to it. Controls are aligned with OWASP ASVS Level 2, the OWASP Top 10, Brunei's Personal Data Protection Order 2025 (PDPO) and AMBD's technology risk expectations for Takaful operators. We claim alignment, not certification. IIFT's policies are mapped in detail in the Solution Architecture (DEL-06).

11.1 Controls

Control	Implementation (defaults configurable)	RFP reference
Authentication	Argon2id password hashing; rate-limited login; MFA by e-mail OTP or authenticator app for back-office (optional for portal); Active Directory single sign-on	AP-01, BO-01, NFR-09, INT-06
Password policy	12 characters minimum with complexity; last 5 passwords blocked; 90-day expiry; change at first login. All configurable	AP-02
Sessions	Sessions held on the server; HttpOnly, Secure, SameSite=Strict cookie; CSRF token; new session id at login; 15-minute idle and 8-hour absolute timeout; optional single session; revoked at logout	AP-03
Lockout	Locked for 30 minutes after 5 failed attempts (configurable); alert on repeated lockouts	AP-04
Authorisation	Permissions per role plus data scoping by agency, bank, branch and hierarchy, checked by the API on every call; deny by default	AP-59, BO-02, COM-02, NFR-10
Encryption in transit	TLS 1.2 or later with HSTS; TLS also for database, SMTP, LDAPS and APIs	COM-10, NFR-08
Encryption at rest	IC and passport numbers encrypted per field with AES-256-GCM, keys held outside the database; HMAC-SHA256 blind index for exact search; masked on screen; encrypted document storage and backups	COM-01, COM-10, NFR-08
Input and output	Whitelist validation of every request body; parameterised queries only; output encoding; CSP and security headers	NFR-14, NFR-30
File uploads	Type allow-list with magic-byte check, size limit, SHA-256 checksum, ClamAV malware scan (mandatory in production); files served only through authorised API calls	AP-46, COM-06
Audit	Append-only table; a database trigger rejects UPDATE and DELETE; the application role can only insert and read; before and after values, user, time, IP, correlation id	BO-26–28, COM-03, NFR-11
Errors and secrets	Generic error messages with correlation id; secrets in environment or vault, rotated on staff change and yearly	NFR-15, INT-10
System APIs	mTLS or OAuth2 client credentials for integrations; rate limiting	INT-09, INT-10

Table 45: Security controls

The audit trigger, field-level encryption, encrypted document storage and the ClamAV upload scan already run in the working application; the API will not start in production without the scanner configured. MFA, Active Directory sign-on and the remaining parameters are set up during implementation. Personal data is limited to what products and AML/KYC rules need; a data inventory with purpose and retention is delivered with the Solution Architecture.

11.2 Secure development lifecycle

Stage	Practice	Mechanism
Design	Threat model for login, payment, approval and e-signature flows	Workshop, ASVS checklist
Build	Secure coding standard aligned with OWASP ASVS Level 2 and Top 10; peer review of every change	Pull-request review
Pipeline	Static analysis, dependency, secret and container image scans; a failed scan blocks release	CI security gates
Testing	Authorisation tests per role, negative tests, DAST baseline in SIT	API tests, DAST

Stage	Practice	Mechanism
Pre-production	Independent VAPT, remediation and re-test	Independent tester
Operation	Monthly patching, vulnerability monitoring, log review	Maintenance service

11.3 Penetration testing and remediation

An independent tester assesses the portal, back-office, APIs and hosting set-up before production (NFR-13, DEL-17). The cost is in the fixed price. Findings are fixed and re-tested, and no Critical or High finding may be open at go-live. After go-live, vulnerabilities are fixed within these limits (NFR-12, MNT-12):

Severity	Description	Timeline
Critical	Remotely exploitable; data exposure or system compromise	Contained within 24 hours; fixed within 7 days
High	Significant impact, exploitation likely	Within 14 days
Medium	Limited impact or hard to exploit	Within 30 days
Low	Minimal impact; hardening advice	Within 90 days or next release

Table 46: Vulnerability remediation timelines

11.4 Regulatory alignment

For PDPO we provide consent and purpose notices at participant capture, access and correction support, retention rules and breach-notification support. For AMBD expectations we cover access control, change management, outsourcing and cloud considerations (infrastructure options), DR testing and audit trails. AML/CFT record keeping is met by the screening history and review decisions. Screens and documents use Shariah-appropriate terms: participant, contribution, Takaful operator, wakalah and tabarru'.

11.5 Security incident response

Phase	Action
Detect	Alert, log review or user report; logged as a P1 security incident.
Contain	Within 30 minutes: assess scope, isolate the component, revoke sessions or credentials.
Notify	IIFT IT security told immediately; iorta supports regulatory and PDPO notifications.
Recover	Patch, restore from a clean backup if needed, verify integrity, monitor.
Review	Root-cause report within 5 business days; actions tracked to closure.

12. Deployment & Release Management

12.1 Environments

Environment	Purpose	Hosted by	Data	Users
DEV	Development and sprint demos	iorta (both options)	Synthetic	iorta team
SIT	Integration and performance testing	A: IIFT data centre (iorta-hosted until the IIFT VM is ready). B: cloud	Synthetic and masked samples	iorta, IIFT IT
UAT	Acceptance testing and training	A: IIFT data centre. B: cloud	Masked or migrated test data	IIFT users
PROD	Live operation	A: IIFT/IITH data centre. B: cloud, Malaysia region	Production	IIFT users; iorta under change control
DR	Disaster recovery	A: IIFT/IITH DR site. B: second cloud region	Replicated production	Activated per DR plan

Table 47: Environments

12.2 CI/CD pipeline

Every environment runs the same versioned container images, so what passes UAT is what runs in production. The pipeline can run on IITH's GitLab or on GitHub Actions; the repository is held for IIFT.

Stage	Activities
Commit and review	Branch, pull request, mandatory peer review
Build	Type check, lint, compile front end and API, build container images
Test	Unit and API tests against PostgreSQL; web component tests; coverage report
Scan	CodeQL static analysis, npm audit, gitleaks secret scan and Trivy image scan
Package	Versioned, signed images with release notes and SBOM
Deploy to SIT/UAT	Automated deployment and migration; Playwright smoke tests
Approve	Test sign-off and IIFT change approval
Deploy to PROD	Scripted deployment in the approved window; health checks; post-deployment checks

12.3 Release management and rollback

Every production change follows the agreed procedure (MNT-21): change request, impact assessment, SIT and UAT, release notes, IIFT approval and deployment in an approved window.

Release type	Content	Cadence
Emergency fix	P1/P2 defect or security patch	As needed, expedited approval
Maintenance release	Fixes, minor enhancements, patches	Monthly
Feature release	Change requests	Quarterly or as agreed
Platform upgrade	Node.js, PostgreSQL, framework and SalesVerse 2.0 updates	Typically yearly

The previous container images are kept, so the application rolls back in minutes. Database migrations follow the expand-and-contract pattern, which lets the previous version run on the new schema. A verified backup and a recovery marker are taken before each release. Rollback criteria and decision owners are set in the deployment plan (DEL-20).

12.4 Go-live checklist

Area	Readiness criteria
Business	UAT signed off; users trained; procedures and contacts issued; agents and banks informed
Technical	Production built and hardened; performance test passed; backups and monitoring running
Security	VAPT findings fixed and re-tested; certificates installed; privileged access reviewed
Data	Two migration rehearsals; reconciliation signed; master data and rates verified
Integration	All interfaces tested end to end; credentials in the vault
Operations	Operations manual, runbooks and DR plan; support rota and escalation agreed
Decision	Steering Committee go/no-go recorded with open items and rollback criteria

12.5 Cut-over plan

When	Activities
T-10 days	Scope frozen; deployment plan approved; agents and banks informed
T-5 days	Deployment rehearsal; final migration dry-run and reconciliation
T-2 days	Legacy data frozen; manual process continues where needed
T-1 day	Final extract, migration load, reconciliation, business check
T-0	Deployment, smoke tests, integration checks, go/no-go, users enabled
T+1 to T+20 business days	Hypercare: daily health checks and stand-ups, priority fixes

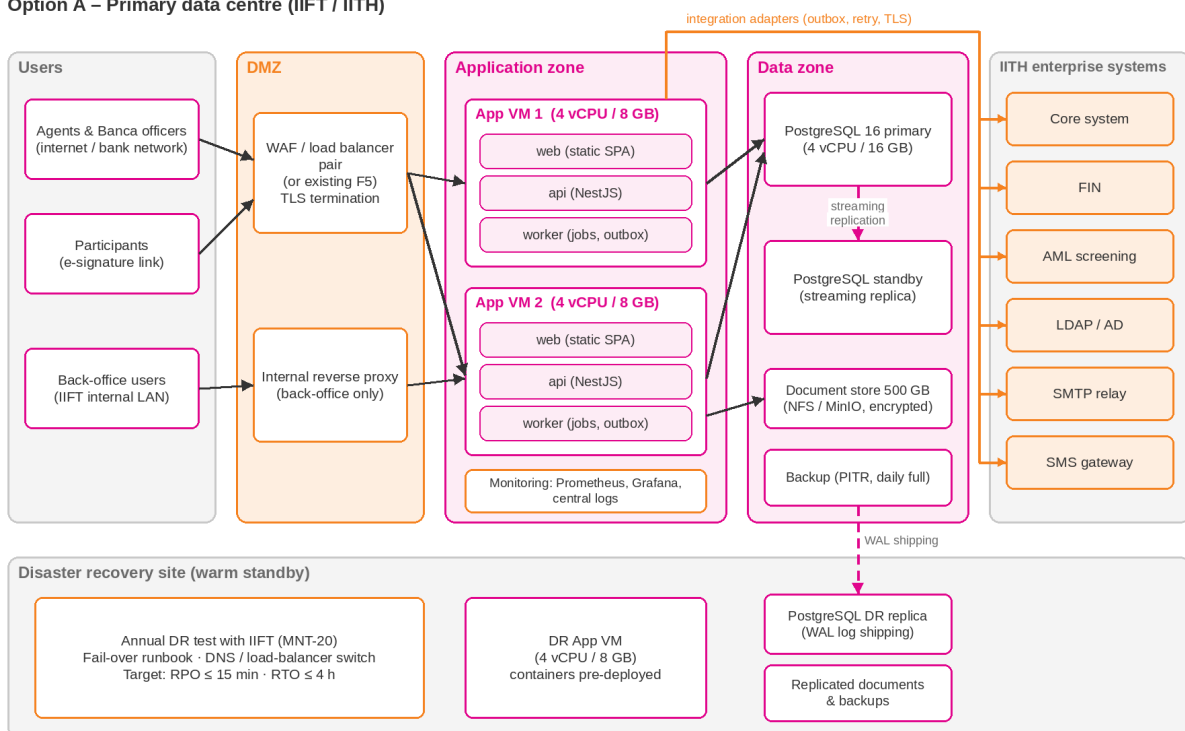
13. Infrastructure Options

SalesVerse 2.0 ships as standard containers and runs unchanged on-premise or in the cloud. The two deployment models match the commercial options: on-premise in the IIFT/IITH data centre under Option A, with infrastructure procured by IIFT to iorta's sizing, and an iorta-hosted cloud under Option B, with cloud infrastructure at cost and managed services by iorta. A hybrid variant of Option A is also possible. Costs are in Section 17 and the Bill of Materials.

13.1 Option A: on-premise in the IIFT/IITH data centre (recommended)

Two application VMs run behind the load balancer. PostgreSQL runs as a primary with a streaming standby, and an asynchronous streaming replica at the DR site stands ready for promotion. The solution uses IITH's network security, backup and monitoring services where capacity allows. Production needs two 4 vCPU / 8 GB application servers, two 4 vCPU / 16 GB database servers and 500 GB of document storage.

Option A – Primary data centre (IIFT / IITH)



Sizing per environment is in the infrastructure chapter. Hardware, OS and network are IIFT/IITH-provided; iorta installs, configures and operates the application stack.

Figure 113: Option A: on-premise deployment

Server	Qty	vCPU	RAM	Storage	Software
Production					
WAF / load balancer	2 (or existing)	2	4 GB	–	IITH appliance or HAProxy/nginx pair
Application VM	2	4	8 GB	100 GB	Linux, Docker Engine; web, api, clamav containers
Database VM	2	4	16 GB	200 GB SSD	PostgreSQL 16 primary and hot standby, pgBackRest
Document store	1 share	–	–	500 GB	NFS export, encrypted volume
Backup repository	1	–	–	1 TB	pgBackRest repository, document backups
Monitoring	1 (or existing)	2	4 GB	100 GB	Prometheus, Grafana, log collector
Non-production					
SIT	1	4	16 GB	200 GB	Compose: web, api, clamav, PostgreSQL

Server	Qty	vCPU	RAM	Storage	Software
UAT	1	4	16 GB	200 GB	Compose: web, api, clamav, PostgreSQL
DEV	Not required	–	–	–	Hosted by iorta during the project and support
DR site					
Application VM	1	4	8 GB	100 GB	Cold standby, same images
Database VM	1	4	16 GB	200 GB SSD	Asynchronous replica
Document and backup copies	–	–	–	500 GB + 1 TB	rsync target, backup copy

Table 48: Option A sizing per environment (procured by IIFT; indicative costs in Annex C)

Operating system: Ubuntu Server 24.04 LTS or Red Hat Enterprise Linux 9, hardened to the IITH baseline. DEV stays with iorta, so IIFT provides no development environment.

Item	IIFT / IITH	iorta TechNXT
Servers, storage, network, firewall, load balancer	Provide	Specify, review
Linux OS (RHEL 9 or Ubuntu 24.04 LTS) and licences	Provide, patch	Specify hardening
Container runtime, application, proxy configuration	Approve	Install, configure, operate
PostgreSQL install, replication, backup set-up	Approve; run backup infrastructure	Install, configure, tune
Monitoring and alerting	Infrastructure monitoring; receive alerts	Application monitoring
Certificates, DNS, SMTP relay, SMS account, AML subscription	Provide	Configure; support renewal
DR site and DR tests	Provide site; lead test	Replication; take part in tests

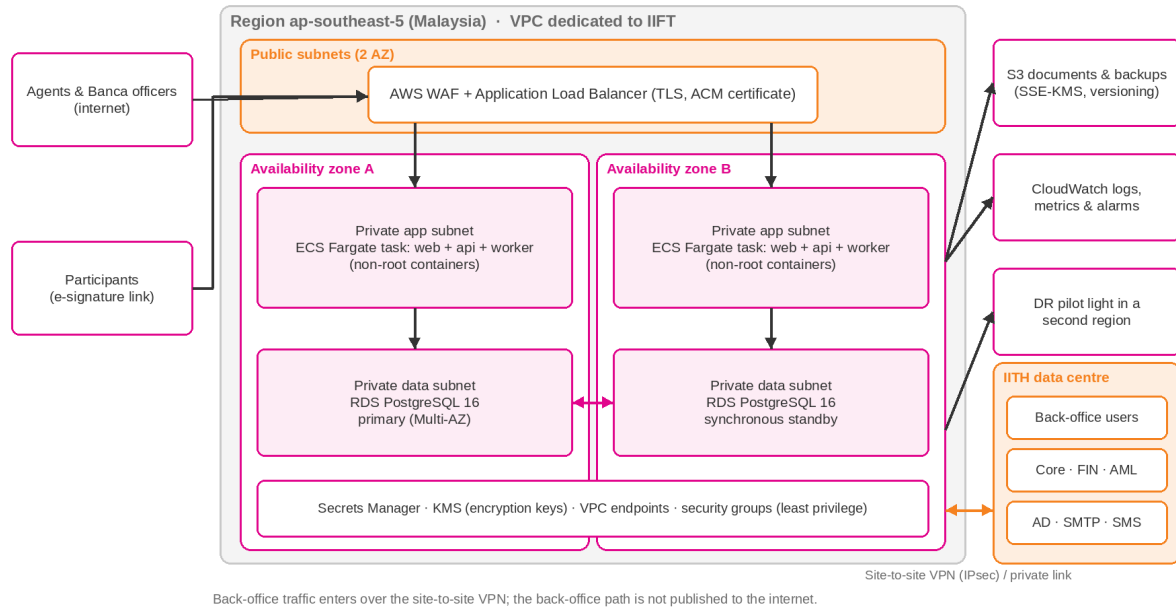
Table 49: Option A responsibilities

13.2 Option B: iorta-hosted cloud

iorta hosts and operates the solution in a cloud account dedicated to IIFT in AWS Asia Pacific (Malaysia) or Azure Malaysia West. Cloud use depends on IIFT's data-residency decision and on AMBD's outsourcing and cloud notification, which IIFT makes and iorta supports with the required documents. A Brunei-hosted alternative can be priced on request.

Service	AWS reference (Azure equivalent)
Region	AWS Asia Pacific (Malaysia), ap-southeast-5 (Azure Malaysia West); DR copies in a second region
Compute	ECS Fargate across two availability zones (AKS on Azure), or two EC2 instances
Database	RDS for PostgreSQL 16 Multi-AZ (Azure Database for PostgreSQL, zone-redundant)
Storage	S3 and EFS with encryption, versioning and lifecycle rules (Azure Blob and Files)
Edge	Application Load Balancer with AWS WAF (Azure Application Gateway with WAF)
Monitoring	CloudWatch logs, metrics and alarms; GuardDuty threat detection (Azure Monitor, Defender)
Connectivity	Site-to-site VPN to the IITH data centre for Core, FIN, AD, SMTP and back-office users
DR	Automated backups with point-in-time recovery; pilot-light DR in a second region

Option B – iorta-hosted cloud (AWS Asia Pacific – Malaysia; Azure Malaysia West equivalent)



Cloud infrastructure at cost, indicative B\$ 1,330 per month, plus iorta managed services. Subject to AMBD outsourcing / cloud notification and IIFT approval.

RPO ≤ 15 min (PITR) · RTO ≤ 4 h (restore in secondary AZ or region)

Figure 114: Option B: iorta-hosted cloud deployment (AWS reference)

Service	Reference sizing	B\$ / month
Container compute, 2 tasks across 2 availability zones	2 tasks × (1 vCPU, 2 GB) incl. ClamAV sidecar; scales to 4 tasks	B\$ 260
Managed PostgreSQL, Multi-AZ, 100 GB, automated backups	db.t4g.large (2 vCPU, 8 GB), Multi-AZ, 100 GB gp3, 35-day backups and PITR	B\$ 420
Application load balancer and web application firewall	1 load balancer; WAF with managed OWASP rule sets; managed TLS certificate	B\$ 130
NAT gateway and data transfer	NAT gateway, site-to-site VPN to the IITH data centre, outbound data	B\$ 90
Object storage (encrypted) and cross-region backup copies	Shared file storage up to 100 GB; about 200 GB object storage; cross-region copy	B\$ 70
Logging, metrics, alarms and threat detection	Log retention 90 days, metrics, alarms, threat detection	B\$ 70
Secrets and key management	Secrets store and customer-managed encryption keys	B\$ 20
UAT environment, scaled down, stopped out of hours	1 task and a single-AZ db.t4g.medium; stopped outside test periods	B\$ 120
DR environment, pilot light in a second region	Cross-region database replica or snapshots and standby images	B\$ 150
Total per month		B\$ 1,330
Total per year (at cost, estimate)		B\$ 15,960

Table 50: Option B cloud infrastructure estimate (recharged at cost)

iorta's managed services (monitoring, patching, backups, DR drills, security monitoring and cost management) cost B\$ 1,100 a month in Year 1. Cloud charges move with usage and exchange rates; they are recharged at the provider's cost without mark-up, or IIFT holds the account and pays the provider directly. The one-off cloud set-up fee is B\$ 15,000.

13.3 Hybrid variant of Option A

The back-office, database and documents stay in the IITH data centre. Only the portal's web tier (static application and reverse proxy) sits in the DMZ or at a cloud edge and forwards API calls over an encrypted

private link. Personal data never leaves the data centre. This suits IIFT if internet-facing capacity in the data centre is limited; it is priced as Option A, with any edge hosting at actuals.

13.4 Comparison and recommendation

Criterion	A: On-premise	B: iorta-hosted cloud	Hybrid variant
Commercial option	Option A: perpetual licence and AMC	Option B: subscription and managed services	Option A
Data residency	Data stays in Brunei	Malaysia region; DR copies in a second region; needs IIFT approval and AMBD notification	Personal data on-premise; portal web tier in DMZ or cloud
Regulatory steps	Internal approval only	AMBD outsourcing and cloud notification by IIFT	Depends on what sits in the cloud
Infrastructure cost	IIFT procures: indicative B\$ 20,500–42,000 one-time (less where IITH capacity is reused)	Set-up B\$ 15,000 one-off; cloud at cost, no mark-up: about B\$ 1,330 a month	Mostly as Option A
Operations	IITH infrastructure team; iorta application support under the AMC	iorta managed services, B\$ 1,100 a month in Year 1	Shared
Provisioning time	Depends on IITH VM provisioning	Days	Mixed
Resilience	Two nodes, standby database, DR site	Multi-AZ; DR in a second region	Per component

Table 51: Deployment models compared

Recommendation: Option A, on-premise in the IITH/IIFT data centre

Participant and agent data stays in Brunei and no extra regulatory step is needed.

IITH's existing data-centre services and operating model are reused, as for other group systems, which keeps cost and support effort down.

Option A has the lowest five-year cost (Section 17). Option B remains available if IIFT prefers an operating-expense model with iorta running the infrastructure.

14. Technical Document Pack

The technical documents below accompany this proposal in the folder docs/technical. They describe SalesVerse 2.0 as it stands today and are updated for IIFT during the project. This chapter summarises them; it does not repeat their content.

Document	Content	RFP references
SalesVerse-2.0-Solution-Architecture.docx	Logical, functional and deployment architecture; integration and security architecture; on-premise and cloud infrastructure sizing	DEL-06, DEL-07, DEL-09 (baseline)
SalesVerse-2.0-Data-Dictionary.xlsx and .docx	Every table and field with type, constraints, encryption and retention	DEL-07, DEL-26
SalesVerse-2.0-Code-Standards-and-Quality-Report.docx	Coding standards, review process, automated test coverage and static analysis results	NFR-14, NFR-20
SalesVerse-2.0-Security-Assessment-Report.docx	Internal vulnerability assessment and security test results; the independent third-party penetration test follows before go-live	NFR-12 to NFR-14; DEL-17 (precursor)
SalesVerse-2.0-Production-Support-Handover.docx	Support process, runbooks, knowledge transfer and release management	DEL-27, MNT-01 to MNT-04, MNT-21, MNT-28

Table 52: Technical document pack

The security assessment report covers iorta's internal testing. It does not replace the independent penetration test, which is performed by a third party on the IIFT environment before go-live (DEL-17).

15. Implementation Methodology

15.1 Delivery approach: configure and integrate

SalesVerse 2.0 already runs IIFT's products and rules, so the project configures and integrates an existing platform instead of building one. Discovery starts with fit-gap workshops on the working system: IIFT users walk through each journey with their own products, confirm what fits, and record the gaps. The gaps become the configuration backlog.

Governance follows formal gates, so IIFT signs off scope, design, test exit and go-live at fixed points and pays against them. Configuration and integration run in two-week sprints, each ending with a demonstration to IIFT key users. SIT, an independent VAPT, UAT, two migration rehearsals and a rehearsed cut-over follow, then four weeks of hypercare.

The Steering Committee reviews each gate: design sign-off (M2), build complete (M3), SIT exit and security clearance (M4), UAT sign-off (M5), go-live (M6) and hypercare exit (M7).

15.2 Phases and activities

#	Phase	Weeks	Key activities	Outputs
0	Mobilisation	1–2	Kick-off, governance set-up, environment and data requests, detailed plan	Project charter, project plan (DEL-01, 02)
1	Discovery and fit-gap confirmation	2–6	Workshops per product and process on the working system; collect IIFT rates, wording, questionnaires, templates; confirm gaps	BRS, FRS, RTM, SAD, TDD, UI/UX, interface specification, test strategy (DEL-03 to 09, 12)
2	Configuration sprints	6–16	Five two-week sprints configuring products, rates, workflows, roles, reports and templates; demo every sprint	Configured solution, source code (DEL-10, 11)
3	Integration with IIFT systems	8–17	Adapters to core, FIN, AML provider, SMS, SMTP and Active Directory; joint tests with system owners	Working interfaces, test evidence
4	SIT, performance and VAPT	16–19	SIT, regression, performance test, independent VAPT, remediation, re-test	SIT results, performance report, security assessment (DEL-13, 14, 17, 18)
5	UAT, migration rehearsals, training	19–23	UAT support, two migration rehearsals, train-the-trainer and end-user training, manuals	UAT results, migration plan, training, manuals (DEL-15, 16, 19, 22 to 27)
6	Cut-over and go-live	24	Production deployment, final migration, go/no-go	Deployment plan, go-live checklist, production deployment (DEL-20, 21, 28)
7	Hypercare and closure	25–28	Hypercare, stabilisation, handover to support, closure	Go-live support, closure report (DEL-29, 30)

Table 53: Implementation phases

Configuration sprints

Sprint	Scope
Sprint 1 (wk 6–7)	Access set-up to IITH policy, MFA, roles and data scopes; agent, agency and bank master data loaded
Sprint 2 (wk 8–9)	Onboarding and approval matrix; participant rules; AML connection to IIFT's screening service
Sprint 3 (wk 10–11)	IIFT actuarial rates, questionnaires and document templates for all seven products
Sprint 4 (wk 12–13)	Referral and QC rules, policy servicing, e-Policy and e-Receipt wording and branding
Sprint 5 (wk 14–15)	Payment rules, EOD and FIN file sign-off, BRR reconciliation, remaining reports, notifications
Hardening (wk 16)	Performance tuning, security hardening, defect fixing, documentation; build complete (M3)

All 30 deliverables in RFP section 7 are provided; Section 5 gives the due week and acceptance evidence of each.

15.3 Project governance

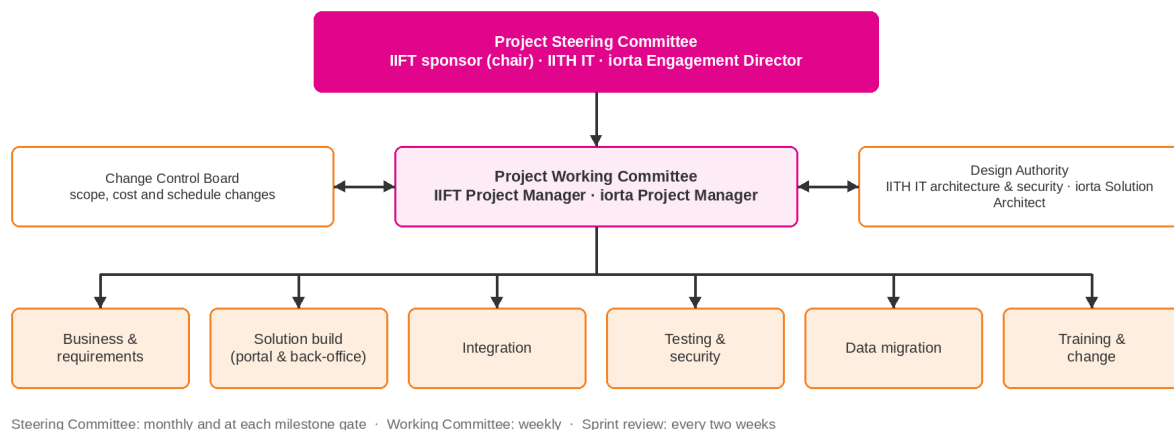


Figure 115: Project governance structure

Forum	Members	Frequency	Purpose
Steering Committee	IIFT sponsor (chair), IITH IT head, IIFT business heads, iorta Engagement Director, both PMs	Monthly and at milestone gates	Direction, milestone approval, escalated risks and changes, go/no-go
Project Working Committee	IIFT PM, iorta PM, business and IT leads	Weekly	Progress, RAID log, dependencies, decisions
Sprint review & planning	Product owner, key users, iorta team	Every two weeks	Demo of completed features, backlog priorities
Design Authority	IITH IT architecture & security, iorta Solution Architect	As needed in phases 1–4	Architecture, security and integration decisions
Change Control Board	IIFT PM, sponsor delegate, iorta PM	As needed	Assess and approve change requests

Progress is reported weekly (status, milestones and the RAID log of risks, assumptions, issues and dependencies) and monthly to the Steering Committee. Issues that cannot be resolved by the Working Committee within five business days are escalated to the Steering Committee.

RACI matrix

R = Responsible, A = Accountable, C = Consulted, I = Informed.

Activity	IIFT Sponsor	IIFT PM	IIFT business	IITH IT	iorta PM	iorta team
Project governance & reporting	A	R	I	I	R	C
Requirements & process sign-off	A	R	R	C	C	R
Solution design & architecture	I	C	C	A	R	R
Build & configuration	I	I	C	C	A	R
Infrastructure provisioning (Option A)	I	C	I	A/R	C	C
Interface availability (Core, FIN, AML, AD)	I	C	I	A/R	C	R
SIT & performance testing	I	I	I	C	A	R
VAPT (independent)	I	C	I	C	A	R
UAT execution & sign-off	A	R	R	C	C	C
Data extraction from legacy sources	I	C	R	A	C	C
Data transformation, load & reconciliation	I	C	R	C	A	R
Training delivery	I	C	R	C	A	R

Activity	IIFT Sponsor	IIFT PM	IIFT business	IITH IT	iorta PM	iorta team
Go-live decision	A	R	C	C	C	I
Hypercare & support	I	C	C	C	A	R

15.4 Quality assurance

A feature is done when its code is reviewed, its automated tests and security scans pass, its documentation is updated and the product owner has seen it working. The traceability matrix (DEL-05) links every RFP requirement ID to design, build, test case and acceptance evidence. Each major document is reviewed internally by someone outside the delivery team before it reaches IIFT.

15.5 Testing strategy

Level	Scope	Responsibility	Environment	Tools / exit criteria
Unit	Business rules, rating engines, validators, web components	iorta developers	DEV / CI	Vitest and Testing Library; coverage ≥ 70% on core modules
API / integration	Endpoints, permissions per role, database constraints	iorta	CI with PostgreSQL	Vitest e2e; every endpoint has positive and negative tests
SIT	End-to-end flows across portal, back-office and interfaces	iorta QA, IITH system owners	SIT	No open Critical/High defects; ≥ 95% pass
Regression	Automated suite re-run every sprint and release	iorta QA	SIT	Playwright; 100% of critical paths automated
Performance	Load (100 concurrent), stress and soak	iorta	SIT (production-like)	k6; non-functional targets met
Security	SAST/SCA/DAST in pipeline; independent VAPT and re-test	iorta, independent tester	SIT / pre-prod	No open Critical/High findings
UAT	Business scenarios per product and role	IIFT users with iorta support	UAT	Signed UAT certificate; no open Critical/High defects
Migration	Dry-runs, reconciliation, sample verification	iorta, IIFT data owners	UAT	100% record count and control-total match

Table 54: Test levels

Defect severity	Definition	Resolution in test phases
Critical	System unusable or data integrity at risk; no workaround	Fix before exit of current test phase
High	Key function fails; workaround difficult	Fix before exit of current test phase
Medium	Function impaired; workaround exists	Fix before go-live or agreed deferral
Low	Cosmetic or minor	Planned release

15.6 Data migration approach

Migration covers existing agent, agency, bank and branch records, participant records needed for servicing, and reference data. Sources and volumes are confirmed in the Data Migration Plan (DEL-19).

Step	Description
Discovery & profiling	Identify sources (spreadsheets, existing agent records, IIGT/core extracts), profile quality, agree scope
Mapping & rules	Field mapping to target model, transformation and cleansing rules, ownership of corrections
Tooling	Repeatable scripts (TypeScript/SQL) with validation reports; no manual re-keying
Dry-run 1 & 2	Full loads in UAT; exceptions corrected at source; reconciliation reports

Step	Description
Reconciliation	Record counts, control totals, mandatory-field completeness, sample verification by business owners
Production load	Executed during cut-over; signed reconciliation before users are enabled

15.7 Training plan

Audience	Format	Duration	Participants	Materials
Agents & bank officers (Portal)	Train-the-trainer plus two hands-on sessions	Half day per session	Up to 21 users (12 bank, 4 agent, 5 IIFT portal)	Quick reference guide, video walkthroughs, user manual
Back-office users	Hands-on workshop per role	1 day	IIFT admin, Finance, Compliance, Underwriting users	Role-based user manual, exercises
System administrators	Workshop	1 day	IITH/IIFT administrators	Administrator manual, configuration guide
Technical team	Technical handover sessions	2 days	IITH IT support, infrastructure, security	Technical and operations manuals, runbooks
Management	Briefing and dashboard demo	1 hour	IIFT management	Dashboard guide

Table 55: Training plan

15.8 Change management

- Stakeholder map and communication plan covering IIFT teams, bank partners and agencies.
- Early involvement of key users in design workshops and sprint demos to build ownership.
- Bank and agent onboarding pack: invitation e-mail, account activation guide, FAQ and support contacts.
- Super-user network within IIFT to provide first-line help after go-live.
- Adoption metrics on the management dashboard (active users, digital submissions, turnaround times).

15.9 Hypercare

- Four weeks immediately after go-live (weeks 25–28) with the core delivery team on standby.
- Daily health checks of application, jobs, EOD and integrations; daily stand-up with IIFT during the first two weeks.
- Priority fixing of go-live defects; daily defect and usage report.
- Onsite presence for go-live and the first month of support: a Senior Developer for the month and the Project Manager in go-live week; the rest of the team supports remotely.
- Formal handover to the maintenance team and hypercare exit criteria: no open P1/P2 issues, EOD stable for ten consecutive business days, all interfaces reconciled.

16. Delivery & Implementation Timeline

Production go-live is in week 24 counted from kick-off, followed by four weeks of hypercare. The six-month warranty and Year 1 of maintenance (the AMC under Option A, the subscription under Option B) both start at go-live; defects found under warranty are fixed free of charge.

Delivery plan – 24 weeks to go-live, 4 weeks hypercare

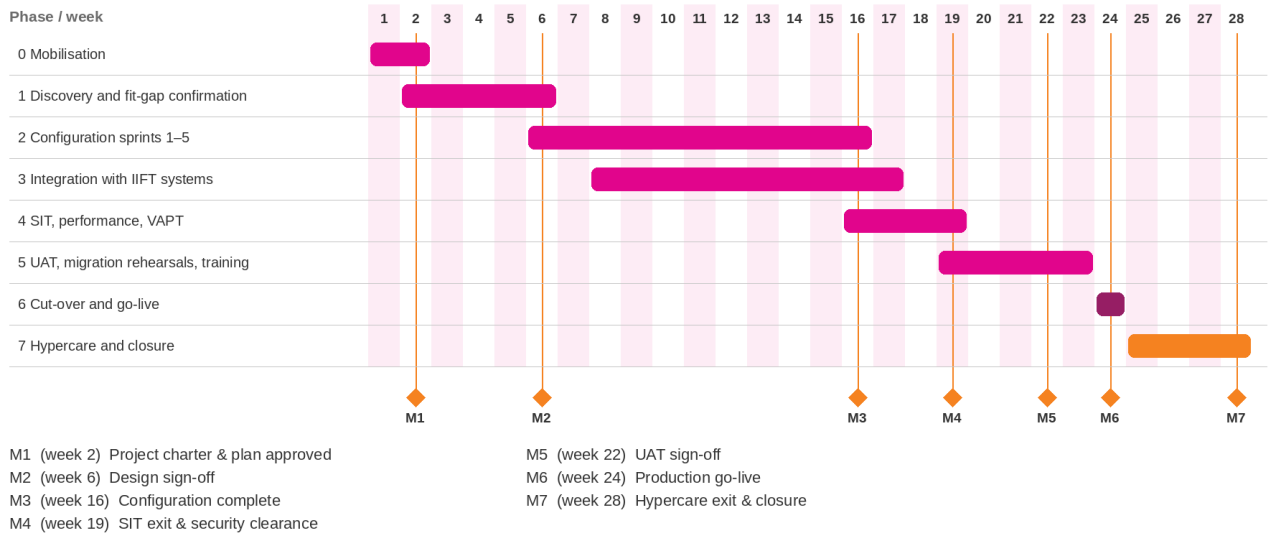


Figure 116: Delivery plan (weeks from kick-off)

16.1 Key milestones

Milestone	Description	Target	Services fee
M1	Project charter & plan approved	Week 2	15%
M2	Design sign-off	Week 6	15%
M3	Configuration complete	Week 16	–
M4	SIT exit & security clearance	Week 19	25%
M5	UAT sign-off	Week 22	25%
M6	Production go-live	Week 24	15%
M7	Hypercare exit & closure	Week 28	5%

Table 56: Milestones and linked payments

Under Option A the licence fee is paid separately: 50% at contract signing, 30% on installation in SIT and 20% at go-live. Amounts are in Section 17.

16.2 Dependencies on the critical path

- Contract signature and kick-off date (week 1 starts at kick-off).
- Availability of IIFT business owners for workshops in weeks 2–5 and sign-off of design by week 6.
- SIT and UAT environments (VMs, network, database access) available by week 8 and week 16 respectively.
- Interface specifications, test endpoints and credentials for Core, FIN, AML and AD available by week 8.
- IIFT actuarial rate tables and product documents (proposal forms, PDS, questionnaires) by week 6.
- UAT participants available in weeks 19–22; legacy data extracts available by week 12.

17. Commercial Proposal

This chapter sets out the commercial offer: the options, the fees in IIFT's format and in the RFP section 10 breakdown, the terms of each option, out-of-pocket expenses, third-party charges, the rate card and the payment schedules. The pricing workbook (IIFT-Commercial-Pricing-iorta-TechNXT.xlsx) holds the same figures with live formulas, and the Bill of Materials (IIFT-Bill-of-Materials-iorta-TechNXT.xlsx) lists every infrastructure and third-party item.

17.1 Commercial summary

iorta TechNXT offers SalesVerse 2.0 under two commercial models and one add-on. All three are priced on the same scope, team and 24-week plan; they differ in who holds the licence, who runs the infrastructure and whether IIFT takes the platform source code.

	Option A On-premise perpetual licence	Option B Subscription, iorta-hosted cloud	Option C Source code handover (add-on)
What IIFT gets	Perpetual enterprise licence, implementation and annual maintenance (AMC). Runs in the IIFT/IITH data centre.	Right to use for the term, implementation, maintenance and application management. iorta hosts and operates the solution on cloud.	Core platform source code with an 8-week knowledge transfer and transition, after go-live and hypercare.
Infrastructure	IIFT procures to iorta's sizing (Bill of Materials).	Cloud at cost (disbursement, no mark-up) or paid by IIFT directly; set-up and managed services by iorta.	Unchanged.
One-time fees	B\$ 190,000 Licence 60,000; services 130,000	B\$ 145,000 Services 130,000; cloud set-up 15,000	B\$ 88,000 Source code licence 60,000; knowledge transfer 28,000
Recurring, Year 1	AMC B\$ 33,990 (22% of B\$ 154,500)	Subscription B\$ 40,800 (B\$ 3,400 a month) Managed services B\$ 13,200 Cloud at actuals ≈ B\$ 15,960	Optional L3 support B\$ 15,450 (10% of the AMC base) in place of the AMC
Yearly increase	5% on the AMC	5% on subscription and managed services; cloud at actuals	5% on post-handover support
Term	Licence perpetual; AMC for five years, renewable yearly	Minimum term 60 months from go-live	Exercisable after hypercare
iorta fees, five years	B\$ 377,816	B\$ 443,404	Add-on to A, or to B on conversion
Five-year total with OPE and cloud estimates	B\$ 409,256 USD 314,812	B\$ 556,144 USD 427,803	+ B\$ 88,000 and OPE B\$ 8,070

Table 57: Commercial options at a glance (B\$)

Recommendation: Option A, with the Option C decision taken after go-live

Data stays in Brunei. The solution runs in the IITH data centre next to IIFT's core and finance systems, under IITH's existing security and DR arrangements; no cloud outsourcing notification to AMBD is needed.

Lowest five-year cost. B\$ 409,256 against B\$ 556,144 for Option B, a difference of B\$ 146,888 (26%). Even with IIFT's own infrastructure added (indicative B\$ 20,500–42,000), Option A is the cheaper option from Year 2.

IIFT owns a perpetual licence. After go-live IIFT can take the platform source code (Option C) and replace the AMC with the 10% L3 support, or keep the full AMC.

Option B suits IIFT if it prefers a lower upfront payment (B\$ 145,000 instead of B\$ 190,000) and an operating-expense model with iorta running the infrastructure. It can be converted to Option A during the term.

Fee basis

Currency	Brunei Dollars (B\$). USD amounts are indicative at USD 1 = B\$ 1.30, rounded to the nearest dollar; the B\$ amounts are binding.
Withholding tax (WHT)	Inclusive. Where Brunei law requires IIFT to withhold tax from iorta's fees, IIFT deducts and remits it and pays the balance. iorta bears the tax; invoices are not grossed up. Cloud infrastructure and OPE are not fees: they are recharged at cost as disbursements with the provider and travel invoices attached.
Out-of-pocket expenses (OPE)	Exclusive. Recharged at cost as disbursements for the onsite phases in Section 17.8, with the per diem fixed at USD 50 (B\$ 65) a day. Estimate B\$ 31,440, plus B\$ 8,070 if Option C is taken.
Third-party charges	Exclusive. SMS, AML data service, certificates, e-mail relay and optional items are listed in the Bill of Materials (Section 17.9) and are paid by IIFT or passed through at cost without mark-up.
Infrastructure	Exclusive. Option A: IIFT procures servers, storage, OS and network to iorta's sizing. Option B: cloud infrastructure is recharged at the provider's cost as a disbursement with no mark-up, or IIFT holds the cloud account in its own name and pays the provider directly. The one-off cloud set-up and iorta's managed services are fixed fees.
Indirect taxes	Brunei Darussalam has no GST or VAT at present. If an indirect tax is introduced, it is added to invoices at the statutory rate.
Price basis	One-time fees are fixed prices for the scope of this proposal. Recurring fees rise by 5% a year, which is also the cap (COM-18).

Table 58: What the fees include and exclude

17.2 Fees in IIFT's requested format

The tables follow the format in the RFP (Description, Fee, WHT, OPE, Total). Agent/Banca Portal and Back-office solution are the two modules of SalesVerse 2.0, and every fee is split between them. Amounts in B\$ with indicative USD.

Fee statement

All fees are **inclusive of WHT**, **exclusive of OPE** (recharged at cost as disbursements; estimates shown) and **exclusive of third-party charges and infrastructure** (listed in the Bill of Materials).

Description	Fee (B\$)	Fee (USD)	WHT	OPE	Total (B\$)	Total (USD)
One-time fees (licence and services)						
Agent/Banca Portal	104,000	80,000	Inclusive	Excluded	104,000	80,000
Back-office solution	86,000	66,154	Inclusive	Excluded	86,000	66,154
Total one-time fees	190,000	146,154	Inclusive	Excluded	190,000	146,154
Annual maintenance (AMC), Year 1 – 22% of licence and customisation, +5% a year						
Agent/Banca Portal	18,920	14,554	Inclusive	Excluded	18,920	14,554
Back-office solution	15,070	11,592	Inclusive	Excluded	15,070	11,592
Total AMC, Year 1	33,990	26,146	Inclusive	Excluded	33,990	26,146
Out-of-pocket expenses – disbursements at cost, estimate for the onsite phases						
OPE (travel, accommodation, per diem, insurance, transfers, visas)	–	–	–	31,440	31,440	24,185

Table 59: Option A – on-premise perpetual licence, in IIFT's format

Description	Fee (B\$)	Fee (USD)	WHT	OPE	Total (B\$)	Total (USD)
One-time fees (services)						

Description	Fee (B\$)	Fee (USD)	WHT	OPE	Total (B\$)	Total (USD)
Agent/Banca Portal	71,500	55,000	Inclusive	Excluded	71,500	55,000
Back-office solution	58,500	45,000	Inclusive	Excluded	58,500	45,000
Cloud set-up, platform (landing zone, environments, monitoring, backups, DR)	15,000	11,538	Inclusive	Excluded	15,000	11,538
Total one-time fees	145,000	111,538	Inclusive	Excluded	145,000	111,538
Recurring fees, Year 1 – +5% a year						
Agent/Banca Portal subscription (B\$ 1,850 a month)	22,200	17,077	Inclusive	Excluded	22,200	17,077
Back-office solution subscription (B\$ 1,550 a month)	18,600	14,308	Inclusive	Excluded	18,600	14,308
Managed services, platform (B\$ 1,100 a month)	13,200	10,154	Inclusive	Excluded	13,200	10,154
Total recurring fees, Year 1	54,000	41,538	Inclusive	Excluded	54,000	41,538
Out-of-pocket expenses – disbursements at cost, estimate for the onsite phases						
OPE (travel, accommodation, per diem, insurance, transfers, visas)	–	–	–	31,440	31,440	24,185

Table 60: Option B – subscription, in IIFT's format

- **WHT – inclusive.** The fee is the full amount payable; any withholding tax is deducted from it and borne by iorta.
- **OPE – exclusive.** The OPE line is an estimate for the onsite phases in Section 17.8; actual costs are recharged as disbursements with the travel invoices attached, and the per diem is fixed.
- **Third-party charges and infrastructure – exclusive.** Under Option B, cloud infrastructure is recharged at cost as a disbursement (or paid by IIFT directly), estimated at B\$ 1,330 a month (B\$ 15,960 a year) plus B\$ 1,500 for the project environments.

17.3 Commercial pricing breakdown (RFP section 10)

All 21 items of RFP section 10, for both options. Amounts in B\$; the portal and back-office split of every amount is in the pricing workbook (IIFT-Commercial-Pricing-iorta-TechNXT.xlsx).

No	Commercial component	Basis	Option A (B\$)	Option B (B\$)	Remarks
1	Software / application licence	One-off / licence	60,000	Included in subscription	A: perpetual, unlimited IIFT named users, all environments. B: right to use during the term.
2	Implementation	One-off	64,000	64,000	PM, BA, design, configuration, customisation, deployment (COM-06)
3	Integration	Per interface	30,500	30,500	Six interfaces, priced per interface (COM-05)
4	Data migration	One-off	7,500	7,500	Two dry runs, load and reconciliation (COM-07)
5	Infrastructure / cloud	Annual	IIFT procures (BOM)	15,000 set-up (one-off) + managed services 13,200 (Year 1) + cloud at cost	A: IIFT's own procurement to iorta's sizing. B: set-up at signing; cloud recharged at cost as a disbursement; managed services +5% a year
6	Database	Annual	0 (PostgreSQL)	In cloud actuals	No database licence fee; support in AMC or managed services
7	Security	One-off / annual	8,500	8,500	Independent VAPT and re-test; certificates in the BOM
8	Testing	One-off	10,000	10,000	SIT, performance, regression and UAT support (COM-08)
9	Training	One-off	4,500	4,500	End-user, administrator and technical training
10	Documentation	One-off	5,000	5,000	User, administrator, technical and operations manuals

No	Commercial component	Basis	Option A (B\$)	Option B (B\$)	Remarks
11	Warranty	Included	Included (6 months)	Included (6 months)	Defect correction from production go-live
	Subtotal one-time (items 1–11)		190,000	145,000	Fixed price; B includes the cloud set-up
12	Year 1 maintenance	Annual	33,990	40,800	A: 22% of B\$ 154,500, +5% a year. B: subscription, +5% a year
13	Year 2 maintenance	Annual	35,690	42,852	Operational Support
14	Year 3 maintenance	Annual	37,474	44,988	Optimisation
15	Year 4 maintenance	Annual	39,347	47,232	Technology Refresh
16	Year 5 maintenance	Annual	41,315	49,596	Sustainability & Transition
	Subtotal Years 1–5 (items 12–16)		187,816	225,468	AMC (A) / subscription (B)
	Managed services Years 1–5 (item 5)		–	72,936	Option B only
	iorta fees, five years		377,816	443,404	Excludes OPE, cloud actuals and third-party charges
17	Enhancement rate	Hour / day rate	680 / man-day 90 / hour	680 / man-day 90 / hour	Year 1 rates, +5% a year
18	Onsite support	Per day	850 / day + OPE	850 / day + OPE	Unplanned visits; planned phases per the OPE schedule
19	After-hours support	Per incident / included	P1 included 120 / hour	P1 included 120 / hour	Other after-hours work only at IIFT's request
20	Third-party charges	Annual / usage	Excluded (BOM)	Excluded (BOM)	SMS, AML data, certificates, e-mail relay: B\$ 600–4,720 a year (A), B\$ 300–3,720 a year (B)
21	Exit / transition	Fixed / included	Included (Year 5 AMC)	Included (end of term)	Transition, data extraction, knowledge transfer; core source code is Option C

Table 61: RFP section 10 commercial pricing breakdown

17.4 Option A – On-premise perpetual licence

IIFT buys a perpetual licence and a fixed-price implementation, and runs the solution in the IIFT/IITH data centre on infrastructure it procures to iorta's sizing (Section 13). Maintenance and support are charged as an AMC from go-live.

Licence terms (COM-02)

Licence type	Perpetual, non-exclusive, non-transferable enterprise licence to SalesVerse 2.0 (Agent/Banca Portal and Back-office modules), in object code, for IIFT's internal business.
Quantity	Unlimited named and concurrent users for IIFT, its agents and bank partners (Appendix 1 lists 26). No per-user, per-policy or per-transaction fees.
Environments	Production, DR and non-production (SIT, UAT, training) without further licence fees.
Duration and renewal	Perpetual. No renewal fee. The licence continues if the AMC ends.
Fee	B\$ 60,000 one-off (Agent/Banca Portal 32,500; Back-office 27,500), paid against milestones L1–L3.
Transfer	Not transferable, except to a successor of IIFT's business in a merger or group reorganisation, with written notice to iorta. IITH may operate and support the solution on IIFT's behalf.
Other IITH companies	Use by another group company, such as IIGT, needs an additional entity licence, priced separately.
Updates	New SalesVerse 2.0 releases, patches and upgrades are included while the AMC is in force.
Source code	Source code of IIFT-specific components is delivered under every option (DEL-11). The core platform source code is Option C (Section 17.6) or escrow.

Table 62: Option A licence terms

One-time fees

No	Component	Portal (B\$)	Back-office (B\$)	Total (B\$)	In AMC base
1	SalesVerse 2.0 software licence (Agent/Banca Portal and Back-office)	32,500	27,500	60,000	Yes
2	Implementation and customisation (project management, business analysis, design, configuration, customisation, deployment)	36,000	28,000	64,000	Yes
3	Integration (per interface; see integration breakdown)	17,500	13,000	30,500	Yes
4	Data migration	3,000	4,500	7,500	No
7	Security (independent VAPT and re-test)	4,500	4,000	8,500	No
8	Testing (SIT, performance, regression, UAT support)	5,500	4,500	10,000	No
9	Training	2,500	2,000	4,500	No
10	Documentation	2,500	2,500	5,000	No
Total one-time fees, Option A		104,000	86,000	190,000	154,500

Table 63: Option A one-time fees (RFP section 10 numbering)

Item	What the fee covers
1. SalesVerse 2.0 software licence	Perpetual, non-exclusive, non-transferable enterprise licence for IIFT. Unlimited named users; production, DR and non-production environments included. Option A only.
2. Implementation and customisation	Fixed price for the functional scope in RFP section 4 and Appendix 3, configured for IIFT's seven products, workflows, documents and reports.
3. Integration	Core system, FIN, AML screening, LDAP/AD, SMS gateway and Email/SMTP.
4. Data migration	Extraction support, transformation, cleansing, two dry runs, load and reconciliation of agreed agent, agency, bank and participant data.
7. Security	Independent vulnerability assessment and penetration test before go-live, remediation and re-test.
8. Testing	Test strategy, SIT cases and execution, performance test, automated regression suite, UAT support and defect fixing.
9. Training	End-user, administrator and technical training including materials.
10. Documentation	User, administrator, technical and operations manuals; maintained throughout the contract.

Table 64: Scope of each one-time fee (COM-06 to COM-10)

Integration per interface (COM-05)

Interface	Scope	Portal (B\$)	Back-office (B\$)	Total (B\$)
Core system	Agent, agency, bank and policy master synchronisation; agent information exchange	4,250	4,250	8,500
Financial system (FIN)	End-of-day postings, receipts, commission / referral fee information, BRR reconciliation	4,250	4,250	8,500
AML screening service	Participant and agent screening, result capture, re-screening	2,500	2,500	5,000
LDAP / Active Directory	Single sign-on for back-office users; optional directory look-up for portal administration	2,000	2,000	4,000
SMS gateway	OTP and alert delivery through IIFT-approved SMS provider	2,500	0	2,500
Email / SMTP relay	Notification and document e-mail delivery through IIFT mail relay	2,000	0	2,000
Total integration (item 3)		17,500	13,000	30,500

Table 65: Integration cost per interface (same under Option B)

Annual maintenance charge (AMC)

The AMC is **22% of the licence and customisation fees** (items 1, 2 and 3): $22\% \times \text{B\$ } 154,500 = \text{B\$ } 33,990$ **in Year 1**, rising by 5% a year. Year 1 starts at production go-live. The 6-month warranty runs inside Year 1: defects are fixed free of charge under the warranty, while the AMC pays for the service desk, SLA, monitoring, patches, upgrades and 60 enhancement hours a year.

Year	Focus (RFP section 9)	Portal (B\$)	Back-office (B\$)	AMC (B\$)	Per quarter (B\$)
Year 1	Stabilisation & Hypercare	18,920	15,070	33,990	8,497.50
Year 2	Operational Support	19,866	15,824	35,690	8,922.50
Year 3	Optimisation	20,859	16,615	37,474	9,368.50
Year 4	Technology Refresh	21,902	17,445	39,347	9,836.75
Year 5	Sustainability & Transition	22,997	18,318	41,315	10,328.75
Total	Five years	104,544	83,272	187,816	

Table 66: AMC by year (COM-12, COM-18)

AMC inclusions:

- Helpdesk during IIFT business hours with 24x7 coverage for P1 (Critical) incidents
- Incident, problem and service request management against the agreed SLA
- Corrective and preventive maintenance; defect fixing
- Security patches, dependency updates and vulnerability remediation
- Platform and version upgrades (Node.js, PostgreSQL, frameworks) within the supported lifecycle, including SalesVerse 2.0 platform releases
- 60 hours per year of minor enhancements
- Monthly service report and quarterly service review
- Backup verification and annual DR test participation
- Documentation updates and ongoing knowledge transfer
- Exit and transition support in Year 5

Billing: quarterly in advance; year 1 starts at go-live. After Year 5 the AMC is renewable yearly, with the increase capped at 5%. The services are described in Section 18.

Warranty (COM-11)

Both options carry a 6-month warranty from production go-live: defects against the approved specifications, including IIFT-specific configuration, adapters, reports and templates, are corrected at no charge within the SLA. The warranty does not cover changes made by others, misuse, or infrastructure and third-party products outside iorta's scope.

17.5 Option B – Subscription, hosted and managed by iorta

IIFT pays the implementation fee and a monthly subscription that includes the right to use SalesVerse 2.0, maintenance and support, and application management. iorta hosts the solution on cloud in AWS Asia Pacific (Malaysia) or Azure Malaysia West. A one-off set-up fee covers building the cloud environments. Cloud infrastructure is recharged at cost as a disbursement (or paid by IIFT directly), and iorta's managed services for the infrastructure are a separate fixed fee.

Implementation fee

B\$ 130,000 (Agent/Banca Portal 71,500; Back-office 58,500): the same scope and fees as Option A items 2 to 10 (Section 17.4), without the licence. It is paid against the service milestones in Section 17.11.

Cloud set-up fee

B\$ 15,000 one-off, payable at contract signing: set-up of the cloud landing zone (account structure, network, VPN to the IITH data centre, identity and key management), the PROD, DR and UAT environments,

monitoring and alerting, backups and the DR arrangement, with security baselines applied. It is RFP section 10 item 5 (one-off part) and is not part of the AMC base. Option B one-time fees therefore total **B\$ 145,000**.

Subscription (COM-03)

Contract year	Portal per month (B\$)	Back-office per month (B\$)	Total per month (B\$)	Per year (B\$)
Year 1	1,850	1,550	3,400	40,800
Year 2	1,943	1,628	3,571	42,852
Year 3	2,040	1,709	3,749	44,988
Year 4	2,142	1,794	3,936	47,232
Year 5	2,249	1,884	4,133	49,596
Total				225,468

Table 67: Subscription by year (+5% a year)

- **Billing:** monthly in advance from go-live (quarterly in advance on request).
- **Term:** minimum term of 60 months from go-live (B\$ 225,468 of subscription). If IIFT ends the subscription earlier other than for iorta's breach, the subscription for the rest of the minimum term is payable; conversion to Option A ends it without that charge. After the minimum term it renews yearly, with the increase capped at 5%.
- **Users:** unlimited named users for IIFT; no per-user or per-policy charges.

The subscription includes:

- Right to use SalesVerse 2.0 (Agent/Banca Portal and Back-office) for unlimited IIFT named users
- Everything in the Option A annual maintenance: SLA support, P1 24x7, patches, upgrades, 60 enhancement hours a year, monthly reports and quarterly reviews
- Application management: releases, configuration changes, job and integration monitoring
- Production, DR and UAT environments operated by iorta (infrastructure charged separately on actuals)
- Data export in open formats at any time and at exit; data remains IIFT's property

Cloud infrastructure at cost (COM-04)

Service	Reference sizing	B\$ / month	B\$ / year
Container compute, 2 tasks across 2 availability zones	2 tasks × (1 vCPU, 2 GB) incl. ClamAV sidecar; scales to 4 tasks	260	3,120
Managed PostgreSQL, Multi-AZ, 100 GB, automated backups	db.t4g.large (2 vCPU, 8 GB), Multi-AZ, 100 GB gp3, 35-day backups and PITR	420	5,040
Application load balancer and web application firewall	1 load balancer; WAF with managed OWASP rule sets; managed TLS certificate	130	1,560
NAT gateway and data transfer	NAT gateway, site-to-site VPN to the IITH data centre, outbound data	90	1,080
Object storage (encrypted) and cross-region backup copies	Shared file storage up to 100 GB; about 200 GB object storage; cross-region copy	70	840
Logging, metrics, alarms and threat detection	Log retention 90 days, metrics, alarms, threat detection	70	840
Secrets and key management	Secrets store and customer-managed encryption keys	20	240
UAT environment, scaled down, stopped out of hours	1 task and a single-AZ db.t4g.medium; stopped outside test periods	120	1,440
DR environment, pilot light in a second region	Cross-region database replica or snapshots and standby images	150	1,800
Total, production run		1,330	15,960
Project environments (DEV, SIT, UAT) for 5 months before go-live	Scaled-down environments	300	1,500 (one-off)

Table 68: Cloud infrastructure estimate (recharged at cost as a disbursement)

- **Recharged at cost as a disbursement.** iorta recharges the cloud provider's charges monthly in arrears, with the provider invoice attached and no mark-up, separately from its fees and outside the fees to which withholding tax applies. The estimate above is for budgeting; IIFT pays the actual charges.
- **Or paid by IIFT directly.** IIFT may hold the cloud account in its own name and pay the provider directly; iorta then operates the account under the managed services with delegated access. Either way the account is dedicated to IIFT.
- **Currency.** Providers bill in USD; recharged amounts are converted to B\$ at the bank selling rate on the provider's invoice date, shown on iorta's disbursement invoice.
- **Cost control.** Budget alerts at 80% and 100% of the monthly estimate, a monthly cost report and a quarterly right-sizing review. Reserved capacity is bought only with IIFT's approval, and the saving passes to IIFT.

Managed services by iorta

Contract year	Per month (B\$)	Per year (B\$)
Year 1	1,100	13,200
Year 2	1,155	13,860
Year 3	1,213	14,556
Year 4	1,273	15,276
Year 5	1,337	16,044
Total		72,936

Table 69: Managed services fee (+5% a year), billed monthly in advance from go-live

Managed services include:

- 24x7 infrastructure and availability monitoring with alerting
- Operating system, container and database patching in agreed windows
- Backups, monthly restore tests and the annual DR drill
- Security monitoring: WAF, threat detection, vulnerability scanning of the cloud estate
- Capacity, performance and cloud cost management with a monthly report
- Infrastructure incident response and liaison with the cloud provider

Hosting region and data residency

The reference design uses AWS Asia Pacific (Malaysia) or Azure Malaysia West, with production across two availability zones and DR copies in AWS Asia Pacific (Singapore) or Azure Southeast Asia as the DR region, subject to IIFT's data-residency approval. Data is encrypted at rest with keys held in IIFT's account and in transit with TLS; back-office traffic and integrations with Core, FIN, AD and SMTP run over a site-to-site VPN to the IITH data centre.

Cloud hosting of personal data is an outsourcing arrangement for IIFT. IIFT makes the AMBD outsourcing and cloud notification; iorta provides the supporting material (architecture, controls, exit plan, provider certifications). A Brunei-hosted alternative, such as a local data-centre provider, can be priced on request.

Exit and data return

- Data belongs to IIFT. A full export (database dump, CSV extracts, documents and audit records) is available on request during the term and is delivered at exit.
- Exit assistance is included at the end of the term: transition plan, knowledge transfer, data export and up to 30 days of parallel support.
- After IIFT confirms receipt of the data, iorta deletes it from the cloud account and backups and issues a deletion certificate. An account held in IIFT's name stays with IIFT.

Conversion to Option A

IIFT may convert to Option A at any point in the term. The perpetual licence fee of B\$ 60,000 applies **[bid owner to confirm any credit of subscription fees already paid]**; from the conversion date the subscription, including any remaining minimum term, is replaced by the Option A AMC for that contract year. The solution

moves to the IIFT/IITH data centre, or stays in the cloud account with managed services and cloud at actuals continuing. Option C becomes available once converted.

17.6 Option C – Source code handover (add-on)

Source code under every option

RFP DEL-11 (Must) asks for the source code of agreed custom-developed components. Under **every option** iorta delivers, at go-live and with every release, the source code of IIFT-specific components: configuration, integration adapters, reports, document templates, migration scripts and build scripts.

Option C adds the source code of the SalesVerse 2.0 core platform, with a structured knowledge transfer so that IIFT, IITH or a provider IIFT appoints can maintain and extend it.

What is delivered

- Full source code of SalesVerse 2.0 core and IIFT customisations in a Git repository with history
- Build scripts, CI/CD pipeline definitions, infrastructure and container definitions
- Database schema, migrations, reference data and seed scripts
- Automated test suites (unit, end-to-end, security checks) and test data
- Developer guide, architecture document, data dictionary, runbooks and coding standards
- Third-party component inventory and software bill of materials (SBOM)

Licence rights

Rights	Use, modify, compile, maintain and extend the source code for the internal business of IIFT and the IITH group. A group company that deploys the solution needs its own licence.
Restrictions	No sale, sublicensing, distribution or provision to third parties as a product or service. Contractors may work on the code for IIFT under confidentiality obligations.
Ownership	The SalesVerse 2.0 core remains iorta TechNXT's intellectual property. Changes IIFT makes after handover belong to IIFT.
Confidentiality	The source code is iorta's confidential information and is kept in IIFT/IITH-controlled repositories.
Third-party code	Open-source components remain under their own licences, listed in the SBOM.
Assurance	The delivered code builds the production release; this is verified by IIFT before payment C1. iorta does not warrant changes made by others.

Table 70: Option C licence rights

Timing and transition plan

Handover starts after go-live and hypercare exit (week 29 at the earliest) and takes 8 weeks. It can also start later in the contract. The plan below assumes up to six IIFT/IITH or nominated contractor staff with Node.js and React experience.

Week	Activity	Outcome
1	Transition plan, roles, access to repositories and pipelines	Signed transition plan
2	Architecture, data model and module walkthroughs	Architecture walkthrough sign-off
3	Code walkthroughs: sales, policy, billing and claims modules	Module checklists
4	Code walkthroughs: workflow, integration, reports, security	Module checklists
5	Build, test, CI/CD, release and database migration practice	IIFT team produces a release
6	Shadow support: IIFT developers observe live tickets	Ticket log
7	Reverse shadow: IIFT developers resolve tickets, iorta reviews	Ticket log, code reviews
8	Competency assessment, documentation handover, closure	Handover certificate

Table 71: Knowledge transfer and transition plan

Acceptance. The handover certificate is signed when IIFT's team has built and deployed a release to SIT from the delivered repository without iorta's help, resolved tickets in the reverse-shadow weeks, and received the documentation set; the competency assessment in week 8 records the result for each participant.

Onsite: the Solution Architect and a Senior Developer spend the code walkthrough weeks onsite (OPE estimate B\$ 8,070, at actuals); the rest is delivered remotely.

Price and payment

Milestone	Payment trigger	Component	Share	Amount (B\$)
C1	Delivery of the source code repository and a verified build by IIFT	Source code licence	100%	60,000
C2	Start of knowledge transfer	Knowledge transfer	50%	14,000
C3	Handover certificate signed	Knowledge transfer	50%	14,000
Total Option C				88,000

Table 72: Option C: source code licence B\$ 60,000 and knowledge transfer B\$ 28,000

The price holds if Option C is exercised within 12 months of go-live; later exercise is priced at this fee plus 5% for each further contract year. OPE for the onsite weeks is charged at actuals.

Support after handover

Choice	What iorta provides	Price
1. Continue the full AMC	All AMC services: service desk, SLA, patches, upgrades, enhancement hours.	AMC unchanged (B\$ 33,990 in Year 1, +5% a year)
2. Platform updates and L3 support	SalesVerse 2.0 core releases and security patches, L3 advice and defect fixes in the core under the SLA. IIFT's team handles L1/L2 and its own changes.	10% of the AMC base: B\$ 15,450 in Year 1, +5% a year, in place of the AMC
3. Rate card only	No retainer; work ordered when needed.	Rate card (Section 17.10)

Table 73: Support choices after handover

Contract year	Choice 2 (B\$)	Full AMC (B\$)	Difference (B\$)
Year 1	15,450	33,990	18,540
Year 2	16,223	35,690	19,467
Year 3	17,034	37,474	20,440
Year 4	17,885	39,347	21,462
Year 5	18,780	41,315	22,535
Total	85,372	187,816	102,444

Table 74: Choice 2 compared with the full AMC (pro rata from the month after the handover certificate)

Escrow alternative

- Instead of Option C, the core source code can be deposited with an independent escrow agent chosen with IIFT, under a tripartite agreement.
- Deposits at go-live and after every major release; an optional verification test confirms the deposit builds.
- Escrow agent fees are paid by IIFT at cost (indicative B\$ 2,500–4,000 a year); iorta's deposit work is included.
- Release conditions: iorta's insolvency or winding-up, iorta ceasing to trade or to support SalesVerse 2.0 without a successor, or a material breach of support obligations not remedied within 30 days of notice.
- On release IIFT receives the rights described above for Option C. Knowledge transfer is not part of escrow and can be ordered at the rate card.

17.7 Five-year cost of ownership

Cost element	Option A (B\$)	Option B (B\$)
One-time iorta fees (B: services and cloud set-up)	190,000	145,000
AMC (A) / subscription (B), Years 1–5	187,816	225,468
Managed services, Years 1–5	–	72,936
iorta fees, five years	377,816	443,404
OPE, core onsite phases (estimate, disbursements at cost)	31,440	31,440
Cloud infrastructure at cost (estimate, incl. project environments)	–	81,300
Five-year total (B\$)	409,256	556,144
Five-year total (USD, indicative)	314,812	427,803
Paid by IIFT directly – indicative, not included in the totals above		
Infrastructure procured by IIFT (one-time, Bill of Materials)	B\$ 20,500–42,000	–
Required third-party services, five years (SMS, AML, certificates, e-mail)	B\$ 3,000–23,600	B\$ 1,500–18,600
Optional items, five years (RHEL, PostgreSQL support, escrow)	B\$ 60,000–124,000	B\$ 12,500–20,000
Option C add-on (either option)		
Source code handover with knowledge transfer	88,000	On conversion
OPE for onsite knowledge transfer (estimate)	8,070	–

Table 75: Five-year cost of ownership

Period	A: iorta fees	A: OPE	A: total	B: iorta fees	B: cloud	B: OPE	B: total
Year 0 (implementation)	190,000	31,440	221,440	145,000	1,500	31,440	177,940
Year 1	33,990	0	33,990	54,000	15,960	0	69,960
Year 2	35,690	0	35,690	56,712	15,960	0	72,672
Year 3	37,474	0	37,474	59,544	15,960	0	75,504
Year 4	39,347	0	39,347	62,508	15,960	0	78,468
Year 5	41,315	0	41,315	65,640	15,960	0	81,600
Total	377,816	31,440	409,256	443,404	81,300	31,440	556,144

Table 76: Cash view by contract year (B\$)

Option B needs B\$ 45,000 less in one-time fees. Option A costs less over five years by B\$ 146,888, and becomes the cheaper option in Year 3 even when the mid-point of IIFT's infrastructure estimate (B\$ 31,250) is added to it.

17.8 Out-of-pocket expenses (OPE)

The team works from Malaysia and India. Onsite presence in Bandar Seri Begawan is recommended for requirements gathering, user training and UAT support, go-live with one month of support and, if Option C is taken, knowledge transfer and handover. All other work is delivered remotely and carries no OPE.

Cost item	From Malaysia (B\$)	From India (B\$)	Basis
Return airfare (economy)	500	1,200	Per trip, at actuals
Accommodation	150	150	Per night, at actuals
Per diem (meals, local transport)	65	65	Per day onsite incl. travel days; fixed at USD 50
Travel insurance	50	70	Per trip, at actuals
Airport transfers	80	80	Per trip, at actuals

Cost item	From Malaysia (B\$)	From India (B\$)	Basis
Visa	0 (visa-free)	40	Per trip, at actuals

Table 77: OPE unit rates used for the estimate

Role	From	Nights	Airfare	Hotel	Per diem	Insurance, transfers, visa	Total (B\$)
Requirements gathering and fit-gap (weeks 1–4)							
Project Manager	Malaysia	12	500	1,800	845	130	3,275
Business Analyst	Malaysia	19	500	2,850	1,300	130	4,780
Solution Architect	India	12	1,200	1,800	845	190	4,035
User training and UAT support (weeks 19–23)							
Business Analyst / Trainer	Malaysia	19	500	2,850	1,300	130	4,780
QA Lead	India	12	1,200	1,800	845	190	4,035
Go-live and one month of support (weeks 24–28)							
Senior Developer	India	33	1,200	4,950	2,210	190	8,550
Project Manager	Malaysia	6	500	900	455	130	1,985
Knowledge transfer and source code handover (Option C) – optional							
Solution Architect	India	12	1,200	1,800	845	190	4,035
Senior Developer	India	12	1,200	1,800	845	190	4,035
Core onsite phases		113					31,440
Option C knowledge transfer (optional)		24					8,070
Total including Option C							39,510

Table 78: Onsite plan and OPE estimate

- **Billing:** recharged monthly in arrears at cost as disbursements, with the travel invoices and receipts attached and no mark-up, separately from iorta's fees and outside the fees to which withholding tax applies. The per diem is fixed and needs no receipts.
- **Ceiling:** total OPE will not exceed the estimate by more than 10% (B\$ 34,584 for the core phases) without IIFT's prior written approval.
- **Standards:** economy-class flights; business hotels near IIFT's office. If IIFT provides accommodation or transport, the OPE falls accordingly.
- **Changes:** trips are agreed with IIFT's project manager two weeks ahead. Extra onsite days that IIFT requests outside these phases are charged at the rate card (B\$ 850 a day) plus OPE.

17.9 Third-party charges, licences and infrastructure (COM-04, COM-17)

Every third-party product, licence, service and infrastructure item is listed in the Bill of Materials workbook (IIFT-Bill-of-Materials-iorta-TechNXT.xlsx) and summarised in Annex C. The table gives the totals.

Category	Option A	Option B	Treatment
Runtime software: Node.js, NestJS, React, Ant Design, Prisma, PostgreSQL, nginx, ClamAV, Docker, monitoring stack	B\$ 0 licence	B\$ 0 licence	Open source; included
Infrastructure	B\$ 20,500–42,000 one-time (indicative)	≈ B\$ 1,330 a month at cost	A: IIFT procures. B: disbursement at cost or paid by IIFT directly
Operating system	Ubuntu LTS B\$ 0; RHEL optional B\$ 4,500–10,800 a year	In cloud actuals	A: IIFT procures

Category	Option A	Option B	Treatment
Required services: SMS, AML data, certificates, e-mail relay	B\$ 600–4,720 a year	B\$ 300–3,720 a year	IIFT procures or pass-through at cost
Optional: commercial PostgreSQL support, RHEL, escrow agent	B\$ 12,000–24,800 a year	B\$ 2,500–4,000 a year	Only if IIFT elects them
Cloud set-up (landing zone, environments, monitoring, backups, DR)	Not applicable	B\$ 15,000 one-off	iorta fee (Option B one-time fees)
Independent VAPT and re-test	Included (item 7)	Included (item 7)	Included in iorta fee

Table 79: Third-party charges and licences (detail in Annex C and the BOM workbook)

The CI pipeline produces a CycloneDX software bill of materials (SBOM) of the production dependencies on every build. It is delivered with each release and lists every open-source package with its licence.

17.10 Rate card (COM-16)

Service	Unit	Year 1	Year 2	Year 3	Year 4	Year 5
Enhancements and change requests	Per man-day	680	714	750	787	827
Enhancements and change requests	Per hour	90	95	99	104	109
Onsite support, unplanned visit (plus OPE)	Per day	850	893	937	984	1,033
After-hours support other than P1	Per hour	120	126	132	139	146
After-hours support for P1 (Critical) incidents	Per incident	Included	Included	Included	Included	Included
Exit and transition at contract end	Fixed	Included	Included	Included	Included	Included

Table 80: Rate card in B\$ (RFP section 10 items 17–19 and 21; +5% a year)

The man-day rate is blended across roles. 60 hours of minor enhancements a year are included in the AMC and the subscription. Change requests follow a written request, an impact assessment within five business days, a quotation at these rates and IIFT's written approval before work starts; quotations remain valid for 90 days.

17.11 Payment schedules

Option A licence fee

Milestone	Payment trigger	Share	Amount (B\$)
L1	Contract signing	50%	30,000
L2	Installation in the SIT environment	30%	18,000
L3	Production go-live	20%	12,000
	Total licence fee	100%	60,000

Table 81: Licence milestones

Services fee (Options A and B)

Milestone	Payment trigger	Target	Share	Amount (B\$)
M1	Contract signing and project kick-off	Week 2	15%	19,500
M2	Design sign-off (BRS, FRS, SAD, TDD, UI/UX, interface specification)	Week 6	15%	19,500
M4	SIT exit and security clearance (VAPT remediated)	Week 19	25%	32,500
M5	UAT sign-off	Week 22	25%	32,500
M6	Production go-live	Week 24	15%	19,500

Milestone	Payment trigger	Target	Share	Amount (B\$)
M7	Hypercare exit and project closure	Week 28	5%	6,500
	Total services fee		100%	130,000

Table 82: Service milestones (M3, build complete, carries no payment)

Recurring charges, OPE and Option C

Charge	Option	Billing	Year 1 invoice (B\$)
Annual maintenance (AMC)	A	Quarterly in advance; Year 1 starts at go-live	8,497.50 per quarter
Cloud set-up fee	B	At contract signing	15,000 once
Subscription	B	Monthly in advance from go-live	3,400 per month
Managed services	B	Monthly in advance from go-live	1,100 per month
Cloud infrastructure	B	Monthly in arrears, disbursement at cost with provider invoices (or paid by IIFT directly)	≈ 1,330 per month (300 during the project)
OPE	A and B	Monthly in arrears, disbursement at cost with travel invoices	As incurred
Source code handover	C	Milestones C1–C3	88,000 in total
Post-handover support (choice 2)	C	Quarterly in advance	3,862.50 per quarter

Table 83: Billing of recurring charges

All invoices are payable within 30 days of receipt.

17.12 Commercial terms

Payment terms	30 days from invoice.
Currency	B\$. USD amounts are indicative at USD 1 = B\$ 1.30. Cloud charges are converted at the rate on the provider's invoice date.
Taxes	Fees inclusive of WHT, borne by iorta. Cloud infrastructure and OPE are recharged at cost as disbursements with the provider and travel invoices attached, outside the fees to which WHT applies. No GST or VAT applies in Brunei Darussalam at present; any indirect tax introduced by law is added at the statutory rate.
Quotation validity	90 days from submission (Section 22).
Price protection	One-time fees are fixed. AMC, subscription, managed services and the rate card rise by 5% a year, capped at 5% (COM-18). Cloud and third-party charges follow actual provider prices.
User growth	Unlimited named users for IIFT under both options; more agents, banks or branches do not change the fees.
Additional entities	Use by another IITH group company, such as IIGT, is priced separately (licence or subscription, configuration and integration).
Term	Implementation to hypercare exit; maintenance or subscription for five years from go-live. The subscription's minimum term is 60 months from go-live.
Changes	Through change requests at the rate card (Section 17.10).

Table 84: Commercial terms

17.13 Commercial assumptions and exclusions

Assumptions

- Scope is the RFP (sections 3–8, Appendices 1–4) as described in this proposal; changes follow the change request process.
- The contract is signed within the quotation validity and the project starts within four weeks of signature.

- IIFT meets the dependencies in Section 16. A delay of more than four weeks caused by dependencies outside iorta's control is re-planned with IIFT; extra effort is agreed as a change request.
- Option A: IIFT provides SIT and UAT by week 8 and week 16 and production and DR by week 20, to the sizing in Section 13.
- Option B: IIFT approves the cloud region and completes the AMBD notification before production data is loaded; the project environments run in the cloud from kick-off; the cloud set-up fee is paid at contract signing.
- Cloud infrastructure and OPE are recharged at cost as disbursements with the provider and travel invoices attached, outside the fees to which withholding tax applies. IIFT may instead hold the cloud account in its own name and pay the provider directly.
- Data migration covers agent, agency, bank, branch, participant and reference data: below 10,000 records from no more than three source extracts.
- Onsite presence follows the OPE schedule; all other work is remote.

Exclusions

- Infrastructure, operating system licences, network and data-centre services under Option A.
- Cloud infrastructure under Option B (recharged at cost or paid directly) and third-party charges in the Bill of Materials.
- OPE, recharged at cost as set out above.
- Changes to IIFT's core system, FIN, AML service or other third-party systems.
- Use by other IITH group companies; Malay translation of screens; payment gateway; native mobile apps.
- Penetration tests after go-live beyond the pre-go-live VAPT and re-test (quoted on request).

General assumptions, dependencies and exclusions are in Section 20.

18. Five-Year Maintenance & Support

Maintenance and support start at go-live and run for five years. Under Option A they are charged as the AMC, B\$ 33,990 in Year 1 (22% of licence and customisation) rising by 5% a year; under Option B they are part of the subscription. The service is the same under both options. In the first six months the warranty covers defect fixes; the AMC or subscription pays for the service desk, SLA, monitoring, patches, upgrades, 60 enhancement hours a year and the services below. Under Option B, iorta's managed services also cover the cloud infrastructure.

18.1 Support model

Element	Description
Channels	Issue module in the portal/back-office (preferred), dedicated support e-mail and telephone hotline; P1 incidents must be reported by telephone in addition to the ticket.
Support hours	IIFT business hours as defined in the 'Insurans Islam TAIB Business Hours (Family Takaful)' schedule (COM-14); P1 incidents 24 x 7, including weekends and public holidays.
Incident management	Record, classify, prioritise, investigate, resolve and close per ITIL-aligned procedure; IIFT confirms closure (MNT-02).
Problem management	Root-cause analysis for every P1 and P2 and for recurring incidents, with report within five business days and tracked corrective actions (MNT-03).
Service requests	User support, configuration changes, data corrections and operational assistance (MNT-04).
Preventive maintenance	Monthly health check: logs, error trends, job runs, capacity, slow queries, certificate expiry, backup status (MNT-10, 16, 17).
Security maintenance	Monthly patch cycle for application dependencies and container images; emergency patches per remediation timelines; certificate management support (MNT-11 to 13).
Technology currency	Compatibility with supported OS, browser, Node.js and PostgreSQL versions; upgrades included (MNT-14, 15, 22).
Minor enhancements	60 hours per year included; further changes via change request at the rate card (MNT-23, 24).
Onsite support	Remote by default; onsite for major incidents, upgrades or critical activities at the rate card day rate plus OPE at actuals (MNT-29).

18.2 Service levels

Priority	Definition	Response	Resolution / restoration	Coverage
P1 Critical	Production down or critical function unavailable for all users; data integrity or security breach	30 minutes	4 hours (restore / workaround)	24 x 7
P2 High	Major function impaired for many users; no reasonable workaround	2 hours	1 business day	IIFT business hours
P3 Medium	Function impaired with workaround; limited users affected	4 business hours	3 business days	IIFT business hours
P4 Low / Cosmetic	Cosmetic issue, question or minor defect	1 business day	Next scheduled release	IIFT business hours

Table 85: Incident service levels (MNT-05 to MNT-07, COM-13)

Availability target: 99.5% per month overall and 99.9% during IIFT business hours, excluding approved maintenance windows and infrastructure outside iorta's responsibility. Response is measured from ticket logging; resolution may be a workaround with a permanent fix scheduled. Service credits for repeated breaches are set out in Section 24.

18.3 Escalation matrix

Level	Role	Contact	Escalation trigger
Level 1	Service Desk / Support Engineer	[Name, phone, e-mail]	Immediately on logging
Level 2	Support Lead / Senior Engineer	[Name, phone, e-mail]	P1 not restored in 1 h; P2 not responded in 2 h
Level 3	Account / Project Manager	[Name, phone, e-mail]	P1 not restored in 2 h; P2 not resolved in 1 business day
Level 4	Engagement Director	[Name, phone, e-mail]	P1 not restored in 4 h; repeated SLA breach

Table 86: Critical incident escalation (COM-15)

18.4 Year-by-year focus

Year	Focus	Activities
Year 1	Stabilisation & Hypercare	Intensive post-go-live support, defect resolution, performance tuning, monitoring and stabilisation of integrations. Includes the 6-month defect warranty.
Year 2	Operational Support	BAU incident management, preventive maintenance, security patches, performance monitoring and minor enhancements.
Year 3	Optimisation	System optimisation, reporting improvements, integration improvements and technology review.
Year 4	Technology Refresh	Platform compatibility review, security hardening, version upgrades and capacity assessment.
Year 5	Sustainability & Transition	Comprehensive health assessment, documentation refresh, knowledge transfer, technology roadmap and exit/renewal preparation.

Table 87: Five-year maintenance structure (RFP section 9)

18.5 Service reporting and reviews

- Monthly service report (MNT-25): incidents and service requests by priority, SLA performance, availability, changes and releases, security patches and vulnerabilities, capacity trends, outstanding issues.
- Quarterly service review (MNT-26): service performance, improvement plan, roadmap, upcoming platform changes.
- Annual DR test participation (MNT-20) and quarterly backup restore verification (MNT-19) with written results.
- Documentation updated after every material change (MNT-27) and ongoing knowledge-transfer sessions for IIFT/IITH support staff (MNT-28).

18.6 Exit and transition plan

Activity	Description
Transition planning	Exit plan agreed in Year 5 (or within 30 days of a termination notice): scope, timeline, receiving party.
Knowledge transfer	Structured sessions on architecture, code, configuration, operations and known issues.
Documentation	Final refresh of technical, operations, administrator and user manuals; runbooks.
Source code & tooling	Final source code of IIFT-specific components (and of the core platform if Option C or escrow applies), pipelines, infrastructure scripts and credentials handed over.
Data extraction	Full export of business data, documents and audit records in open formats (SQL dump, CSV, original files).
Parallel support	Shadow support to the new provider or IIFT team for up to 30 days.
Confirmation	Signed transition acceptance; secure deletion of IIFT data held by iorta, with certificate.

18.7 iorta business continuity

Source code, documentation and infrastructure scripts are kept in version-controlled repositories with off-site backup. At least two engineers know each module, support can run from more than one location, and every support role has a named backup (COM-24). **[Confirm or extend with iorta's corporate BCP details.]**

19. Project Team & Organisation

The team pairs Takaful business analysts with engineers who know the platform. CVs are in Annex B. Key personnel are not replaced without IIFT's consent, and any replacement has equivalent experience.

Role	Name	Responsibilities
Engagement Director	[Name]	Executive sponsor; Steering Committee member; commercial and escalation owner
Project Manager	[Name]	Plan, governance, RAID, reporting, change control; single point of contact
Solution Architect	[Name]	Architecture, security and integration design; Design Authority member
Business Analyst (Takaful)	[Name]	Workshops, BRS/FRS, product rules, RTM, UAT preparation, training
UI/UX Designer	[Name]	Screen designs, user journeys, accessibility
Technical Lead	[Name]	Code quality, technical design, reviews, deployment
Software Engineers	[Names]	Portal, back-office and API development
Integration Engineer	[Name]	Core, FIN, AML, AD, SMS and e-mail adapters; interface testing
QA Lead / Test Engineer	[Name]	Test strategy, SIT, regression automation, performance testing
DevOps & Security Engineer	[Name]	Environments, CI/CD, monitoring, hardening, VAPT coordination
Support Manager	[Name]	Hypercare, maintenance service, SLA reporting

Table 88: Proposed team

19.1 Involvement by phase

● = lead / high involvement, ○ = supporting / part-time, – = not involved.

Role	0 Mob.	1 Design	2 Build	3 Integr.	4 SIT	5–6 UAT & go-live	7 Hypercare
Engagement Director	●	○	○	○	○	●	○
Project Manager	●	●	●	●	●	●	●
Solution Architect	●	●	○	●	○	○	–
Business Analyst	●	●	○	○	●	○	○
UI/UX Designer	–	●	○	–	–	–	–
Technical Lead	○	●	●	●	●	●	●
Software Engineers	–	○	●	●	●	○	●
Integration Engineer	–	●	●	●	○	○	○
QA Lead / Test Engineer	–	○	●	●	●	○	○
DevOps & Security Engineer	○	○	●	●	○	●	○
Support Manager	–	–	–	○	○	●	●

19.2 Onsite and remote delivery

The team works from Malaysia and India. Onsite presence in Bandar Seri Begawan is planned for the phases where it matters most; everything else is remote, with video conferencing and shared tools.

Phase	Role	From	Nights onsite
Requirements gathering and fit-gap (weeks 1–4)	Project Manager	Malaysia	12
Requirements gathering and fit-gap (weeks 1–4)	Business Analyst	Malaysia	19

Phase	Role	From	Nights onsite
Requirements gathering and fit-gap (weeks 1–4)	Solution Architect	India	12
User training and UAT support (weeks 19–23)	Business Analyst / Trainer	Malaysia	19
User training and UAT support (weeks 19–23)	QA Lead	India	12
Go-live and one month of support (weeks 24–28)	Senior Developer	India	33
Go-live and one month of support (weeks 24–28)	Project Manager	Malaysia	6
Knowledge transfer and source code handover (Option C)	Solution Architect	India	12
Knowledge transfer and source code handover (Option C)	Senior Developer	India	12

Table 89: Planned onsite presence (OPE in the Commercial Proposal)

19.3 Expected IIFT roles

IIFT role	Responsibility
Project Sponsor	Chairs Steering Committee; final decisions
Project Manager	Coordinates IIFT resources, approvals and logistics
Business owners / SMEs	Banca, Underwriting, Finance and Compliance: requirements, rules, UAT
IITH IT	Infrastructure, network, security, interfaces, environments
UAT testers / super-users	Execute UAT; support colleagues after go-live

20. Assumptions, Dependencies & Exclusions

20.1 Assumptions

- Scope is as described in the RFP (sections 3–8 and Appendices 1–4) and this proposal; changes follow the change request process.
- IIFT provides timely access to business owners and decisions; document reviews are completed within five business days.
- Each external system (Core, FIN, AML, AD, SMS gateway, SMTP) offers a usable interface (API, file or protocol) with test environments; iorta does not modify those systems.
- Rate tables, product rules, document templates and questionnaires are provided by IIFT and signed off by week 6.
- Data migration covers agent, agency, bank, branch, participant and reference data, estimated below 10,000 records from no more than three source extracts; legacy data cleansing at source is IIFT's responsibility.
- Under Option A, hardware, OS licences, network, firewall, load balancer, backup infrastructure and DR site are provided by IIFT/IITH to the sizing in Section 13 and the Bill of Materials.
- Under Option B, IIFT approves the cloud region and makes the AMBD outsourcing and cloud notification before production data is loaded.
- The user interface and documents are in English; Malay translation of screens can be added as a change request.
- Delivery is remote from Malaysia and India, with onsite presence for requirements gathering, user training and UAT support, and go-live with one month of support (and knowledge transfer if Option C is taken). Travel and subsistence for these phases are recharged as OPE at cost (Section 17).
- IIFT business hours are as published in the 'Insurans Islam TAIB Business Hours (Family Takaful)' schedule.

20.2 Dependencies

- Contract signature and kick-off date; project timeline counts from kick-off.

- SIT and UAT environments by weeks 8 and 16; production and DR environments by week 20.
- Interface specifications, endpoints and credentials by week 8.
- Legacy data extracts by week 12; UAT testers available weeks 19–22.
- Independent VAPT provider access to the SIT/pre-production environment in weeks 17–19.

20.3 Exclusions

- Hardware, operating system licences, network equipment and data-centre services (Option A).
- Cloud infrastructure charges (Option B), recharged at cost without mark-up or paid by IIFT directly.
- Third-party charges listed in the Bill of Materials: SMS messages, AML data service, SSL certificates, e-mail relay services and optional items such as commercial PostgreSQL support or RHEL subscriptions.
- Out-of-pocket expenses, recharged at cost.
- Changes to the Core system, FIN, AML service or other third-party systems.
- Native mobile applications (the portal is responsive and works on tablets and mobile browsers).
- Payment gateway / online card payments (can be added later as a change request).
- Full claims administration and policy administration in the core system beyond claim notification and the portal/back-office functions described.
- Onsite visits outside the planned phases (rate card plus OPE) and work outside IIFT business hours other than P1 incidents (rate card).

21. Risk Register

The initial risk register is below. It is maintained by the project managers, reviewed weekly by the Working Committee and monthly by the Steering Committee.

ID	Risk	Impact	Likelihood	Mitigation
R1	Delayed availability or incomplete specification of Core/FIN interfaces	High	Medium	Early interface workshops (week 3); mock services so build continues; interface specification sign-off by week 6
R2	Late provision of SIT/UAT/production environments	High	Medium	Environment request at kick-off; iorta-hosted SIT as interim; containerised deployment reduces set-up time
R3	Rate tables or product rules not finalised in time	High	Medium	Configurable rating with placeholder rates; rate validation as an SIT exit criterion
R4	Scope growth during design	Medium	Medium	Workshops on working screens; scope baseline at M2; formal change control
R5	Legacy data quality lower than expected	Medium	Medium	Early profiling; two dry-runs; cleansing rules owned by IIFT data owners
R6	Limited availability of IIFT users for UAT	Medium	Low	UAT plan agreed at M2; scenario scripts prepared by iorta; super-user model
R7	VAPT reveals significant findings close to go-live	High	Low	Secure SDLC with SAST/SCA/DAST from sprint 1; VAPT in weeks 17–19 leaves time for remediation
R8	AML provider integration constraints (rate limits, formats)	Medium	Low	Adapter with queueing and retry; manual screening fallback with audit
R9	Low adoption by bank officers and agents	Medium	Low	Change management, train-the-trainer, simple UX, onboarding pack, adoption KPIs
R10	Regulatory feedback on cloud hosting (Option B)	Medium	Low	Recommend Option A; early AMBD engagement if Option B is chosen
R11	Key personnel unavailability	Medium	Low	Named backups; shared code ownership; documentation as we go
R12	Performance issues with document-heavy transactions	Low	Low	Streaming uploads, object storage, performance tests on production-like data
R13	Cloud charges above the estimate (Option B)	Low	Medium	Recharged at cost with budget alerts, monthly cost report, quarterly right-sizing; reserved capacity only with IIFT approval
R14	Exchange-rate movement on USD-billed cloud and third-party services	Low	Medium	B\$ fees for iorta services; pass-through items converted at invoice-date rates and reported monthly
R15	Knowledge transfer (Option C) without enough IIFT/IITH developers	Medium	Low	Skills profile agreed in week 1 of the transition; competency assessment; support choice 2 as a safety net

Table 90: Risk register

22. Quotation Validity

This proposal and the prices quoted are valid for **90 days from the date of submission** ([submission date] to [expiry date]).

On award, the one-time fees are fixed for the implementation. The AMC, subscription, managed services and rate card follow the yearly schedule in the Commercial Proposal, with increases capped at 5% a year. Cloud infrastructure and third-party charges follow the providers' prices at the time they are incurred, and OPE is recharged at cost within the stated ceiling.

23. Supporting Information

23.1 Value-added items beyond the RFP

Item	Benefit to IIFT
Requirements compliance matrix	All 234 requirement IDs answered line by line, with fitment (Annex A).
Working application before contract	IIFT can test SalesVerse 2.0 with its own products before signing.
Choice of commercial model	Perpetual licence on-premise or subscription on iorta-managed cloud, priced on the same scope, with conversion from subscription to licence.
Source code handover option	Core platform source code with an eight-week knowledge transfer (Option C), or escrow; IIFT-specific source code under every option.
Bill of Materials	Every infrastructure item, licence and third-party service with indicative cost and who pays, in a separate workbook.
Multi-factor authentication	E-mail OTP or authenticator app for back-office users, delivered during implementation.
Remote participant e-signature	Single-use e-mail link, so participants can sign without visiting a branch.
Integration monitor	One screen for outbox messages, retries, dead-letters and reconciliation results.
End-of-day run screen	Run, check and sign off EOD, the FIN file and reconciliation, with run history.
Retention and archival policy	Retention period per data class, with archival of audit records and documents.
PDPO 2025 and AMBD alignment	Controls mapped to Brunei data protection law and AMBD expectations (Section 11).
Shariah-appropriate terminology	Contribution, participant, Takaful operator, wakalah and tabarru' throughout.
Accessibility	WCAG 2.1 AA target.
Exit plan	Transition, data extraction and knowledge transfer defined now (Section 18).
Hardware sizing	Sizing for every environment for the on-premise option (Section 13).
Unlimited users	More banks, branches and agents at no extra licence or subscription cost.

Table 91: Value-added items included at no extra cost

Further supporting material: the technical document pack (Section 14), the screen index (Section 7.15), the pricing workbook and the Bill of Materials workbook.

24. Terms & Conditions

We propose the key terms below and are willing to work from IIFT's standard contract in negotiation.

24.1 Basis of contract. This proposal, the RFP and agreed clarifications form the basis for a definitive agreement to be negotiated in good faith. In case of conflict, the signed agreement prevails.

24.2 Commercial model. IIFT selects Option A (perpetual licence) or Option B (subscription) at award; Option C may be exercised later on the terms in Section 17.6.

24.3 Prices and taxes. Prices are in Brunei Dollars as set out in Section 17. Fees are inclusive of withholding tax, which is borne by iorta TechNXT. Out-of-pocket expenses and cloud infrastructure (Option B) are recharged at cost as disbursements with the provider and travel invoices attached, outside the fees to which withholding tax applies; IIFT may instead hold the cloud account and pay the provider directly. Third-party charges are excluded and payable at cost. Any indirect tax introduced by law after submission is added at the statutory rate.

24.4 Payment terms. One-time fees are invoiced on the milestones in Section 17.11. Recurring fees are invoiced as stated there. Invoices are payable within 30 days.

24.5 Acceptance. Deliverables and milestones are accepted by written sign-off against agreed acceptance criteria. IIFT will review within ten business days; a deliverable is deemed accepted if no material non-conformity is notified within that period or if it is used in production.

24.6 Intellectual property. The SalesVerse 2.0 core platform remains the intellectual property of iorta TechNXT. Under Option A IIFT receives the perpetual licence in Section 17.4; under Option B a right to use for the subscription term. Configurations, reports, documentation and other work products produced specifically for IIFT belong to IIFT. Open-source components remain under their own licences.

24.7 Source code and escrow. Source code of IIFT-specific components (configuration, integration adapters, reports, templates, migration and build scripts) is delivered at go-live and with every release under both options. Source code of the core platform is delivered under Option C, or deposited in escrow at IIFT's cost on the release conditions in Section 17.6.

24.8 Data ownership and protection. All business data, including participant, agent and transaction data and documents, remains the property of IITH/IIGT/IIFT. iorta TechNXT processes personal data only on IIFT's instructions and for the purpose of the contract, applies the security controls in Section 11, supports IIFT's obligations under the Personal Data Protection Order 2025 and returns or securely deletes data at contract end.

24.9 Confidentiality. Each party keeps the other's confidential information confidential, uses it only for the contract and discloses it only to personnel and approved subcontractors bound by equivalent obligations. These obligations survive termination for [five] years, and indefinitely for personal data and source code.

24.10 Warranty. iorta TechNXT warrants that the solution will perform materially in accordance with the approved specifications for 6 months from production go-live and will correct reported defects at no charge within the SLA. The warranty does not cover issues caused by changes made by others, misuse, infrastructure or third-party products outside iorta's scope.

24.11 Service levels and service credits. Service levels are as defined in Section 18. Where a P1 restoration target is missed, service credits apply: [e.g. 5% of the quarterly AMC or of the quarterly subscription per breach, capped at 20% of that quarter's fee]. Service credits are IIFT's financial remedy for SLA failure without prejudice to termination rights for persistent failure.

24.12 Limitation of liability. Each party's total aggregate liability under the contract is limited to [the total fees paid and payable in the twelve months before the claim, or the one-time fees if higher]. Neither party is liable for indirect or consequential loss. The limitation does not apply to liability for fraud, wilful misconduct, breach of confidentiality or infringement of intellectual property.

24.13 Change requests. Changes to scope follow the change request process: written request, impact assessment (scope, cost, schedule, risk) within five business days, quotation using the rate card, written approval before work starts.

24.14 Price protection. The AMC, subscription, managed services fee and rate card increase by 5% a year and by no more than 5% a year (COM-18). One-time fees are fixed.

24.15 Personnel and subcontracting. Key personnel are not replaced without IIFT's consent. The independent VAPT is performed by a qualified third-party tester engaged by iorta with IIFT's approval; iorta remains responsible for its subcontractors.

24.16 Term and termination. Either party may terminate for material breach not remedied within 30 days of notice. IIFT may terminate for convenience on [90] days' notice, paying for work performed and accepted up to the termination date and, under Option B, the subscription for any remaining part of the 60-month minimum term. On any termination iorta provides the exit assistance in Section 18, hands over work in progress and the source code of IIFT-specific components, and returns IIFT's data.

24.17 Force majeure. Neither party is liable for delay or failure caused by events beyond its reasonable control (including natural disasters, pandemics, war, government action or widespread utility or network failure), provided it notifies the other party promptly and uses reasonable efforts to mitigate. If force majeure continues for more than 60 days, either party may terminate the affected services.

24.18 Governing law and disputes. The contract is governed by the laws of Brunei Darussalam. Disputes are first escalated through the governance structure; unresolved disputes are referred to [mediation / arbitration in Brunei Darussalam] and, failing that, to the courts of Brunei Darussalam.

24.19 Validity. This proposal is valid for 90 days from the submission date.

Annex A – Requirements Compliance and Fitment Matrix

Every requirement ID in the RFP is listed. **Compliance:** Fully Compliant is a standard capability of the delivered solution; Compliant – configuration is met by setting rules, parameters or master data; Compliant – custom is met by IIFT-specific work inside the fixed price. **Fitment** uses the four categories in Section 6. Priority follows RFP Appendix 4. The RFP uses the prefix COM for both shared platform (section 4.4) and commercial (section 6) requirements; each set is listed under its section, and commercial items carry no fitment.

ID	Requirement	Priority	Compliance	Fitment	How addressed
RFP 4.1 – Agent/Banca Portal functional requirements					
AP-01	Secure login	Must	Fully Compliant	Available today	Username/password with Argon2id hashing, rate-limited login, optional MFA, server-side sessions.
AP-02	Password management	Must	Compliant – configuration	Configuration	Self-service change and reset; length, complexity, history (last 5) and 90-day expiry are parameters.
AP-03	Session management	Must	Compliant – configuration	Configuration	15-min idle and 8-h absolute timeout, logout revokes server session, ID regenerated at login, single-session option.
AP-04	Account lockout	Must	Compliant – configuration	Configuration	Lock after 5 failed attempts for 30 min (configurable); manual release by an administrator is audited.
AP-05	View agent/banca profile	Must	Fully Compliant	Available today	My Profile page: personal, contact, agency/bank, licence, hierarchy and document status.
AP-06	Maintain profile (approval)	Must	Fully Compliant	Available today	Profile change request with before/after values routed through maker-checker workflow.
AP-07	Agent/banca registration	Must	Fully Compliant	Available today	Online registration wizard with AML/KYC data, document upload (IC copy, or passport copy for passport holders) and back-office approval.
AP-08	Agent ID	Must	Compliant – configuration	Configuration	Unique agent/banca code produced on approval from a configurable numbering pattern.
AP-09	Main/sub-agent relationships	Must	Fully Compliant	Available today	Hierarchy view of main agent, sub-agents and bank officers scoped to the user's position.
AP-10	Agency/banca information	Must	Fully Compliant	Available today	Agency/bank details, branches, contacts and status shown on dashboard and profile.
AP-11	Cross-agency participant sharing	Must	Compliant – custom	Available today	Single participant master with de-duplication by IC/registration no.; sharing governed by RBAC and policy rules.
AP-12	Agency-wide participant visibility	Must	Compliant – custom	Available today	Consolidated participant 360 view (policies, payments, claims, interactions) for authorised roles.
AP-13	Participant registration	Must	Fully Compliant	Available today	Individual and corporate participant forms with validation and duplicate check.
AP-14	Participant search	Must	Compliant – configuration	Available today	Search by name, IC/passport (blind index), phone, policy no.; criteria configurable.
AP-15	Participant maintenance (approval)	Must	Fully Compliant	Available today	Update requests go through maker-checker; approved values applied with audit history.
AP-16	AML screening	Must	Compliant – custom	Available today	Screening on registration and update via AML adapter; hits routed to compliance review queue.
AP-17	Product selection	Must	Compliant – configuration	Configuration	Product catalogue from configuration showing products the agent/bank is authorised to sell.
AP-18	New quotation	Must	Compliant – custom	Available today	Product-specific quotation wizard for all seven Appendix 3 products.
AP-19	Automatic contribution calculation	Must	Compliant – custom	Configuration	Rating engines per product using configurable rate tables supplied by IIFT actuarial.

ID	Requirement	Priority	Compliance	Fitment	How addressed
AP-20	Draft quotation	Must	Fully Compliant	Available today	Quotations saved as DRAFT at any step and resumed later.
AP-21	Policy status	Must	Fully Compliant	Available today	Draft, Pending Approval, Pending Payment, Active, Rejected, Expired and Cancelled statuses.
AP-22	Policy listing	Must	Fully Compliant	Available today	Paginated policy list with status chips, scoped to agent, agency or bank.
AP-23	Policy search	Must	Compliant – configuration	Available today	Filter by policy no., participant, product, status, date range; criteria configurable.
AP-24	Policy details	Must	Fully Compliant	Available today	Coverage, participant, nominees, contribution, payment status, documents and history.
AP-25	Policy issuance	Must	Fully Compliant	Available today	Submit eligible quotation for issuance after validation; e-Policy produced on approval/payment.
AP-26	Policy renewal	Must	Compliant – configuration	Configuration	Renewal-due list and one-click renewal quotation inside the renewal window (notice period before expiry to 30 days after); renewal enable/disable per product (e.g. Khairat).
AP-27	Policy endorsement	Must	Fully Compliant	Available today	Endorsement request with change details and documents, subject to approval rules.
AP-28	Policy cancellation	Must	Fully Compliant	Available today	Cancellation request with reason and supporting documents, subject to approval.
AP-29	Policy documents	Must	Fully Compliant	Available today	View/download e-Policy, schedule, receipts, certificates and correspondence.
AP-30	Payment status	Must	Fully Compliant	Available today	Payment status and outstanding amount shown on policy and billing screens.
AP-31	Policy history	Should	Fully Compliant	Available today	Timeline of every transaction and status change with user and timestamp.
AP-32	Policy validation	Must	Compliant – configuration	Available today	Server-side rules for product, eligibility, documents, payment and AML before submission.
AP-33	Approval workflow	Must	Compliant – configuration	Available today	Transactions needing approval routed by workflow definition to authorised checkers.
AP-34	Notifications	Should	Compliant – configuration	Configuration	In-portal, email and SMS notifications for issuance, approval, rejection, expiry and renewal.
AP-35	Audit trail	Must	Fully Compliant	Available today	Append-only audit of policy creation, updates and approvals with before/after values.
AP-36	Policy restrictions	Must	Compliant – configuration	Available today	Issuance blocked when payment, AML or document requirements are not met.
AP-37	Payment dashboard	Must	Fully Compliant	Available today	Billing dashboard with collected, outstanding and overdue amounts and ageing.
AP-38	Bulk payment (per policy)	Must	Fully Compliant	Available today	Single or bulk payment submission with amounts allocated per policy.
AP-39	Upload payment proof	Must	Fully Compliant	Available today	Upload bank slip / transfer evidence against one or many policies.
AP-40	Payment verification	Must	Fully Compliant	Available today	Payment proof routed to Finance for verification (approve/reject with remarks).
AP-41	Receipt generation	Must	Fully Compliant	Available today	PDF e-Receipt with unique number produced on approval and e-mailed.
AP-42	Payment rules (7-day grace, agency block)	Must	Compliant – configuration	Available today	Daily job blocks all agents of an agency when any policy is unpaid > 7 days; auto-lifted on settlement.
AP-43	Claims notification	Must	Fully Compliant	Available today	Claim notification: validate policy, capture details, upload documents, submit to IIFT.
AP-44	Policy schedule / receipt	Must	Fully Compliant	Available today	Branded PDF policy schedule and receipt produced after issuance.
AP-45	Document download / e-mail	Must	Fully Compliant	Integration with IIFT systems	Download and e-mail documents to participant from the portal.
AP-46	Document upload	Must	Fully Compliant	Available today	Upload during registration and maintenance with type allow-list, size limit, checksum and ClamAV malware scan (mandatory in production).
AP-47	Document status	Should	Fully Compliant	Available today	Document status, validity and expiry dates displayed with reminders.
AP-48	AML information	Must	Compliant – configuration	Configuration	AML/KYC fields captured in onboarding (configurable questionnaire).

ID	Requirement	Priority	Compliance	Fitment	How addressed
AP-49	Application submission	Must	Fully Compliant	Available today	Save, submit and track applications and requests.
AP-50	Application status	Must	Fully Compliant	Available today	Draft, Submitted, Pending Approval, Approved, Rejected and Completed statuses.
AP-51	Rejection remarks	Must	Fully Compliant	Available today	Rejection reasons displayed; maker can amend and resubmit.
AP-52	Agent dashboard	Must	Fully Compliant	Available today	Dashboard with profile, applications, pending actions, production and payments.
AP-53	Pending actions	Must	Fully Compliant	Available today	Action list of incomplete drafts, rejected items, overdue payments and expiring documents.
AP-54	Notifications (applications)	Should	Compliant – configuration	Configuration	Notification templates per workflow event; channels configurable.
AP-55	Report issue	Must	Fully Compliant	Available today	Issue form with category, priority, description and attachments.
AP-56	Issue tracking	Must	Fully Compliant	Available today	Unique issue reference and status tracking with comments.
AP-57	Issue attachment	Should	Fully Compliant	Available today	Screenshots, documents and evidence attached to issues.
AP-58	Agency/agent/banca reports	Should	Compliant – configuration	Configuration	Role-scoped reports (production, payments, renewals) with export.
AP-59	Role-based access	Must	Compliant – configuration	Configuration	Permissions by agent type, hierarchy and role; data scoping enforced in the API.
AP-60	Authority limit	Must	Compliant – configuration	Available today	Financing above B\$150,000 is referred to IIFT Sales; further thresholds per product and role are configurable.
AP-61	Approval history	Must	Fully Compliant	Available today	Approval actions with remarks retained and displayed; immutable audit.
AP-62	Electronic signature	Must	Compliant – custom	Available today	On-screen signature by agent or participant, signature by single-use e-mail link, or upload of signed documents.
RFP 4.2 – Back-office (Admin) functional requirements					
BO-01	Secure login	Must	Fully Compliant	Available today	Internal login with optional AD/LDAP SSO and MFA (email OTP / TOTP).
BO-02	Role-based access control	Must	Compliant – configuration	Available today	Roles, permissions and data scopes configured by administrators.
BO-03	User administration	Must	Fully Compliant	Available today	Create, update, deactivate and reset back-office users and release lockouts; changes audited.
BO-04	Role & permission management	Must	Compliant – configuration	Available today	Role builder with fine-grained permissions grouped by module; maker-checker optional.
BO-05	Agent administration	Must	Fully Compliant	Available today	Create, view, update, activate, suspend and terminate agent records.
BO-06	Agent search	Must	Compliant – configuration	Available today	Search by agent/banca ID, name, agency/bank, status and configurable criteria.
BO-07	Agent status	Must	Fully Compliant	Available today	Pending, Active, Inactive, Suspended and Terminated with controlled transitions.
BO-08	Main/sub-agent management	Must	Fully Compliant	Available today	Maintain hierarchy (agency → main agent → sub-agent / bank → branch → officer).
BO-09	Agency administration	Must	Fully Compliant	Available today	Maintain agency/bank details, branches, contacts and associated agents.
BO-10	Agent approval (maker-checker)	Must	Compliant – configuration	Configuration	Registration and profile changes approved through configurable workflow levels.
BO-11	Agent documents	Must	Fully Compliant	Available today	View, validate, approve, reject and manage agent documents.
BO-12	Document validation	Should	Compliant – configuration	Configuration	Mandatory document checklist per entity type; expiry tracking and reminders.
BO-13	AML screening	Must	Compliant – custom	Available today	Watch-list screening with fuzzy name matching for agents and participants; manual re-screen.
BO-14	AML status	Must	Fully Compliant	Available today	Screening result, score, status, reviewer and dates recorded and displayed.

ID	Requirement	Priority	Compliance	Fitment	How addressed
BO-15	Compliance review	Must	Fully Compliant	Available today	Compliance queue to clear, escalate or reject flagged cases with remarks; a confirmed match rejects the subject's pending approval requests.
BO-16	Approval workflow	Must	Compliant – configuration	Configuration	Configurable levels and amount thresholds per transaction type (e.g. second approval from B\$300,000 sum covered).
BO-17	Maker-checker	Must	Fully Compliant	Available today	Server-side rule: maker cannot approve own transaction; enforced for every workflow.
BO-18	Workflow status	Must	Fully Compliant	Available today	Tracking of every request through Submitted, Pending, Approved, Rejected, Completed.
BO-19	Rejection & remarks	Must	Fully Compliant	Available today	Mandatory remarks on rejection; visible to maker and in audit.
BO-20	Management dashboard	Should	Fully Compliant	Available today	KPIs, trends, production by product/channel, outstanding actions and SLA.
BO-21	Pending action dashboard	Must	Fully Compliant	Available today	Approvals inbox, incomplete applications and outstanding actions per user/role.
BO-22	Standard reports	Must	Compliant – custom	Available today	12 standard reports in the working application; remaining IIFT reports added from the catalogue agreed in design.
BO-23	Report filtering	Must	Fully Compliant	Available today	Filters by date, agent, agency/bank, product, status and other criteria.
BO-24	Report export	Must	Fully Compliant	Available today	Export to Excel, CSV and PDF.
BO-25	Scheduled reports	Should	Compliant – configuration	Available today	Report schedules (daily/weekly/monthly) e-mailed to authorised recipients.
BO-26	Audit trail	Must	Fully Compliant	Available today	Records user activities, transactions, approvals, rejections and system events.
BO-27	Change history	Must	Fully Compliant	Available today	Previous and new values, user, date/time and IP retained for critical changes.
BO-28	Audit search	Must	Fully Compliant	Available today	Audit search by user, entity, action and date with export for authorised roles.
BO-29	Issue assignment	Must	Compliant – configuration	Available today	Assignment to support personnel or teams with assignment rules.
BO-30	Issue priority	Must	Compliant – configuration	Configuration	Critical, High, Medium, Low priorities (configurable).
BO-31	Issue SLA monitoring	Should	Compliant – configuration	Configuration	Response/resolution timers per priority with breach alerts and SLA report.
BO-32	Master data management	Must	Fully Compliant	Available today	Maintain products, plans, rates, branches, banks, occupations, document types, reasons, etc.
BO-33	System configuration	Must	Compliant – configuration	Configuration	Business rules, statuses, approval levels and parameters maintainable without redevelopment.
RFP 4.3 – Integration functional requirements					
INT-01	Core system integration	Must	Compliant – custom	Integration with IIFT systems	Core system adapter to retrieve/update agent and policy information per interface specification.
INT-02	Agent/banca master data synchronisation	Must	Compliant – custom	Integration with IIFT systems	Scheduled and event-driven synchronisation of agent, agency, bank and status data.
INT-03	Agent information exchange	Must	Compliant – custom	Integration with IIFT systems	Agent-related events exchanged between portal, back-office and core via outbox.
INT-04	Financial system integration	Must	Compliant – custom	Integration with IIFT systems	EOD postings, receipts and commission data exchanged with FIN.
INT-05	Commission / referral fee information	Must	Compliant – custom	Integration with IIFT systems	Commission/referral statements retrieved from FIN and shown to authorised agents.
INT-06	Enterprise authentication	Should	Compliant – custom	Integration with IIFT systems	LDAP/Active Directory SSO for back-office; SAML/OIDC possible if IITH IdP exists.

ID	Requirement	Priority	Compliance	Fitment	How addressed
INT-07	Email integration	Should	Compliant – configuration	Integration with IIFT systems	SMTP relay integration with TLS; templates configurable.
INT-08	SMS integration	Should	Compliant – custom	Integration with IIFT systems	SMS gateway adapter for OTP and alerts via IIFT-approved provider.
INT-09	API gateway / API services	Must	Fully Compliant	Available today	Versioned REST APIs (OpenAPI 3) exposed and consumed through the integration layer.
INT-10	API security	Must	Fully Compliant	Available today	Mutual TLS or API keys/OAuth2 client credentials, RBAC, input validation, rate limiting.
INT-11	API monitoring	Should	Fully Compliant	Available today	Integration monitor with availability, volumes, failures and response times; metrics endpoint.
INT-12	Data validation	Must	Fully Compliant	Available today	Schema validation of inbound and outbound payloads; rejects logged with reason.
INT-13	Error handling & retry	Must	Fully Compliant	Available today	Transactional outbox, exponential back-off retry, dead-letter and alerting.
INT-14	Integration logging	Must	Fully Compliant	Available today	Every attempt logged with correlation id, status, duration and payload hash.
INT-15	Data reconciliation	Should	Fully Compliant	Integration with IIFT systems	Receipt reconciliation in the working application; BRR and Core reconciliation added during implementation.
RFP 4.4 – Shared / common platform requirements					
COM-01	Sensitive data protection	Must	Fully Compliant	Available today	TLS 1.2+, AES-256-GCM field encryption for IC/passport, masking, encrypted storage.
COM-02	Access control	Must	Fully Compliant	Available today	Least-privilege RBAC with data scoping by agency/bank/hierarchy.
COM-03	Comprehensive audit	Must	Fully Compliant	Available today	Common append-only audit service used by all modules.
COM-04	Configurable workflow	Must	Compliant – configuration	Configuration	Workflow definitions by transaction type, role, level and business rule.
COM-05	Notification framework	Should	Fully Compliant	Available today	Common notification service for email, SMS and in-portal channels with templates.
COM-06	Document repository	Must	Fully Compliant	Available today	Encrypted document store with metadata, versioning, checksum and access control.
COM-07	Global search	Should	Fully Compliant	Available today	Global search across participants, policies, agents and requests for authorised users.
COM-08	Reporting framework	Must	Fully Compliant	Available today	Common report engine with filters, scheduling and Excel/CSV/PDF export.
COM-09	Parameter management	Must	Compliant – configuration	Configuration	System parameters and master data maintained in the back-office without code change.
COM-10	Data encryption	Must	Fully Compliant	Available today	Encryption in transit (TLS) and at rest (volume/DB/object storage + field level).
RFP 5 – Non-functional requirements					
NFR-01	Response time	Must	Fully Compliant	Delivered during implementation	Targets: page < 3 s, standard API p95 < 1 s, reports < 10 s for 12 months of data.
NFR-02	Concurrent users	Must	Fully Compliant	Delivered during implementation	Sized for 100 concurrent / 500 named users against 26 named users in Appendix 1.
NFR-03	Transaction volume	Must	Fully Compliant	Available today	Sized for > 10x the Appendix 2 five-year projection (~628 policies per year).
NFR-04	System availability	Must	Fully Compliant	Delivered during implementation	99.5% overall, 99.9% during IIFT business hours, excluding approved maintenance.
NFR-05	High availability	Should	Fully Compliant	Delivered during implementation	Two application nodes, PostgreSQL primary + standby, redundant load balancer.
NFR-06	User scalability	Must	Fully Compliant	Available today	Stateless API nodes scale horizontally; no per-user licences.
NFR-07	Transaction scalability	Must	Fully Compliant	Available today	Indexed relational model, partition-ready audit tables, archival policy.
NFR-08	Encryption	Must	Fully Compliant	Available today	TLS 1.2+ in transit; encrypted volumes, backups and field-level encryption at rest.

ID	Requirement	Priority	Compliance	Fitment	How addressed
NFR-09	Authentication	Must	Compliant – configuration	Delivered during implementation	Password policy, lockout, MFA and AD integration aligned to IITH security policy.
NFR-10	Authorisation	Must	Fully Compliant	Available today	Least privilege RBAC with deny-by-default API guards.
NFR-11	Audit logging	Must	Compliant – configuration	Available today	Security and business events logged; retention period configurable (default 7 years).
NFR-12	Vulnerability management	Must	Fully Compliant	Available today	CodeQL static analysis, dependency audit, secret and container scanning in CI; remediation within severity-based timelines.
NFR-13	Penetration testing	Must	Fully Compliant	Delivered during implementation	Independent VAPT before go-live with remediation and re-test (included in price).
NFR-14	Secure development	Must	Fully Compliant	Available today	OWASP ASVS Level 2 aligned SDLC, OWASP Top 10 controls, peer review.
NFR-15	Error handling	Must	Fully Compliant	Available today	Generic user messages with correlation id; no stack traces or internals exposed.
NFR-16	Transaction integrity	Must	Fully Compliant	Available today	ACID transactions in PostgreSQL; issuance, receipt, audit and outbox commit atomically.
NFR-17	Data backup	Must	Fully Compliant	Delivered during implementation	Daily full backup, continuous WAL archiving (PITR), configuration backup, restore tests.
NFR-18	DR capability	Must	Fully Compliant	Delivered during implementation	Warm standby DR with RPO ≤ 15 min and RTO ≤ 4 h.
NFR-19	DR testing	Must	Fully Compliant	Delivered during implementation	Annual DR test with IIFT, documented results and improvement actions.
NFR-20	Modular architecture	Should	Fully Compliant	Available today	Modular monolith with module boundaries per functional area; modules can be split later.
NFR-21	Documentation	Must	Fully Compliant	Delivered during implementation	Technical, configuration, operations and user documentation maintained for 5 years.
NFR-22	User interface	Must	Fully Compliant	Available today	Consistent Ant Design based UI themed to Insurans Islam TAIB branding.
NFR-23	Browser compatibility	Must	Fully Compliant	Available today	Current and previous major versions of Chrome, Edge, Firefox and Safari.
NFR-24	Accessibility	Should	Fully Compliant	Delivered during implementation	Designed to WCAG 2.1 AA (keyboard navigation, contrast, labels).
NFR-25	API standards	Must	Fully Compliant	Available today	REST/JSON over HTTPS, OpenAPI 3 contracts, ISO 8601 dates, UTF-8.
NFR-26	System monitoring	Must	Fully Compliant	Delivered during implementation	Health endpoints, Prometheus metrics, integration monitor and infrastructure checks.
NFR-27	Alerting	Must	Fully Compliant	Delivered during implementation	Alert rules for availability, errors, latency, job failures and dead-letters to support teams.
NFR-28	Centralised logs	Should	Fully Compliant	Delivered during implementation	Structured JSON logs with correlation id shipped to central log store.
NFR-29	Data retention	Must	Compliant – configuration	Configuration	Retention and archival rules per data class, configurable to IIFT policy.
NFR-30	Data quality	Must	Fully Compliant	Available today	Validation on every DTO plus database constraints (FK, CHECK, UNIQUE).
RFP 6 – Commercial requirements					
COM-01	Implementation cost	Must	Fully Compliant	Commercial	Fixed implementation price: Option A B\$190,000 (licence B\$60,000, services B\$130,000); Option B B\$145,000 (services B\$130,000, cloud set-up B\$15,000).
COM-02	Software licence	Must	Fully Compliant	Commercial	Option A: perpetual, non-exclusive enterprise licence for IIFT, unlimited named users, all environments, B\$60,000 one-off, no renewal fee. Option B: right to use within the subscription.

ID	Requirement	Priority	Compliance	Fitment	How addressed
COM-03	Subscription	Must	Fully Compliant	Commercial	Option B: subscription B\$3,400 a month and managed services B\$1,100 a month in Year 1 (+5% a year), minimum term 60 months; cloud recharged at cost. Option A has no subscription.
COM-04	Infrastructure	Must	Fully Compliant	Commercial	Option A: IIFT procures to iorta's sizing (BOM, indicative B\$20,500–42,000). Option B: one-off cloud set-up B\$15,000; cloud recharged at cost, about B\$1,330 a month, or paid by IIFT directly. PostgreSQL B\$0; third-party costs in the BOM.
COM-05	Integration cost	Must	Fully Compliant	Commercial	Cost per interface itemised (six interfaces, B\$30,500).
COM-06	Implementation services	Must	Fully Compliant	Commercial	PM, BA, design, configuration, customisation and deployment: B\$64,000 (item 2).
COM-07	Data migration	Must	Fully Compliant	Commercial	Scope, assumptions, tools and cost (B\$7,500) stated.
COM-08	Testing	Must	Fully Compliant	Commercial	SIT, UAT support and performance testing (B\$10,000) and security testing (B\$8,500) itemised.
COM-09	Training	Must	Fully Compliant	Commercial	User, administrator and technical training included (B\$4,500).
COM-10	Documentation	Must	Fully Compliant	Commercial	Preparation and maintenance of documentation included (B\$5,000).
COM-11	Warranty period	Must	Fully Compliant	Commercial	6 months from production go-live under both options, included.
COM-12	5-year maintenance	Must	Fully Compliant	Commercial	Detailed five-year plan. Option A AMC B\$33,990 in Year 1 (22% of licence and customisation, +5% a year; B\$187,816 over five years). Option B: included in the subscription.
COM-13	Service level agreement	Must	Fully Compliant	Commercial	Response, restoration and availability targets committed (Maintenance and Support chapter).
COM-14	Support hours	Must	Fully Compliant	Commercial	Aligned to IIFT business hours with 24x7 for P1 incidents.
COM-15	Critical incident escalation	Must	Fully Compliant	Commercial	Four-level escalation matrix with time-based triggers.
COM-16	Change request	Must	Fully Compliant	Commercial	Rate card (B\$680 per man-day, B\$90 per hour in Year 1, +5% a year) and formal CR process.
COM-17	Third-party dependencies	Must	Fully Compliant	Commercial	All third-party products, licences, services and infrastructure disclosed in the Bill of Materials (Annex C and BOM workbook); runtime components open source, B\$0; SBOM with every build.
COM-18	Annual increase	Should	Fully Compliant	Commercial	AMC, subscription, managed services and rate card rise 5% a year, capped at 5%. Cloud and third-party items at actual cost.
COM-19	Exit assistance	Must	Fully Compliant	Commercial	Transition, data extraction in open formats and knowledge transfer included in Year 5, at the end of the subscription or on termination.
COM-20	IP ownership	Must	Fully Compliant	Commercial	SalesVerse 2.0 core remains iorta IP (perpetual licence under Option A, right to use under Option B); IIFT-specific configuration, reports, documentation and data belong to IIFT.
COM-21	Source code	Should	Fully Compliant	Commercial	Source code of IIFT-specific components delivered under every option (DEL-11); core platform source via Option C (B\$88,000 with 8-week knowledge transfer) or escrow.
COM-22	Data ownership	Must	Fully Compliant	Commercial	All business data remains the property of IITH/IIGT/IIFT.
COM-23	Vendor liability	Must	Fully Compliant	Commercial	Responsibilities, liability and service credits defined in Terms and Conditions.
COM-24	Vendor BCP	Must	Fully Compliant	Commercial	iorta business continuity arrangements: code repositories, backup staff, remote support.
RFP 7 – Project deliverables					
DEL-01	Project charter	Must	Fully Compliant	Delivered during implementation	Week 2: objectives, scope, governance, stakeholders, assumptions, milestones.
DEL-02	Project plan	Must	Fully Compliant	Delivered during implementation	Week 2: activities, dependencies, resources, timeline; updated fortnightly.
DEL-03	Business requirement specification	Must	Fully Compliant	Delivered during implementation	Week 4: agreed business requirements and processes (as-is / to-be).

ID	Requirement	Priority	Compliance	Fitment	How addressed
DEL-04	Functional requirement specification	Must	Fully Compliant	Delivered during implementation	Week 5: detailed functional behaviour, rules, screens and reports.
DEL-05	Requirements traceability matrix	Must	Fully Compliant	Delivered during implementation	Week 6, maintained to closure: requirement → design → build → test → acceptance.
DEL-06	Solution architecture	Must	Fully Compliant	Delivered during implementation	Week 6: application, infrastructure, security, integration and deployment architecture.
DEL-07	Technical design	Must	Fully Compliant	Delivered during implementation	Week 6: components, database, APIs, interfaces and configuration.
DEL-08	UI/UX design	Must	Fully Compliant	Delivered during implementation	Week 5: clickable screen designs, navigation and user journeys.
DEL-09	Interface specification	Must	Fully Compliant	Delivered during implementation	Week 6: APIs, data mappings, methods, security and error handling per interface.
DEL-10	Configured solution	Must	Fully Compliant	Delivered during implementation	Week 16: complete portal and back-office, demonstrated sprint by sprint.
DEL-11	Source code	Must	Fully Compliant	Delivered during implementation	Week 24 and every release, all options: source code of IIFT-specific components (configuration, adapters, reports, templates), build scripts and migrations. Core platform via Option C or escrow.
DEL-12	Test strategy	Must	Fully Compliant	Delivered during implementation	Week 6: levels, responsibilities, environments, entry/exit and acceptance criteria.
DEL-13	SIT test cases	Must	Fully Compliant	Delivered during implementation	Week 14: scenarios and expected results traced to requirements.
DEL-14	SIT results	Must	Fully Compliant	Delivered during implementation	Week 19: execution evidence, defect log and closure.
DEL-15	UAT test cases	Must	Fully Compliant	Delivered during implementation	Week 18: business scenarios prepared with IIFT users.
DEL-16	UAT results	Must	Fully Compliant	Delivered during implementation	Week 22: execution evidence and business sign-off.
DEL-17	Security assessment	Must	Fully Compliant	Delivered during implementation	Week 19: VAPT report, remediation evidence and re-test results.
DEL-18	Performance test report	Must	Fully Compliant	Delivered during implementation	Week 19: load, stress and soak results against NFR targets.
DEL-19	Data migration plan	Must	Fully Compliant	Delivered during implementation	Week 8: extraction, transformation, validation, migration and reconciliation.
DEL-20	Deployment plan	Must	Fully Compliant	Delivered during implementation	Week 21: deployment sequence, responsibilities, rollback and contingency.
DEL-21	Go-live checklist	Must	Fully Compliant	Delivered during implementation	Week 23: technical, operational, security, business and support readiness.
DEL-22	User training	Must	Fully Compliant	Delivered during implementation	Weeks 21–23: agents, banca officers, back-office users and stakeholders.
DEL-23	Administrator training	Must	Fully Compliant	Delivered during implementation	Weeks 22–23: system administrators and support personnel.

ID	Requirement	Priority	Compliance	Fitment	How addressed
DEL-24	User manual	Must	Fully Compliant	Delivered during implementation	Week 21: portal and back-office user guides with screenshots.
DEL-25	Administrator manual	Must	Fully Compliant	Delivered during implementation	Week 21: configuration, user administration and operational support.
DEL-26	Technical manual	Must	Fully Compliant	Delivered during implementation	Week 23: architecture, database, API, integration and deployment.
DEL-27	Operations manual	Must	Fully Compliant	Delivered during implementation	Week 23: monitoring, backup, incident handling and routine operations.
DEL-28	Production deployment	Must	Fully Compliant	Delivered during implementation	Week 24: production deployment per approved deployment plan.
DEL-29	Go-live support	Must	Fully Compliant	Delivered during implementation	Weeks 25–28: hypercare with daily checks and priority defect fixing.
DEL-30	Project closure report	Must	Fully Compliant	Delivered during implementation	Week 28: final status, outstanding items, lessons learned, formal acceptance.
RFP 8 – Five-year maintenance & support requirements					
MNT-01	Helpdesk support	Must	Fully Compliant	Delivered during implementation	Service desk via portal issue module, e-mail and phone for five years.
MNT-02	Incident management	Must	Fully Compliant	Delivered during implementation	Record, classify, investigate, resolve and close incidents per ITIL-aligned procedure.
MNT-03	Problem management	Must	Fully Compliant	Delivered during implementation	Root-cause analysis for P1/P2 and recurring incidents with corrective actions.
MNT-04	Service request	Must	Fully Compliant	Delivered during implementation	User support, configuration and operational assistance requests handled.
MNT-05	Critical incident SLA	Must	Fully Compliant	Delivered during implementation	P1: 30-min response, 4-h restoration, 24x7, immediate escalation.
MNT-06	High-priority SLA	Must	Fully Compliant	Delivered during implementation	P2: 2-h response, resolution within 1 business day.
MNT-07	Medium/Low/Cosmetic SLA	Must	Fully Compliant	Delivered during implementation	P3: 4 business hours / 3 business days; P4: 1 business day / next release.
MNT-08	Bug fixing	Must	Fully Compliant	Delivered during implementation	Defects fixed under warranty (6 months) and maintenance thereafter.
MNT-09	Corrective maintenance	Must	Fully Compliant	Delivered during implementation	Corrective changes to maintain functionality included.
MNT-10	Preventive maintenance	Should	Fully Compliant	Delivered during implementation	Monthly health checks, log and capacity review, proactive fixes.
MNT-11	Security patches	Must	Fully Compliant	Delivered during implementation	Application, framework and dependency patches; monthly cycle, emergency as needed.
MNT-12	Vulnerability remediation	Must	Fully Compliant	Delivered during implementation	Critical ≤ 7 days, High ≤ 14 days, Medium ≤ 30 days, Low ≤ 90 days.
MNT-13	Certificate management	Must	Fully Compliant	Delivered during implementation	Certificate inventory, expiry alerts and renewal support.

ID	Requirement	Priority	Compliance	Fitment	How addressed
MNT-14	OS/platform compatibility	Must	Fully Compliant	Delivered during implementation	Compatibility with supported OS, container runtime, Node.js and PostgreSQL versions.
MNT-15	Browser compatibility	Must	Fully Compliant	Delivered during implementation	Compatibility maintained with supported browser versions.
MNT-16	Database maintenance	Must	Fully Compliant	Delivered during implementation	Health checks, vacuum/index optimisation, query tuning, upgrade support.
MNT-17	Performance monitoring	Should	Fully Compliant	Delivered during implementation	Monthly performance review and optimisation recommendations.
MNT-18	System monitoring	Must	Fully Compliant	Delivered during implementation	Availability and critical component monitoring with alerting.
MNT-19	Backup verification	Must	Fully Compliant	Delivered during implementation	Quarterly restore test and backup job verification.
MNT-20	Disaster recovery	Must	Fully Compliant	Delivered during implementation	Participation in DR planning and annual DR test.
MNT-21	Release management	Must	Fully Compliant	Delivered during implementation	Releases follow agreed testing, CAB approval and deployment procedures.
MNT-22	Version upgrade	Must	Fully Compliant	Delivered during implementation	Application and platform upgrades during the maintenance period included.
MNT-23	Minor enhancement	Should	Fully Compliant	Delivered during implementation	60 hours per year of minor enhancements included; beyond that per rate card.
MNT-24	Change request	Must	Fully Compliant	Delivered during implementation	Formal CR: impact assessment, quotation, approval, delivery.
MNT-25	Monthly service report	Must	Fully Compliant	Delivered during implementation	Incidents, SLA performance, availability, changes and outstanding issues.
MNT-26	Quarterly review	Should	Fully Compliant	Delivered during implementation	Quarterly service review with improvement plan.
MNT-27	Documentation updates	Must	Fully Compliant	Delivered during implementation	System, technical and operational documents updated after material changes.
MNT-28	Knowledge transfer	Must	Fully Compliant	Delivered during implementation	Ongoing knowledge transfer sessions to IIFT/IITH support staff.
MNT-29	Onsite support	Should	Fully Compliant	Delivered during implementation	Onsite support for major incidents or upgrades at the rate card plus OPE; planned onsite phases per the OPE schedule.
MNT-30	Transition support	Must	Fully Compliant	Delivered during implementation	Year 5 knowledge transfer, documentation and technical assistance for transition.

Table 92: Requirements compliance and fitment matrix

Annex B – Curricula Vitae of Key Personnel

Key personnel from Section 19 are summarised below; full CVs are attached. **[Complete the table and attach one CV per person.]**

Proposed role	Name	Qualifications	Years of experience (insurance/Takafu I)	Relevant projects (project, client, role, year)
Project Manager	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]
Solution Architect	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]
Business Analyst (Takafu)	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]
Technical Lead	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]
QA Lead	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]
DevOps & Security Engineer	[Full name]	[Degrees, certifications]	[Years]	[3 to 5 projects]

Table 93: Key personnel summary

Annex C – Bill of Materials, Software Components and Network Flows

This annex summarises the Bill of Materials workbook (IIFT-Bill-of-Materials-iorta-TechNXT.xlsx). Indicative costs are for budgeting; items marked IIFT procures are bought by IIFT and are not part of iorta's fees.

On-premise infrastructure (Option A, IIFT procures)

Server / item	Qty	vCPU	RAM	Storage	Indicative cost (B\$)
Production					
WAF / load balancer	2 (or existing)	2	4 GB	–	B\$ 0–2,000
Application VM	2	4	8 GB	100 GB	B\$ 3,000–6,000
Database VM	2	4	16 GB	200 GB SSD	B\$ 5,000–9,000
Document store	1 share	–	–	500 GB	B\$ 1,000–2,000
Backup repository	1	–	–	1 TB	B\$ 1,500–3,000
Monitoring	1 (or existing)	2	4 GB	100 GB	B\$ 0–1,500
Non-production					
SIT	1	4	16 GB	200 GB	B\$ 2,000–3,500
UAT	1	4	16 GB	200 GB	B\$ 2,000–3,500
DEV	Not required	–	–	–	B\$ 0
DR site					
Application VM	1	4	8 GB	100 GB	B\$ 1,500–3,000
Database VM	1	4	16 GB	200 GB SSD	B\$ 2,500–4,500
Document and backup copies	–	–	–	500 GB + 1 TB	B\$ 2,000–4,000
Total, indicative one-time					B\$ 20,500–42,000

Table 94: Option A infrastructure, sized as in the Solution Architecture

Operating system: Ubuntu Server 24.04 LTS (no licence fee) or RHEL 9 (B\$ 4,500–10,800 a year for 9 VMs). Virtualisation, network, firewall, backup platform and SIEM are existing IITH services. Costs fall where existing IITH capacity is used.

Software components

Component	Version	Role	Licence	Fee
Node.js	22 LTS	API runtime (inside the container image)	MIT	B\$ 0
NestJS	12	API framework	MIT	B\$ 0
Prisma ORM and PostgreSQL driver adapter	7	Data access and migrations	Apache-2.0	B\$ 0
React and React DOM	19	Portal and back-office user interface	MIT	B\$ 0
Ant Design and Ant Design Icons	6	UI component library	MIT	B\$ 0
TanStack Query, React Router, Recharts, Day.js	5 / 7 / 3 / 1	UI data, routing, charts, dates	MIT	B\$ 0
pdfkit, exceljs	0.20 / 4.4	PDF documents and Excel reports	MIT	B\$ 0
Argon2 (@node-rs/argon2), helmet, express-session	2 / 8 / 1	Password hashing, security headers, sessions	MIT	B\$ 0
pino, prom-client	10 / 15	Structured logs and metrics	MIT / Apache-2.0	B\$ 0
Idapts, nodemailer	9 / 10	Active Directory and e-mail integration	MIT / MIT-0	B\$ 0
PostgreSQL	16	Database	PostgreSQL Licence	B\$ 0
pgBackRest	2	Database backup and point-in-time recovery	MIT	B\$ 0
nginx (unprivileged image)	1.29	Web server for the user interface	BSD-2-Clause	B\$ 0
ClamAV	stable	Malware scan of uploaded documents (separate container)	GPL-2.0	B\$ 0
Docker Engine, containerd	27+ / 1.7+	Container runtime	Apache-2.0	B\$ 0

Component	Version	Role	Licence	Fee
Linux operating system	Ubuntu Server 24.04 LTS or RHEL 9	Server OS	Ubuntu: free; RHEL: subscription	B\$ 0 / RHEL optional
Prometheus, Grafana, Loki	current	Monitoring, dashboards, log aggregation	Apache-2.0 / AGPL-3.0	B\$ 0

Table 95: Runtime software components

Third-party services and optional items

Item	Option A	Option B	Indicative cost (B\$)
SSL/TLS certificates	IIFT procures	Pass-through at cost	300–1,000 a year
SMS gateway	IIFT procures	IIFT procures	300–600 a year
AML screening data / service	IIFT procures	IIFT procures	0–3,000 a year
E-mail relay	IIFT procures	IIFT procures	0–120 a year
Commercial PostgreSQL support (EDB or similar) (optional)	IIFT procures	Not required	5,000–10,000 a year
Red Hat Enterprise Linux subscriptions (optional)	IIFT procures	Not required	4,500–10,800 a year
Source code escrow agent (optional)	Pass-through at cost	Pass-through at cost	2,500–4,000 a year
Independent VAPT and re-test	Included in iorta fee	Included in iorta fee	Included
Backup software	IIFT procures	Pass-through at cost	–
SIEM / security monitoring	IIFT procures	Pass-through at cost	–
Virtualisation, network, firewall, DNS	IIFT procures	IIFT procures	–
Source code repository and CI/CD	Included in iorta fee	Included in iorta fee	Included
Monitoring stack (Prometheus, Grafana, Loki)	Included in iorta fee	Included in iorta fee	Included
PKI digital signature certificates (optional)	Not required	Not required	–
Payment gateway (optional)	Not required	Not required	–

Table 96: Third-party services and treatment

Network flows and storage growth

From	To	Port / protocol	Purpose
Internet / bank network	WAF / load balancer	443 (HTTPS)	Portal and e-sign
IIFT LAN	Internal reverse proxy	443 (HTTPS)	Back-office
Load balancer / proxy	Application VMs	8080 (HTTP, internal only) or 443	Application traffic
Application VMs	PostgreSQL	5432 (TLS)	Database
Application VMs	Active Directory	636 (LDAPS)	Back-office SSO
Application VMs	SMTP relay	587 (STARTTLS)	E-mail
Application VMs	Core / FIN / AML / SMS	443 (HTTPS) or 22 (SFTP)	Integrations
Monitoring server	Application VMs	9100 / 9464 (metrics)	Monitoring
PostgreSQL primary	Standby / DR replica	5432 (TLS)	Replication

Table 97: Network flows

Data	Year 1	Year 5	Notes
Structured data (PostgreSQL)	< 1 GB	< 5 GB	Policies, participants, payments, audit
Documents	10–20 GB	50–100 GB	Uploads and system-produced PDFs
Logs (online)	About 20 GB	About 20 GB	Rolling retention; archived to backup
Backups	About 150 GB	About 600 GB	35 days online plus monthly copies

Table 98: Storage growth estimate

Annex D – Glossary

Term	Meaning
Takaful	Islamic insurance based on mutual co-operation and shared responsibility, in which participants contribute to a common fund to help one another against defined risks.
Takaful operator	The company (here IIFT) that manages the Takaful fund on behalf of participants.
Participant	The person or entity that joins a Takaful plan and makes contributions (the Takaful equivalent of a policyholder).
Contribution	The amount paid by the participant for Takaful cover (the Takaful equivalent of a premium).
Tabarru'	The portion of the contribution donated to the participants' risk fund to help fellow participants who suffer a covered loss.
Wakalah	An agency contract under which the Takaful operator manages the fund for a fee (wakalah fee).
Participants' risk fund	The fund made up of tabarru' contributions from which claims are paid.
Nominee / beneficiary / executor	Persons designated to receive Takaful benefits or administer them on the participant's behalf.
Mortgage Takaful	Decreasing-term Takaful protecting a financing facility (hire purchase or property).
Banca / bancassurance	Distribution of Takaful products through bank partners and bank officers.
Main agent / sub-agent	Hierarchical agency relationship in which a main agent supervises sub-agents.
e-Policy / e-Receipt	Electronic policy document and receipt issued to the participant.
Grace period	Period (seven days) within which payment for an issued policy must be submitted before the agency's issuance is blocked.
Maker-checker	Control in which a transaction prepared by one user (maker) must be approved by another (checker) before it takes effect.
Quality check (QC)	Review of an application by IIFT before contract issuance.
End-of-day (EOD)	Daily batch that produces issuance reports, interfaces postings to FIN and reconciles with BRR.
Transactional outbox	Integration pattern in which outbound messages are stored in the same database transaction as the business change and delivered asynchronously, guaranteeing no loss.
Dead-letter	Holding area for messages that could not be delivered after the maximum number of retries.
Blind index	Keyed hash of an encrypted value that allows exact-match search without decrypting data.
Modular monolith	Single deployable application internally divided into independent modules with clear boundaries.
Hypercare	Period of intensified support immediately after go-live.
Point-in-time recovery	Ability to restore the database to any moment using base backups and transaction logs.
Perpetual licence	A licence with no end date, paid once; maintenance (AMC) is charged separately.
Subscription	A right to use software for a term, paid periodically, with maintenance included.
Annual maintenance charge (AMC)	Yearly fee for support, maintenance, patches and upgrades of licensed software, here 22% of the licence and customisation fees.
At actuals / pass-through	A cost re-charged at the amount the provider invoices, without mark-up.
Per diem	Fixed daily allowance for meals and local transport during onsite work.
Source code escrow	Deposit of source code with an independent agent, released to the customer only on agreed events such as the supplier's insolvency.